MIL-HDBK-600
1 SEP 1976

# MILITARY STANDARDIZATION HANDBOOK

## GUIDELINES FOR IDENTIFICATION, MARKING, LABELING, STORAGE, AND TRANSPORTATION OF RADIOACTIVE COMMODITIES



GP 59

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C.

MIL-HDBK-600
Guidelines For Identification, Marking, Labeling, Storage, and Transportation of Radioactive Commodities

1. This standardization handbook was developed by the Department of Defense with the assistance of the military departments, federal agencies, and industry.

2. This publication was approved on 1 September 1976 for printing and inclusion in the military standardization handbook series.

3. This document supplements departmental manuals, directives, military standards, etc., and provides basic and fundamental information on radioactive commodities. It contains listings of radioactive commodities, conversion tables and charts, and should provide valuable information and guidance to personnel responsible for the packaging, handling, labeling, and transportation of radioactive commodities.

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: Director, Directorate of Engineering Standardization, DESC-E, Dayton, OH 45444, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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# FOREWORD

The use of radioactive materials in the manufacture of certain commodities has introduced a potential hazard within the Department of Defense supply system. Therefore, all personnel should be constantly aware of the potential dangers and of the need for observing extreme caution while handling these commodities. It is important, therefore, that each radioactive commodity be properly packaged, identified, and labeled.

The current status of the private segment of the nuclear industry in the U.S.A. is such that the vast majority of the current shipments of radioactive materials involves small or intermediate quantities of material in relatively small packages. The external radiological hazard resulting from small radioactive commodities is usually insignificant. However, if the commodity is broken or damaged, the hazard is increased by release of the enclosed radioactive material An internal radiation hazard exists when radioactive materials enter the body through eating, drinking, inhalation, or an open wound. Any of these media for internal radiation are made possible when a radioactive commodity is broken.

Since the beginning of the relatively young atomic energy industry, i.e., about three decades, there has been an excellent record of safety in the transportation of millions of packages of radioactive materials. Recent estimates indicate that current shipments involve approximately 800,000 packages of radioactive materials per year in the U.S.A. Thus far, based on best available information, there have been no known deaths or serious injuries to the public or to the transportation industry personnel as a result of the radioactive nature of any radioactive material shipment. This fact can generally be attributed to the close attention which has been given by the shippers to the proper packaging of radioactive materials, and to the effectiveness of the safety standards and regulations applicable to their transportation.

In spite of the excellent past record of safety, the term radioactive unfortunately conveys to the average person an extremely confusing aura of personal concern. The memory of the devastating destruction and violent deaths which were the aftermath of the use of the atomic bomb near the end of World War II is still very vivid in the minds of many. The more recent confusion regarding the effects of nuclear power plants on the environment and the ecology have also added to this aura of concern. This concern, however, has proven to be completely unwarranted as related not only to the safe uses of nuclear energy in industry, but also to the safe transportation of radioactive materials. The record of safety in transportation of these materials clearly exceeds that for any other type of regulated hazardous commodity.



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## 1. SCOPE

This handbook provides general information for shippers, carriers, and handlers of radioactive commodities for use and storage, and for shipment by any mode of transportation, to insure that such materials are properly packaged, packed, marked, and labeled, when offered for storage or shipment. These instructions, in general, apply to radioactive commodities usually encountered in GP 59.

## 2. PURPOSE

This handbook does not replace official Federal regulations (CFRs), laws, or military regulation. It implements and provides guidance in determining the requirements of the pertinent regulations, military standards, specifications and manuals which provide the marking and labeling requirements for identification in use, storage and transportation of radioactive commodities, their packages and shipping containers. It is intended to provide, in simple, concise and convenient form, factual data to familiarize personnel and agencies concerned with operational and logistical problems involving radioactive commodities.

2.1 Application. The military departments are presently utilizing radioisotopes in many applications, including those within and without the jurisdiction of the Nuclear Regulatory Commission. Many commodities employ such radioisotopes as Cobalt 60, Cesium 137, Strontium 90, etc., and include electron tubes, spark gaps, switches, light sources, luminous markers, and nuclear batteries.

2.2 Precautions. The amount of radioactive material in many commodities is such that no significant external radiation hazard is present when handled individually or in small numbers. However, extremely large quantities of radioactive commodities may present an external hazard and breakage of one or more will definitely present a potential internal hazard to personnel working in and around the area in which the breakage occurs. Therefore, needless exposure to any radiation, no matter how slight, should be avoided.

Materials that emit alpha and beta particles and gamma rays may be absorbed and deposited within the body. These particles and rays act as damaging agents that injure or destroy blood-forming organs and other tissue. Their removal is limited by the rate of excretion of the elements from the body and the natural radioactive decay of the element. In severe cases, clinical evidence of injury may be apparent in a few weeks; in less severe cases, evidence may not appear for years. Internal radiation effects as a result of broken radioactive tubes are possible in the latter category. It is, therefore, necessary that the instructions, procedures, and precautions contained herein be observed if the hazards involved are to be reduced to a minimum.

## 3. INCLUSIONS

This handbook provides tables that list the known commodities containing radioactive materials, including the isotopes, the quantity in microcuries, and the manufacturers of these commodities. This handbook has summarized in the tables the marking and labeling (where required) requirements for radioactive commodities for transportation by:

| | |
|---|---|
| a. Rail | (Title 49 CFR 174) |
| b. Highway carrier | (Title 49 CFR 177) |
| c. Water carrier | (Title 49 CFR 176) |
| d. Mail | (Title 39 CFR 124) |
| e. Aircraft | (Title 49 CFR 175) |
| f. Aircraft (military) | (DSAM 4145.3, AFM 71-4, TM38-250, NAVSUPPUB 505, and MCO P 4030.19) |



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## 4. ARRANGEMENT

4.1 Composition. This handbook presents general information and technical information in sections pertaining to the technical and operational characteristics of radioactive commodities. The sections are:

| | |
|---|---|
| Section A | Radioisotopes By Transport Group |
| Section B | Activity Mass Relationship - Specific Activity Microcuries Per Gram Conversion Factors |
| Section C and D | Provide listing by part number sequence and NSN sequence of commodities known to contain radioactive material. Some commodities are listed for reference only since the specific activity is less than 0.002 microcurie per gram. This data, including isotopes and quantities, has been provided voluntarily by the manufacturers listed. These commodities have been classified in accordance with the data included in this table and the marking and labeling requirements indexed. |

4.2 Appendix. The appendix provides general information and procedures for radiation monitoring, personnel protection, decontamination, and disposition of radiological commodities. DoD personnel should be in a position to determine if shippers and carriers have conformed to the requirements specified in this handbook and other applicable directives.

## 5. REFERENCED DOCUMENTS

5.1 The following documents form a part of this handbook to the extent specified herein.

STANDARDS AND SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-STD-1458 | Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use, Storage and Transportation |
| MIL-E-75 | Electron Tubes, Packaging of, General Specification for |
| MIL-E-1 | Electron Tubes, General Specification for |
| MIL-HDBK-758 | Special Handling Data/Certification for shipment via Military Aircraft |

JOINT MILITARY

DSAM 4145.3, AFM 71-4, TM 38-250, NAVSUPPUB 505, MCO P 4030.19
Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft

DSAM 4145.8, AR 700-64, NAVSUPPUB 5012 AFM 67-8, MCO P 4400.105
Radioactive Commodities in the DoD Supply System

DSAR 4145.24, AR 40-14 Control and Recording Procedures for Occupational Exposure to Ionizing Radiation

CODE OF FEDERAL REGULATIONS (See 7.2 and table I)



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## 6. DEFINITIONS

For purposes of this handbook, the following definitions and abbreviations shall apply:

6.1 Activity. The number of nuclear transformations (disintegrations) occurring in a given mass of radioactive material per unit time. (See Curie.)

6.2 Alpha particle. One of the three primary forms of radioactive emissions from radioactive atoms. A charged particle emitted from the nucleus of an atom having a mass and charge equal in magnitude of a helium nucleus (2 protons + 2 neutrons). Alpha particles have very little penetrating ability and therefore are chiefly internal radiation hazards. They travel very short distances in air and are shielded very easily.

6.3 Atom. The smallest (or ultimate) particle of an element which is capable of entering into a chemical reaction. Every atom consists of a positively charged central nucleus, which carries nearly all the mass of the atom, surrounded by a number of negatively charged electrons, so that the whole system is electrically neutral (see nucleus).

6.4 Atomic mass. The mass of a neutral atom of a nuclide, usually expressed in terms of "atomic mass units." The "atomic mass unit" is one-twelfth the mass of one neutral atom of carbon-12; equivalent to $1.6604 \times 10^{-24}$ gm. (Symbol: u).

6.5 Atomic number. The number of protons in the nucleus of a neutral atom of a nuclide. The "effective atomic number" is calculated from the composition and atomic numbers of a compound or mixture. An element of this atomic number would interact with photons in the same way as the compound or mixture. (Symbol: Z).

6.5.1 Atomic weight. The weighted mean of the masses of the neutral atoms of an element expressed in atomic mass units.

6.6 Beta particles. One of three primary forms of radioactive emissions from radioactive atoms. Beta particles are negatively charged particles emitted from the nucleus of an atom and have a mass and charge equal to that of an electron. They travel greater distances in air than alpha particles, have an intermediate penetrating ability, but still are relatively easily shielded.

6.7 Byproduct material. Any radioactive material (except special nuclear material) yielded in or made radioactive by exposure to the radiation incident to the process of producing or utilizing special nuclear material (10 CFR 20).

6.8 CFR. The approved citation of the Code of Federal Regulations. Example: 49 CFR 389 refers to Title 49 Section 389 of the Code of Federal Regulations.

6.9 Curie. The special unit of activity. One curie equals $3.700 \times 10^{10}$ nuclear transformations per second. (Abbreviated Ci.) Several fractions of the curie are in common usage.

Microcurie: One-millionth of a curie ($3.7 \times 10^{4}$ disintegrations per sec.). Abbreviated μCi.

Millicurie: One-thousandth of a curie ($3.7 \times 10^{7}$ disintegrations per second). Abbreviated mCi.

Picocurie: One-millionth of a microcurie ($3.7 \times 10^{-2}$ disintegrations per second or 2.22 disintegrations per minute). Abbreviated pCi; replaces the term μμc.



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6.10 Decay, radioactive. Disintegration of the nucleus of an unstable nuclide, due to the spontaneous emission from the atomic nuclei of either alpha or beta particles, sometimes accompanied by gamma radiation. (See half-life, radioactivity.)

6.11 Dose (10 CFR 20). The quantity of radiation absorbed, per unit of mass, by the body or any portion of the body. Several units of dose are in current use. Definitions of units used in this handbook are specified in 6.11.1 and 6.11.2. (Also, see exposure.)

6.11.1 Rad. A measure of the dose of any ionizing radiation to body tissues in terms of the energy absorbed per unit mass of the tissue. One rad is the dose corresponding to the absorption of 100 ergs per gram of tissue (10 CFR 20). Also defined as the unit of absorbed dose equal to 0.01 Joules/kg in any medium.

6.11.2 Rem. A measure of the dose of any ionizing radiation to body tissues in terms of its estimated biological effect, relative to a dose of one roentgen (R) of X-rays. (One millirem (mrem) = 0.001 rem.) The relation of the rem to other dose units depends upon the biological effect under consideration and upon the conditions of irradiation. For the purpose of this handbook, any of the following is considered to be equivalent to a dose of one rem (10 CFR 20):

a. A dose of 1 R due to X - or gamma radiation;
b. A dose of 1 rad due to X -, gamma, or beta radiation;
c. A dose of 0.1 rad due to neutrons or high energy protons;
d. A dose of 0.05 rad due to particles heavier than protons and with sufficient energy to reach the lens of the eye.

6.12 Dose rate. As a general rule, the amount of radiation to which an individual would be exposed or which would be absorbed per unit of time. It is usually expressed as roentgens, rads, or rems per hour or in multiples or submultiples of these units, such as a milliroentgens per hour. The dose rate is commonly used to indicate the level of radioactivity in a contaminated area.

6.13 DOT. The Department of Transportation.

6.14 Electron tubes. Electron tubes include spark gap tubes, power tubes, gas tubes, pickup tubes, radiation detection tubes, and any other completely sealed tube that is designed to conduct or control electrical currents (see 10 CFR 30).

6.15 Encapsulation. A term used to denote an additional fabrication technique often used in preparation of radiation sources, wherein the basic material is physically placed within sealed, high physical integrity capsules or envelopes to provide further assurance that in the event a package breaks and the capsule escapes, there would be little possiblity of a spread of particulate contamination (49 CFR 173).

6.16 Exclusive use. Exclusive use (also referred to as sole use or full load as used in IAEA regulations) means any shipment (49 CFR 173):

a. From a single consignor having the exclusive use of a transport vehicle or of an aircraft, or of a hold or compartment of an inland watercraft, or of a hold, compartment, or defined deck area of a seagoing vessel; and
b. For which all initial, intermediate, and final loading and unloading is carried out by or under the direction of the consignor, consignee, or his designated agent.

6.17 Exposure: A measure of the ionization produced in air by X or gamma radiation. It is the sum of the electrical charges on all ions of one sign produced in air when all electrons liberated by photons in a volume element of air are completely stopped in air, divided by the mass of the air in the volume element. The special unit of exposure is the roentgen.



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6.18 Fissile. Of a nuclide, capable of undergoing fission by interaction with slow neutrons.

6.18.1 Fission, nuclear. A nuclear transformation characterized by the splitting of a nucleus into at least two other nuclei and the release of a relatively large amount of energy.

6.18.2 Fission products. Elements or compounds resulting from fission.

6.18.3 Fission yield. The percentage of fissions leading to a particular nuclide.

6.18.4 Fissionable. Of a nuclide, capable of undergoing fission by any process.

6.19 Fissile radioactive material. This term refers to plutonium-238, plutonium-239, plutonium-241, uranium-233, or uranium-235, or any material containing any of these materials (see 11.3.7.1 for exclusions), (see 11.3.9):

a. Uranium-235. Exists only in combination with various percentages of uranium-234 and uranium-238. Fissile radioactive material as applied to uranium-235 refers to the amount actually contained in the total quantity of uranium being transported.
b. Radioactive material. May consist of a mixture of fissile and nonfissile radionuclides.
   (1) Fissile radioactive materials refers to the amount of materials (see 2 below) or any combination thereof actually contained in the mixture.
   (2) Radioactivity of the mixture consists of the total activity of both the fissile and nonfissile radionuclides.

6.19.1 Fissile radioactive material classes. Fissile radioactive material package are classified according to the controls needed to provide nuclear criticality safety during transportation as follows (see 11.3.9):

a. Fissile Class I. Packages which may be transported in unlimited numbers and in any arrangement, and which require no nuclear criticality safety controls during transportation. For purposes of nuclear criticality safety control, a transport index is not assigned to Fissile Class I packages. However, the external radiation levels may require a transport index number.
b. Fissile Class II. Packages which may be transported together in any arrangement but in numbers which do not exceed an aggregate transport index of 50. For purposes of nuclear criticality safety control, individual packages may have a transport index of not less than 0.1 and not more than 10. However, the external radiation levels may require a higher transport index number but not to exceed 10. Such shipments require no nuclear ctiticality safety control by the shipper during transportation.
c. Fissile Class III. Shipments of packages which do not meet the requirements of Fissile Class I or II and which are controlled to provide nuclear criticality safety in transportation by special arrangements between the shipper and the carrier.

6.20 Gamma rays. One of the three primary forms of radioactive emissions from radioactive atoms. Gamma rays are not particulate (as opposed to alpha and beta particles) but are short wave length electromagnetic radiations from the nucleus of radioactive atoms. Except for their origin (the nucleus of the atom rather than the outer shell) they are identical in characteristics to X-rays. Gamma rays are the most penetrating form of radiation and travel great distances in air. They require heavy shielding materials such as lead to attenuate the radiation.



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6.21 Half-life. The time required for the activity of a given radioactive species to decrease to half of its initial value due to radioactive decay. The half-life is a characteristic property of each radioactive species and is independent of its amount or condition. The biological half-life of a given isotope is the time in which the quantity in the body will decrease to half as a result of both radioactive decay and biological elimination.

6.22 Isotopes. Forms of the same element having identical chemical properties but differing in their atomic masses (due to different numbers of neutrons in their respective nuclei) and in their nuclear properties, e.g., radioactive, fission, etc.

6.23 Large quantity radioactive materials. A quantity of radioactive materials in which the aggregate radioactivity exceeds: (See 11.3.8.3 and 11.3.7.4.)

a. Transport Groups I and II radionuclides: 20 curies.
b. Transport Groups III and IV radionuclides: 200 curies.
c. Transport Group V radionuclides: 5,000 curies.
d. Transport Groups VI and VII radionuclides: 50,000 curies.
e. Special form materials: 5,000 curies.

6.24 Licensed material (10 CFR 20). Source material, special nuclear material, or byproduce material received, possessed, used, or transferred under a general or specific license issued by the NRC.

6.25 License. A license issued under 10 CFR 30, 40, or 70. "Licensee" means the holder of such a license.

6.26 Low specific activity radioactive material. Radioactive material in which the activity is essentially uniformly distributed. This means any of the following: (See 11.3.8.)

a. Uranium or thorium ores and physical or chemical concentration of those ores.
b. Unirradiated natural or depleted uranium or unirradiated natural thorium.
c. Tritium oxide in aqueous solutions, provided the concentration does not exceed 5.0 millicuries per milliliter.
d. Radioactive material in which the activity is uniformly distributed in which the estimated average concentration per gram does not exceed:
   (1) 0.0001 millicuries of Transport Group I radionuclides.
   (2) 0.005 millicuries of Transport Group II radionuclides.
   (3) 0.3 millicuries of Transport Groups III and IV radionuclides.
e. Items externally contaminated with radioactive material provided such radioactive material is not readily dispersible and the surface contamination (when averaged over an area of one square meter) does not exceed:
   (1) 0.0001 millicurie (220,000 disintegrations per minute) per square centimeter of Transport Group I radionuclides.
   (2) 0.001 millicurie (2,200,000 disintegrations per minute) per square centimeter of other radionuclides.

6.27 Mass. A measure of the quantity of matter. The material equivalent of energy. Mass and energy are different forms of the same thing. Different from weight in that it neither increases nor decreases with gravitational force.

6.28 Microcurie (μCi). A one-millionth ($10^{-6}$) part of a curie. (See curie.)

6.29 Millirem, milliroentgen (mR). A one-thousandth ($10^{-3}$) part of a rem or roentgen (see dose and roentgen).



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6.30 Normal form radioactive materials. Those materials which do not meet the requirements of special form radioactive materials. Normal form radioactive materials are grouped into transport groups (see section A), (49 CFR 389).

6.31 NRC. Nuclear Regulatory Commission (formerly: AEC. Atomic Energy Commission).

6.32 Nucleus (or atomic nucleus). The small, central, positively charged region of an atom which carries essentially all the mass. Except for the nucleus of ordinary (light) hydrogen, which is a single proton, all atomic nuclei contain both protons and neutrons. The number of protons determines the total positive charge, or atomic number; this is the same for all the atomic nuclei of a given chemical element. The total number of neutrons and protons, called the mass number, is closely related to the mass of the atom. The nuclei of isotopes of a given element contain the same number of protons, but different numbers of neutrons. They thus have the same atomic number, and so are the same element, but they have different mass numbers (and masses). The nuclear properties, e.g., radioactivity, fission, neutron capture, etc., of an isotope of a given element are determined by both the number of neutrons and the number of protons (see atom, isotope).

6.33 Nuclide: A species of atom characterized by the constitution of its nucleus. The nuclear constitution is specified by the number of protons (Z), number of neutrons (N), and energy content; or, alternatively, by the atomic number (Z), mass number $A = (N + Z)$, and atomic mass. To be regarded as a distinct nuclide, the atom must be capable of existing for a measurable time. Thus, nuclear isomers are separate nuclides, whereas promptly decaying excited nuclear states and unstable intermediates in nuclear reactions are not so considered.

6.34 NSN. National Stock Number. (Formerly FSN. Federal Stock Number.)

6.35 Person (10 CFR 20, 30). Any individual, corporation, partnership, firm, association, trust, estate, public or private institution, group, Government agency other than the NRC, any State, any foreign government or nation or any political subdivision of any such government or nation, or other entity; and any legal successor, representative, agent, or agency of the foregoing.

6.36 Radiation. The emission and propagation of energy through space or through a medium in the form of waves. This includes the following: Alpha rays, beta rays, gamma rays, X-rays, neutrons, high-speed electrons, high-speed infrared, or ultraviolet light.

6.37 Radioactivity. The spontaneous disintegration of unstable nuclei with the resulting emission of nuclear radiation (see curie for units of measurements).

a. Artificial radioactivity. Man made radioactivity produced by particle bombardment or electromagnetic irradiation, as opposed to natural radioactivity.
b. Induced radioactivity. Radioactivity produced in a substance after bombardment with neutrons or other particles. The resulting activity is "natural radioactivity" if formed by nuclear reactions occurring in nature, and "artificial radioactivity" if the reactions are caused by man.
c. Natural radioactivity. The property of radioactivity exhibited by more then fifty naturally occurring radioisotopes.



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6.38 Radioactive material. Radioactive material is any material, or combination of materials, which spontaneously emits ionizing radiation. For the purposes of transportation, materials in which the estimated specific activity is not greater than 0.002 microcuries per gram of material, and in which the radioactivity (see specific activity) is essentially uniformly distributed, are not considered to be radioactive materials (see 49 CFR 173.389).

NOTE: Radioactive materials, as refrenced above, include natural elements such as radium and accelerator-produced radioisotopes whether or not subject to licensing control by the NRC.

6.39 Radioactive (commodity) device. Radioactive (commodity) device (see 11.3.10.2b) means any manufactured article such as an instrument, clock, electronic tube or apparatus, or similar device having radioactive material (other than liquid) in a nondispersible form as a component part.

6.40 Rad. rem. See dose and exposure.

6.41 Radioisotope and radionuclide. For the purpose of transportation, these items are synonymous with radioactive materials.

6.42 Radiotoxicity. A term used to denote the relative hazards of the various radionuclides, that is, their internal radioactive effect within the body.

6.43 Roentgen (R). A unit of radiation exposure. One roentgen is the exposure of X- or gamma radiation such that the associated corpuscular emission per kilogram of air produces in air, ions carry $2.58 \times 10^{-4}$ coulombs of electrical charge of either sign. (See exposure.)

6.44 Small quantity radioactive materials. A quantity of radioactivity which does not exceed the limits specified in table IX. Small quantities and certain radioactive devices as specified in 11.3.10.2 are exempt from specification packaging, marking and labeling requirements, but still are subject to certain requirements, such as shipping paper requirements.

6.45 Source material (10 CFR 20). Uranium or thorium, or any combination thereof, in any physical or chemical form; or ores which contain by weight one-twentieth of one percent (0.05%) or more of (a) uranium, (b) thorium or (c) any combination thereof. Source material does not include special nuclear material (see 10 CFR 40).

6.46 Special form radioactive materials. Radioactive materials which, if released from a package, present some radiation hazard, but little hazard due to radiotoxicity and possibility of contamination. To qualify as a special form material, the radioactive material must either be in massive solid form or encapsulated (see 11.3.7 for a discussion of radioactive material in special form). Plutonium, uranium 233, uranium enriched in the isotope 233 or in the isotope 235, and any other material which the NRC determines to be special nuclear material, but does not include source material (see 10 CFR 70).

6.47 Specific activity. Specific activity (S.A.) is the rate of decay of the radioactive element or isotope expressed in curies per gram or microcuries per gram of the radioactive element. Specific activity is a natural constant for each radioactive element whether the amount involved in the radio-active material is a microgram or 1,000 grams. For specific activity constants of particular radioisotopes. See section B, table B-1 column VI, to determine whether the isotope of interest exceeds 0.002 μCi/gm and therefore, falls within the classification of radioactive material.

6.48 Transport group. Any one of seven groups into which normal form radionuclides are classified according to their radiotoxicity and their relative potential hazard in transportation (see section A), (49 CFR 173).



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6.49 Transport index number (see 11.3.9.1). The number placed on a package to designate the degree of control to be exercised in the transportation cycle. To assign a transport index number, a package of radioactive materials shall be assigned the higher (larger) number, rounded up to the next higher tenth, when determined by one of the following methods:

a. The highest radiation dose rate in millirems per hour, at three feet from any accessible external surface of the package.
b. For Fissile Class II packages only, the transport index number is calculated by dividing the number "50" by the number of similar packages which may be transported together on a single aircraft.

6.50 Type A packaging. Packaging which is designed in accordance with the general packaging requirements of 49 CFR 173.24 and 173.393 and which is adequate to prevent the loss or dispersal of the radioactive contents and to retain the efficiency of its radiation shielding properties if the package is subjected to the test prescribed in 49 CFR 173.398 (b) (Normal conditions of transport) (see 11.3.3).

6.51 Type A quantity radioactive material. That material which may be transported in Type A packaging (see 49 CFR 173.389), (11.3.5).

6.52 Type B packaging. Packaging which meets the standards for Type A packaging, and in addition, meets the standards for the hypothetical accident conditions of transpo t as prescribed in Part 49 CFR 173.398 (c), (see 49 CFR 173.398) (see 11.3.4).

6.53 Type B quantity radioactive material. That material which may be transported in Type B packaging (see 49 CFR 173.398), (11.3.5).

6.54 Unrestricted area. Any area access to which is not controlled by the licensee for purposes of protection of individuals from exposure to radiation and radioactive materials, and any area used for residential quarters.

## 7. REGULATORY ORGANIZATIONS AND REGULATIONS

### 7.1 ORGANIZATIONS

7.1.1 Department of Transportation (DOT). Under the Department of Transportation Act of 1966, the U.S. Department of Transportation (DOT) has regulatory responsibility for safety in the transportation of radioactive materials by all modes of transport in interstate or foreign commerce (rail, road, air, water), and by all means (truck, bus, auto, ocean vessel, airplane, river barge, railcar, etc.), except postal shipments. Postal shipments come under the jurisdiction of the U.S. Postal Service, formerly known as the Post Office Department. Shipments not in interstate or foreign commerce are subject to control by a state agency in most cases.

7.1.1.1 Materials Transportation Bureau (MTB). The Materials Transportation Bureau was established on July 7, 1975, as a line organization element reporting to the Secretary of Transportation and having responsibility for those hazardous materials and pipeline safety operational functions previously addigned to the Office of the Secretary and certain new responsibilities vested in the Secretary by the Hazardous Materials Transportation Act (Title I of the Transportation Safety Act of 1974., Public Law 93-633). The MTB will be responsible for operational functions relating to regulations and exemptions; functions of an intermodal nature; and, compliance functions pertaining to packaging manufacturers and, in certain instances, shippers. The MTB will consist of the Office of Hazardous Material Operations (OHMO), and the Office of Pipeline Safety Operations.



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7.1.2 Interstate Commerce Commission (ICC). The Interstate Commerce Commission (ICC) formerly had the jurisdiction over both the safety and economic aspects of the transport of radioactive materials by land, but the jurisdiction over safety was transferred to the Department of Transportation when DOT was formed in April, 1967. The ICC (for land shipments) and the Civil Aeronautics Board (for air shipments) still exercise jurisdiction over the economic aspects of radioactive materials transport through the issuance of operation authorities to carriers and control of shipping costs (freight rates).

The Bureau of Explosives (B of E) of the Association of American Railroads was for many years the principal technical advisor to the ICC. The B of E no longer is directly involved in the development or administration of the U. S. radioactive materials safety regulatory program.

7.1.3 Nuclear Regulatory Commission (NRC). The Nuclear Regulatory Commission has responsibility for safety in the possession and use, including transport, of byproduct, source and special nuclear materials. Except for certain small quantities and specific products for which the possession and use are exempted, a license is required from the Nuclear Regulatory Commission (NRC) for possession and use of such materials. The NRC has established, in 10 CFR 71, requirements which must be met for licensees to deliver licensed material to a carrier for transport if fissile material or large radioactive sources are involved. (Rule making is pending which will revise these requirements to include Type B quantities of radioactive material as well as large quantities.) The NRC also assists and advises DOT in the establishment of national safety standards and in the review and evaluation of packaging designs.

7.1.3.1 Agreement states. Several states have entered into formal agreements with the NRC whereby the regulatory authority over byproduct, source and less than critical quantities of special nuclear material has been transferred to the states from the NRC. These "agreement states" have adopted uniform regulations pertaining to intrastate transportation of radioactive materials which require the shipper to conform to the packaging, labeling, and marking requirements of the U.S. Department of Transportation to the same extent as if the transportation were subject to the rules and regulations of that agency.

7.1.4 International Atomic Energy Agency (IAEA). The international standards, as promulgated by IAEA are included in the revised edition of "Safety Series No. 6, Regulations for the Safe Transport of Radioactive Materials," and have now been accepted and adopted either wholly or in part by most nations as a standard for both national and international regulations. The Office of Hazardous Materials Operations (OHMO), and the Nuclear Regulatory Commission (NRC) are currently developing proposed changes to the USA regulations in 49 CFR 100-199 and 10 CFR 71 in order to maintain compatibility of domestic and international standards.

7.1.4.1 Application of IAEA Transport Regulations. Concurrently with the issuance of Revised Safety Series No. 6, the IAEA has also issued a companion document entitled "Advisory Material for the Application of the IAEA Transport Regulations, Safety Series No. 37," STI/PUB/324. The intent of the IAEA Regulations is to define in general what has to be achieved in terms of regulatory performance, the new advisory document is intended to assist in determining "how" to achieve such performance. In order that this advisory document would adequately reflect the most current developments in procedures and techniques, the IAEA has stated its intention to review it, if necessary, by a panel of experts at intervals of every two to three years. The first such review was conducted by an IAEA Panel in December 1974. Based on the decisions of the Panel and the current review being carried out, the first revision of Safety Series No. 37 should be available in 1976.

7.1.5 International Air Transport Association (IATA). The regulations of IATA are essentially identical to the IAEA regulations referenced above.



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## 7.2 FEDERAL REGULATIONS

7.2.1 Principal regulations. The principal regulations pertaining to the use and transport of radioactive materials and commodities are summarized in table I. The regulations of the United States of America are published primarily by three agencies--the U.S. Department of Transportation, Nuclear Regulatory and the U.S. Postal Service. This Handbook is concerned primarily with those regulations of the U.S. Department of Transportation, as published in Title 49, Code of Federal Regulations, Parts 100 through 199. Persons involved as actual shippers, package designers, or carriers, are advised to procure a copy of these regulations (all in one bound volume) which are republished each year as of January 1, by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Changes to DOT regulations, in the form of amendments or notices of proposed rule making are published in the daily Federal Register, as issued by the Materials Transportation Bureau (MTB). Another means of keeping abreast of changes would be to request in writing to be placed on the mailing list maintained by the DOT (MTB), Office of Hazardous Materials Operations (OHMO), Washington, D.C. 20590.

7.2.2 Revoked federal regulations. DOT regulations were also published by the Federal Aviation Administration in Title 14, CFR,103, and by the U. S. Coast Guard in Title 46, CFR, 146-149. The U.S. Coast Guard regulations, although using different paragraph and section numbers, were essentially identical with those as published in Title 49, with the exception of some additional specific requirements peculiar to shipments by water. The regulations of the Federal Aviation Administration as codified in Title 14, CFR, Part 103, in addition to specifying certain air carrier requirements, had the principal effect of making reference to the Title 49 standards for the packaging requirements. The hazardous materials regulations of these agencies (14 CFR 103 and 46 CFR 146.01-146.28 and 146.30) were revoked effective July 1, 1976, and consolidated with DOT's hazardous material regulations in 49 CFR.

TABLE I. Sources of federal regulations.

| | |
|---|---|
| Title 49 | Transportation<br>(Department of Transportation's Hazardous Materials Regulations, Parts 100-189 (formerly called the ICC Regulations)).<br><br>Main Headings<br><br>49 CFR 102 - Rulemaking procedures<br>49 CFR 107 - Exemption procedures<br>49 CFR 171 - General Information, Regulations and definitions<br>49 CFR 172 - Hazardous Materials Table and Hazardous Materials Communications Regulations<br>49 CFR 173 - Shippers General Requirements for Shipment and packagings<br>49 CFR 174 - Carriage by rail<br>49 CFR 175 - Carriage by aircraft<br>49 CFR 176 - Carriage by vessel<br>49 CFR 177 - Carriage by public highway<br>49 CFR 178 - Shipping Container Specifications. |
| Title 46 | Shipping<br>46 CFR 146 - Transportation or Stowage of Explosives or Other Dangerous Articles or Substances and Combustible Liquids Aboard Vessels<br>(Revoked 1 July 1976) See 49 CFR 176 |



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TABLE I. Sources of federal regulations - Continued.

| | |
|---|---|
| Title 39 | United States Postal Service<br>39 CFR 123 - Nonmailable Matter<br><br>Publication 6 - Radioactive Matter<br>(Postal Regulations for Transport of Radioactive Matter are published in U.S. Postal Service Publication 6, April 1971 and in the U.S. Postal Manual.) |
| Title 29 | Labor<br>29 CFR 1910.96 - Occupational Safety and Health Standards (Ionizing Radiation) |
| Title 14 | Aeronautics and Space<br>14 CFR 103 - Transportation of Dangerous Articles and Magnetized Materials (Revoked 1 July 1976) See 49 CFR 175 |
| Title 10 | Energy<br>10 CFR 20 - Standards for Protection Against Radiation<br>10 CFR 30 - Rules of General Applicability to Licensing of Byproduct Material<br>10 CFR 31 - General Licenses for Byproduct Material<br>10 CFR 32 - Specific Licenses of Broad Scope for certain item containing Byproduct Material<br>10 CFR 40 - Licensing of Source Material<br>10 CFR 70 - Special Nuclear Material<br>10 CFR 71 - Packaging of Radioactive Materials for Transport and Transportation of Radioactive Material Under Certain Conditions |

7.3 Other sources of regulations and tariffs*. Other agencies or organizations which publish regulations or tariffs on the transportation of radioactive materials are as follows:

a. "IATA Restricted Articles Regulations," - International Air Transport Association (IATA). Geneva, Switzerland.

b. "Regulations for the Safe Transportation of Radioactive Materials," Revised Safety Series #6 - International Atomic Energy Agency (IAEA). Vienna, Austria.

*c. "Official Air Transport Restricted Articles Tariff" - Airline Publishers, Inc., Washington, D.C.

*d. "R.M. Graziano's Tariff," - Hazardous Materials Regulations of the Department of Transportation, Including Specifications for Shipping Containers," - Bureau of Explosives, Association of American Railroads, Washington, D.C.

*e. "Dangerous Articles Tariff," - Department of Transportation Regulations for Governing Transportation of Hazardous Materials by Motor, Rail, and Water, Including Specifications for Shipping Containers, - American Trucking Associations, Inc.

*f. "R.M. Graziano's Tariff," Department of Transportation Regulations for Governing the Transportation or Storage of Explosives or Other Dangerous Articles or Substances and Combustible Liquids on Board Vessels.

*CAUTION: A tariff should not be construed to be an official regulation per se. A tariff is merely a publication by some originating association, which reprints certain Federal regulations, showing the application and acceptance of those regulations by the carriers who participate in the tariff. As such, tariffs are binding only on the organization's or association member carriers.



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## 8. LICENSING REQUIREMENTS

8.1 The Nuclear Regulatory Commission (NRC). The NRC under the Energy Reorganization Act of 1974 has the responsibility for safety in the possession, use, and transport of byproducts, source and special nuclear materials. A license is required from the NRC for possession and use of such materials, except for certain specific products and small quantities, which are exempt from these requirements. The requirements for licensing have been established in 10 CFR (See table I). The regulations establish standards for protection against radiation hazards arising out of activities under licenses issued by the Nuclear Regulatory Commission and are issued pursuant to the Atomic Energy Act of 1954 and apply to all persons who receive, possess, use or transfer byproduct material, source material, or special nuclear material under a general or specific license issued by the Commission pursuant to the regulations in 10 CFR 30, 40, or 70. (See 7.1.3.1, Agreement States.)

8.1.1 Types of licenses and requirements. The NRC regulations provides for two types of licenses: General and specific. General licenses are effective without the filing of applications with the NRC or the issuance of licensing document to a particular person. A general license provides for the possession of certain limited quantities of radioactive materials. The generally licensed material and quantities are included in 10 CFR 30, 31, 40, and 70. Specific licenses are those for which an application must be submitted to NRC, and are issued to a named person.

8.1.1.1 Specific license and requirements. The general requirements for approval of an application are that the applicant has (1) proposed a use authorized by the NRC Regulations; (2) equipment and facilities adequate to protect health and minimize danger to life or property; and (3) the applicant is qualified by training and experiences to use the material for the proposed use in such a manner as to protect health and minimize danger to life or property; (4) The applicant satisfies any special requirements contained in 10 CFR 32-36. These are rather broad requirements and not specific for any particular type of program.

8.1.1.2 General requirements (10 CFR 30). In accordance with recommendations of the Federal Radiation Council, approved by the President, persons engaged in activities under licenses issued by the Nuclear Regulatory Commission pursuant to the Atomic Energy Act of 1954, as amended, should, in addition to complying with the requirements set forth by NRC, make every reasonable effort to maintain radiation exposures, and releases of radioactive materials in effluents to unrestricted areas as far below the limits specified in 10 CFR 20 as practicable. The term "as far below the limits specified in this part as practicable" means as low as is practicably achievable taking into account the state of technology and the economics of improvements in relation to benefits to the public health and safety and in relation to the utilization of atomic energy in the public interest.

8.1.2 Exemptions to NRC licensing requirements (10 CFR 50). The NRC may, upon application of any interested person or upon its own initiative, grant such exemptions from the requirements of 10 CFR 31-36 as it determines are authorized by law and will not endanger life or property or the common defense and security and are otherwise in the public interest (see 10 CFR 30).

8.1.2.1 Exempt commodities (item) (electron tubes). Any person is exempt from the licensing requirements of 10 CFR 20, 30-36, and 81 to the extent that such person receives, possesses, uses, transfers, owns or acquires electron tubes, and provided that each tube does not contain more than one of the specified quantities of radioactive material (except for persons who apply or incorporate byproduct material to or into, or import for sale or distribution):

a. 150 millicuries of tritium (1H3) per microwave receiver protector tube (TR, ATR, PTR, TRL, etc.) or 10 millicuries of tritium per any other electron tube (see 6.14).



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b. 1 microcurie of cobalt 60 (27 $Co^{60}$)

c. 5 microcuries of nickel 63 (28 $Ni^{63}$)

d. 30 microcuries of krypton 85 (36 $Kr^{85}$)

e. 5 microcuries of cesium 137 (55 $Cs^{137}$)

f. 30 microcuries of promethium 147 (61 $Pm^{147}$)

g. And provided further, that the levels of radiation from each electron tube containing byproduct material do not exceed 1 millirad per hour at 1 centimeter from any surface when measured through 7 milligrams per square centimeter of absorber.

8.1.2.2 Exempt carriers and personnel. Common and contract carriers, freight forwarders, warehousemen and the U. S. Postal Service are exempt from the licensing requirements of 10 CFR 30, 31-36, and 81 to the extent they transport or store radioactive (byproduct) material in the regular course of carriage for another or storage incident thereto.

8.1.2.3 Exempt concentrations. Any person is exempt from the licensing requirements of 10 CFR 31-36, and 81 to the extent that such person receives, possesses, uses, transfers, owns or aquires products or materials containing byproduct material in concentrations not in excess of those listed, table A-1, column II and III of this handbook. (Except as otherwise provided in 10 CFR 30).

8.1.2.4 Exempt quantities. Any person is exempt from the licensing requirements of 10 CFR 30-34 and 81 to the extent that such persons receives, possesses, transfers, uses, owns, or aquires byproduct material in individual quantities which does not exceed the limits in Schedule B of 10 CFR 30 (see table A-1, column IV, "Exempt quantities", of this handbook). Certain exceptions to the above exemption are listed in 10 CFR 30).

8.1.2.5 Exempt radiation measuring instruments. Ionizing radiation measuring instruments containing a source or byproduct material for internal calibration or standardization which does not exceed the limits in Schedule B of 10 CFR 30.71 (See table A-1, column IV, "Exempt quantities," of this handbook) are exempt from the licensing requirements of 10 CFR 20, 30-36 and 81.

NOTE: Do not confuse exempt commodities, exempt quantities, and exempt concentrations with radioactive material that does not require a license from the NRC. (See 8.1.3 of this handbook.)

8.1.3 Unlicensed radioactive material (10 CFR 20). The use of radioactive material or other sources of radiation not licensed by the NRC is not subject to the NRC regulations. However, it is the purpose of the NRC regulations to control the possession, use, and transfer of licensed material by any licensee in such manner that exposure to such material and to radiation from such material, when added to exposures to unlicensed radioactive material and to other unlicensed sources of radiation in the possession of the licensee, and to radiation therefrom, does not exceed the standards of radiation protection prescribed in the NRC regulations (also see 8.1.1.2 of this handbook).

8.2 Service licenses and authorizations. Certain radioactive material is not licensed by the NRC and therefore, not subject to the NRC regulations. In such cases, a service license or authorization shall be obtained. Service authorizations or licenses are controlled in much the same manner as Nuclear Reg. Commission licensure, the authorization serving the same purpose as the license, with various DoD agency and single, bi- or tri- Military Service directives being used as regulations (see DSAM 4145.8, AR 700-64, NAVSUP INST 4000.34, AFM 67-8, MCO P4400.105).



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8.2.1 Applications for licenses. Applications for licenses or authorizations will be forwarded from the applicant only through the channels prescribed in the following directives:

a. Defense Supply Agency. DSAR 4145.23, Licensing and Control of Radio-Active Materials.
b. Department of the Army. AR 700-52, Licensing and Control of Sources of Ionizing Radiation.
c. Department of the Navy. NAVMED P-5055, Radiation Health Protection Manual.
d. Department of the Air Force. AFR 160-124, Radioisotope Licenses and Permits.
e. Marine Corps. Marine Corps Order P4400.19, Marine Corps Supply Manual, Volume I.

NOTE: Applications for authorization for use of radioisotopes that are not controlled by the Nuclear Regulatory Commission will not be prepared on Commission forms.

## 9. SPECIAL REQUIREMENTS

9.1 Guidelines for shippers and carriers of radioactive materials. This guide is primarily intended to serve as an aid in reviewing the marking, labeling, packaging, and transportation requirements and procedures of the CFRs shown in table I. The use and storage requirements of 10 CFR will be discussed in the general requirements section, and appendix 1. Shippers and carriers of radioactive materials are legally obligated to comply with the applicable DOT regulations, including those that might not be listed herein. DOT regulations are usually minimum requirements; state and local requirements are frequently more stringent. Therefore, it is important that shipping and carrier personnel be familiar with all applicable regulations. (See general requirements.) A brief resume of some of the more important requirements are listed under 9.1.1 and 9.1.2.

9.1.1 Shippers requirements. Personnel involved with shipping radioactive material should be familiar with DOT radioactive materials definitions, and have available for reference, the marking, labeling, and packaging regulations for these materials. (Also see 11.3.13.)

9.1.1.1 Shippers personnel. The shippers technical, packaging and related shipping personnel shall be capable of performing the following minimum requirements when shipping radioactive material:

a. Shall determine the proper DOT classification, shipping name, class, label, and assure that outside packages are properly labeled and marked (see 10.2 and 11.3.11).
b. Shall determine and assure that the proper authorized DOT containers and closures are being used for the quantities and chemical characteristics of each radioactive material being shipped, and that no leaks are evident (see 11.3.7 and 11.3.8).
c. Shall assure that the shipping documents include the proper DOT shipping name, class, signed certificate, proper count and weight. Do not use abbreviations for required entries (see 10.2.1).
d. Shall assure that the carrier's driver is aware that the shipment contains a radioactive material. Give the driver the proper DOT shipping name, class, quantity, and prepare any specific instructions needed to handle transportation emergencies involving the radioactive material being shipped.
e. Shall apply the same checking procedures used in originating a shipment, when repackaging or breaking-down shipments received from others for further distribution. Do not assume that the original shipments were correct.



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f. Shall not tender non-compatible materials in the same shipment. When non-compatible shipments are tendered to the same carrier, assure that the carrier recognizes the situation. Check the loading and storage charts.
g. When radioactive materials are loaded by shipper personnel, assure that a carrier representative has an opportunity to approve the placement, securing and blocking the material before sealing the load, when seals are required.
h. Shall not permit a carrier to transport the shipment unless placards are properly displayed, when placards are required (see 11.3.12).

9.1.2 Carriers requirements. Personnel involved with the transportation of radioactive material should have a broad knowledge of the radioactive material regulations involving proper DOT shipping names, classes, packaging requirements, compatibility, labeling, marking requirements, placarding, and shipping documents. The following requirements shall be implemented by the carrier and carrier personnel (see 11.3.14):

a. Determine and maintain a list of shippers of radioactive materials, including classes, quantities, type of materials, and any special problem associated with the material.
b. The dispatcher or traffic department shall be kept informed relative to radioactive materials to be received or picked-up from the shippers.
c. Work with the shippers concerning shipping requirements, and act as liaison with the shipper when discrepancies are discovered.
d. Ascertain that the drivers have been informed (written or verbal) when they are scheduled to pick-up radioactive materials. The drivers shall have adequate placards and means of securing the radioactive material.
e. The drivers should have specific instructions for use when radioactive material incidents occur and when suspect shipments are encountered.
f. Maintain a current list of sources of help and information to be used when contaminations and other radioactive material emergencies occur.

9.1.2.1 Road, delivery, and pick-up drivers. The following requirements are in addition to those listed under 9.1.2 above:

a. Shall have in his possession, available for immediate use, proper shipping papers covering all radioactive materials loaded on his vehicle.
b. Shall inspect all radioactive material shipments prior to loading, and be able to recognize discrepancies in shipping documents, packaging, labeling, and compatibility.
c. Shall refuse to accept radioactive material shipments from shippers or interline carriers unless in compliance with prescribed regulations. (Shippers must certify that articles are properly packaged and in compliance with the DOT regulations). Do not accept leaking or damaged containers.
d. Shall assure that radioactive material containers will not be damaged by other freight or the vehicle, and that it is properly blocked and secured in the vehicle before movement.
e. Shall understand and execute the proper procedures for decontamination, in case of accident or indicent involving radioactive materials. Report all descrepancies and irregularities observed during the trip (see 10.4).
f. Shall be capable of reporting full details concerning any radioactive material accident or incident including detailed information as to cause, container damage, specific container identification and corrective action taken (see 9.3).
g. Shall understand their responsibilities as to attendance requirements when transporting a hazardous material.



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9.1.2.2 Traffic and freight personnel. The following requirements are in addition to those listed under 9.1.2 above.:

a. Shall check all radioactive material against the shipping documents. Flag all improper documents and questionable radioactive material shipments for verification. Do not prepare bills or process suspect shipments until all discrepancies are resolved (see 10.2.1).
b. Shall refuse to accept radioactive material freight from shippers or interline carriers if the shipping documents are improperly prepared, or cannot be reconciled with the freight involved. Do not accept leaking, damaged, or otherwise improper containers.
c. Shall assure that the Special Permit number is included on the freight bill, if the radioactive material is moving under a DOT Special Permit. If the DOT Special Permit includes any carrier conditions, the shipper must give the carrier a copy of that Special Permit.
d. Shall assure that all required information is included on freight bills. Special emergency instructions supplied by the shipper shall also be included on the freight bill. Do not abbreviate.
e. Shall inspect radioactive material freight for leakage or damage each time it is handled. When damaged containers are discovered - isolate and make certain they are not moved until they are in proper condition for further transportation. Make certain that all container information is obtained for use in preparing the required report to the Department of Transportation. (NOTE - In some instances an immediate telephone notification to DOT is also required.)
f. Assure that qualified personnel supervises any decontamination or disposal operation, should they be required (see Appendix 1).
g. Shall assure that radioactive material will not be damaged by other freight or the vehicle and that it is properly blocked and secured before closing the vehicle.
h. Shall not load noncompatible materials in the same vehicle.
i. Shall assure that proper placards are placed on vehicles when required and that placards are removed or covered when not required (see 11.3.14.b).
j. Shall be familiar with, and have available for reference, provisions of 49 CFR Part 177, Subpart B, titled "Loading and Unloading" and Subpart C, titled "Loading and Storage Chart of Hazardous Materials."

9.2 Indicators of hazardous materials shipment violations (see 11.3.15). This partial checklist is intended to be used in checking shipments of radioactive commodities or materials which are shipped by carriers subject to DOT regulations. (NOTE: DOT Special Permits may permit deviations from the normal packaging regulations.) Typical violations involves the following:

9.2.1 Shipping papers.

a. No proper shipping name or classification.
b. Proper shipping name or classification abbreviated.
c. Lack of wording "No Label Required" on shipments exempt from specification packaging, marking, and labeling.
d. No special permit number on shipments moving under DOT special permit.
e. Color of label used in lieu of classification.

9.2.2 Marking of containers.

a. No commodity description on container not of "exempt" size (commodity description per shipping name in Title 49 CFR 172.101).
b. No special permit number on container shipped under DOT special permit.
c. No name and address of consignee on package which exceeds "exempt" size.



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9.2.3 Labeling.

a. No label on container of hazardous materials which exceeds "exempt" size.
b. Label on container not compatible with classification on shipping papers.
c. Surface label on containers destined for air shipment.
d. Less than two Radioactive Material Labels on container, other than "small quantities."

9.2.4 Containers.

a. Steel:

(1) Labeled containers with no DOT specification marking, or the markings are illegible.
(2) Labeled containers that are dented, rusted, corroded, or temporary repairs sealed with tape, putty, or screws.
(3) Labeled reused containers marked "NRC" - (Look for old date of manufacture and evidence of reuse such as dents, rust and paint layers).
(4) Labeled reused containers marked "STC" and 17-C, 17-E and 17-H with no reconditioner's marking.

b. Corrugated fiber board:

(1) Labeled box with no DOT specification marking when inside containers are larger than "exempt" size.
(2) Labeled box with DOT specification marking constructed of less than 175 lb. test singlewall fiber board (check boxmaker's certificate).
(3) Labeled box with DOT specification marking which is poorly constructed (i.e., gaps and uneven closures, seams and joints separating).
(4) Water damaged boxes.
(5) Improperly closed boxes (i.e., masking tape, scotch tape, string).

9.3 Hazardous materials incident reporting. Hazardous materials transportation incidents have resulted in injuries, property damage, and highly unfavorable publicity to the transportation industry. Much of this could have been avoided or minimized had the causes of previous incidents been identified and appropriate preventative measures taken. In 1971 a nationwide intermodal system for the reporting of hazardous materials incidents was established to provide the Department of Transportation with the factual data necessary to comply with the Hazardous Materials Transportation Control Act of 1970.

9.3.1 Regulations requiring incident reports. The regulations requiring reporting of hazardous materials incidents during transportation are contained in the Code of Federal Regulations, viz:

a. Title 49, Parts 100 to 199 (Governing the transport of hazardous materials by highway, rail, water, and air).
171.15 Immediate notice of certain hazardous materials incidents
171.16 Detailed hazardous materials incident reports
b. Title 49, Part 175 (Governing the transport of hazardous materials by aircraft)
175.45 Reporting certain dangerous article incidents
c. Title 49, Part 176 (Covering the transport of hazardous materials by water vessel
176.48 Situation requiring report



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9.3.2 Reporting system. This system is two-fold in that an immediate telephone notice is required under certain conditions and a detailed written report is required whenever there is any unintentional release of a hazardous material during transportation or temporary storage related to transportation. Any time there is an unintentional release of hazardous materials, the carrier must submit a report on Form DOT F 5800.1 within 15 days from the date of the incident. While carriers are required to report, any interested party may report. In order to include all pertinent information, other reporting parties are encouraged to also utilize this form. Generally, most of this required information is available at the time of the incident, but since leaking and damaged containers are destroyed and spills are cleaned up, some investigation is often necessary in order to obtain all of the facts. Much of this information is also required by carriers for other purposes: insurance records, damage claims, etc. In view of this, carriers may find it to their advantage to incorporate reporting requirements into standard company procedures, thereby making the needed details for the report more readily available and enabling the company to more easily comply with the reporting regulations.

## 10. GENERAL REQUIREMENTS

10.1 Identification for transportation. For transportation purposes, radioactive material or commodity has been defined as any material emitting ionizing radiation with a specific activity greater than 0.002 microcurie per gram (μCi/gm). This computation involves variables which leads to inconsistent results among individuals making the computations. Therefore, section B has been provided to establish uniform microcurie per gram conversion factors for the isotopes referenced herein.

NOTE: Radioactive material as defined above includes natural elements such as radium and accelerator-produced radioisotopes.

10.2 Marking and labeling. Commodities and their containers containing radioactive materials are divided into groups according to the degree of hazard associated with their handling, storage, and transportation (see section A). These commodities shall be marked and labeled as specified herein and by the applicable referenced documents. The markings shall be placed on the radioactive commodity itself and on each surrounding container so that the presence of the radioactive material can be readily recognized and located. Where it is impractical to mark the radioactive material enclosure itself because of small size, the marking may be placed on an appropriate permanent tag attached to the enclosure, or an alternate method as specified by the applicable commodity specifications (see 11.3.11).

10.2.1 Prohibited labeling. No person shall offer for transportation and no carrier shall transport a package or container bearing any marking or label which by its color, design, or shape could be confused or conflict with a label prescribed by DOT (49 CFR 172.400).

10.2.2 Additional marking requirements. Additional markings on packages containing radioactive materials are required as follows (see 11.3.11 and 49 CFR 172).

a. Each package of radioactive materials in excess of 50 kilograms (110 pounds) must have its gross weight plainly and durably marked on the outside of the package.
b. Each package of radioactive materials which conforms to the requirements for Type A or Type B packaging must be plainly and durably marked on the outside of the package in letters at least 13 millimeters (1/2-inch) high, with the words "TYPE A" or "TYPE B" as appropriate. A packaging which is not in compliance with these requirements must not be so marked. Each package of radioactive materials destined for export shipment must also be marked "USA" in conjunction with the specification marking, special permit, or other package certificate identification (see 173.393).



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10.2.3 Shipping papers (49 CFR 172.200). Each shipper offering for transportation a radioactive commodity or material subject to DOT regulations shall describe same on the shipping paper by the proper shipping name (49 CFR 172.101) and by the prescribed classification R.A.M. The total quantity by weight, volume, or as otherwise appropriate shall be shown. When applicable, include the exemption notation "DOT-E" followed by the assigned exemption number. The shipping papers shall also include;

a. The name of each radioisotope in the radioactive material (listed in section A herein). Abbreviations are authorized.
b. A description of the physical and chemical form if the material is not in special form.
c. The activity of the radioactive material in each package in units of curies or submultibles thereof. Abbreviations are authorized.
d. The type of label applied to the package: i.e., radioactive white - I, radioactive yellow - II, radioactive yellow - III.
e. Where the regulations exempt the commodity, package, or container from labeling, the exemption must be indicated by the words "No Label Required" immediately following the description on the shipping paper.
f. The transport index assigned to each package or container bearing the radioactive yellow - II or III labels.
g. For transportation of fissile radioactive materials, the following criteria shall apply:
   (1) If the package or container is "exempt", the words "Fissile Exempt" shall be included (see 11.3.7.1), or
   (2) If not exempt, the fissile class of each package or container shall be included (see definitions), and
   (3) For a Fissile Class III shipment, the following statement shall be added:
      "Warning - Fissile Class III Shipment. Do not Load More Than *** Packages per Vehicle. In Loading and Storage areas, Keep at Least 20 Feet (6 Meters) from Other Packages Bearing Radioactive Labels."
      (Asterisks to be replaced by the appropriate number)
   (4) For Fissile Class III shipment by water, add the following statement to the statement in g.(3) above:
      "For shipment by water, only one Fissile Class III shipment is permitted in a hold."
h. Transportation by air. When a hazardous material is offered for transportation by air and the regulations prohibits its transportation by passenger-carrying aircrate, the description described above. must be followed by the words "Cargo-only aircraft," (see 11.6.6).
   NOTE: When the DOT regulations prohibit a hazardous material aboard a cargo-only aircraft, this material is also prohibited aboard a passenger-carrying aircraft.
j. Transportation by rail. When a hazardous material is offered for transportation by rail, the description described must also include the notation "Placarded" followed by the name of the required placard, and the required placard endorsement (see figure 10 and 49 CFR 174).
k. Package(s) approved by the U. S. Energy Research and Development Administration (ERDA) or the U. S. Nuclear Regulatory Commission (USNRC) requires a notation of the package identification marking (see 11.3.6.2).
l. Export shipments or shipments in foreign made package(s) require a notation of the package identification marking (see 11.3.6.3 c.).

10.2.3.1 Shipper's certification (49 CFR 172). Each shipper offering for transportation a radioactive commodity or material subject to DOT regulations shall include



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the following certification on the shipping paper containing the description: (For transportation by air, the alternate statement may be used)

"This is to certify that the above-named materials are properly classified, described, packaged, marked and labeled and are in proper condition for transportation according to the applicable regulations of the Department of Transportation."

Alternate statement for transportation by air

"I hereby certify that the contents of this consignment are fully and accurately described above by proper shipping name and are classified, packed, marked and labeled, and in proper condition for carriage by air according to the applicable national governmental regulations."

a. For passenger and cargo aircraft, add the following statement to the preceding statement (see 11.6):

"This shipment is within the limitations prescribed for passenger aircraft/cargo-only aircraft." (delete nonapplicable)

b. Shipping papers for air shipments must be made out in duplicate and the shipper's certificate must be executed on both copies.

c. The shipper may also add the words: "*** and to the IATA Restricted Articles Regulations."

d. Shipper certification is not required for shipments to be transported by the shipper except for shipments which are to be reshipped or transferred from one carrier to another, or for bulk shipments in cargo tanks supplied by the carrier.

10.3 Transportation requirements. Radioactive materials or commodities may be offered to carriers for transportation by rail, highway or water, provided the materials or commodities are properly packaged, described and certified as specified herein and in 49 CFR 100-189.

10.3.1 United States Government shipments.

a. Shipments or radioactive materials offered by or consigned to the Departments of the Army, Navy, and Air Force of the United States Government, must be packed, including limitations of weight in accordance with the regulations in 49 CFR 100-189, or in containers of equal or greater strength and efficiency as required by their regulations.

b. Shipments of radioactive materials, made by or under the direction or supervision of the U. S. Nuclear Regulatory Commission or the Department of Defense, and which are escorted by personnel specially designated by or under the authority of those agencies, for the purpose of national security, are exempt from the regulations in 49 CFR 100-189.

10.4 Contamination control (49 CFR 173.397).

a. Removable (non-fixed) radioactive contamination is considered significant if the level of contamination, when averaged over any area of 300 square centimeters of any part of the package surface, exceeds any of the limits specified in table II.



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TABLE II. Radioactive material contamination limits.

| "SIGNIFICANT" REMOVABLE CONTAMINATION IS DEFINED AS THAT WHICH, AS MEASURED BY WIPE TEST, EXCEEDS |
|---|
| 1. For all materials except uranium or thorium:<br>Alpha $10^{-5}$ $\mu Ci/Cm^2$ (22 dis/min/$Cm^2$)<br>Beta-Gamma $10^{-4}$ $\mu Ci/Cm^2$ (220 dis/min/$Cm^2$) |
| 2. For natural or depleted uranium and natural thorium:<br>Alpha $10^{-4}$ $\mu Ci/Cm^2$ (220 dis/min/$Cm^2$)<br>Beta-Gamma $10^{-3}$ $\mu Ci/Cm^2$ (2200 dis/min/$Cm^2$) |

In assessing the surface contamination of a package, a sufficient number of measurements must be taken in the most appropriate locations so as to yield a representative assessment of the contamination situation. The average amount of removable (nonfixed) radioactive contamination may be determined by wiping the external surface of the package with an absorbent material, using moderate pressure, and then measuring the activity on the wiping material. If the measured activity per square centimeter does not exceed 10 percent of the levels prescribed above, it may be assumed that those levels have not been exceeded. Other measurement methods of equal or greater efficiency may also be utilized.

b. When radioactive materials packages are consigned as exclusive use, as defined herein, removable (nonfixed) radioactive contamination may not exceed 10 times that as specified in paragraph a. of this section.

c. Each transport vehicle used for transporting radioactive materials as exclusive use, as defined herein, must be surveyed with appropriate radiation detection instruments after each use. A vehicle may not be returned to service until the ratiation dose rate at any accessible surface is 0.5 millirem per hour or less, and there is no significant removable radioactive surface contamination, as defined in paragraph a. of this section.

## 11. DETAIL REQUIREMENTS

11.1 Arrangement. The requirements for marking, labeling, and packaging of radioactive materials and commodities are enumerated herein according to the requirements of the applicable referenced documents listed below.

a. Use and storage (MIL-STD-1458) (See 11.2)
b. Transportation by rail or highway carrier (49 CFR 174 or 176) (See 11.3)
c. Transportation by water carrier (49 CFR 176) (See 11.4)
d. Transportation by mail (39 CFR) (See 11.5)
e. Transportation by aircraft (49 CFR 175) (See 11.6)
f. Transportation by military aircraft (AFM 71-4, DSAM 4145.3, TM 38-250, NAVSUPPUB 505, MCO P-4030.19) (See 11.7)



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11.2 Use and storage label. Any commodity or item containing radioactive material in excess of the values listed in table III shall be marked and labeled as illustrated by figure 6. This requirement (figure 6, and table III), was established by MIL-STD-1458.

NOTE: Commodities requiring radioactive labeling and marking shall be as specified. Where physical size preclude printing all required data on the commodity itself, the standard radiation symbol (see figure 1-4) and the words "accountability", where required may be adopted.

TABLE III. Radioactivity requirements for use and storage label.

| Radioisotope | Microcuries |
|---|---|
| Radioactive materials not listed below | 0.1 |
| Americium 241 | 0.01 |
| Plutonium 239 | 0.01 |
| Radium 226 | 0.01 |
| Uranium 233, 234, and 235 | 0.01 |
| Mixtures of alpha emitters of unknown composition | 0.01 |

11.2.1 Shipping container labels (10 CFR 20). Each shipping container of licensed radioactive material shall bear a durable, clearly visible label identifying the radioactive contents. The required label shall bear the radiation caution symbol and the words "CAUTION, RADIOACTIVE MATERIAL" or "DANGER, RADIOACTIVE MATERIAL" (see figures 6 and 1-4). The label shall also include sufficient information (radiation levels, types of materials, estimates of the activity, date of estimate, etc.) to permit individuals using, handling, or in the vicinity thereof to take proper precautions to minimize exposures (see 11.2.2 for exemptions).

11.2.2 Exempt shipping containers (10 CFR 20). Shipping containers are exempt from the labeling requirements of 11.2.1, provided the following conditions are satisfied:

a. The container does not contain licensed radioactive material in excess of the limits shown in table A-1, column IV, "Exempt quantities."
b. Containers of natural uranium or thorium do not contain more than 10 times the limits shown in table A-1, column IV.
c. Containers of licensed materials in concentrations not greater than the limits shown in column 2, table I, appendix B of 10 CFR 20.
d. Containers are attended by an individual and precautions taken to prevent exposure of any individual to radiation or radioactive material in excess of the limits established by the NRC.
e. Containers are in transport and packaged and labeled in accordance with the regulations of the Department of Transportation (DOT).
f. Containers available or accessible to authorized individuals, provided that the contents are identified to such individuals by a readily available written record.

11.3 Transportation by rail or highway carrier (49 CFR 174 and 176).

11.3.1 Purpose. The purpose of this section is to summarize the basic requirements of the Department of Transportation (DOT) regulations governing the packaging and shipment of radioactive materials and commodities, and to provide guidance toward more easily and correctly applying the regulation in actual practice. The applicable DOT regulations will be stated formally and then paraphrased where practicable.



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11.3.1.1 The best approach to using the regulations. Perhaps persons that have made shipments of radioactive materials may feel like the person illustrated in figure 1. The primary consideration for the achievement of safety in the transportation of radioactive materials is the use of proper packaging for the specific radioactive materials to be transported. Listed below are three basic questions that a prospective shipper or package designer must address himself to, in determining his packaging requirements. These questions are as follows:

(a) What radioistope is being shipped? Section A contains a listing of over 250 specific radionuclides plus certain "ground rules" for dealing with unlisted or unknown radionuclides, or with mixtures of radionuclides.

(b) What quantity of the radionuclides is being shipped? The packaging requirements relative to quantity are essentially structured about the aggregate quantity in a package, in terms of curies, millicuries, or microcuries.

(c) What is the form of the radioistope?

(1) Is the material in special form?; or
(2) Is it in normal form?

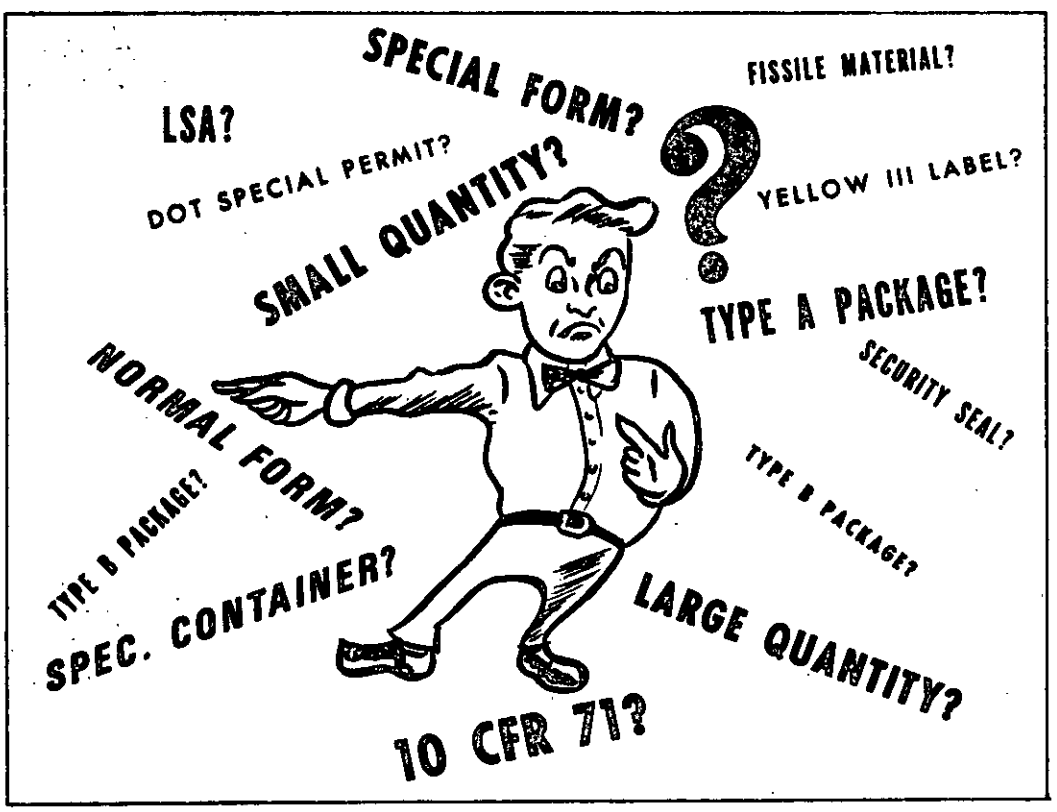



FIGURE 1. What must I do to make a safe and legal shipment of radioactive materials?



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11.3.2 Transport groups. Radionuclides are listed in transport groups I through VII according to their radiotoxicity and relative potential hazard in transportation, and are categorized as specified in section A of this handbook.

11.3.3 Type A packaging. Type A packaging shall be designed in accordance with the general packaging requirements specified in 11.3.6 (49 CFR 173.393), and shall be adequate to prevent loss or dispersal of the radioactive contents and to retain the efficiency of its radiation shielding properties if the package is subjected to the defined "normal" conditions of transport specified in 49 CFR 173.398 b.

11.3.3.1 Selection criteria. Figure 2 illustrates typical type A package schemes. Typically, as is seen, type A packaging in 49 CFR 178, including certain specification steel drums, wooden boxes, fiberboard boxes, etc. Also, provided for is a purely performance-based DOT Spec. 7A, type A general package, for which a shipper may make his own assessment of his particular package design against the performance requirements. The regulatory framework therefore provides for the use of all type A packaging without specific regulatory approvals of the package designs, either via the use of the various prescribed specification containers or the DOT Spec. 7A performance specification. Additionally, foreign-made type A packages are acceptable internationally, provided they are so marked as type A, without specific approval by the competent national authority of either country.

11.3.4 Type B packaging. Type B packaging shall meet all requirements for type A packaging, and the standards for hypothetical accident conditions of transportation as prescribed in 49 CFR 173.398 c.

11.3.4.1 Selection criteria. Figure 3 illustrates type B packaging. Except for a limited number of specification type B packagings (i.e., DOT-6M), all type B package designs require prior approval of the DOT. This involves packaging which must be designed to withstand, in addition to the general packaging requirements and the performance standards for normal conditions of transport, certain serious accident damage test conditions with resultant limited loss of shielding capability and essentially no loss of containment. The performance criteria which the package designer must use to assess type B packaging against these empirically established accident damage test conditions of the transport are prescribed in 49 CFR 173.398 c. and include the following:

a. A 30 foot free drop onto an unyielding surface.
b. A puncture test which is a free drop over 40 inches onto a six inch diameter steel pin.
c. Thermal exposure at 1475°F. for 30 minutes.
d. Water immersion for 8 hours (for fissile materials packaging only).

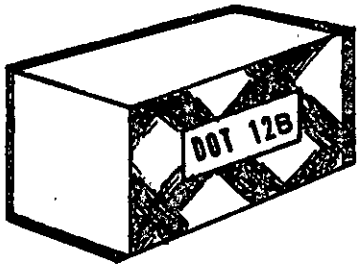
11.3.4.2 Large quantity packaging. "Large quantities" of radioactive material are defined as those quantities which involve greater than type B quantities. The most common materials involved as large quantities are the high-curie irradiator sources and irradiated fuel materials. Packaging requirements for large sources involve all of the type B packaging requirements plus other provisions for such things as decay heat dissipation, potential leakage of contaminated heat transfer medium, heavier shielding and the like, which in some cases, may involve a requirement for certain administrative controls peculiar to the specific package design during shipment. The materials are additionally controlled by the packaging standards as promulgated by the U. S. Nuclear Regulatory Commission in 10 CFR 71.

11.3.4.3 Small quantities, devices, and low-specific activity materials. See "Exemptions" 11.3.10 (49 CFR 173.391 and 173.392).

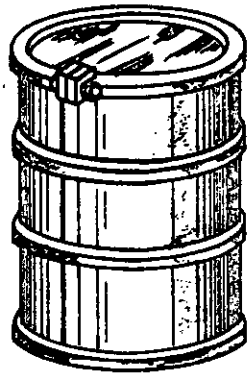


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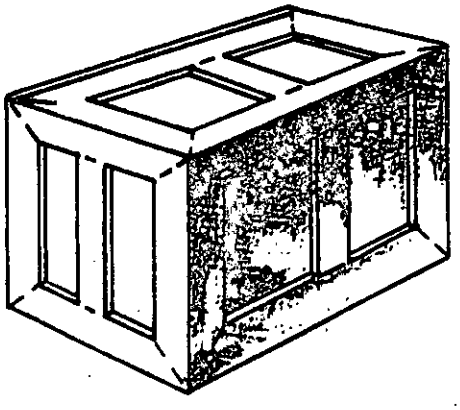
PACKAGE MUST WITHSTAND NORMAL CONDITIONS OF TRANSPORT ONLY WITHOUT LOSS OR DISPERSAL OF THE RADIOACTIVE CONTROL CONTENTS.



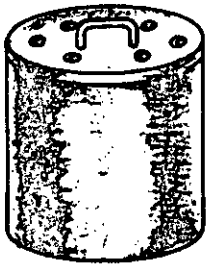

DOT SPECIFICATION
FIBERBOARD BOX



DOT SPECIFICATION
STEEL DRUM



DOT SPECIFICATION
WOODEN BOX



DOT SPECIFICATION 7A
TYPE "A" PACKAGE

FIGURE 2. Typical type "A" packaging.



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PACKAGE MUST STAND BOTH NORMAL (49CFR 173.398b) AND ACCIDENT (49CFR 173.398c) TEST CONDITIONS WITHOUT LOSS OF CONTENTS.

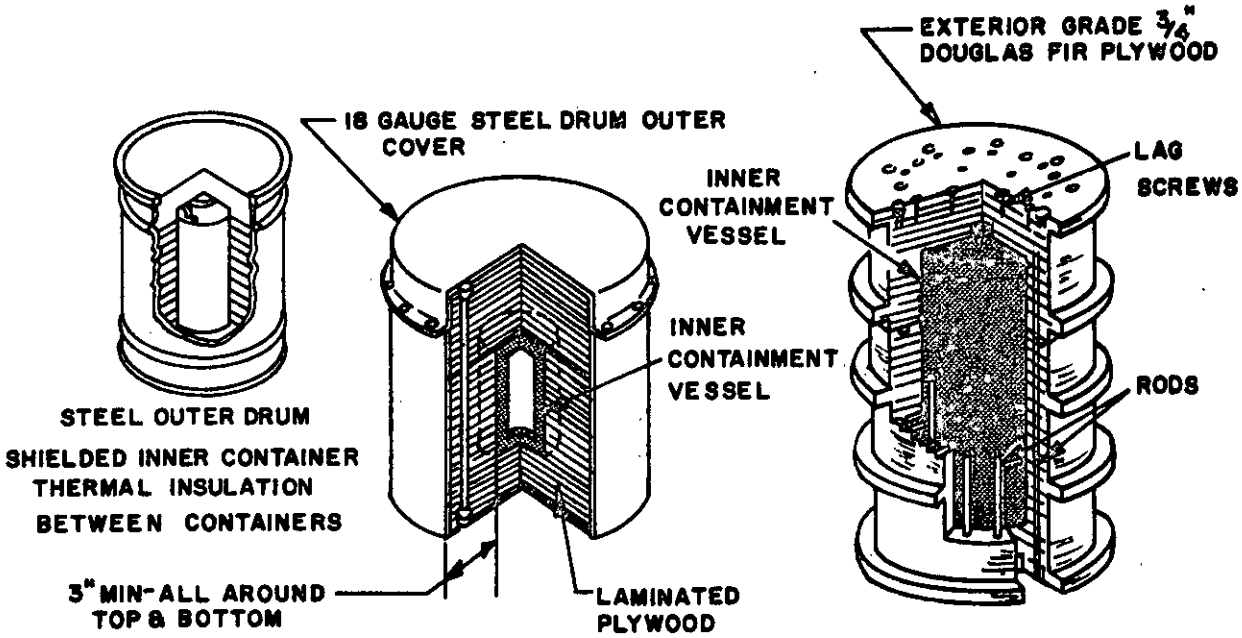



FIGURE 3. Typical type "B" packaging schemes.



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11.3.5 Type A and type B quantities. The quantities of radioactive materials which can be shipped in type A or type B packaging are listed in table IV. The aggregate radioactivity for each type shall not exceed the limits specified. Note here the extreme importance of the special form group. For normal form materials, note that the increasing transport group numbers range from group I, which includes the highly radiotoxic alpha emitters, such as plutonium, americium, and radium, to the less radiotoxic, higher transport groups which have a higher allowable aggregate quantity per package. As indicated in a footnote to this table, a large quantity of radioactive material would be any quantity which exceeds the specified type B quantity limit.

TABLE IV. Type A and type B quantities by transport group.

| Transport group | Type A quantity (curies) | Type B quantity (curies) 1/ |
|---|---|---|
| I | 0.001 | 20 |
| II | 0.05 | 20 |
| III | 3 | 200 |
| IV | 20 | 200 |
| V | 20 | 5,000 |
| VI and VII | 1,000 | 50,000 |
| Special form | 20 2/ | 5,000 |

1/ Quantities exceeding type B are classified as "large quantities" (large radioactive sources).
2/ Except the limit is 2 curies for Californium-252.

11.3.6 General packaging requirements. Unless otherwise specified, all shipments of radioactive materials or commodities shall be packaged as prescribed in 49 CFR 173.391 through 173.396, and shall meet all of the following requirements:

a. The exterior of each package or container must incorporate a feature such as a seal, which is not readily breakable and while intact will be evidence that the package or container has not been opened.
b. The smallest outside dimension of any exterior or container must be four inches or greater.
c. Radioactive material must be packaged in packaging which has been designed to maintain shielding efficiency and leak tightness to assure that there will be no release of radioactive material under normal transportation conditions. If necessary, additional suitable inside packaging must be used. Each package or exterior container must be capable of meeting the standards for type A packaging as specified in 49 CFR 173.398 and 173.24.
d. Internal bracing or cushioning, where used, must be adequate to assure that, under typical transportation conditions, the distance from the inner container or radioactive material to the outside wall of the package remains within the design limits and the radiation dose rate external to the package does not exceed the transport index number shown on the label. Inner shield closures must be positively secured to prevent loss of the contents.
e. The packaging must be designed, constructed, and loaded, so that during transport:
   (1) The heat generated within the package because of the radioactive materials present will not, at any time during transportation affect the efficiency of the package under normal transportation conditions.



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(2) The temperature of the accessible surfaces of the package, when fully loaded, will not exceed 122°F in the shade assuming still air at ambient temperature. If the package is transported in a transport vehicle consigned for the sole use of the consignor, the maximum accessible external surface tamperature shall be 180°F.

f. *Liquid radioactive material in type A quantities* must be packed in (or within) a leak-resistant and corrosion-resistant inner containment vessel. In addition:
   (1) The packaging must be adequate to prevent loss or dispersal of the radioactive contents from the inner containment vessel if the package were subjected to the 9 meter (30-foot) drop test prescribed in 49 CFR 173.398 c. (2) i.; and either
   (2) Enough absorbent material must be provided to absorb at least *twice the volume of radioactive* liquid contents. The absorbent material may be located outside the radiation shield only when it can be shown that, if the radioactive liquid contents were taken up by the absorbent material, the resultant dose rate at the surface of the package would not exceed 1,000 millirem per hour; or
   (3) A secondary leak-resistant and corrosion-resistant containment vessel must be provided to retain the radioactive contents *under the normal conditions of transport* as prescribed in 49 CFR 173.398 b., assuming the failure of the inner primary containment vessel.

g. There must be no significant removable radioactive surface contamination on the exterior of the package (see 10.4,49 CFR 173.397).

h. Packages consigned for export are also subject to the regulations of the foreign governments involved in the shipment (see 49 CFR 173.8, 173.9 and 173.393 b).

i. *The shipper shall determine, by examination or test, prior to shipment,* that the package is proper for the contents, the packaging is unimpared and meets the specified design and construction criteria (see 9.1.1 "shippers requirements").

j. Exeept for shipments described in paragraph k., all radioactive materials must be packaged in suitable packaging (shielded, if necessary) so that at any time during the normal conditions incident to transportation the radiation dose rate does not exceed 200 millirem per hour at *any* point on the external surface of the package and the transport index does not exceed 10.

k. Packages for which the radiation dose rate exceeds the limits specified in paragraph j. of this section, but does not exceed at any time during transportation any of the limits specified in Table V of this section may be transported in a transport vehicle which has been consigned as exclusive use (except aircraft). Specific instructions for maintenance of the exclusive use (sole use) shipment controls must be provided by the shipper to the carrier. Such instructions must be included with the shipping paper information.

11.3.6.1 Control of radiation during transport (transport index). The regulations prescribe that the maximum permissible dose rate at contact with the accessible exterior surface of any package of radioactive materials offered for transport shall not exceed 200 mR/hr, or 10 mR/hr at 3 feet (this latter value being equal to the maximum "Transport Index"). Higher dose rates are also prescribed, which are allowable provided that "sole use" of the transport vehicle is assured by the shipper. The radiation level limitations are summarized in table V. To control the radiation levels of accumulations of multiple numbers of packages once in the transportation



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environment, the regulations require that carriers shall maintain certain prescribed separation distances between radioactive materials packages and other areas which are continuously occupied by persons or photographic film. These separation distances relate the storage time against the "Transport Index" which was defined previously as the dose rate in mrem/hr at three feet from the accessible exterior surface of the package. No package offered for transport (in other than sole-use vehicles) may have a transport index exceeding 10. The establishment of this maximum transport index of 10 is based on considerations for prevention of "fogging" of "fast" photographic film. The total transport index of any aggregate number of packages in any single transport vehicle (other than an exclusive use vehicle) or storage location may not exceed 50. The regulations provide graded tables of stowage distances index. These tables are found in the carrier sections of (49 CFR 174.586 (h) and 177.842 (b), 46 CFR 146.19-35, and 14 CFR 103.23 (a)).

TABLE V. Radioactive materials - maximum radiation level limitations (see 49 CFR 173.393(i) and (j)).

| |
|---|
| Radiation level (dose) rate at any point on external surface of any package of R.A.M. may not exceed:<br><br>A. 200 millirem per hour<br>B. 10 millirem per hour at three feet (transport index may not exceed 10).<br><br>Unless the packages are consigned to a "sole use" or "exclusive use" closed transport vehicle (except aircraft) then the maximum radiation levels may be:<br><br>A. 1000 millirem per hour at 3 feet from external package surface.<br>B. 200 millirem per hour at external surface of the vehicle.<br>C. 10 millirem per hour at 2 meters (6 feet) from the vertical planes projected by the outer lateral surface of the car or vehicle; or if the load is transported in an open transport vehicle, at any point 2 meters (six feet) from the vertical planes projected from the outer edges of the vehicle.<br>D. 2 millirem per hour in any position of the vehicle which is occupied by a person. |

11.3.6.2 U. S. Nuclear Regulatory Commission approved packages; standard requirements and conditions. In addition to the applicable requirements of USNRC approval and 49 CFR 170-189, each shipper of a package containing radioactive material, which has been approved by the USNRC in accordance with 49 CFR 173.394, 173.395, 173.396(b)(2), (c)(2), 173.396(b)(4), or 173.396(c)(3), shall comply with the following:

a. Before the first shipment in a package approved by the Nuclear Regulatory Commission for use by another person, each shipper shall register in writing with the NRC Division of Materials Licensing, his name and address, the name of the person to whom the USNRC approval was issued, and the approval number assigned to the package. Each shipper shall have a copy of the NRC approval and the document referred to in the approval in his possession. Each shipment must be made in compliance with the terms and conditions of the approval;
b. The outside of each package must be durably and legibly marked with the package identification marking indicated in the USNRC approval;



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c. Each shipping paper related to the shipment of this package must bear a notation of the package identification marking indicated in the USNRC approval;
d. Before the first export shipment of the package, the shipper shall submit a copy of the applicable competent authority certificate applying to that package design to the competent national authority of each country into or through which the package will be transported, unless a copy has already been furnished to this party by another person. (Detailed requirements for the issuance and content of competent authority certificates are provided in marginal C-6 of the IAEA "Regulations for the Safe Transport of Radioactive Materials, Safety Series No. 6, 1967 edition," hereinafter referred to as the "IAEA Regulations." A list of the national competent authorities of each country is published annually by the IAEA.);
e. Each package of fissile radioactive material must be marked with the numerical value for the transport index if the shipment is fissile class II. Any vehicle limitation indicated in the USNRC approval applies if the shipment is fissile class III; and
f. For a fissile class III shipment the statement prescribed in 49 CFR 173.427(a)(5)(v) must be included with the shipping papers.

11.3.6.3 International shipments and foreign-made packages; standard requirements and conditions. In addition to the other applicable requirements of 49 CFR 170-189, each shipper of a package containing radioactive material, for which a foreign competent authority certificate has been issued and revalidated pursuant to the IAEA regulations and 49 CFR 173.394(b)(4), 173.394(c)(3), 173.395(b)(3), 173.396(c)(3), 173.396(b)(5), or 173.396(c)(4), also shall comply with the following:

a. Before the first shipment of the package, each shipper shall register in writing his identity and type of package with the Office of Hazardous Materials Operations, U. S. Department of Transportation, Washington, DC 20590, furnishing a copy of the foreign certificate or revalidation thereof which is applicable to that package, unless a copy has already been furnished by another person;
b. The outside of each package must be durably and legibly marked with the competent authority identification marking indicated on the certificate of revalidation;
c. Each shipping paper related to the shipment of the package must bear a notation of the package identification marking indicated in the certificate of revalidation;
d. Before the first export shipment of the package, the shipper shall furnish a copy of the applicable competent authority certificate applying to that package design and any required revalidation, to the competent national authority of each country through or into which the package will be transported, unless a copy has already been furnished by another person;
e. The applicable competent authority certificates need not accompany the packages to which they relate. However, the shipper shall supply them to the carrier upon request; and
f. For a fissile class III shipment, the statement prescribed in 49 CFR 173.427(a)(5)(v) must be included with the shipping papers.
g. The designated competent authority in the USA responsible for administering the requirements (Marginal C-6) of the International Atomic Energy Agency's (IAEA) "Regulations for the Safe Transport of Radioactive Materials," Safety Series No. 6; Office of Hazardous Materials Operations, Department of Transportation, Washington, DC 20590



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h. Any request for a competent authority certificate required by the IAEA regulations must be submitted in writing to the address given in paragraph g. of this section. This request should be in duplicate and must contain all the information required by the applicable subsection of Marginal C-6 of the IAEA regulations. Unless there is good reason for priority treatment, each request will be considered in the order in which it is received. To permit timely consideration, requests should be submitted at least 45 days prior to the requested effective date.

11.3.7 Special form radioactive material. What is meant by "special form" radioactive material? As illustrated in figure 4, we see that "special form" materials are defined as materials, which, if released from a package, might present a hazard due to direct, external radiation, but due to their high physical integrity, would present very little hazard, if any, due to radiotoxicity as a result of spread of contamination. This high physical integrity could be the result of inherent property of the material, such as its being in massive, solid form or an acquired characteristic such as encapsulation as a sealed source. Note a word of caution at this point: The DOT Specification 2R containment is not necessarily synonymous with "special form." In other words, each specific DOT Specification 2R encapsulation must be evaluated against the "special form" performance criteria as prescribed in the regulations (49 CFR 173.398 (a)) in just the same fashion as any other prospective "special form" material. This determination, however, is left to the responsibility of the shipper or package designer, as appropriate, and is not made by the DOT (except where required for export). Recognizing that "special form" materials are not very likely to get "scattered around" in the event of package failure, the regulations allow substantially larger quantities of such materials to be placed in given packagings, than if the materials were in "normal form".

11.3.7.1 Exempted fissile material. The following fissile radioactive materials (exempted from subparagraph e. below) are considered as a special form for packaging purposes: (See 49 CFR 173.396.)

a. Not more than 15 grams of fissile material.
b. Thorium, or uranium containing not more than 0.72 percent by weight of fissile material.
c. Uranium compounds other than metal (for example, UF4, UF6, or uranium oxide in bulk form, not pelleted or fabricated into shapes), and aqueous solutions of uranium in which the total amount of uranium -233 and plutonium present does not exceed 1.0 percent by weight of the uranium -235 content, and the total fissile content does not exceed 1.00 percent by weight of the total uranium content.
d. Homogeneous hydrogenous solutions (or mixtures) containing not more than:
   (1) 500 grams of any fissile material, provided the atomic ratio of hydrogen to fissile material is greater than 7,600; or
   (2) 800 grams of uranium -235, if the atomic ratio of hydrogen to fissile material is greater than 5,200, and the content of other fissile material is not more than 1.0 percent by weight of the total uranium -235 content; or
   (3) 500 grams of uranium -233 and uranium -235, if the atomic ratio of hydrogen to fissile material is greater than 5,200, and the content of plutonium is not more than 1.0 percent by weight of the total uranium -233 and uranium -235 content.
e. A package containing less than 350 grams of fissile material, if there is not more than 5 grams of fissile material in any cubic foot within the package.

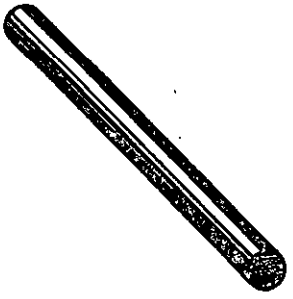


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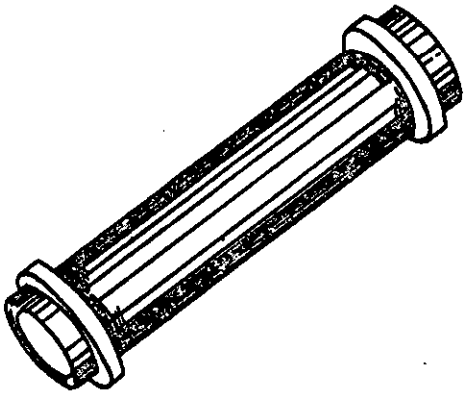
MAY PRESENT A DIRECT RADIATION HAZARD IF RELEASED FROM PACKAGE, BUT LITTLE HAZARD DUE TO CONTAMINATION

"SPECIAL FORM" R.A.M. MAY BE "NATURAL" CHARACTERISTIC, I.E., MASSIVE SOLID METAL, OR "ACQUIRED" THROUGH HIGH INTEGRITY ENCAPSULATION

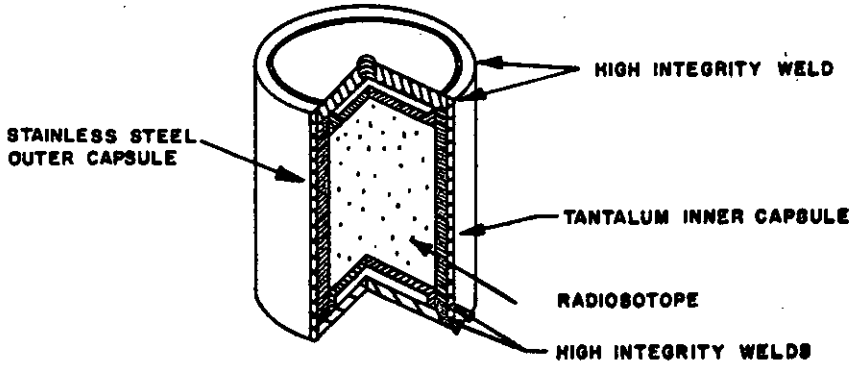
MASSIVE
SOLID METAL



DOT SPEC 2R *
(49 CRF 178.34)



HIGH INTEGRITY ENCAPSULATION
AS A SEALED SOURCE




* SPEC. 2R CONTAINMENT DOES NOT AUTOMATICALLY QUALIFY AS "SPECIAL FORM" SPECIFIC EVALUATION IS NECESSARY AGAINST 173.398(A)

FIGURE 4. Special form radioactive material.



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11.3.7.2 Type A quantities. In addition to the general packaging requirements (see 11.3.6), Type A quantities must be packaged as follows:

a. Specification 7A, Type A general packaging (49 CFR 178.350). Each shipper of Specification 7A packaging must maintain on file for at least one year after the latest shipment, and be prepared to provide the DOT a complete certification and supporting safety analysis demonstrating that the construction methods, packaging design, and materials of construction are in compliance with the DOT specification.
b. Specification 55, metal encased shielded container (49 CFR 178.250). Use of existing containers authorized. Construction not authorized after March 31, 1975.
c. Any Type B packaging authorized by DOT.
d. Foreign-made packaging which bear the marking "TYPE A."

11.3.7.3 Type B quantities. Must be packed as follows (also see 11.3.4):

a. Metal packaging (drum), DOT Specification 6M. Authorized only for solid or gaseous materials which will not decompose at temperatures up to 250°F.
b. Metal-encased shielded container, DOT specification 55. Authorized for not more than 300 curies per container (domestics shipments only). Use of existing containers authorized. Construction not authorized after March 31, 1975.
c. DOT SP 4888.
d. DOT SP 5426.
e. DOT SP 5447.
f. DOT SP 5637.
g. DOT SP 5758.
h. DOT SP 5800.
i. DOT SP 5890.

11.3.7.4 Large quantities of radioactive material in special form. Must be packed in metal packaging (drum), DOT Specification 6M. Radioactive thermal decay energy must not exceed 10 watts. Any other type B packaging which meet the pertinent requirements for large quantities of radioactive materials in the U. S. Nuclear Regulatory Commission (10 CFR 71).

11.3.8 Radioactive materials in normal form and low specific activity (see table IX). Figure 5 illustrates "normal form" radioactive materials which do not qualify as special form, as specified in 11.3.7. Normal form radioactive materials are classified into one of the seven transport groups specified in Section A of this handbook.

11.3.8.1 Type A quantities. Must be packed in specification containers as specified in 11.3.7.2 (also see 11.3.3):

11.3.8.2 Type B quantities. Shall be packaged as follows (also see 11.3.4):

a. Specification 55 (See 11.3.7.3 b.)
b. Metal packaging (drum), DOT Specification 6M. Authorized only for solid or gaseous materials which will not decompose at temperatures up to 250°F.
c. Any other type B packaging for which DOT has issued a certificate of compliance.

11.3.8.3 Large quantities of radioactive materials in normal form. Must be packed as follows:

a. Metal packaging (drum), DOT Specification 6M. Authorized only for solid or gaseous materials which will not decompose at temperatures up to 250°F. Radioactive thermal decay energy must not exceed 10 watts.

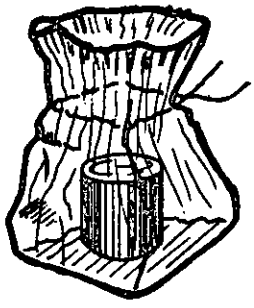


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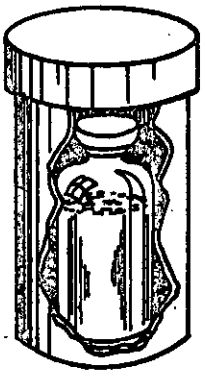
# "NORMAL FORM" R.A.M. (RADIOACTIVE MATERIAL-N.O.S.)

INCLUDES ANY MATERIAL WHICH DOES NOT QUALIFY AS "SPECIAL FORM".

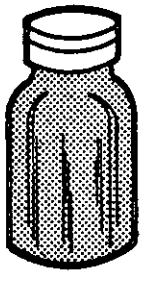
NORMAL FORM MATERIALS ARE CLASSIFIED INTO EITHER OF SEVEN TRANSPORT GROUPS.



WASTE MATERIAL IN PLASTIC BAG



LIQUID IN BOTTLE WITHIN METAL CONTAINER, NO FIRE RESISTANCE



POWDER IN GLASS OR PLASTIC BOTTLES

LOW-MELTING, COMBUSTIBLE AND/OR FRANGIBLE SOLID

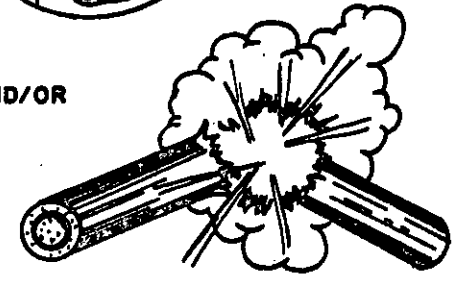


FIGURE 5. Normal form radioactive material.



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b. Any other type B packaging which meets the pertinent requirements for large quantities of radioactive materials in the regulations of the U. S. Nuclear Regulatory Commission (10 CFR 71).

11.3.9 Fissile radioactive materials. See 11.3.7.1 for exempted fissile material. In addition to considerations for the radioactive content, shippers of fissile radioactive material must also take into account certain other packaging and shipment requirements to ensure against nuclear criticality due to the fissile (fissionable) nature of the materials. The design of the packaging for fissile radioactive material, the transport index to be assigned (if Fissile Class II), and any special procedures for packaging are prescribed in 49 CFR 173.396 (see 11.3.6) and in 10 CFR 71. Each fissile radioactive materials package design (except for the DOT Spec. 6L and 6M) must be reviewed and approved by the USNRC prior to its first use. The packaging must be such to ensure against nuclear criticality (an unplanned nuclear chain reaction) under both normal and accident damage test conditions, and to prevent loss of contents of transport. For purposes of control in transportation, fissile radioactive material packages are classified with one of three groups, according to the degree of control which must be exercised to assure nuclear criticality safety, as shown in table VI. (See 6.18 and 6.19.)

TABLE VI. Fissile radioactive material classes.

| SHIPMENT CONTROLS FOR FISSILE RADIOACTIVE MATERIALS PACKAGES |
|---|
| 1. Fissile Class I - Packages may be transported in unlimited numbers (Transport Index is based only on external radiation levels). |
| 2. Fissile Class II - Number of packages limited by aggregate maximum of Transport Index of 50 (50 unit rule). No single package may exceed a Transport Index of 10. Transport Index will have been calculated either on criticality or external radiation level basis. |
| 3. Fissile Class III - Packages which do not meet the requirements of Fissile Classes I or II. Controlled by special arrangements between the shipper and carrier. (See 49 CFR 173.396 (g).) |

11.3.9.1 Transport index system. The transport index (TI) system with the separation tables controls the exposure of personnel handling, transporting, or casually exposed to radiation from radioactive material packages. The TI system has also been adapted for limiting aggregations of packages containing fissile radioactive materials to avoid assembling in one location an amount of fissile material which, under credible conditions, would support a chain reaction, i.e., "go critical." As for radiation levels, the shipper determines in accordance with the regulations of the NRC (10 CFR 71), a TI (49 CFR 173.389 (i) (2)) which is to be assigned to the fissile material package (see 6.45). The shipper assigns to each package of fissile material the nuclear safety TI as calculated or the radiation level TI (as described earlier), whichever is greater. The transport worker adds these radiation level TI's and by complying with the TI limitation in any one vehicle or location, limits the amount of fissile material in fissile material packages.



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TABLE VII. Clarification of classification terminology.

| CAUTION - - - DO NOT CONFUSE THE FOLLOWING: |
|---|
| Transport Groups I through VII (6.47 and Section A) with: Radioactive materials package labels (11.3.11) RADIOACTIVE WHITE - I RADIOACTIVE YELLOW - II RADIOACTIVE YELLOW - III or with: Fissile Classes I, II, or III (6.17 and 11.3.9) |

11.3.9.2 Type A quantities. Fissile radioactive materials containing not more than type A quantities of radionuclides, in either normal form or special form, must be packed as follows:

a. Metal container (drum), DOT Specification 6L. Authorized contents shall be as specified in 11.3.9.3 (a).
b. Metal package (drum), DOT Specification 6M. Authorized only for solid radioactive material as provided for in 11.3.9.3 (b).
c. Any container provided for in 11.3.8, provided the following conditions are met:

(1) Not more than 500 grams of uranium -235 as Fissile Class III, or not more than 40 grams of uranium -235 as Fissile Class II shipment, the transport index to be assigned to each package shall be 0.4 for each gram of uranium -235 above 15 grams, up to a maximum of 40 grams (transport index of 10).
(2) Not more than 320 grams of plutonium -239 as plutonium-beryllium neutron sources in special form. Total radioactivity content must not exceed 20 curies. The transport index to be assigned to each package shall be 0.5 for each 20 grams, or fraction thereof, of fissile plutonium.

d. Any other type A or B packaging which meets the standards for packaging for fissile radioactive material in the NRC regulations (10 CFR 71), and for which DOT has issued a certificate of compliance.

11.3.9.3 Type B quantities. Fissile radioactive materials containing type B quantities of radionuclides, in either normal form or special form, must be packaged as follows:

a. Metal container (drum), DOT Specification 6L. Authorized only for uranium-235, plutonium-239 or 241, as metal, oxide, or compounds which will not decompose at temperatures up to 149°C (300°F.) Radioactive thermal decay energy output shall not exceed 5 watts. Large quantity radioactive materials in normal form must be packaged in one or more sealed and leak tight metal cans or



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polyethylene bottles within the Specification 2R containment vessel. The following quantities of fissile radioactive materials are authorized under the Fissile Class II and III conditions listed:

TABLE VIIIA. Maximum authorized contents. 1/

| Uranium-235 2/ | | Plutonium 2/4/ | | Fissile class II transport index | Fissile class III maximum number of packages per transport vehicle |
|---|---|---|---|---|---|
| H/X≤3 | 3<II/X≤20 | H/X≤10 | 10≤H/X≤20 | | |
| 14 | 3.6 3/ | --- | --- | 1.3 | 80 |
| --- | --- | 2.5 | 2.4 | 1.8 | 55 |

1/ Quantity in kilograms.
2/ All sources of hydrogen within the inner containment vessel must be considered in determining the H/X ratio of inner containment vessel.
3/ Volume not to exceed 3.6.1.
4/ Plutonium solutions are not authorized.

b. Metal package (drum), DOT Specification 6M. Authorized only for solid radioactive materials which will not decompose at temperatures up to 250°F. Radioactive thermal decay energy output shall not exceed 10 watts. Large quantity radioactive materials in normal form must be packaged in one or more sealed and leak-tight metal cans or polyethylene bottles within the Specification 2R containment vessel.

11.3.9.4 Fissile classes to containers. Classification of Fissile Class I, II, or III to containers (see table VI):

a. Fissile Class I packages. The following quantities of fissile radioactive material are authorized for Fissile Class I packages: 1.6 kilograms uranium -235; 0.9 kilograms of plutonium (because of the 10-watt thermal decay heat limitation, the limit for plutonium -238 is only 0.02 kilograms); and 0.5 kilograms of uranium -233. The maximum ratio of hydrogen to fissile material must not exceed three, all sources of hydrogen within the Specification 2R containment vessel being considered.
b. Fissile Class II and III packages. Quantities of fissile radioactive material as shown in table VIIIB are authorized for Fissile Class II and III packages. When the table specifies a maximum ratio of hydrogen to fissile material, this refers only to the hydrogen interspersed with the fissile material.

(1) For Fissile Class II packages, the minimum Transport Index Number to be assigned is also shown in table VIIIB.
(2) For Fissile Class III packages, the maximum number of similar packages per transport vehicle is shown. Fissile Class III shipments are also subject to the restrictions outlined in 11.3.9.4 (c). For a uranium-233 shipment, the maximum inside diameter of the inner containment vessel must not exceed 12 centimeters (4.75 inches). Where necessary, a tight fitting steel insert must be used to reduce a larger diameter inner containment vessel specified in (49 CFR 178.104-3(b) to the 12 centimeters (4.75 inches) limit.



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TABLE VIIIB. Maximum authorized contents (in kilograms).

| Uranium-233 4/ | | | Uranium-235 3/6/ | | | Plutonium 1/2/3/ | | | Fissile class II transport index | Fissile class III maximum number of packages per transport vehicle |
|---|---|---|---|---|---|---|---|---|---|---|
| Metal or alloy H/X=0 | Compounds | | Metal or alloy H/X=0 | Compounds | | Metal or alloy H/X=0 | Compounds | | | |
| | H/X=0 | H/X=3 | | H/X=0 | H/X≤3 | | H/X=0 | H/X≤3 | | |
| 3.6 | 4.4 | 2.9 | 7.2 | 7.6 | 5.3 | 3.1 | 4.1 | 3.4 | 0.1 | 1,250 |
| 4.2 5/ | 5.2 | 3.5 | 8.7 | 9.6 | 6.4 | 3.4 | 4.5 | 4.1 | .2 | 625 |
| 5.2 | 6.8 | 4.5 | 11.2 | 13.9 | 8.3 | 4.2 | --- | 4.5 | .5 | 250 |
| --- | --- | --- | 13.5 | 16.0 | 10.1 | 4.5 | --- | --- | 1.0 | 125 |
| --- | --- | --- | --- | 26.0 | 16.1 | --- | --- | --- | 5.0 | 25 |
| --- | --- | --- | --- | 32.0 | 19.5 | --- | --- | --- | 10.0 | 12 |

1/ Minimum percentage of plutonium-240 is 5 weight percent.
2/ 4.5 kg limitation of plutonium due to 10 w decay heat limitation.
3/ For a mixture of uranium-235 and plutonium an equal amount of uranium-235 may be substituted for any portion of plutonium authorized.
4/ Maximum inside diameter not to exceed 12 cm (4.75 in.) (see 11.3.9.4(c)(2)(ii)).
5/ Granulated or powdered metal with any particle less than 8 mm (0.25-in) in the smallest dimension is not authorized.
6/ Maximum uranium-235 enrichment is 93 percent.





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c. Fissile Class II packages shipped under the exclusive use provisions (49 CFR 173.393(j)) shall provide for packages with high radiation dose rates. The transport index number which is calculated for nuclear criticality control purposes must not exceed 10 for any single package or a total of 50 for the full load, unless specifically authorized by the DOT for Fissile Class III shipments.

(1) Fissile Class II packages may be shipped with a transport index greater than 10, and combined with other packages of the same or different designs in a Fissile Class III shipment, under the conditions prescribed in paragraph d. of this section, Provided:

(a) The transport index which has been assigned in the package approval for nuclear criticality control purposes does not exceed 10 for any single package;
(b) The total transport index for nuclear criticality control purposes for all packages in the shipment does not exceed 100;
(c) The shipment satisfies the provisions of (49 CFR 173.393(j)) if any package has radiation dose rates exceeding 10 millirem per hour at 1 meter (three feet) from any accessible external surface of the package; and
(d) The shipment will not be transported by water.

(2) Fissile Class III packages, which have been assigned a transport index for nuclear criticality control purposes in accordance with Fissile Class II criteria, may be combined with Class II criteria, may be combined with other Fissile Class III packages of the same or different design for which a transport index has been so assigned for nuclear criticality control purposes, and may be combined with Fissile Class II packages in a Fissile Class III shipment under the conditions prescribed in paragraph d. of this section, provided:

(a) The transport index which has been assigned in the package approval for nuclear criticality control purposes does not exceed 50 for any single package;
(b) The total transport index for nuclear criticality control purposes for all packages in the shipment does not exceed 100;
(c) The shipment satisfies the provisions of (49 CFR 173.393(j)) if any package has radiation dose rates exceeding 10 millirem per hour at 1 meter (three feet) from any accessible external surface of the package; and
(d) The shipment will not be transported by water.

d. Fissile Class III shipments may be made only in accordance with subparagraph (1) or (2) of this paragraph, or in accordance with other procedures authorized by the Department. The transport controls must provide nuclear criticality safety and shall be carried out by the shipper or carrier, as appropriate, to protect against loading, transporting, or storing of that shipment together with other fissile material.



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(1) Transportation in a transport vehicle assigned for the sole use of that consignor, with a specific restriction for such sole use to be provided in the special arrangements, and with instructions to that effect issued with the shipping papers; or

(2) Transportation under escort by a person in a separate vehicle, with the escort having the capability, equipment, authority, and instructions to provide administrative control adequate to assure compliance with this paragraph.

11.3.10 Small quantities of radioactive materials, commodities, and low specific activity materials (49 CFR 391 and 392). The transport group and special form package limits are also used as a basis for defining the package quantity limits for small or "exempt" specific activity materials, as illustrated in table IX. As a matter of interest, the "small" or exempt quantities and "radioactive device" categories are those which are allowable for shipment by U. S. Mail provided that certain other limitations are fulfilled (see 11.5). Note in table IX that the limitations for low specific activity materials are based on concentration, i.e., specific activity, rather than on the aggregate quantity per package.

a. Exemption from all labeling. No labeling, marking or specified packaging is required where the radioactive material (RAM) used has an activity level within the limits defined in 11.3.10.1.

b. Exemption from DOT labeling.

(1) No Department of Transportation (DOT) labeling, marking or specified packing is required where the radioactivity of devices and their packaging is within the limits defined in 11.3.10.2.

(2) The "Use and Storage" labels, figure 6, are required for DOT exempt shipments when the RAM material used exceeds the limits defined in 11.3.10.1. The information required is indicated on the figure 6 label and is as follows:

(a) AEC license number and service organization if required by section 8. (See 8.1.2 for exemption for electronic tubes.)
(b) Name of radioactive material (RAM)
(c) Activity of RAM. See section B, table B-1.
(d) Manufacturing lot number. See MIL-STD-1458.

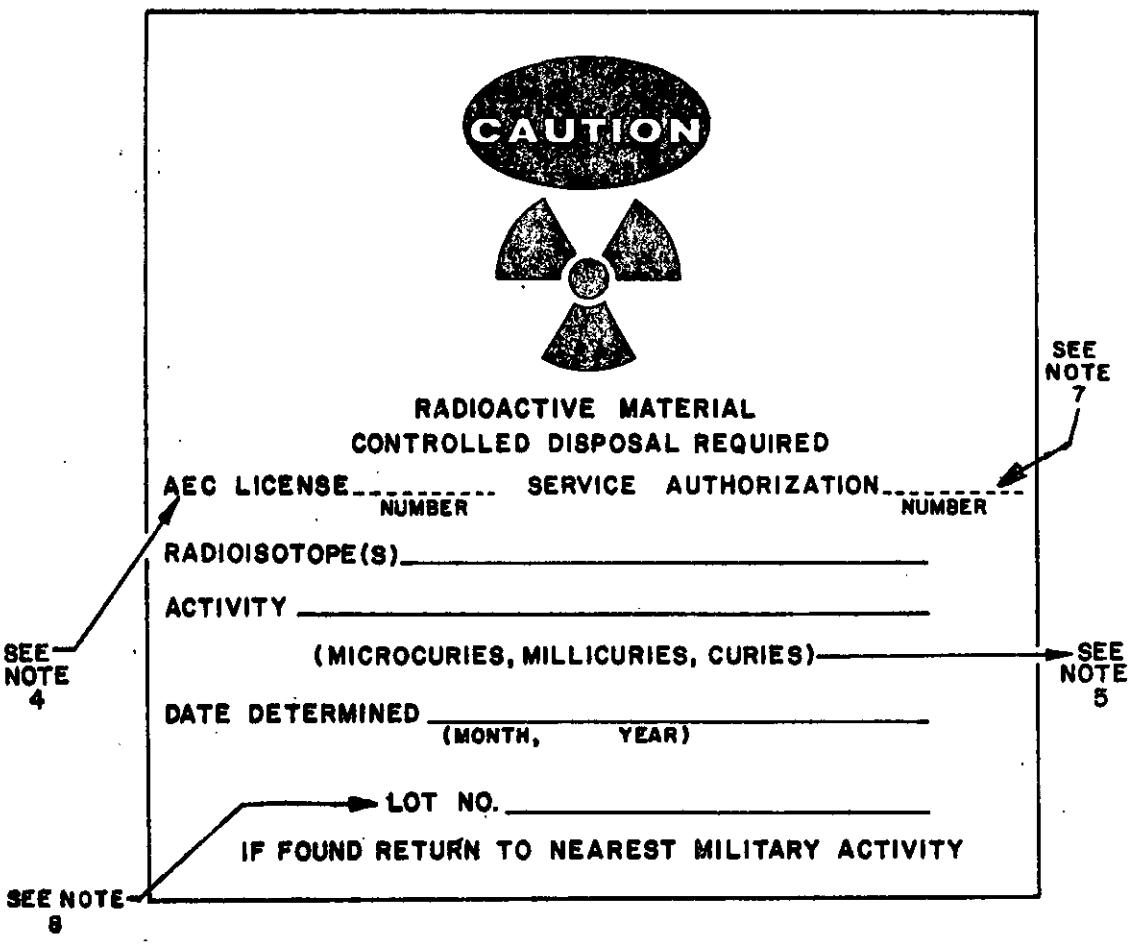
(3) Shipping papers for DOT exempt shipments require the following information where the RAM material used exceeds the limits defined in 11.3.10.1:

(a) The statement "No Label Required" (see 10.2.2 g.).
(b) Statement of certification (see 10.2.2 h.).
(c) Name of radioisotope (see 10.2.2 b.).
(d) Activity of radioisotope (see 10.2.2 c.).
(e) Proper name of RAM (see 11.3.13).
(f) Transport group (see 10.2.2 a.).

11.3.10.1 Non-radioactive material (minimum requirements). Materials and commodities in which the calculated specific activity is not greater than 0.002 microcuries per gram of material (see section B), and in which the radioactivity is essentially uniformly distributed, are not considered to be radioactive materials and marking and labeling requirements are not applicable (49 CFR 173.389(e)).



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NOTES:
1. Modification of the figure above is authorized as provided for in the applicable commodity specification and technical data package.
2. Symbol, words, and background for "caution" to be magenta or purple.
3. Background and "caution" to be yellow.
4. Cross out the one that is not applicable.
5. Cross out the two terms that are not applicable.
6. AEC license number is that issued to agency having logistical responsibility by Atomic Energy Commission.
7. For an explanation of this requirement, see 8.2.
8. For an explanation of this requirement, see MIL-STD-1458.

FIGURE 6. Use and storage label.



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11.3.10.2 Exemptions. The following radioactive materials and commodities shown in table IX are exempt from specification packaging, marking, and labeling and are exempt from 11.3.6 (49 CFR 173.393), except for the name of contents as required by paragraph 11.3.13, and provided the following conditions are met:

a. Radioactive materials in normal form.

(1) The limits in table IX for small or exempt quantities are not exceeded:
(2) The materials are packaged in strong, tight shipping containers such that there will be no leakage of radioactive materials under conditions normally incident to transportation.
(3) The shipping container shall be such that the radiation dose rate at any point on the external surface of the shipping container does not exceed 0.5 millirem per hour.
(4) There shall be no significant removable radioactive surface contamination on the exterior of the shipping container. (See 10.4.)
(5) The outside of each inner container must be marked "Radioactive." or "Radioactive - LSA" as appropriate.

TABLE IX. Activity limits for small quantities, devices, and low specific activity materials.

| Transport group | Small or exempt quantities | Radioactive Commodities (Devices) maximum quantities | | Low specific activity materials |
|---|---|---|---|---|
| | | Per device | Per package | |
| I | 0.01 mCi | 0.0001 Ci | 0 0.001 Ci | 0.0001 mCi/gm |
| II | 0.1 mCi | 0.001 Ci | 0.05 Ci | 0.005 mCi/gm |
| III | 1 mCi | 0.01 Ci | 3 Ci | 0.3 mCi/gm |
| IV | 1 mCi | 0.05 Ci | 3 Ci | 0.3 mCi/gm |
| V, VI | 1 mCi | 1 Ci | 1 Ci | --- |
| VII | 25 Ci | 25 Ci | 200 Ci | --- |
| Special form | 1 mCi | 0.05 Ci | 20 Ci | --- |
| Tritium oxide (in aqueous solution) | 0.5 mCi/ml (3 Ci/package limit) | --- | --- | --- |
| Fissile Material | 15 grams (U-235) | --- | 15 grams | --- |

b. Radioactive commodities (devices): This includes manufactured articles such as instruments, clocks, electronic tubes or apparatus, or similar devices having radioactive materials (other than liquids) in a nondispersible form as a component part, are exempt from specification packaging, marking, and labeling, provided the following conditions are met. (49 CFR 173.391b).

NOTE: For radioactive gases, the requirement for the radioactive material to be in a nondispersible form does not apply.

(1) The limits in table IX for Radioactive Commodities (devices) are not exceeded.



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(2) Radioactive materials are securely contained within the devices, or are securely packaged in strong, tight shipping containers, so that there will be no leakage of radioactive materials under conditions normally incident to transportation.
(3) The radiation dose rate at four inches from any unpackaged device does not exceed 10 millirem per hour.
(4) The radiation dose rate at any point on the external surface of the package may not exceed 0.5 millirem per hour. However, for exclusive use shipments only, the radiation at the external surface of the package or the item may exceed 0.5 millirem per hour, but must not exceed 2 millirem per hour.
(5) There must be no significant removable radioactive surface contamination on the exterior of the shipping container. (See 10.4. and 49 CFR 173.397).
(6) The outside of the inner container must bear the marking "Radioactive."

c. Radioactive commodities containing uranium. A manufactured article, other than reactor fuel elements, in which the only radioactive material is metallic natural or depleted uranium or natural thorium or alloys thereof, is exempt from DOT specification packaging, marking, and labeling, and is exempt from 11.3.6 (49 CFR 173.393) if the following conditions are met:

(1) The radiation dose rate at any point on the external surface of the shipping container does not exceed 0.5 millirem per hour.
(2) There shall be no significant radioactive surface contamination of the exterior of the package (see 10.4 or 49 CFR 173.397).
(3) The total radioactivity content of each article must not exceed 3 curies.
(4) The outer surface of the uranium or thorium is enclosed in a non-radioactive, sealed, metallic sheet.

11.3.11 Radioactive material label requirement. Each package or shipping container of radioactive material, unless exempted under paragraph 11.3.10.2 (49 CFR 173.391 - 49 CFR 173.392) of this handbook, shall be labeled with distinctive warning labels affixed to opposite sides of the package. The labels are described below and shown in figures 6 and 7. The labels alert persons handling packages that the package may require special handling. If the background color of the label is all white, the radiation is minimal and nothing special is required for that package. If, however, the background of the upper half of the label is yellow, a radiation label requiring consideration may exist at the outside of the package and an indication of what controls must be exercised for that package is related to the transport index concept. If the package bears a yellow label, with three stripes, the rail or highway vehicle in which it is carried must be placarded. Placarding is discussed in more detail in paragraph 11.3.12. The requirements for the use and storage label (figure 6) are specified under paragraph 11.2. The requirements for the three DOT radioactive material labels (figure 7) shall be determined by the transport index or other considerations as follows (summarized in table X). See 8.1.2.6 for NRC requirements.

11.3.11.1 Radioactive white-I label (figure 7). Each shipping container or package not exceeding 0.5 millirem per hour at any point on the external surface of the shipping container and provided it does not contain a large quantity of radioactive material (see definitions), shall bear a radioactive white-I label. This label is not authorized for Fissile Classes II and III shipping containers.

11.3.11.2 Radioactive yellow-II label (figure 7). Radioactive yellow-II labels shall be affixed to:

a. Each shipping container or package measuring more than 0.5 but not exceeding 50 millirem per hour at each point on the external surface of the shipping container and not exceeding 1.0 millirem per hour at 3 feet from each point on the external surface of the shipping container; or



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b. Each fissile class II package or shipping container for which the transport index does not exceed 1.0 at any time during transportation.

11.3.11.3 Radioactive yellow-III label (figure 7). Radioactive yellow-III labels shall be used when either of the limits in 11.3.11.2 is exceeded. In addition, the following types of packages or shipping containers shall also bear this label:

a. Each Fissile Class III shipping container.
b. Each package or shipping container containing a large quantity of radioactive material as defined herein.
c. Each package or shipping container being transported under a permit.

TABLE X. Radioactive material label requirements.

| Label | Exposure rate limits | |
|---|---|---|
| | At any point on accessible surface of package (millirem/hour) | At 3 feet from external surface of package (millirem/hour) |
| "RADIOACTIVE-WHITE I" | 0.5 | 0 |
| "RADIOACTIVE-YELLOW II" | 0.5-50 | 1.0 |
| "RADIOACTIVE-YELLOW III"* | >50 | >1.0 |

*Requires vehicle placarding
(This label mandatory for any fissile Class III or large quantity package regardless of dose rate levels.)

11.3.11.4 Mixed hazardous materials. Radioactive materials having other hazardous characteristics must also be labeled with other labels as required according to the hazards of the commodity. For example:

a. Packages containing the solid nitrates of uranium or thorium must bear both a "radioactive" label and a "yellow" oxidizing materials label.
b. Packages containing nitric acid solutions of radioactive materials must bear both a "radioactive" label and a "white" corrosive acid label.

11.3.11.5 Empty container label. The "empty label" as illustrated in figure 8 shall be applied to shipping containers which have been emptied and on which the old labels have not been removed, obliterated, or destroyed. They shall be so placed on the shipping container as to completely cover the old labels. Labels for empty containers must be not less than 6 inches on each side, white in color, and printed in letters not less than 1 inch high in black ink.

11.3.11.6 Placarding. The shipper or carrier must apply the appropriate placard to the transport vehicle (rail or highway) if a radioactive material package or container on board bears a "Radioactive Yellow-III" label (figure 7). The format for these placards is illustrated in figures 9 and 10. (See 9.1 and 49 CFR 172.500,)

11.3.12 Shipper's labeling requirements. The following shipper's requirements pertaining to labels are in addition to special requirements listed under 9.1.1 herein:

a. Shippers must furnish and attach the labels prescribed for their packages. Labels should be applied to that part of the package bearing consignee's name and address.



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b. Labels must not be applied to packages containing articles which are not subject to 49 CFR 170-189 or are exempted therefrom. (See 11.3.10.2.). However, this paragraph does not prohibit the use of labels required on packages destined for transportation by air (11.6.4).

c. Shippers must not use labels which by their size, shape, and color, may readily be confused with the standard caution labels prescribed in this part.

d. Labels must conform to standards as to size, printing, and color, and samples will be furnished, on request, by the Bureau of Explosives.

e. A combination diamond-shaped label-tag of proper size and color, bearing on one side the shipping information and on the reverse side the wording prescribed in this part, will be permitted.

f. The carrier's name and stationery form number, or the shipper's name and address, may be printed on the labels, in type not larger than 10 point, if placed within the blackline border and in the upper or lower corner of the diamond.

g. Each package of radioactive material (except when exempted) requiring a radioactive (white I, yellow II, or yellow III) label shall be labeled with two such labels, affixed to opposite sides of the package. The method of determining which label to use is discussed in 11.3.11.

h. Labels which conform to the model prescribed in the regulations of the International Atomic Energy Agency, and which are similar in appearance to the labels prescribed herein (although the inscriptions on the labels may be in a foreign language) are authorized in place of the labels prescribed herein for import or export shipments only.

i. A person who offers for transportation a package containing hazardous material shall conspicuously label it in compliance with the requirements of 49 CFR 172. The applicable information as required in any blank spaces on the label must be inserted by legible printing, using a durable, waterproof means of marking. Labels should be applied to that part of the package bearing the consignee's name and address.

j. The following requirements apply to completion of the items of information in the blank spaces of the labels specified in this section:

(1) "Contents". The name of the radionuclide, as taken from the listing of radionuclides in Section A (symbols which conform to established radiation protection terminology are authorized, i.e., Mo. Co, etc.). For mixtures of radionuclides, the most restrictive radionuclides on the basis of radiotoxicity must be listed as space on the label allows.

(2) "Number of curies". Units shall be expressed in appropriate curie units, i.e., curies (Ci), millicuries (mCi) or microcuries (μCi) (abbreviations are authorized). For a fissile material, the weight in grams or kilograms of the fissile radioisotope may also be inserted.

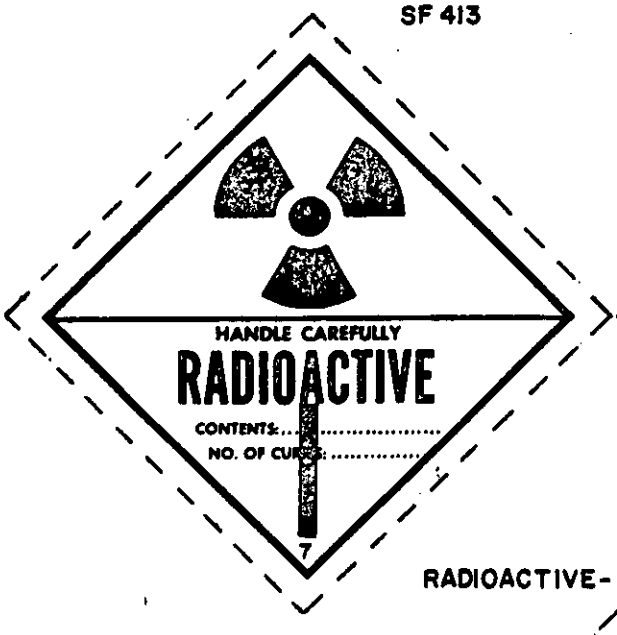
(3) "Transport index". (See definitions and Section A.)



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RADIOACTIVE-WHITE I

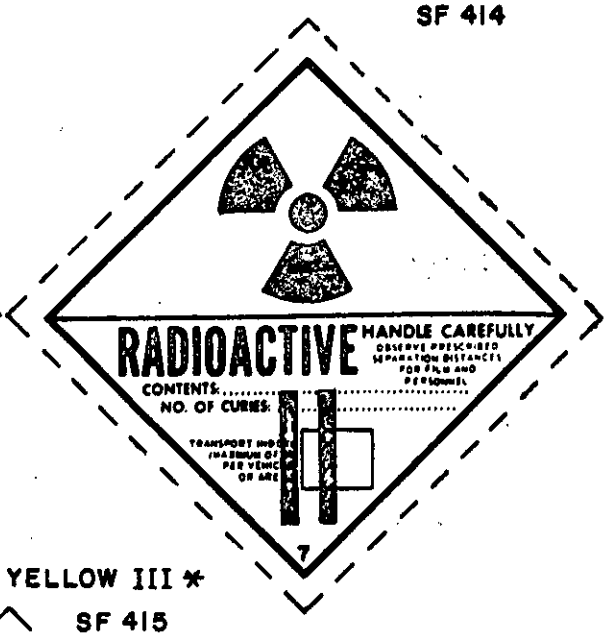
SF 413




BACKGROUND--WHITE
VERTICAL BAR--BRIGHT RED

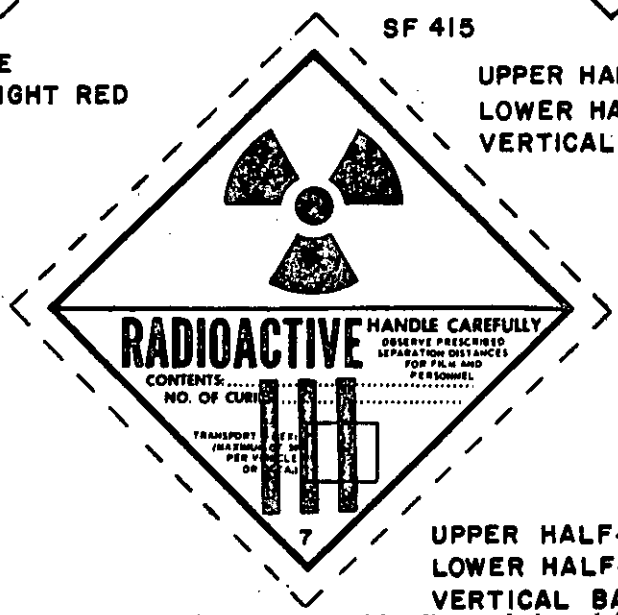
RADIOACTIVE-YELLOW II

SF 414




RADIOACTIVE-YELLOW III *

SF 415




UPPER HALF--BRIGHT YELLOW
LOWER HALF--WHITE
VERTICAL BARS--BRIGHT RED

UPPER HALF--BRIGHT YELLOW
LOWER HALF--WHITE
VERTICAL BARS--BRIGHT RED

LABEL REQUIREMENTS: All labels must be diamond shaped 4.00 inches (101.6 mm) long on each side with black printing inside a black border at least 3.50 inches (88.9 mm) on each side.

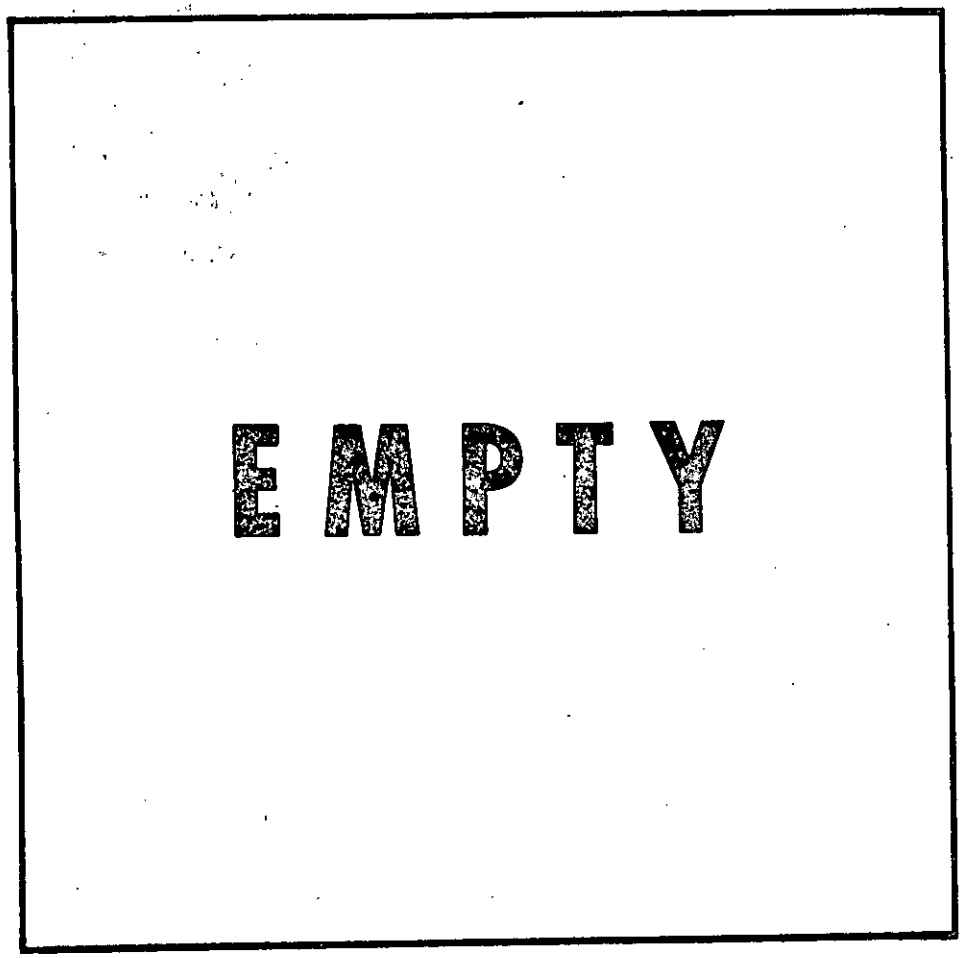
* Vehicle placarding (rail or highway) requirement is keyed to this label.

NOTES:
1. The bottom half of the above labels may be printed in the language of the country of origin.
2. The United Nations hazard class number "7" must be black, approximated .25 inch (6.4 mm) high.
3. The above described labels are mandatory for use after January 1, 1975.

FIGURE 7. Radioactive material labels (UN class 7).



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SF 417

Labels for empty containers must be not less than 6.00 inches (152.4 mm) on each side, white in color, and printed in letters not less than 1.00 inch (25.4 mm) high in black ink.

FIGURE 8. Empty container label.



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MIL-HDBK

PG. 249

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50

Pg. 50

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11.3.13 Additional shipper's requirements. These requirements summarize the shipper's requirements specified herein (see 9.1.1 and 11.3.11.6). After selecting the proper packaging for the specific contents, and having considered the radiation level limits, contamination limits, and the label requirements, the shipper must also check for compliance with the following:

a. Other package markings - The outside of the package must also be marked with the appropriate specification number (see 49 CFR 173.24 (c)(i)) or special permit number, if applicable, and also with the proper shipping name as shown in the commodity list (see 49 CFR 172.101. The most commonly used proper shipping names for radioactive materials are listed in table XI:

TABLE XI. Principal commodity descriptions (proper shipping names) for radioactive materials (R.A.M.) (from 49 CFR 172.100).

| | |
|---|---|
| Radioactive materials, small quantities | (see 49 CFR 173.391(a)) |
| Radioactive devices | (see 49 CFR 173.391(b)) |
| Fissile radioactive materials | (see 49 CFR 173.396) |
| Radioactive materials, low specific activity (L.S.A.) | (see 49 CFR 173.392) |
| Radioactive materials, special form | (see 49 CFR 173.394) |
| Radioactive materials, N.O.S. | (see 49 CFR 173.395) |

b. Shipping papers - Certain essential elements of information must also be included on the shipping paper description (see 10.2.3). These requirements also apply to packages containing small quantities and radioactive commodities (11.3.10.2) except that the notation "No Label Required" must be entered in lieu of the type of label applied.

c. Shipper's certification - The shipping papers must include a certificate signed by the shipper, which reads as specified in paragraph 10.2.3.1.

d. Security seal - The outside of each radioactive materials package must incorporate a feature such as a seal, which is not readily breakable and which, while intact, will be evidence that the package has not been illicitly opened (49 CFR 173.393 (b)). For this requirement, some ingeniuty may be called for on the part of the package designer, especially on such packages as fiberboard cartons and wooden boxes. The regulations also require that "inner shield closures must be positively closed to prevent loss of contents." In connection with these requirements, the sorry experience of the past has proven that too often an attempt has been made by shippers to utilize a padlock as both a security seal and a closure mechanism. Most padlocks, however, are not even a good security seal, let alone a closure device, because it is usually not possible with most types of padlocks to ascertain if they have been illicitly opened. Such combinations as serially numbered lead wire seals, in combination with such closure mechanisms as slotted screw-in plugs, bolted flanges, and positive-action shutter mechanisms are usually the best approach toward meeting the dual requirements.



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e. Smallest dimension - The smallest outside dimension of any radioactive materials package must be four inches or greater (49 CFR 173.393 (c)).

f. Liquid packaging provisions (49 CFR 173.393 (g)) - Liquid radioactive material must be packaged in or within a leak-resistant and corrosion-resistant inner container. In addition, (1) the packaging must be adequate to prevent loss or dispersal of the radioactive contents from the inner container, if the package were subjected to the 30-foot drop test prescribed in 49 CFR 173.398(c)(2)(i); or (2) enough absorbent material must be provided to absorb at least twice the volume of the radioactive liquid contents. The absorbent material may be located outside the radiation shield only if it can be shown that if the radioactive liquid contents were taken up by the absorbent material the resultant dose rate at the surface of the package would not exceed 1,000 millirem per hour. Care should be exercised by the package designer to assure that the positioning of the absorbent material about the liquid-containing vessel is such that the "absorber will absorb" in the event of leakage from the vessel.

g. Surface temperature of package - Maximum surface temperature limits on packages, resulting from radioactive thermal decay energy of the contents are prescribed in 49 CFR 173.393 (e). The limit is either 122°F. or in the case of full load or sole-use shipments, 180°F.

h. Quality assurance requirements (49 CFR 173.393(m) and (n)) - The regulations also prescribe certain quality assurance requirements before the first shipment of a radioactive materials package and before each shipment of a package. With regard to packages of liquids containing in excess of type A quantity, destined for shipment by air, an additional requirement (49 CFR 173.393(n)(7)) is imposed such that the containment system of each package offered for shipment must be tested to assure that it will remain leak-free in a specified ambient reduced atmosphere (0.5 atmosphere).

11.3.14 Carrier requirements in handling of radioactive materials packages. Up to this point, this handbook has been conterned principally with the regulatory requirements applicable to the shipper, or in some cases indirectly to the package designer. The reason for this is simple. Most of the regulatory requirements for the assurance of safety are through proper packaging. It follows, therefore, that the majority of these requirements will be ones which must be met by the shipper. The principal carrier responsibilities in transport of radioactive materials are as follows: (See 9.1.2 herein.)

a. Certification by shipper - CArriers may not accept for transport any packages of radioactive materials which have not been properly certified by the shipper pursuant to 49 CFR 173.204. This certificate is relied upon by the carrier, as evidence that the packaging is in accordance with the regulatory requirements. In the case of air shipments, one signed copy of the shipper's certificate must accompany the shipment, with the originating air carrier retaining a second copy.

b. Placarding - The carrier must apply the appropriate placard to the transport vehicle (rail or highway) if any RAM package on board bears a "Radioactive Yellow - III" label. The format for these placards is specified under paragraph 11.3.11.6.

c. Stowage control by maximum transport index (see 11.3.6.1) - The carrier must assure that the total transport index does not exceed 50 for any group of "Yellow-labeled" packages in a single transport vehicle or storage location, and further he must assure that such groups of yellow-labeled packages are kept separated from undeveloped film shipments and areas continuously occupied by



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persons, in accordance with a table of storage time vs. the graded total transport index.

d. Reporting of incidents (see 9.3) - The carrier must assure that the shipper and the DOT are notified in the event of fire, accident, breakage, or suspected radioactive contamination involving shipment of RAM and also that vehicles, areas, or equipment in which RAM may have spilled, are not placed in service again until they have been surveyed and decontaminated.

The reporting requirement cited above is not necessarily a means of receiving technical assistance in radiological monitoring in the event of a transportation incident. To obtain such assistance any person may call upon the services of the IRAP (Interagency Radiological Assistance Plan). The map of regional coordinating offices and telephone numbers for radiological assistance and geographical areas of responsibility is available from the DOT. Since its inception, over 10 years ago, the IRAP has served to assist many persons in obtaining technical guidance in coping with radiation emergencies. The IRAP organization works closely with other state, Federal, Military, and regional groups in integrating the technical capabilities of those groups.

e. Shipping papers (10.2.3) - Carriers must prepare and carry with the shipments the appropriate shipping papers based on the information derived from the shipper's shipping papers. For water shipments, a Dangerous Cargo Manifest and storage plan is also required.

f. Notification to pilot (for aircraft shipments) see 11.6.3 - The aircraft operator must notify the pilot in command of the aircraft of the name, type of label, quantity, and location of any "hazardous material," such as radioactive materials packages. The cargo load manifest must be conspicuously marked to indicate the presence of such packages (49 CFR 175.33).

11.3.15 Discussion of more frequently noted discrepancies in radioactive materials shipments. This discussion is intended to serve as an aid to both shippers and carriers by making them more aware of the nature of the more frequently observed items of noncompliance in radioactive materials shipments. Such items are generally either of a safety related nature, i.e., improper packaging, excessive radiation or contamination, or an administrative nature, i.e., improper shipping paper description, illegible labels, etc.

a. By shippers.

(1) Excessive radiation levels (49 CFR 173.389(i)) - Fortunately, this item is not noted frequently. It does, however, rank along with improper packaging, as the most serious type of shipper violation of the safety requirements for transportation of radioactive materials. Excessive radiation levels on packages of radioactive materials are generally the result of either of two causes:

a. An excessive quantity of material in the package relative to the shielding capability of the package design; or
b. A failure to properly secure a shielded closure mechanism, a faulty closure mechanism, etc.

For "special form" radioactive sources the hazard then becomes one of excessive radiation levels. For dispersible, non-"special form" radioactive materials, the



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hazard may then be due to both excessive radiation and possible dispersal of loose contamination. In some cases, it has been noted that suppliers of radioisotopes of very short half-life will place into their packages at the time of loading more than the total quantity allowed for shipment with the view that at the time of actual shipment, the radiation levels will be within the limits, due to radioactive decay. Such a practice can, of course, result in a violation if the package is offered for shipment too soon, with the radiation level not having decayed to the point of being within the limits.

(2) Improper packaging - This, of course, is also a most serious safety item and is closely related to the excessive radiation level in that an improper package may be one which does not incorporate sufficient thickness of shielding for the quantity of material involved. Improper packaging also may result from not using a container as authorized in the regulations or under a DOT Special Permit.

Even when an approved package is utilized, however, if it is not in its proper condition as required by its design, safety may be affected. Good quality control practices by shippers of radioactive materials are paramount. A relevant requirement of 49 CFR 173.22 reads as follows:

"... the shipper shall be responsible to determine that shipments of hazardous materials are made in containers which..., have been made, assembled with all their parts or fittings in their proper place, and marked in accordance with applicable specifications..."

On some types of packages, i.e., steel drums, hinged lid boxes, etc., provision for a security seal is fairly simple. On many other types, i.e., wooden boxes, fiberboard cartons, much more thought and ingenuity in designing of a seal to meet the requirements will be necessary. The use of padlocks as a security seal may not, in all cases, be appropriate since many types of padlocks may be illicitly opened and closed again without knowledge of the consignee.

(3) Improper labels - Incorrect labeling of radioactive materials packages is a common deficiency. The most frequent error is "overlabeling," i.e., the use of a yellow-III label where a white-I or yellow-II label would have been adequate. When done too often, such "overlabeling" can cause a loss in the distinction which is intended to be implied by the graded series of the three labels which are used to indicate the degree of control the package requires.

(4) Illegible/incorrect label notations - This item should speak for itself. Needless to say, shippers should exercise care to insert legible, durable entries on the labels. These entries call for noting the "contents," "number of curies," and "transport index." In future rule making, more precise guidance is planned on this aspect. In the meantime, the entry of "n.o.s." for "not otherwise specified" under "contents" is totally inappropriate. The name of the isotope must be entered, as taken from the list of radionuclides in 49 CFR 172.101. Also, care should be taken to clearly indicate in the "number of curies" entry whether the quantity is in terms of curies, millicuries or microcuries. Finally, the "transport index must be rounded up to the next highest tenth as prescribed in 49 CFR 173.389(i).



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(5) *Improper or incomplete shipping paper description* - The basic requirements for the shipping paper description are prescribed in 49 CFR 172.200 (see 10.2.2). The first entry on the shipper's bill of lading should always be the applicable "proper shipping name" as taken from the commodity list in 49 CFR 172.101. Any other description or information not inconsistent therewith may follow. However, there are at present only eight different proper shipping names for radioactive materials in the commodity list. The six most commonly used are listed in table XI. The appropriate one of these must be used. Care should be exercised to properly enter the other information as required by 49 CFR 172.200). In the shipper's certification, care should further be exercised to add the notation that the shipment meets the requirements for passenger-carrying aircraft, when applicable.

(6) Inadequate provision for liquid contents - The regulations prescribe certain additional packaging requirements for liquid radioactive materials. As required by 49 CFR 173.393(g), these are basically either of two options, i.e., a "performance" requirement wherein the package must withstand a 30-foot drop test without loss of liquid contents, or absorbent material must be used to absorb the liquid contents in event of breakage of the primary liquid container.

In meeting the overall provision, if the liquid absorbent option is not utilized, the shipper is then responsible to properly evaluate the liquid "contaminant system" of his specific packaging against the "performance" requirement option. Such shipper should be prepared to demonstrate the method in which he has determined that his packaging complies with this requirement.

b. By carriers:

(1) Acceptance of consignments without shipper's certification - Each of the carrier regulations (49 CFR 172.204, 176.27) prescribe a requirement that shipments of regulated hazardous materials may not be accepted by a carrier unless accompanied by the appropriate certification by the shipper that the packages have been packaged, marked, and labeled in accordance with the regulations. This certification must be signed by the shipper and it is his legal representation to the carrier that the safety requirements of the shipment are in order. Needless to say, carriers must refuse acceptance of hazardous materials shipments which are tendered to them without this certification.

(2) Failure to prepare proper shipping paper description - Each of the carrier regulations (49 CFR 172.200, 176.24) require that carrier's manifests, waybills, etc., carry the appropriate proper shipping name and an indication of the type of label applied or the notation "no label applied," when applicable. In many cases, carriers in preparing their shipping papers, do not properly transpose these essential items of information from the shipper's papers.



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(3) Acceptance of radioactive materials consignments exceeding the 50 transport index maximum per vehicle - In many cases carriers either do not appear to be aware of this limitation or fail to follow it. Concurrent with this requirement, each of the carrier regulations contain a table which prescribes certain stowage distances and in certain cases stowage times for accumulations of radioactive materials packages, based on the total transport index. These stowage controls are intended to provide the necessary segregation distance of packages from areas occupied by persons or photographic film. An increased effort is needed by carriers to more completely train and educate their personnel in this requirement.

(4) Failure to placard transport vehicles - For any rail or highway vehicle transporting any quantity of radioactive materials packages bearing the radioactive yellow-III label, the carrier is required to display the appropriate placard (49 CFR 172.203 (49 CFR 172.203(j). Intentional failure to placard vehicles so as to avoid certain bridge, turnpike, tunnel or other travel restrictions is a very serious offense.

11.4 Transportation by water carrier (49 CFR 176). The regulations and requirements for transportation of radioactive materials and commodities by water carrier are in general identical to the requirements specified herein, with the following additions or exceptions: (These were formerly included in 46 CFR 146.19, but were revoked effective July 1, 1976.)

11.4.1 Shipping papers. Shipping paper requirements for shipments by water are almost identical to the requirements for shipments by rail or highway (see 10.2.3). The regulations are divided into requirements for domestic shipments, export shipments, and import shipments. The two significant differences are as follows: First, for other than domestic shipments, when the proper shipping name of a hazardous material is an "N.O.S." (Not Otherwise Specified) entry in the list of hazardous materials (49 CFR 172.10), this description must be qualified by the "chemical name of the commodity" in parenthesis, e.g. "Corrosive liquid, n.o.s." (caprylyl chloride). Secondly, in connection with the entry (on the shipping documents) of each hazardous material, the description is required to include the kind and color of label applied to the package when the label is required.

11.4.2 General stowage requirements aboard vessels. The following requirements for water carriers are in addition to those listed under 11.3 for rail or highway carriers (49 CFR 176.700):

a. Packages or containers bearing the radioactive white or radioactive yellow labels shall be kept separated from living accommodations and from spaces that may be continually occupied by persons, except those exclusively reserved for couriers specially authorized to accompany such shipments. Packages bearing the radioactive yellow label shall be separated from persons and undeveloped photographic and radiographic films in accordance with table XII. (Mail bags shall be assumed to contain undeveloped film and separated from radioactive materials as for film.) The sum of the transport indexes for the shipment is the arithmetic total of the transport indexes of of each package as stated on the label.



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b. The number of packages bearing the radioactive-yellow label shall be so limited that the sum of the transport indexes does not exceed 200 in any one ship unless authorized by the Commandant. In addition to complying with the segregation distance in paragraph (a) of this section, no single group of packages shall have a total Transport Index of more than 50 and each such group shall be handled and stowed not closer to any other group than 20 feet. The requirements of this paragraph do not apply to low specific activity materials as defined in a compact stack nor need they apply in case of a full load where the consignor has the exclusive use of the whole vessel, providing that the transport indexes of Fissile Class II packages aboard the vessel do not exceed 50.

c. Packages of radioactive materials which are significant heat sources shall not be overstowed with any other cargo. If stowed below decks, the hold or compartment in which stowed must be ventilated.

d. For radioactive materials shipments requiring supplementary operational procedures, the package shall be stowed in an accessible location and the necessary operational instructions furnished to the vessel's master.

e. Fissile Class III shipments shall be separated by at least 20 feet from other packages bearing radioactive-yellow labels during handling and stowage.

f. All containers of radioactive materials shall be carried by the handles when handles are provided.

g. When "On Deck in open" stowage is permitted for any substances by DOT regulations, it shall apply only to the substances when packaged in authorized, waterproof packagings.

h. No person shall remain unnecessarily in a hold or compartment or in the immediate vicinity of a deck cargo space containing radioactive materials. The shipper shall furnish the carrier with such information and equipment as is necessary for the protection of the carrier's employees, stevedores, or other persons engaged in the handling of such cargo. In no instance shall any person who must necessarily remain in a hold, compartment, or deck cargo space containing radioactive material be exposed to a total of more than 100 millirem in any 7-day period. Provided further, that a maximum whole body dose of 500 millirem per year is not exceeded. The radiation level in any space or area on board continuously occupied by passengers, crew, or shipments of animals shall not exceed 0.5 millirem per hour at any time during transportation.

11.4.3 Segregation from other cargoes.

a. Radioactive materials shall not be stowed in the same hold or compartment with Class A, Class B, or Class C explosives, flammable gases, flammable liquids, oxidizion materials, flammable solids, corrosive liquids, or cotton.

b. Radioactive materials shall be stowed away from (i.e., with intervening cargo or ship's structure) nonflammable gases and foodstuff.

c. Radioactive materials shall be separated from mail and undeveloped films in accordance with 11.4.2.a.

11.4.4 Care following leakage or shifting of radioactive materials.

a. In case of fire, collision, or breakage involving shipments of radioactive materials, other than the materials of low specific activity, the package or material shall be segregated from unnecessary contact



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TABLE XII. Safe distance for persons and undeveloped films.

| Sum of transport indexes of the packages | Minimum distance in feet from living accommodation or regularly occupied working space | | | | Minimum distance 1/ in feet from undeveloped films or plates | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Less than 24-hour voyage | 24 up to 48-hour voyage | Over 2 up to 4-day voyage | | | | Over 4 up to 9-day voyage | | | | | |
| | A | B | C | D | A | A | A | B | C | D | A | B | C | D | | |
| 0 through 0.5 | 10 | 10 | 10 | 10 | 3 | 5 | 10 | 10 | 10 | 10 | 15 | 10 | 10 | 10 | | |
| Over 0.5 through 1.0 | 10 | 10 | 10 | 10 | 5 | 7 | 15 | 10 | 10 | 10 | 20 | 10 | 10 | 10 | | |
| Over 1.0 through 2.0 | 10 | 10 | 10 | 10 | 7 | 12 | 20 | 10 | 10 | 10 | 25 | 15 | 10 | 10 | | |
| Over 2.0 through 5.0 | 10 | 10 | 10 | 10 | 12 | 16 | 35 | 10 | 10 | 10 | 40 | 15 | 10 | 10 | | |
| Over 5.0 through 10.0 | 15 | 10 | 10 | 10 | 16 | 25 | 40 | 15 | 10 | 10 | 55 | 20 | 10 | 10 | | |
| Over 10.0 through 25.0 | 25 | 10 | 10 | 10 | 25 | 35 | 65 | 20 | 10 | 10 | 100 | 30 | 10 | 10 | | |
| Over 25.0 through 50.0 | 40 | 15 | 10 | 10 | 35 | 50 | 90 | 30 | 10 | 10 | 145 | 45 | 10 | 10 | | |
| Over 50.0 through 100 | 60 | 20 | 10 | 10 | 50 | 70 | 140 | 45 | 10 | 10 | 210 | 65 | 10 | 10 | | |
| Over 100 through 200 | 80 | 25 | 10 | 10 | 70 | 100 | 190 | 55 | 10 | 10 | 285 | 85 | 15 | 10 | | |
| Over 200 through 300 2/ | 110 | 30 | 10 | 10 | 90 | 125 | 230 | 70 | 10 | 10 | 320 | 105 | 15 | 10 | | |
| Over 300 through 400 2/ | 130 | 35 | 10 | 10 | 100 | 145 | 260 | 80 | 15 | 10 | 360 | 120 | 20 | 10 | | |

| Sum of transport indexes of the packages | Minimum distance 1/ in feet from undeveloped films or plated - Con. | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Over 9 up to 16-day voyage | | | | Over 16 up to 25-day voyage | | | | Over 25 up to 36-day voyage | | | | Over 36 up to 49-day voyage | | | |
| | A | B | C | D | A | B | C | D | A | B | C | D | A | B | C | D |
| 0 through 0.5 | 20 | 10 | 10 | 10 | 20 | 10 | 10 | 10 | 25 | 15 | 10 | 10 | 30 | 15 | 10 | 10 |
| Over 0.5 through 1.0 | 25 | 15 | 10 | 10 | 30 | 15 | 10 | 10 | 35 | 20 | 10 | 10 | 45 | 20 | 10 | 10 |
| Over 1.0 through 2.0 | 35 | 15 | 10 | 10 | 45 | 20 | 10 | 10 | 55 | 25 | 10 | 10 | 65 | 30 | 10 | 10 |
| Over 2.0 through 5.0 | 60 | 20 | 10 | 10 | 70 | 25 | 10 | 10 | 90 | 30 | 10 | 10 | 100 | 35 | 10 | 10 |
| Over 5.0 through 10.0 | 85 | 30 | 10 | 10 | 110 | 35 | 10 | 10 | 135 | 40 | 10 | 10 | 155 | 45 | 10 | 10 |
| Over 10.0 through 25.0 | 140 | 45 | 10 | 10 | 170 | 50 | 10 | 10 | 200 | 60 | 10 | 10 | 230 | 70 | 10 | 10 |
| Over 25.0 through 50.0 | 195 | 60 | 10 | 10 | 240 | 70 | 10 | 10 | 275 | 85 | 15 | 10 | 315 | 105 | 15 | 10 |
| Over 50.0 through 100 | 270 | 85 | 15 | 10 | 320 | 95 | 15 | 10 | 360 | 115 | 20 | 10 | 400 | 145 | 20 | 10 |
| Over 100 through 200 | 325 | 110 | 20 | 10 | 400 | 145 | 20 | 10 | 450 | 180 | 25 | 10 | 500 | 170 | 30 | 10 |
| Over 200 through 300 2/ | 400 | 145 | 25 | 10 | 460 | 175 | 25 | 10 | 500 | 220 | 30 | 10 | (3/) | 240 | 35 | 10 |
| Over 300 through 400 2/ | 450 | 165 | 30 | 10 | 500 | 200 | 30 | 10 | (3/) | 235 | 35 | 10 | (3/) | 270 | 40 | 10 |

See notes on following page.





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1/ Column A applies when no intervening cargo or bulkheads screen the radioactive material from the living accommodation or undeveloped photographic film or plate. Column B applies when the radioactive material is to be surounded by at least 2 feet of cargo of unit density and at least 1 steel bulkhead between the radioactive material and the living accommodation or undeveloped photographic film or plate.

Column C applies when radioactive material is to be surrounded by at least 6 feet of cargo of unit density and at least 2 steel bulkheads between the radioactive material and the living accommodation or undeveloped photographic film or plate.

Column D applies when the radioactive material is to be surrounded by at least 14 feet of cargo of unit density and at least 2 steel bulkheads between the radioactive material and the living accommodation or undeveloped photographic film or plate.

"Cargo of unit density" means cargo stowed at a density of 1 ton per 36 cubic feet (1 ton metric per cubic meter). Where the density of the cargo is less than this, the depth of the cargo specified in this note for columns B, C, and D (i.e., 2 feet, 6 feet, and 14 feet) must be increased in proportion.

"Minimum distance" means the least distance in any direction, whether vertical or horizontal.

2/ The total consignment on board at any time must not exceed transport indexes totaling 200 without prior authorization by the Commandant (see 46 CFR 146.19-35(b)).

3/ Not to be carried unless screening by other cargo and bulkheads can be arranged in accordance with columns B, C, or D.



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with personnel. In case of obvious leakage, or if the inside container appears to have been damaged, the section (hold, deck area, or compartment) containing this cargo must be isolated as much as possible, and care should be taken to prevent radioactive material from entering the body through contact, inhalation, or ingestion. No person shall be allowed to handle the material or to remain in the vicinity until qualified personnel are present to supervise. In any incident in which radioactive materials are involved in fires or are damaged, the shipper and the District Commander of the U.S. Coast Guard, or his authorized representative as prescribed by(see 9.3.1) 49 CFR 171.15 , of this chapter, having supervision over the port or place where the vessel is located or bound, shall be notified immediately.

(b) Holds and compartments used for the transport of low activity materials as full loads or in which leakage of radioactive materials has occurred because of fire, collision or breakage, shall not be used for other goods until decontaminated as specified in 10.4 and Appendix 1 of this handbook.

11.5 Transportation by the United States Postal Service (39 CFR 124.2e and Publication 6). The regulations and requirements for transportation of radioactive materials and commodities by parcel post or mail carrier are in general identical to and conform to the requirements 49 CFR and 11.3 herein, under the conditions of 11.5.1 and 11.5.2.

11.5.1 Nonmailable matter. Any package of radioactive matter bearing any of the Department of Transportation's Radioactive white-I, Radioactive yellow-II, or Radioactive yellow-III labels (see 11.3.11) or which contains quantities of radioactive material in excess of those authorized under 11.3.10 is nonmailable. These labels are illustrated in figure 7.

11.5.2 Mailable matter. Authorized mailable radioactive materials include only those which are classified as "small quantities" of radioactive materials or radioactive devices, as prescribed in 49 CFR 173.391. These authorized materials, the maximum quantities mailable, and the conditions under which they may be mailed are described in table IX and 11.3.10.2 herein.

NOTE: Radioactive Commodities shall not be entered into the U. S. Postal System except when approved by the cognizant transportation officer of the specifiec shipping activity.

11.6 Transportation by aircraft (49 CFR 175). The regulations and requirement for transportation of radioactive materials and commodities by aircraft are, in general identical to and conform to the requirements specified herein for highway or rail carrier, with the following additions or exceptions:

11.6.1 Exceptions. These regulations do not apply to the following:

a. Shipments of radioactive materials via cargo-only aircraft, made by or under the direction or supervision of the U. S. Atomic Energy Commission or the Department of Defense, which are escorted by personnel especially designated by or under the authority of that Commission of Department for the purpose of national security.

b. Shipments of radioactive materials that meet those requirements in 49 CFR 173.391 (see 11.3.10) that exempt them from packaging, marking, and labeling requirements for shipment by rail. (Note: This exception is not valid after May 3, 1977.)

c. Human beings and animals implanted with medical devices (heart pacemaker, etc) that contain radioactive material.



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11.6.2 Certification and shipping paper requirements.

a. No shipper may offer, and no person operating an aircraft may knowingly accept, any hazardous material for shipment in an aircraft unless there is accompanying the shipment a clear and visible statement that the shipment complies with the requirements of 49 CFR 172 (see 10.2.3) and (11.3.14). In the case of shipments in passenger-carrying aircraft, the shipper shall also state that the shipment complies with the requirements 49 CFR 172 for carrying hazardous materials (see 11.6.6) in passenger-carrying aircraft. The shipper or his authorized agent shall sign the statement or stamp it with a facimile of his signature. The person operating an aircraft may rely on the shipper's statement as prima facie evidence that the shipment complies with the requirements of DOT.
b. The shipper shall execute the required certificates in duplicate. One signed copy accompanies the shipment and the originating air carrier retains the other signed copy.
c. For the purposes of this Handbook, a passenger-carrying aircraft is an aircraft that carries any person other than a crewmember, company employee, an authorized representative of the United States, or a person accompanying the shipment.

11.6.3 Notification of pilot in command. Whenever articles subject to the provisions of 49 CFR 175 are carried in an aircraft, the operator of the aircraft shall inform the pilot in command, before takeoff, in writing, of the shipping name and the classification of each hazardous material as prescribed in 49 CFR 172.11, the quantity in terms of weight, volume or as otherwise appropriate, and the location of the hazardous materials in the aircraft. The person marking the cargo-load manifest shall mark it conspicuously to indicate the hazardous materials (see 11.3.14), and the results of the inspections required by 49 CFR 175(b), 175.700.

11.6.4 Labeling requirements. Except as otherwise provided the shipper shall label each hazardous material that is acceptable for transportation in air commerce, with the appropriate label required by 11.3.11 herein, even though that article is exempt from from those labeling requirements because of quantity and packing limitations (see 11.3.10.2). Label with "Cargo-aircraft only" label where required.

11.6.5 Quantity limitations. No person may carry aboard an aircraft a number of packages of radioactive materials that make the total transport index number (determined by adding together the transport index numbers shown on the labels of the individual packages) more than 50.

11.6.6 Passenger-carrying aircraft. No person shall carry (or offer for transportation) any radioactive item in a passenger-carrying aircraft (see 11.6.2.c) unless there is a signed printed certificate stating that the radioactive materials involved are intended for use in, or incident to, research, or medical diagnosis or treatment, so long as such materials as prepared for and during transportation do not pose an unreasonable hazard to health and safety and meets the requirements of 49 CFR 175.

11.6.7 Cargo-only aircraft. No person shall carry any radioactive item in a cargo-only aircraft except those items permitted on passenger-carrying aircraft under 11.6.6 above and except items that:

a. Are specified in 49 CFR 172 (see 11.3) as acceptable for shipment by rail express.
b. Do not exceed the maximum quantity for each outside container specified in 49 CFR 172 for rail.
c. Are packaged, marked, and labeled as specified in 49 CFR 173 (11.3) herein for shipment by rail.
d. For the purposes of this part, a cargo-only aircraft is any aircraft that is not a passenger-aircraft.



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11.6.8 Special requirements for radioactive materials and commodities. No person may place any package of radioactive materials bearing "radioactive yellow-II" or "radioactive yellow-III" labels in aircraft closer than the distances shown in table XIII to a space (or dividing partition between spaces) which may be continuously occupied by people, or shipments of animals, or closer than the distances shown in table XIII to any package containing undeveloped film (if so marked). If more than one of these packages is present, the distance shall be computed from table XIII on the basis of the total transport index numbers shown on the labels of the individual packages in the aircraft.

TABLE XIII. Radioactive cargo items - Temporary storage requirements.

| Total transport index | Minimum separation distances in feet to nearest undeveloped film for various times of transit | | | | | Minimum distance in feet to area of persons, or minimum distance in feet from dividing partition of cargo compartment |
|---|---|---|---|---|---|---|
| | Up to 2 hours | 2-4 hours | 4-8 hours | 8-12 hours | Over 12 hours | |
| None - - - - - - | 0 | 0 | 0 | 0 | 0 | 0 |
| 0.1 to 1.0 - - - | 1 | 2 | 3 | 4 | 5 | 1 |
| 1.1 to 5.0 - - - | 3 | 4 | 6 | 8 | 11 | 2 |
| 5.1 to 10.0 - - | 4 | 6 | 9 | 11 | 15 | 3 |
| 10.1 to 20.0 - - | 5 | 8 | 12 | 16 | 22 | 4 |
| 20.1 to 30.0 - - | 7 | 10 | 15 | 20 | 29 | 5 |
| 30.1 to 40.0 - - | 8 | 11 | 17 | 22 | 33 | 6 |
| 40.1 to 50.0 - - | 9 | 12 | 19 | 24 | 36 | 7 |

11.6.9 Special requirements for fissile class III radioactive materials. No person may carry aboard any aircraft any package of fissile class III radioactive material (as defined herein), except as follows:

a. On a cargo-only aircraft which has been assigned for the sole use of the consignor for the specific shipment of fissile radioactive material. Instructions for such sole use must be provided for in special arrangements between the consignor and carrier, with instructions to that effect issued with shipping papers; or
b. On any aircraft on which there are no other packages of radioactive material required to bear one of the "radioactive" labels described in 11.3.11 herein. Specific arrangements must be effected between the shipper and carrier, with instructions to that effect issued with the shipping papers; or
c. In accordance with any other procedure specifically approved by the DOT.

11.7 Requirements for transportation of radioactive commodities by military aircraft (DSAM 4145.3, AFM 71-4, TM38-250, NAVSUP PUB 505, MCO P 4030.19). The regulations and requirements for marking and labeling of radioactive materials and commodities for transportation by military aircraft shall, in general, conform to the requirements of 49 CFR 175 (see 11.6), with the following additional requirements.



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11.7.1 Transport index number. (See definitions and 11.3.9.1.) Unless otherwise exempted by 11.3.10.2 herein, the total transport index number of radioactive packages shipped in a single shipment must be noted on all copies of the shipping document. DD Form 1348-1, "DoD Single Line Item Release/Receipt Document", and signed copies of DD Form 1387-2, "Special Handling Data/Certification".

11.7.2 Special handling-data certification form. Unless otherwise exempted by 11.3.10.2 herein, DD Form 1387-2 shall be prepared as shown below for each exempt radioactive item listed herein (see MIL-HDBK-758).

SPECIAL HANDLING DATA/CERTIFICATION PARA 13-4 PAGE 13-5

| TRANSPORTATION CONTROL NUMBER | NOMENCLATURE OF ITEM | NET EXPLOSIVE WEIGHT | GROSS WEIGHT |
|---|---|---|---|
| * | RADIOACTIVE DEVICES<br>DOT CLASS – RAM<br>DOT LABEL – EXEMPT | N/A | * |
| DESTINATION<br>* | | | |

HANDLING INSTRUCTIONS

PROPERTIES: CONTAINS RAM IN NON-DISPERSIBLE FORM.

S/H: HANDLE CAREFULLY TO AVOID BREAKAGE. IF DAMAGE IS SUSPECTED CHECK FOR RADIATION AND HANDLE ACCORDINGLY.

L/S GP 18 NOTE 5

SHIPPER CERTIFICATION: This is to CERTIFY that the contents of the packages in this shipment are properly described by name and are packed, marked, and in proper condition for transportation in accordance with:

☒ SUBPARAGRAPH 13-4d ARM 71-4; TM 38-250, NAVWEPS 16-03-500 AND MCO P4030.19

☐ OFFICIAL AIR TRANSPORT RESTRICTED ARTICLES TARIFF 6, CAB NO. 83

☐ OTHER (Specify)

☒ SHIPMENT WITHIN PASSENGER/CARGO AIRCRAFT LIMITATIONS

| SIGNATURE | DATE |
|---|---|
| * | * |

DD FORM 1387-2, 1 APR 68 GPO 891-342 REPLACES EDITION OF 1 APR 68, WHICH MAY BE USED.

*SHIPPER MUST ENTER APPROPRIATE DATA IN BLOCKS INDICATED WITH ASTERISK. This label is applicable when shipping manufactured articles such as instruments, Clocks, Electronic Tubes or Apparatus, or other similar devices, having RADIOACTIVE materials (other than liquid) in a nondispersible form as a component part. The outside of each inner and exterior container must be marked "RADIOACTIVE".

FIGURE 11. Special handling data/certification form.



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Custodians:
Army - EL
Navy - YD
Air Force - 85

Review activities:
Army - AV, EA
Navy - AS, MC, EC, SA
Air Force - 03

User activities:
Army - ME, SM
Navy - CG, MC, OS
Air Force - 80

Preparing activity:
DSA - ES

(Project MISC 0A05)



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# SECTION A

## RADIOISOTOPE DATA

## TRANSPORT GROUPS - EXEMPT QUANTITIES AND CONCENTRATIONS

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Section A. This section contains radioisotope (radionuclide) data for marking and labeling radioactive materials or commodities for transportation, use, and storage in accordance with the requirements of the DOT (49 CFR 173.390) and the NRC (10 CFR 20 and 10 CFR 30).

A-1 Transport group requirements. Radioactive materials or commodities that are not exempt from specification packaging, marking, and labeling requirements under the conditions of 11.3.10.2, shall include the appropriate transport group when presented for transportation.

A-1.1 Applicability. The transport groups shown in table A-1 are applicable to the following modes of transportation:

| | | |
|---|---|---|
| (1) | Rail carrier | (Title 49 CFR 174) (See 11.3) |
| (2) | Highway carrier | (Title 49 CFR 177) (See 11.3) |
| (3) | Water carrier | (Title 49 CFR 176) (See 11.4) |
| (4) | Mail carrier | (Title 39 CFR and Pub. No. 6) (See 11.5) |
| (5) | Air carrier | (Title 49 CFR 195) (See 11.6) |
| (6) | Air carrier (military) | (DSAM 4145.3, AFM 71-4, TM38-250, NAVSUPPUB 505, and MCO P 4030.19) (See 11.7) |

A-2 Classification by transport groups of radioisotopes. Radioisotopes are listed in transport groups I through VII according to their radiotoxicity and relative potential hazard in transportation.

A-2.1 Transport group I. This group is the most hazardous. Any radioisotope not listed in the table A-1, and which has an atomic number of 82 or over, shall be assigned this group, provided the radioactive half-life is 10 years or less.

A-2.2 Transport group II. Any radioisotope not listed in the table A-1, and which has an atomic number from 1 to 81, shall be assigned this group, provided the radioactive half-life is between 1,000 days and 10 years.

A-2.3 Transport group III. Any radioisotope not listed in the table A-1 and which has:

(1) An atomic number from 1 to 81, shall be assigned this group provided the radioactive half-life is 0 to 1,000 days or over 10 years; or

(2) An atomic number of 82 or over, shall be assigned this group provided the radioactive half-life is over 10 years.

A-2.4 Transport groups IV, V, and VI. No unlisted radioisotope shall be assigned to these groups.

A-2.5 Transport group VII. This group is the least hazardous. No unlisted radioisotope shall be assigned this group.



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A-2.6 Transport group for a mixture of radioisotopes. To determine the transport group for a mixture of radioisotopes the following shall apply:

(1) If the identity and respective activity of each radioisotope is known, the permissible activity of each radioisotope shall be such that the sum, for all groups present, of the ratio between the total activity for each group to the permissible activity for each group will not be greater than unity.

(2) If the groups of the radioisotopes are known but the amount in each group cannot be reasonably determined, the mixture shall be assigned to the most restrictive group present.

(3) If the identity of all or some of the radioisotopes cannot be reasonably determined, each of those unidentified radioisotopes shall be considered as belonging to the most restrictive group which cannot be positively excluded.

(4) Mixtures consisting of a single radioactive decay chain where the radioisotopes are in the naturally occurring proportions shall be considered as consisting of a single radioisotope. The group and activity shall be that of the first member present in the chain, except if a radioisotope "x" has a half-life longer than that of the first member and an activity greater than that of any other member including the first at any time during transportation; in that case, the transport group of the isotope "x" and the activity of the mixture shall be the maximum activity of that isotope "x" during transportation.

A-3 Unlisted radioisotopes or byproduct material (concentrations). For beta and/or gamma emitting byproduct material not listed in table A-1 with half-lives less than 3 years---The column II value shall be $1 \times 10^{-10}$, and the column III value shall be $1 \times 10^{-6}$. (Also see 8.1.2.3 herein.)

A-4 Unlisted radioisotopes (exempt quantities). Any radioisotopes not listed in column IV of table A-1 shall be assigned the following values. (Also see 8.1.2 herein.)

(1) Any alpha emitting radioisotope or mixtures of alpha emitters of unknown composition, the column IV value is-----0.01.

(2) Any radioisotope other than alpha emitting radioisotopes or mixtures of beta emitters of unknown composition, the column IV value shall be----------0.1.

NOTE: Radioisotopes without entries or dashes in columns II, III, and IV of table A-1 were not listed in 10 CFR 20 and 10 CFR 30.



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TABLE A-1. RADIOISOTOPE DATA

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| Actinium (89) | Ac 227 | I | | | |
| | Ac 228 | I | | | |
| Americium (95) | Am 241 | I | | | (.01) |
| | Am 243 | I | | | |
| Antimony (51) | Sb 122 | IV | --- | $3 \times 10^{-4}$ | 100 |
| | Sb 124 | III | --- | $2 \times 10^{-4}$ | 10 |
| | Sb 125 | III | --- | $1 \times 10^{-3}$ | 10 |
| Argon (18) | A 37 | VI | $1 \times 10^{-3}$ | --- | |
| | A 41 | II | $4 \times 10^{-7}$ | --- | |
| | A 41 5/ (uncompressed) | V | --- | --- | |
| Arsenic (33) | Ar 73 | IV | --- | $5 \times 10^{-3}$ | 100 |
| | Ar 74 | IV | --- | $5 \times 10^{-4}$ | 10 |
| | Ar 76 | IV | --- | $2 \times 10^{-4}$ | 10 |
| | Ar 77 | IV | --- | $8 \times 10^{-4}$ | 100 |
| Astatine (85) | At 211 | III | | | |
| Barium (56) | Ba 131 | IV | --- | $2 \times 10^{-3}$ | 10 |
| | Ba 133 | II | | | 10 |
| | Ba 140 | III | --- | $3 \times 10^{-4}$ | 10 |
| Berkelium (97) | Bk 249 | I | | | |
| Beryllium (4) | Be 7 | IV | --- | $2 \times 10^{-2}$ | |
| Bismuth (83) | Bi 206 | IV | --- | $4 \times 10^{-6}$ | |
| | Bi 207 | III | --- | --- | |
| | Bi 210 | II | --- | --- | 1.0 |
| | Bi 212 | III | --- | --- | |
| Bromine (35) | Br 82 | IV | $4 \times 10^{-7}$ | $3 \times 10^{-3}$ | 10 |
| Cadmium (48) | Cd 109 | IV | --- | $2 \times 10^{-4}$ | 10 |
| | Cd 115m | III | --- | $3 \times 10^{-4}$ | 10 |
| | Cd 115 | IV | --- | $3 \times 10^{-4}$ | 100 |
| Calcium (20) | Ca 45 | IV | --- | $9 \times 10^{-5}$ | 10 |
| | Ca 47 | IV | --- | $5 \times 10^{-4}$ | 10 |
| Californium (98) | Cf 249 | I | | | |
| | Cf 250 | I | | | |
| | Cf 252 | I | | | |
| Carbon (6) | C 14 | IV | $1 \times 10^{-6}$ | $8 \times 10^{-3}$ | 100 |
| Cerium (58) | Ce 141 | IV | --- | $9 \times 10^{-4}$ | 100 |
| | Ce 143 | IV | --- | $4 \times 10^{-4}$ | 100 |
| | Ce 144 | III | --- | $1 \times 10^{-4}$ | 1.0 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| Cesium (55) | Cs 131 | IV | --- | $2 \times 10^{-2}$ | 1,000 |
| | Cs 134m | III | --- | $6 \times 10^{-2}$ | 100 |
| | Cs 134 | III | --- | $9 \times 10^{-5}$ | 1.0 |
| | Cs 135 | IV | | | 10 |
| | Cs 136 | IV | | | 10 |
| | Cs 137 | III | | | 10 |
| Chlorine (17) | Cl 36 | III | | | 10 |
| | Cl 38 | IV | $9 \times 10^{-7}$ | $4 \times 10^{-3}$ | 10 |
| Chromium (24) | Cr 51 | IV | --- | $2 \times 10^{-2}$ | 1,000 |
| Cobalt (27) | Co 56 | III | | | |
| | Co 57 | IV | --- | $5 \times 10^{-3}$ | |
| | Co 58m | IV | | | 10 |
| | Co 58 | IV | --- | $1 \times 10^{-5}$ | 10 |
| | Co 60 | III | --- | $5 \times 10^{-4}$ | 1.0 |
| Copper (29) | Cu 64 | IV | --- | $3 \times 10^{-3}$ | 100 |
| Curium (96) | Cm 242 | I | | | |
| | Cm 243 | I | | | |
| | Cm 244 | I | | | |
| | Cm 245 | I | | | |
| | Cm 246 | I | | | |
| Dysprosium (66) | Dy 154 | III | | | |
| | Dy 165 | IV | --- | $4 \times 10^{-3}$ | 10 |
| | Dy 166 | IV | --- | $4 \times 10^{-4}$ | 100 |
| Erbium (68) | Er 169 | IV | --- | $9 \times 10^{-4}$ | 100 |
| | Er 171 | IV | --- | $1 \times 10^{-3}$ | 100 |
| Europium (63) | Eu 150 | III | | | |
| | Eu 152m | IV | | | 100 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | Eu 152 | III | --- | $6 \times 10^{-4}$ | 1.0 |
| | Eu 154 | II | | | 1.0 |
| | Eu 155 | IV | --- | $2 \times 10^{-3}$ | 10 |
| Fluorine (9) | F 18 | IV | $2 \times 10^{-6}$ | $8 \times 10^{-3}$ | 1,000 |
| Gadolinium (64) | Gd 153 | IV | --- | $2 \times 10^{-3}$ | 10 |
| | Gd 159 | IV | --- | 8 x 10-4 | 100 |
| Gallium (31) | Ga 67 | III | | | |
| | Ga 72 | IV | --- | $4 \times 10^{-4}$ | 10 |
| Germanium (32) | Ge 71 | IV | --- | $2 \times 10^{-2}$ | 100 |
| Gold (79) | Au 193 | III | | | |
| | Au 194 | III | | | |
| | Au 195 | III | | | |
| | Au 196 | IV | --- | $2 \times 10^{-3}$ | |
| | Au 198 | IV | --- | $5 \times 10^{-4}$ | 100 |
| | Au 199 | IV | --- | $2 \times 10^{-3}$ | 100 |
| Hafnium (72) | Hf 181 | IV | --- | $7 \times 10^{-4}$ | 10 |
| Holmium (66) | Ho 166 | IV | | | 100 |
| Hydrogen (1) | H3 | (see tritium) | $5 \times 10^{-6}$ | $3 \times 10^{-2}$ | 1,000 |
| Indium (49) | In 113m | IV | --- | $1 \times 10^{-2}$ | 100 |
| | In 114m | III | --- | $2 \times 10^{-4}$ | 10 |
| | In 115m | IV | | | 100 |
| | In 115 | IV | | | 10 |
| Iodine (53) | I 124 | III | | | |
| | I 125 | III | | | 1.0 |
| | I 126 | III | $3 \times 10^{-9}$ | $2 \times 10^{-5}$ | 1.0 |
| | I 129 | III | | | 0.1 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | I 131 | III | $3 \times 10^{-9}$ | $2 \times 10^{-5}$ | 1.0 |
| | I 132 | IV | $8 \times 10^{-8}$ | $6 \times 10^{-4}$ | 10 |
| | I 133 | III | $1 \times 10^{-8}$ | $7 \times 10^{-5}$ | 1.0 |
| | I 134 | IV | $2 \times 10^{-7}$ | $1 \times 10^{-3}$ | 10 |
| | I 135 | IV | | | 10 |
| Iridium (77) | Ir 190 | IV | --- | $2 \times 10^{-3}$ | |
| | Ir 192 | III | --- | $4 \times 10^{-4}$ | 10 |
| | Ir 194 | IV | --- | $3 \times 10^{-4}$ | 100 |
| Iron (26) | Fe 55 | IV | --- | $8 \times 10^{-3}$ | 100 |
| | Fe 59 | IV | --- | $6 \times 10^{-4}$ | 10 |
| Krypton (36) | Kr 85m | III | $1 \times 10^{-6}$ | --- | |
| | Kr 85m 5/ (uncompressed) | V | | | |
| | Kr 85 | III | $3 \times 10^{-6}$ | --- | 100 |
| | Kr 85 5/ (uncompressed) | VI | | | |
| | Kr 87 | II | | | 10 |
| | Kr 87 5/ (uncompressed) | V | | | |
| Lanthanum (57) | La 140 | IV | --- | $2 \times 10^{-4}$ | 10 |
| Lead (82) | Pb 203 | IV | --- | 4 x 10-3 | |
| | Pb 210 | II | | | |
| | Pb 212 | II | | | |
| Lutetium (71) | Lu 172 | III | | | |
| | Lu 177 | IV | --- | $1 \times 10^{-3}$ | 100 |
| Magnesium (12) | Mg 28 | III | | | |
| Manganese (25) | Mn 52 | IV | --- | $3 \times 10^{-4}$ | 10 |
| | Mn 54 | IV | --- | $1 \times 10^{-3}$ | 10 |
| | Mn 56 | IV | --- | $1 \times 10^{-3}$ | 10 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| Mercury (80) | Hg 197m | IV | --- | $2 \times 10^{-3}$ | 100 |
| | Hg 197 | IV | --- | $2 \times 10^{-3}$ | 100 |
| | Hg 203 | IV | --- | $2 \times 10^{-4}$ | 10 |
| Mixed Fission Products | MF-P | II | | | |
| Molybdenum (42) | Mo 99 | IV | --- | $2 \times 10^{-3}$ | 100 |
| Neodymium (60) | Nd 147 | IV | --- | $6 \times 10^{-4}$ | 100 |
| | Nd 149 | IV | --- | $3 \times 10^{-3}$ | 100 |
| Neptunium (93) | Np 237 | I | | | |
| | Np 239 | I | | | |
| Nickel (28) | Ni 56 | III | | | |
| | Ni 59 | IV | | | 100 |
| | Ni 63 | IV | | | 10 |
| | Ni 65 | IV | --- | $1 \times 10^{-3}$ | 100 |
| Niobium (Columbium) (41) | Nb 93m | IV | | | 10 |
| | Nb 95 | IV | --- | $1 \times 10^{-3}$ | 10 |
| | Nb 97 | IV | --- | $9 \times 10^{-3}$ | 10 |
| Osmium (76) | Os 185 | IV | --- | $7 \times 10^{-4}$ | 10 |
| | Os 191m | IV | --- | $3 \times 10^{-2}$ | 100 |
| | Os 191 | IV | --- | $2 \times 10^{-3}$ | 100 |
| | Os 193 | IV | --- | $6 \times 10^{-4}$ | 100 |
| Palladium (46) | Pd 103 | IV | --- | $3 \times 10^{-3}$ | 100 |
| | Pd 109 | IV | --- | $9 \times 10^{-4}$ | 100 |
| Phosphorus (15) | P 32 | IV | --- | $2 \times 10^{-4}$ | 10 |
| Platinum (78) | Pt 191 | IV | --- | $1 \times 10^{-3}$ | 100 |
| | Pt 193 | IV | | | 100 |
| | Pt 193m | IV | --- | $1 \times 10^{-2}$ | 100 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | Pt 197m | IV | --- | $1 \times 10^{-2}$ | 100 |
| | Pt 197 | IV | --- | $1 \times 10^{-3}$ | 100 |
| Plutonium (94) | Pu 238 6/ | I | | | |
| | Pu 239 6/ | I | | | (0.01) |
| | Pu 240 | I | | | |
| | Pu 241 6/ | I | | | |
| | Pu 242 | I | | | |
| Polonium (84) | Po 210 | I | | | 0.1 |
| Potassium (19) | K 42 | IV | --- | $3 \times 10^{-3}$ | 10 |
| | K 43 | III | | | |
| Praseodymium (59) | Pr 142 | IV | --- | $3 \times 10^{-4}$ | 100 |
| | Pr 143 | IV | --- | $5 \times 10^{-4}$ | 100 |
| Promethium (61) | Pm 147 | IV | --- | $2 \times 10^{-4}$ | 10 |
| | Pm 149 | IV | --- | $4 \times 10^{-3}$ | 10 |
| Protactinium (91) | Pa 230 | I | | | |
| | Pa 231 | I | | | |
| | Pa 233 | II | | | |
| Radium (88) | Ra 223 | II | | | |
| | Ra 224 | II | | | |
| | Ra 226 | I | | | (0.01) |
| | Ra 228 | I | | | |
| Radon (86) | Rn 220 | IV | | | |
| | Rn 222 | II | | | |
| Rhenium (75) | Re 183 | IV | --- | $6 \times 10^{-4}$ | |
| | Re 186 | IV | --- | $9 \times 10^{-3}$ | 100 |
| | Re 187 | IV | | | |
| | Re 188 | IV | --- | $6 \times 10^{-4}$ | 100 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | Re Natural | IV | | | |
| Rhodium (45) | Rh 103m | IV | --- | $1 \times 10^{-4}$ | 100 |
| | Rh 105 | IV | --- | $1 \times 10^{-1}$ | 100 |
| Rubidium (37) | Rb 86 | IV | --- | $7 \times 10^{-4}$ | 10 |
| | Rb 87 | IV | | | 10 |
| | Rb Natural | IV | | | |
| Ruthenium (44) | Ru 97 | IV | --- | $4 \times 10^{-4}$ | 100 |
| | Ru 103 | IV | --- | $8 \times 10^{-4}$ | 10 |
| | Ru 105 | IV | --- | $1 \times 10^{-4}$ | 10 |
| | Ru 106 | III | --- | $1 \times 10^{-3}$ | 1 |
| Samarium (62) | Sm 145 | III | | | |
| | Sm 147 | III | | | |
| | Sm 151 | IV | | | 10 |
| | Sm 153 | IV | --- | $8 \times 10^{-3}$ | 100 |
| Scandium (21) | Sc 46 | III | --- | $4 \times 10^{-3}$ | 10 |
| | Sc 47 | IV | --- | $9 \times 10^{-4}$ | 100 |
| | Sc 48 | IV | --- | $3 \times 10^{-4}$ | 10 |
| Selenium (34) | Se 75 | IV | --- | $3 \times 10^{-3}$ | 10 |
| Silicone (14) | Si 31 | IV | --- | $9 \times 10^{-3}$ | 100 |
| Silver (47) | Ag 105 | IV | --- | $1 \times 10^{-3}$ | 10 |
| | Ag 110m | III | --- | $3 \times 10^{-3}$ | 1 |
| | Ag 111 | IV | --- | $4 \times 10^{-3}$ | 100 |
| Sodium (11) | Na 22 | III | | | |
| | Na 24 | IV | --- | $2 \times 10^{-4}$ | 10 |
| Strontium (38) | Sr 85m | IV | | | |
| | Sr 85 | IV | --- | $1 \times 10^{-4}$ | 10 |
| | Sr 89 | III | --- | $1 \times 10^{-4}$ | 1 |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | Sr 90 | II | | | 0.1 |
| | Sr 91 | III | --- | $7 \times 10^{-4}$ | 10 |
| | Sr 92 | IV | --- | $7 \times 10^{-4}$ | 10 |
| Sulphur (16) | S 35 | IV | $9 \times 10^{-8}$ | $6 \times 10^{-4}$ | 100 |
| Tantalum (73) | Ta 182 | III | --- | $4 \times 10^{-4}$ | 10 |
| Technetium (43) | Tc 96m | IV | --- | $1 \times 10^{-1}$ | |
| | Tc 96 | IV | --- | $1 \times 10^{-3}$ | 10 |
| | Tc 97m | IV | | | 100 |
| | Tc 97 | IV | | | 100 |
| | Tc 99m | IV | | | 100 |
| | Tc 99 | IV | | | 10 |
| Tellurium (52) | Te 125m | IV | --- | $2 \times 10^{-3}$ | 10 |
| | Te 127m | IV | --- | $6 \times 10^{-4}$ | 10 |
| | Te 127 | IV | --- | $3 \times 10^{-3}$ | 100 |
| | Te 129m | III | --- | $3 \times 10^{-4}$ | 10 |
| | Te 129 | IV | | | 100 |
| | Te 131m | III | --- | $6 \times 10^{-4}$ | 10 |
| | Te 132 | IV | --- | $3 \times 10^{-4}$ | 10 |
| Terbium (65) | Tb 160 | III | --- | $4 \times 10^{-4}$ | 10 |
| Thallium (81) | Tl 200 | IV | --- | $4 \times 10^{-3}$ | 100 |
| | Tl 201 | IV | --- | $3 \times 10^{-3}$ | 100 |
| | Tl 202 | IV | --- | $1 \times 10^{-3}$ | 100 |
| | Tl 204 | III | --- | $1 \times 10^{-3}$ | 10 |
| Thorium (90) | Th 227 | II | | | |
| | Th 228 | I | | | |
| | Th 230 | I | | | |
| | Th 231 | I | | | |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | Th 232 | III | | | |
| | Th 234 | II | | | |
| | Th Natural | III | | | (100) |
| Thulium (69) | Tm 168 | III | | | |
| | Tm 170 | III | --- | $5 \times 10^{-4}$ | 10 |
| | Tm 171 | IV | --- | $5 \times 10^{-3}$ | 10 |
| Tin (50) | Sn 113 | IV | --- | $9 \times 10^{-4}$ | 10 |
| | Sn 117m | III | | | |
| | Sn 121 | III | | | |
| | Sn 125 | IV | --- | $2 \times 10^{-4}$ | 10 |
| Tritium (1) | H 3 | IV | $5 \times 10^{-6}$ | $3 \times 10^{-2}$ | 1,000 |
| | H 3 (as a gas, as luminous paint, or absorbed on solid material) | VII | | | |
| Tungsten (74) | W 181 | IV | --- | $4 \times 10^{-3}$ | 10 |
| | W 185 | IV | | | 10 |
| | W 187 | IV | --- | $7 \times 10^{-4}$ | 100 |
| Uranium (92) | U 230 | II | | | |
| | U 232 | I | | | |
| | U 233 6/ | II | | | (0.01) |
| | U 234 | II | | | (0.01) |
| | U 235 6/ | III | | | (0.01) |
| | U 236 | II | | | |
| | U 238 | III | | | |
| | U Natural | III | | | (100) |
| | U Enriched 6/ | III | | | |



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TABLE A-1. RADIOISOTOPE DATA - Continued

| Element (Atomic No.) | Radio-isotope 1/ | Column I Transport Group 2/ | Column II Gas Concentration µCi/ml 3/ | Column III Liquid and Solid Concentration µCi/ml 3/ | Column IV Exempt Quantities µCi 4/ |
|---|---|---|---|---|---|
| | U Depleted | III | | | |
| Vanadium (23) | V 48 | IV | --- | $3 \times 10^{-4}$ | 10 |
| | V 49 | III | | | |
| Xenon | Xe 125 | III | | | |
| | Xe 131m | III | $4 \times 10^{-6}$ | --- | 1,000 |
| | Xe 131m 5/ (uncompressed) | V | | | |
| | Xe 133 | III | $3 \times 10^{-6}$ | --- | 100 |
| | Xe 133 5/ (uncompressed) | VI | | | |
| | Xe 135 | II | $1 \times 10^{-6}$ | --- | 100 |
| | Xe 135 5/ (uncompressed) | V | | | |
| Ytterbium (70) | Yb 175 | IV | --- | $1 \times 10^{-3}$ | 100 |
| Ytterium (39) | Y 88 | III | | | |
| | Y 90 | IV | --- | $2 \times 10^{-4}$ | 10 |
| | Y 91m | III | --- | $3 \times 10^{-2}$ | |
| | Y 91 | III | --- | $3 \times 10^{-4}$ | 10 |
| | Y 92 | IV | --- | $6 \times 10^{-4}$ | 100 |
| | Y 93 | IV | --- | $3 \times 10^{-4}$ | 100 |
| Zinc (30) | Zn 65 | IV | --- | $1 \times 10^{-3}$ | 10 |
| | Zn 69m | IV | --- | $7 \times 10^{-4}$ | 100 |
| | Zn 69 | IV | --- | $2 \times 10^{-2}$ | 1,000 |
| Zirconium (40) | Zr 93 | IV | | | 10 |
| | Zr 95 | III | | $6 \times 10^{-4}$ | 10 |
| | Zr 97 | IV | | $2 \times 10^{-4}$ | 10 |
| | Radioisotopes not listed above | See para A-2 | See para A-3 | See para A-3 | See para A-4 |

Footnotes:

1/ Atomic weight shown after the radioisotope symbol.
2/ Transport groups are based upon 49 CFR 173.390 (See paragraph A-2 herein).
NOTE: Radioisotopes without entries or dashes in columns II, III, and IV were not listed in 10 CFR 20 or 10 CFR 30.



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Footnotes: (Continued)

3/ Concentrations are based upon 10 CFR 30, schedule A. Column II values are given for those materials normally used as gasses. The column III unit is µCi/gm for solids. (See 8.1.2 herein.)
4/ Exempt quantities are based upon appendix C of 10 CFR 20 and schedule B of 10 CFR 30. Values in parenthesis apply to 10 CFR 20 only. (See 8.1.2 herein.)
5/ Uncompressed means at a pressure not exceeding 14.7 psi (absolute).
6/ Fissile radioactive material.



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# SECTION B

## ACTIVITY MASS RELATIONSHIP - SPECIFIC ACTIVITY

## MICROCURIE PER GRAM CONVERSION FACTORS

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TABLE B-1. TECHNICAL DATA FOR SELECTED RADIOISOTOPES

| Element (Atomic No.) | I Radio-isotope 1/ | II Type Radiation 2/ | III Half-Life (years) 2/ | IV Transport Group 3/ | V Exempt Quantities (μCi) 4/ | VI Specific Activity (μCi/gm) 5/ |
|---|---|---|---|---|---|---|
| Americium (95) | Am 241 | Alpha and Gamma | 458 | I | (0.01) | $3.24 \times 10^{6}$ |
| Carbon (6) | C 14 | Beta (-) | 5,730 | IV | 100 | $4.46 \times 10^{6}$ |
| Cesium (55) | Cs 137 | Beta, Gamma | 30 | III | 10 | $8.71 \times 10^{7}$ |
| Cobalt (27) | Co 57 | Gamma | 0.739 | III | | $8.48 \times 10^{9}$ |
| | Co 60 | Beta (-), Gamma | 5.27 | III | 1.0 | $1.13 \times 10^{9}$ |
| Krypton (36) | Kr 85 | Beta (-), Gamma | 10.76 | III | 100 | $3.91 \times 10^{8}$ |
| Nickel (28) | Ni 63 | Beta (-) | 92 | IV | 10 | $6.17 \times 10^{7}$ |
| Promethium (61) | Pm 147 | Beta (-) | 2.62 | III | 10 | $9.29 \times 10^{6}$ |
| Radium (88) | Ra 226 | Alpha, Gamma | 1,602 | I | (0.01) | $9.88 \times 10^{5}$ |
| Rhenium (75) | Re 187 | Beta (-) | $4.3 \times 10^{10}$ | IV | | $4.45 \times 10^{-2}$ |
| Strontium (38) | Sr 90 | Beta (-) | 28.1 | II | 0.1 | $1.41 \times 10^{8}$ |
| Thorium (90) | Th 228 | Alpha, Gamma | 1.91 | I | | $8.22 \times 10^{8}$ |
| | Th 232 | Alpha, Gamma | $1.41 \times 10^{10}$ | III | | $1.09 \times 10^{-1}$ |
| | Th (natural) | Alpha, Gamma | $1.41 \times 10^{10}$ | III | (100) | $1.09 \times 10^{-1}$ |
| Tritium (1) | H 3 | Beta (-) | 12.3 | IV | 1,000 | $9.64 \times 10^{9}$ |
| Uranium (92) | U 234 | Alpha, Gamma | $2.47 \times 10^{5}$ | II | (0.01) | $6.19 \times 10^{3}$ |
| | U 235 | Alpha, Gamma | $7.1 \times 10^{8}$ | III | (0.01) | 2.14 |
| | U 238 | Alpha, Gamma | $4.51 \times 10^{9}$ | III | (0.01) | $3.33 \times 10^{-1}$ |
| | U (natural) | Alpha, Gamma | $4.51 \times 10^{9}$ | III | (100) | $3.33 \times 10^{-1}$ |
| | U (enriched) | Alpha, Gamma | 99% U238+ <1% U235 | III | | $>3.33 \times 10^{-1}$ |
| | U (depleted) | Alpha, Gamma | U238 - U235 | III | | $\approx 1.67 \times 10^{-1}$ (assuming one half-life decay) |

1/ Atomic weight shown after the radioisotope symbol.
2/ Data obtained from Radiological Health Handbook, U. S. Department of Health, Education, and Welfare, Public Service Publication N. 2016, January 1970.
3/ Transport groups are based upon 49CFR173.390 (See section A herein for details in determining transport groups.
4/ Exempt quantities are based upon appendix C of 10 CFR 20, and Schedule B of 10 CFR 30. Values in parenthesis apply to 10 CFR 20 only (see 8.1.2 and table A-1 herein).
5/ For details in determining specific activity (S.A.) constants, see section B herein.



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## SECTION B - ACTIVITY MASS RELATIONSHIP - SPECIFIC ACTIVITY

B-1 Introduction. This section contains instructions, information, general data, and a table of pertinent data based on the radioisotopes used in the commodities included herein under section C. The mathematical formulas and derivations of the specific activity (S.A.) in microcurie per gram constants are also included.

B-2 Requirements. Pursuant to the law, Title 49, Code of Federal Regulations, Part 173.389, has defined radioactive (for transportation) material to be any material in which the specific activity is 0.002 (or greater) microcuries per gram (see 6.38, 6.39 and B-3). Therefore, manufacturers of radioactive commodities must comply with these requirements and the additional requirements of MIL-STD-1458. This means then, that for each radioactive commodity, the radioisotope used and the quantity in microcuries shall be identified, unless otherwise exempted (see 11.3.10.2 and 8.1.2.6).

NOTE: Do not confuse specific activity (µCi/gm) with radioactivity (µCi). See definitions, B-3 and B-4.

B-3 Specific activity (µCi/gm). Specific activity (S.A.) is the rate of decay of the radioactive element or isotope expressed in curies per gram or microcuries per gram of the radioactive element. Specific activity is a natural constant for each radioactive element whether the amount involved in the radioactive material is a microgram or 1,000 grams. For specific activity constants of particular radioisotopes see table B-1 specific activity data column VI to determine whether the isotope of interest exceeds 0.002 µCi/gm and therefore, falls within the classification of radioactive material.

B-4 Radioactivity of small radioactive commodities (µCi). In the case of electron tubes, it is usually impractical to measure the radioactivity with any degree of precision for two reasons. The radioisotope is often surrounded by electrodes, spacers, and an envelope such that the geometric considerations preclude a realistic calibration of a measuring device; and the amount of radioisotope intentionally added in the tube is often so extremely small that the amount of radiation is difficult to discern from the normal background radiation that exists in the atmosphere. Therefore, the radioactivity for electron tubes and other small commodities shall be determined by a calculation technique in terms of the known radioisotopes which have been intentionally added to the electron tube to either increase tube life or to improve the overall tube performance.

Radioactivity (µCi/tube) = Specific activity (µCi/gm) times the actual mass in grams of the radioactive element used in the tube.

If the radioactive element or isotope is part of a mixture or a chemical compound, the specific-activity expression may be reduced by an amount corresponding to the fraction of the material which is radioactive since specific activity (S.A.) is expressed per gram of radioactive element rather than the compound.

Several examples of the calculation of specific activity and ultimately the radioactivity in Ci/tube are included



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B-5 Computations of specific activity. The specific activity (S.A.) of a radioisotope (disintegrations per unit time)/(unit mass), is calculated from the base equation:

$$S.A. = \lambda N = \frac{(\ln 2)\ N}{T_{1/2}}$$

WHERE: N = number of radioactive atoms per unit mass, and $T_{1/2}$ = half-life.

This basic equation can be transformed as follows:

by definition: $N = 6.0225 \times 10^{23}$/atomic mass (Avogadro's principal)

$Ci = 3.7 \times 10^{10}$ disintegrations/second,

Substituting:

$$S.A. = \frac{0.69315\ N}{T_{1/2}(secs)} = \frac{0.69315}{T_{1/2}} \times \frac{6.0225 \times 10^{23}}{\text{atomic mass}} \times \frac{1}{3.7 \times 10^{10}}$$

$$\times 10^{6} = \mu Ci/gm$$

This equation is satisfactory when the half-life of the isotope is expressed in seconds. If, however, the half-life is expressed in other units (such as minutes, hours, days, or years), a separate time conversion is required for each. By substituting the appropriate time conversion factors the following five equations can be obtained:

$$\text{microcuries/gram or S.A. } (T_{\frac{1}{2}} \text{ in secs}) = \frac{1.128 \times 10^{19}}{(T_{\frac{1}{2}})(\text{atomic mass})} \quad (1)$$

$$\text{microcuries/gram or S.A. } (T_{\frac{1}{2}} \text{ in mins}) = \frac{1.880 \times 10^{17}}{(T_{\frac{1}{2}})(\text{atomic mass})} \quad (2)$$

$$\text{microcuries/gram or S.A. } (T_{\frac{1}{2}} \text{ in hrs}) = \frac{3.134 \times 10^{15}}{(T_{\frac{1}{2}})(\text{atomic mass})} \quad (3)$$

$$\text{microcuries/gram or S.A. } (T_{\frac{1}{2}} \text{ in days}) = \frac{1.306 \times 10^{14}}{(T_{\frac{1}{2}})(\text{atomic mass})} \quad (4)$$

$$\text{microcuries/gram or S.A. } (T_{\frac{1}{2}} \text{ in yrs}) = \frac{3.578 \times 10^{11}}{(T_{\frac{1}{2}})(\text{atomic mass})} \quad (5)$$

NOTE: For a detailed analysis of the above conversion factors, see section B-6. To obtain radioactivity, see B-6 (23).

B-5.1 Applicability. The specific activities for the isotopes shown in table B-1 are based on the above conversion factors, using the half-lives shown. Integer mass numbers were used rather than actual masses. Also, these specific activities are for pure forms of the isotopes only.

B-6 Basic theory. The number of atoms (dN) in any one radioactive element that will disintegrate in a small interval dt is proportional to the number of atoms (N) present at the beginning of the time period dt.

(1) $-dN = \lambda N dt$



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The fundamental law of nature which describes the rate of disintegration or decay of a radioactive element is therefore as follows:

(2) $-\frac{dN}{dt} = \lambda N dt$

where $\lambda$ = a proportionality constant for the particular radioactive element; often called the decay or disintegration constant.

N = number of radioactive atoms of the element at any time t

$-\frac{dn}{dt}$ = rate of disintegration (the rate change of N is negative because N is decreasing with time)

Rearranging equation (1) gives:

(3) $\frac{dN}{N} = -\lambda dt$

Integrating equation (3) gives:

$\int \frac{dN}{N} = -\int \lambda dt$

(4) $\ln N = -\lambda t + C$

at t = 0; N = No

Therefore, equation (4) can be rewritten:

$\ln No = \lambda (0) + C$

or $C = \ln No$

Referring to equation (4) again and replacing C:

$\ln N = -\lambda t + \ln No$

and $\ln N - \ln No = -\lambda t$

(5) $\ln \frac{N}{No} = -\lambda t$ natural logarithmic form to base $\varepsilon$

Rewriting equation (5):

$\varepsilon^{-\lambda t} = \frac{N}{No}$

(6) or, $N = No\, \varepsilon^{-\lambda t}$ exponential form



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(7) or, $Nt = No\ \varepsilon^{-\lambda t}$ exponential form

Definitions: where No = original number of radioactive atoms of the element at time t = 0

N = Nt = number of radioactive atoms after a given decay time t

$\lambda$ = a proportionality constant (decay constant) for a given radioactive element

t = decay time

Equations 5, 6, and 7 are written in terms of the natural logarithm to the base $\varepsilon$ ( $\varepsilon$= 2.7183), a fundamental constant which derives from other natural phenomena as well as radioactive decay.

The time T required for 1/2 the number of atoms to disintegrate is defined as the half-life of the radioactive element or isotope.

A simplified expression for the half-life time T can thus be derived from equation (7) as follows:

$$Nt = No\ \varepsilon^{-\lambda t}$$

since Nt = 1/2 No in half-life time where t = T

$$1/2\ No = No\ \varepsilon^{-\lambda T}$$

$$0.5 = \varepsilon^{-\lambda t}$$

$$\ln 0.5 = -\lambda T \ln \varepsilon$$

$$-.693 = -\lambda T$$

$$\lambda T = .693$$

(8) $$T = \frac{.693}{\lambda}$$ Half-life time of the particular radioactive element.

Equivalent expressions based on common logarithms to the base 10 are shown for general information. However, the use of natural logarithms results in a simpler and more useable expression for Nt (equation (7) compared to equation (10)).

Repeating equation (7) and taking the common logarithm (base 10):

$$Nt = No\ \varepsilon^{-\lambda t}$$

$$\log Nt = \log No + \log \varepsilon^{-\lambda t}$$

$$\log Nt - \log No = \log (\varepsilon^{-\lambda t})$$

$$\log \frac{Nt}{No} = -\lambda t \log \varepsilon$$



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Since: $\log \varepsilon = .4343$

(9) $\log \frac{Nt}{No} = -.4343\lambda t$ common logarithmic form (base 10)

(10) or $Nt = No\ (10^{-.4343\lambda t})$ common logarithmic form (base 10)

At the half-life time: $N_T$ = 1/2 No due to half the atoms disintegrating

and $t = T$

Substituting in equation (9):

$$\log \frac{No}{2\ No} = -.4343\lambda T$$

$$\log 1/2 = -.4343\lambda T$$

$$\log 1 - \log 2 = -.4343\lambda T$$

$$0 - .3010 = -.4343\lambda T$$

$$\lambda T = \frac{.3010}{.4343} = .693$$

(11) $$T = \frac{.693}{\lambda}$$

This expression for T (equation (11)) is the same result that was derived from the exponential form (equation (8)) using natural logarithms.

For radioactive decay calculations the half-life is a more convenient parameter to use than the decay constant. The rate of disintegration from equation (2) can now be expressed in terms of the half-life T.

$$-\frac{dN}{dt} = \lambda N$$

Substituting for $\lambda$ from equation (8):

(12) $$-\frac{dN}{dt} = \frac{.693}{T} N$$ rate of disintegration

The decay rate for any radioactive substance is referenced to he decay rate of radium. One curie of a radioactive substance is that amount having an activity of $3.7 \times 10^{10}$ disintegrations/second, the disintegration rate of radium.

Therefore,

(13) 1.0 curie = $3.7 \times 10^{10}$ disintegrations/second

(14) 1.0 μCi = $3.7 \times 10^{4}$ disintegration/second

According to Avogadro's principle there are $6.02 \times 10^{23}$ atoms in a gram-atom or in a mass equivalent to the atomic weight expressed in grams. Therefore the number of atoms in one gram is:



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(15) $N = \frac{6.02 \times 10^{23}}{A}$ $\frac{\text{atoms/gram atom}}{\text{grams/gram-atom}} = \frac{\text{atoms}}{\text{gram}}$

where A = atomic weight of the radioactive element

Now the specific activity (S.A) is the rate of decay of one gram of the radioactive element or isotope expressed in curies or microcuries.

Therefore,

(16) $S.A. = \frac{0.693}{T} \times \frac{6.02 \times 10^{23}}{A} \times \frac{1}{3.7 \times 10^{10}}$

(Specific Activity = 1/sec x atoms/gram x $\frac{\text{curies}}{\text{atom/sec}}$ = curies/gram of radioactive element

(17) $S.A. = \frac{0.693}{T} \times \frac{6.02 \times 10^{23}}{A} \times \frac{1}{3.7 \times 10^{10}} \times 10^{6}$ = μcuries/gram of radioactive element

(Specific Activity)

Reducing the specific activity equation (17) gives:

(18) $S.A. = \frac{1.13 \times 10^{19}}{T\,A}$ μCi/gram of radioactive element — for T = half-life in seconds

(19) $S.A. = \frac{1.88 \times 10^{17}}{T\,A}$ μCi/gram of radioactive element — T = half-life in minutes

(20) $S.A. = \frac{3.14 \times 10^{15}}{T\,A}$ μCi/gram of radioactive element — T = half-life in hours

(21) $S.A. = \frac{1.31 \times 10^{14}}{T\,A}$ μCi/gram of radioactive element — T = half-life in days

(22) $S.A. = \frac{3.58 \times 10^{11}}{T\,A}$ μCi/gram of radioactive element — T = half-life in years

To obtain the radioactivity in μCi/tube:

(23) Radioactivity (μCi/tube) = S.A. above in μCi/gm of radioactive element x actual mass in grams of the radioactive element used in the tube

If the radioactive element or isotope is part of a mixture or a chemical compound, the specific-activity expression may be reduced by an amount corresponding to the fraction of the material which is radioactive since specific activity (S.A.) is expressed per gram of radioactive element rather than the compound.

Several examples of the calculation of specific activity and ultimately the radioactivity in μCi/tube are included as follows:

EXAMPLES: RADIOACTIVE CALCULATIONS*

A. For pure radioactive element

* See table B-1 - Half-life values of radioactive elements.



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1. Re187

Known factors

$T = 4.3 \times 10^{10}$ years

A = 187

gms of $Re^{187}$ in tube = 0.1

Reference symbol

T = half-life

A = atomic weight of radioactive element

S.A. = specific activity = rate of decay of 1 gram of pure radioactive element expressed in μCi/gm of the pure radioactive element

Using equation (22):

$$\text{S.A.}_{(\mu Ci/gm)} = \frac{3.58 \times 10^{11}}{T\ A}; \text{ T in years}$$

$$\text{S.A.} = \frac{3.58 \times 10^{11}}{4.3 \times 10^{10} \times 187}$$

S.A. = 0.0445 μCi/gm of $Re^{187}$

From equation (23):

Radioactivity (μCi/tube) = S.A. (μCi/gm) of radioactive element x gm of radioactive element actually used per tube

Radioactivity = 0.0445 x 0.1 gram of $Re^{187}$ used in tube

Radioactivity = 0.00445 μCi/tube of $Re^{187}$

2. $Ni^{63}$

From vendor information of

.002 $\mu Ci/mm^2$ /plated side
compute μCi level/Tab:

$\mu Ci/Tab = .002\ \mu Ci/mm^2 \times 3mm^2 \times 2$ plated sides/Tab

= .012

Known factors

T = 92 years

A = 63

1 Tab of $Ni^{63}$ material used/tube

Area of each side of $Ni^{63}$ tab = 3 mm

$Ni^{63}$ is plated on both sides of each tab

Vendor information: .002 $\mu Ci/mm^2$/plated side



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Since 1 Tab is used/tube, then:

(24) Radioactivity = .012 μCi/tube of $Ni^{63}$

Now using equation (22):

$$\mu Ci/gm \text{ of } Ni^{63} \quad S.A. = \frac{3.58 \times 10^{11}}{T\ A}; \ T \text{ in years}$$

$$S.A. = \frac{3.58 \times 10^{11}}{92 \times 63}$$

(25) S.A. = 61,770,000 μCi/gm of $Ni^{63}$

c/gm of $Ni^{63}$ or 61.77 μCi/gm of $Ni^{63}$

Divide equation (24) = .012 μCi/tube of $Ni^{63}$ by equation (25) where the specific activity = 61,770,000 μCi/gm of $Ni^{63}$ to calculate the grams of $Ni^{63}$ in the tube:

$$\text{gms of } Ni^{63}/\text{tube} = \frac{.012 \ \mu Ci/tube}{61,770,000 \ \mu Ci/gm \ Ni^{63}}$$

Therefore,

(26) gms of $Ni^{63}$ /tube = $1.94 \times 10^{-10}$

Repeating equation (23):

Radioactivity (μCl/tube) = Specific activity (μCi/gm) x gm of radioactive element used/tube

= 61,770,000 μCi/gm of $Ni^{63}$ x $1.94 \times 10^{-10}$ gm of $Ni^{63}$ /tube

(27) Radioactivity = .012 μCi/tube of $Ni^{63}$
μc/tube

This result not only shows the μCi level/tube but also verifies the μc/tube value obtained in equation (24) above. It also certifies that the calculated gm weight of $Ni^{63}$ determined in equation (26) as $1.94 \times 10^{-10}$ gm of $Ni^{63}$/tube is likewise correct.

The small amount of $Ni^{63}$ used/tube ($1.94 \times 10^{-10}$ gms) is nearly infinitesimal and when multiplied by the large specific activity value for $Ni^{63}$ (61,770,000 μCi/gm of $Ni^{63}$) gives an end product having an extremely low radioactivity level of .012 (μCi/tube)



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3. $Co^{60}$

Known factors

T = 5.26 years

A = 60

gms of $Co^{60}$ in tube = $2.64 \times 10^{-10}$

Using equation (22):

$$S.A. = \frac{3.58 \times 10^{11}}{T\ A} = \mu Ci/gm \text{ of } Co^{60}$$

$$= \frac{3.58 \times 10^{11}}{5.26\ (60)}$$

$$S.A. = 1.137 \times 10^{9}\ \mu Ci/gm \text{ of } Co^{60}$$

Radioactivity = $1.137 \times 10^{9}$ µCi/gm of $Co^{60}$ x $2.64 \times 10^{-10}$ gms of $Co^{60}$/tube

Then:

(28) Radioactivity = 0.30 µCi/tube of $Co^{60}$
µCi/tube

4. For several pure radioactive elements or isotopes

When more than one radioactive element or isotope is present in an electron tube, then the radioactivity of each specie is determined (per A1 or A2 or A3 method above) and the radioactivity results can then be added to give the total radioactivity for the electron tube.

B. For radioactive compounds

1. Thorium tungsten wire (ThW)

Using equation (22):

$$S.A. \text{ µCi/gm} = \frac{3.58 \times 10^{11}}{T\ A};\ T \text{ in years}$$

$$S.A. = \frac{3.58 \times 10^{11}}{1.41 \times 10^{10}} \times 232$$

S.A. = 0.1094 µCi/gm thorium

Known factors

$Th^{232}$ (radioactive thorium is the only radioactive compound)

T = $1.41 \times 10^{10}$ years
A = 232

1.2% thoria in thoria tungsten (ThW) wire

Wt of ThW wire used per tube = 0.1 gm

gm of Th/tube = 0.00105
(see calculation of equation (31))



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From equation (23):

Radioactivity = S.A. $\frac{\mu Ci}{gmTH}$ x gm thorium actually used per tube
$\mu c$/tube

= 0.1094 x 0.00105 (from equation (29) and equation (31) below

(30) Radioactivity = 0.000115 µCi/tube of Thorium

2. Alternate approach for B1, above

1 gm of ThW wire contains 1.2% $ThO_2$ (Thoria)

1 gm of ThW wire contains 1 x .012 = .012 gms $ThO_2$

Atomic weight of Th = 232

Molecular weight of $ThO_2$ = 232 + 32 = 264

Therefore, ratio of Th to $ThO_2$ is:

$$\frac{232\ (Th)}{264\ (ThO_2)} = 0.879$$

The specific activity for thorium:
S.A. = 0.1094 µCi/gm thorium (Table B-1)

The specific activity for thoria:
S. A. = 0.1094 µCi/gm (S.A. of Th) x 0.879 (ratio of Th to $ThO_2$)
S. A. = 0.0962 µCi/gm thoria
The S. A. for thoria allows quick calculation of the tube µCi level for any % of thorium in thoriated tungsten wire.

Example:
1 gm of thorium tungsten (ThW) wire contains 1.2% $ThO_2$ (thoria)
(31) Radioactivity = (1.2%) (1 gm/gm ThW) x (0.0962 µCi/gm thoria)
= 0.00115 µCi/gm ThW
0.1 gm used per tube of ThW
(32) Radioactivity = (0.1 gm ThW/tube) x (0.0015 µCi/gm ThW)
= 0.000115 µCi/tube

3. For several radioactive compounds in an electron tube

When more than one radioactive compound is present in an electron tube, then the radioactivity of each compound is determined and the radioactivity results can then be added to give the total radioactivity for the electron tube.



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# SECTION C

## RADIOACTIVE COMMODITIES - GP59

## TYPE OR PART NUMBER SEQUENCE

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## SECTION C RADIOACTIVE COMMODITIES (GROUP 59)

C-1 Introduction. This section contains a list of radioactive commodities by part or type number in alpha-numerical sequence by FSC. The pertinent marking and labeling requirements have been summarized for each entry. Many of the commodities included in table C-1 are not currently in the Department of Defense supply system, but are included for information and future reference.

C-2 Inclusions. The data included in table C-1 applies to commodities in Group 59. Table C-1 includes eight columns of data which provides the essential data for proper marking and labeling for each item listed. These data columns are as follows:

(1) Type or part number of the radioactive commodity
(2) National stock number of the radioactive commodity
(3) Manufacturer of the radioactive commodity
(4) Quantity in microcuries of the radioisotope (see section B)
(5) Radioisotope
(6) Use and storage label requirements (see 11.2)
(7) DOT radioactive label requirements (see 11.3.11).
(8) DOT transport group of the radioisotope (see section A)

C-3 Requirements. The marking and labeling requirements that have been summarized in columns 6, 7, and 8 are consistent with and in accordance with the requirements of the standards and regulations for the modes of transportation, use or storage shown below:

| | |
|---|---|
| (1) Rail freight, express or baggage | (Title 49 CFR173.390) (see 11.3) |
| (2) Highway carrier | (Title 49 CFR173.390) (see 11.3) |
| (3) Water carrier | (Title 46 CFR146.19-5) (see 11.4) |
| (4) Mail carrier | (Title 39 CFR and Pub. No. 6) (see 11.5) |
| (5) Air carrier | (Title 14 CFR103) (see 11.6) |
| (6) Air carrier (military) | (DSAM 4145.3, AFM 71-4, TM38-250, NAVSUPPUB 505, and MCO P 4030.19) (see 11.7) |
| (7) Use and storage | (MIL-STD-1458) |

C-3.1 Use and storage label (see figure 6). The requirements for this label for each listed commodity are shown in column 6, and are in accordance with MIL-STD-1458, and applies to commodities containing radioactive material in excess of the values shown in table I (see 11.2). This label is also applicable to shipping containers containing radioactive commodities that are exempt from DOT (49 CFR) labeling requirements. However, the marking and labeling requirements for shipping containers are not included in the tables herein.

C-3.1.1 Exempt shipping containers (10 CFR 20). See 11.2.2 herein.

C-3.2 DOT radioactive labels (see figure 7). The requirements for these labels are in accordance with Title 49 of the Code of Federal Regulations (see 11.3), but are also applicable to the other modes of transportation shown above. Thus, if the "radioactive white - I" label was listed under the DOT label column of table C-1, it would also be required for any or all other modes of transportation.

C-3.2.1 Exemptions. Exempt commodities shall conform to the requirements of 11.3.10.2.



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C-3.3 DOT transport group (see section A). The transport groups shown in column 8 of table C-1 have been determined in accordance with the applicable standards and federal laws for the modes of transportation shown under C-3. The transport groups for the listed commodities are applicable to all modes of transportation.

C-4 Non-radioactive commodities. Tables C-1 includes many commodities that are not considered radioactive (see 6.38, 6.39, 6.44, and 6.47 herein). The commodities are included for information and reference only. When there is no data shown in columns 6, 7, and 8 for marking and labeling, the listed commodity is not considered radioactive.



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TABLE C-1A

RADIOACTIVE COMMODITIES - FSC 5920

FUSES AND LIGHTING ARRESTORS

TYPE OR PART NUMBER SEQUENCE



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Table C-1A

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5920-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT Transport Group |
| TG101 | 629-2731 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG110 | 856-5965<br>896-5549 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG115 | 953-1924 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG134 | 950-0902 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG14 | 163-4107 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG151 | 789-7979 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG184 | 163-4106 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG191 | 011-4995 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG25 | 578-3088 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG27A | 822-2329 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG31 | 787-7036 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG34 | 738-8074 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG40 | 027-4320 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG42A | 858-0215 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG44 | 892-9060<br>027-8691 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG45 | 578-9265 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG48 | 727-0595 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |



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Table C-1A - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5920-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT Transport Group |
| TG49 | 062-5995<br>062-3993 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG51 | 813-4039 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG52 | 822-2326 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG53 | 827-5751<br>679-3827 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG54 | 682-7301 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG55 | 827-5752 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG56 | 839-5538<br>806-1090 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG57 | 922-4902<br>608-2142 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG58 | 827-5753 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG59 | 613-0114 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG61 | 107-5371 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG76 | 713-4191 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG78 | 020-9745 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG83 | 687-2067 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG84 | 687-2066 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |





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Table C-1A - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5920-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT Transport Group |
| TG86 | 822-2327 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG87 | 087-6337 856-5966 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |



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TABLE C-1B

RADIOACTIVE COMMODITIES - FSC 5925

CIRCUIT BREAKERS

TYPE OR PART NUMBER SEQUENCE



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Table C-1B

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part Number | NSN 5925-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| AM1513F2 | 265-2259 | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| AM1513F35 | 279-1029 | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| AM1614-25 | 279-1031 | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| AM1614M6 | - - - | Heinmann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| AN3161P35 | 175-5222 | Mechanical Prod, Inc. | (20 mi/hr at surface w/toggle out) | Radium 226 | Required | Exempt | I |
| C6363-1-10 | 175-2587 | Matals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C363-1-12-5 | 503-5847 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-15 | 296-8123 | Matals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-2LT | 155-8455 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-20LT | 155-8463 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-25 | 144-0438 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-35 | 583-0590 | Matals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |



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Table C-1B - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part Number | NSN 5925-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| C6363-1-5LT | 258-7121 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-5 | 155-8456 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-6 | 583-8292 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-1-8 | 501-6168 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-2-10 | 144-0443 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| C6363-2-12-1/2LT | 889-0768 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-2-25 | 144-0451 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-2-3LT | 299-1677 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-2-30 | 144-0453 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| C6363-2-5LT | 175-2580 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-100 | 175-5227 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-120 | 144-0459 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-40 | 501-7729 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |





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Table C-1B - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part Number | NSN 5925-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| D6[illegible]4-1-50 | 893-4299 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-60 | 144-0462 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-70 | 144-0464 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-80 | 144-0466 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-1-90 | 279-2930 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-2-100 | 144-0482 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-2-120 | 144-0473 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6364-2-60 | 586-6129 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| D6751-1-50 | 173-7627 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| ET5943 | 679-5834 | ITE Imperial Corp. | - - - | - - - | - - - | - - - | - - - |
| PAM1014MG3 | 581-0376 | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| 1163SM6-5A125V | 279-1350 | Heinemann Elec. Co. | - - - | - - - | - - - | - - - | - - - |



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Table C-1B - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part Number | NSN 5925-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 24-1164 | - - - | Cutler-Hammer | 0.32 per lens | Radium 226 | Required | Exempt | I |
| 24-1939-2 | - - - | Cutler-Hammer | 0.32 per lens | Radium 226 | Required | Exempt | I |
| 24-1997-2 | - - - | Cutler-Hammer | 0.32 per lens | Radium 226 | Required | Exempt | I |
| 271-1289P14 | - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1289P16 | - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1289P18 | 506-4368 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1289P19 | 506-4367 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1889P3 | - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1289P4 | 506-4370 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1289P5 | 506-4365 | Raytheon | - - - | - - - | - - - | - - - | - - - |
| 271-1289P6 | 502-7420 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P19 | 506-4388 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P20 | 506-4389 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P23 | - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P24 | 506-4372 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P25 | - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 271-1290P6 | 506-4382 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |





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Table C-1B - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part Number | NSN 5925-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 271-1290P7 | 506-4384 | Raytheon | - - - | - - - | - - - | - - - | - - - |
| 271-1290P9 | 027-2102 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 3H900-80 | 109-2211 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| 6750-2L-15 | - - - - | Metals and Controls, Inc. | 5.0 | Radium 226 | Required | Exempt | I |
| 6750-2L-25 | - - - - | Metals and Controls, Inc. | 5.0 | Radium 226 | Required | Exempt | I |
| 6750-2-35 | 813-2093 | Metals and Controls, Inc. | 5.0 | Radium 226 | Required | Exempt | I |
| 70-505 | - - - - - | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| 70-505GMG3 | - - - - - | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 70-613 | - - - - | Heinemann Elec. Co. | 0.05 | Radium 226 | Required | Exempt | I |
| 8725K4 | 271-8658 | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8751K4 | 159-0414 | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |



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TABLE C-1C

RADIOACTIVE COMMODITIES - FSC 5930

SWITCHES

TYPE OR PART NUMBER SEQUENCE



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Table C-1C

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or art number | NSN 5930-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT transport Group |
| A9503C28 | 518-3174 | Controls Co. of America | 20 mr/hr at surface | Radium 226 | Required | Exempt | II |
| A9503C30 | 504-3174 | Controls Co. of America | - - - | - - - | - - - | - - - | - - - |
| B7A | 232-6537 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| C6363-12 1/2A | 760-2622 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| C6363-6A | 654-4356 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| 24-1164 | 320-9934 | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 24-1939-2 | - - - | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 24-1997 | - - - | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 8200K12 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8201K11 | 187-3741 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8207K3 | 201-2545 | Cutler-Hammer | 0.2 | Radium 226 | Required | Exempt | I |
| 8208K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |



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Table C-1C - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5930-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT transport Group |
| 8209K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8210K12 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8211K12 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8212K11 | 259-8386 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8213K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8215K9 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8708 | 180-2541 | Cutler-Hammer | - - - | - - - | - - - | - - - | - - - |
| 8715K2 | 296-9017 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8800K12 | 272-1368 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8801K12 | 820-0566 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8802K9 | 669-7711 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8803K9 | 112-5288 655-1514 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |





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Table C-1C - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5930-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT transport Group |
| 8808K12 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8809K12 | 655-1523 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8810K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8811K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8812K12 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8813K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8815K9 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8816K8 | 549-0116 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8817K8 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8818K8 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8819K8 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8820K13 | 991-9287 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8821K9 | 810-6128 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |



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Table C-1C - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5930-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT transport Group |
| 8822K1 | 259-8350 | Cutler-Hammer | - - - | - - - | - - - | - - - | - - - |
| 8822K11 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8823K9 | 296-9664 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8824K11 | 950-7445 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8825K9 | 655-1582 | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 8825K9 | 655-1582 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8826K3 | 486-9058 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8827K9 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8828K3 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8830K3 | 615-7880 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8905K514 | 565-9501 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8905K520 | 518-3135 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8905K526 | 687-1097 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |





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Table C-1C - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Type or Part number | NSN 5930-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and labeling requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and storage label | DOT label (Radioactive) | DOT transport Group |
| 8905K591 | - - - | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 8906K1267 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K1268 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K1269 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K1270 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K1422 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K1665 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K2224 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8906K2428 | - - - | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |
| 8916K504 | 473-9099 | Cutler-Hammer | - - - | - - - | Required | Exempt | I |



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TABLE C-1D

RADIOACTIVE COMMODITIES - FSC 5960

ELECTRON TUBES

TYPE OR PART NUMBER SEQUENCE



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Table C-1D

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| ATR-321 | 549-0045 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR-326 | 842-6660 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR-332 | 262-1696 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR345/5792 | 193-5141 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| ATR345 | 193-5141 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR346/5793 | 193-5142 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR346/5793 | 193-5142 | Sylvania | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| ATR387 | 842-6661<br>248-8504 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR387/6024 | 272-9195<br>842-6661 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR388 | 272-9197 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR407 | 262-0289<br>060-6626 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| ATR407 | 262-0289 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| ATR-427 | 390-5241 | Sylvania | 0.15 | Cobalt 60 | Required | Exempt | III |
| ATR427 | 390-5241 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| A15030 | 847-1319 | RCA | 0.498 | Thorium 232 | Required | Exempt | III |
| A15034 | - - - | RCA | 0.249 | Thorium 232 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| A15039 | - - - | RCA | 0.58 | Thorium 232 | Required | Exempt | III |
| A2519B | - - - | RCA | 0.0276 | Thorium 232 | Not Required | Exempt | III |
| A2548 | - - - | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| A2589 | - - - | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| BES-033 | 070-8080<br>902-0941 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BES-034 | 904-7356 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BES-045 | 434-0229 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BES-046 | 856-8852<br>916-3670 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| BES-048 | 856-9953<br>916-3669 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BES-050 | 222-1388 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| BES-052 | 405-8369 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| EL-A-001 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| EL-A-002 | - - - | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| EL-A-004 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |





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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| BL-A-005 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-006 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-007 | 089-5851 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-A-007-1 | 089-5851 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-008 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-009 | 956-2318 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-011 | 222-1389 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-012 | 222-1397 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-013 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-026 | 903-3308 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-030 | 348-2498 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-A-034 | 615-9618 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-N-003 | 808-9472 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-N-012 | 948-9472 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-N-045 | 913-3128 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| BL-P-027D | 973-7957 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-P-040D | 056-7033 | Varian | 0.8 | Cobalt 60 | Required | Exempt | III |
| BL-P-105D | 916-3369 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| BL-S-001 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-002 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-003 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-004 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-005 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-006 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-012 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-013 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-015 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-S-016 | 809-7515 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-017 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-018 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-019 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-020 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-021 | 880-5258 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-022 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-023 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-024 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-025 | 880-5259 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-027 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-028 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-029 | 968-0500 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-030 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-031 | 880-8713 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-032 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-504 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-505 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-S-506 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-S-508 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-509 | - - - | Varian | 0.20 | Cobalt 60 | Required | Exempt | III |
| BL-S-510 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-511 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-512 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-514 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-515 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-516 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-519 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-S-521 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-521A | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-523 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-524 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-525 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-526 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-527 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-528 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-530 | 027-5015 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-S-531 | 027-5017 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-T-001 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-002 | - - - | Varain | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-003 | 472-9435 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-004 | 754-5431 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-004A/ 7902 | 754-5431 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-005B | 932-3067 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-006 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-007 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-009 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-010 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-011 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-012 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-013 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-T-014 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-015 | 899-8118 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-017 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-018 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-019 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-T-020 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-021 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-022 | 620-1435 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-023 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-024 | 848-8575 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-025 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-026 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-027 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-028 | 617-3129 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-029 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-031 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-032 | 717-7844 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-033 | 472-9435, 817-6181 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-033/ TR610 | 817-6181, 472-9435 | Varian | 0.90 | Cobalt 60 | Required | Exempt | III |
| BL-T-035 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-T-036 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-037 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-038 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-T-039 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-040 | 719-7508 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-041 | 894-7127 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-042 | 844-5183 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-043 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-043A | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-045 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-049 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-050 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-051 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-T-052 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-053 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-054 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-055 | 780-7856 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-056 | 081---- | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-057 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-058 | 781-8963 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-059/8061 | 894-7128 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-060 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-061 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-T-062 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-063 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-064 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-065 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-066 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-T-067 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-068 | 822-8875 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-069 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-070 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-071 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-072 | 897-6637 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-073 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-074 | - - - | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| BL-T-75 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-076 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-077 | 903-5816 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-078 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-079 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-080 | 620-4528 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-081 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| B' -T-082 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-083 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-084 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-085 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-086 | 610-8486<br>852-1514 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-T-087 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-088 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-089 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-090 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-T-091 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-T-092 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-094 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-T-095 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-096 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-T-097 | 849-3494 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| BL-T-098 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-T-099 | 914-6268 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-100 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-101 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-103 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-T-105 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-106 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-107 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-T-108 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-109 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-110 | 847-3752 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-T-112 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-W-004 | 754-5431 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-W-015 | 882-1998 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| BL-W-022 | - - - | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| BL-10 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-11/6282 | 365-2548 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-121 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-137 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-142 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-146 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-15 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-16 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-18/6632 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-20 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-25/6322 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-26 | 390-5250 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-27/6334 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL27/6334 | 578-5483 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL27/6334 | 578-5483 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| BL28/6568 | 538-2793 060-6629 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-29 | 709-2031 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-302 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-303 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-304 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-305 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-306 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-307 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-308 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-309/6592 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-31/6637 | 719-5454 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-310/6593 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-311/6594 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-313/6565 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-314/6614 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-316/6595 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-317/6596 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-320/6597 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-322/6599 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-324/6613 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-326/14 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-327/6609 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-328 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-329/6602 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-331 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-332 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-333 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-335 | 893-1757 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-336 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-337 | 968-0500 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-338 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| F -339 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-34 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-340, H | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-341 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-344 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-345 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-346 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-347 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-348 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-349 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-35/6560 | 501-1066 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-350 | 581-9031 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-351 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-352 | 615-0719 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-352A/7447 | 615-0719 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-354 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-355 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-356 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-357 | 578-8724 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-358 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-359 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-36 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-360H | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-361 | 617-5807 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-362 | 617-3736 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-363 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-365 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-366 | 672-3836 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-367 | 617-5806 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-367A | 060-6825 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-367B | 060-6825 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-367C | 060-6825 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-368 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-369 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-37/6633 | 681-9793 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| BL-371 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-372 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-373 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-374 | 800-3537 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-375 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-377 | 800-3536 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-38 | - - - - | Chatham (Tung-Sol) | 0.055 | Radium 226 | Required | Exempt | I |
| BL-381 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-385 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-386 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-387 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-387A | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-388 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-389 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-39 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-391 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-392 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-392A | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-394 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-395 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-397 | 983-5684 617-5807 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-397A, B, C | 983-5684 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-398 | 990-3077 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-399 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-40 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-41 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-43/6304 | 617-5810 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-44 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-45 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-46/6639 | 295-3589 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-47 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-49 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-509/6604 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-51 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-52 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-54/6629 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-55/6630 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-56 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-57/6635 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-58 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-60/6640 | 543-0143 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL- 00/6642 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-601 | 701-7582 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-601/8048 | 701-7582 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-602 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-603 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-604/6646 | 905-7498 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-604H/6647 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-605 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-606 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-607 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-609 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-61/6455 | 727-5618 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-610 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-611H | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-612/7152 | 807-7374 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| BL-613/6905 | 729-3648 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-614 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-615/6648 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-616/6685 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-617 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | IIW |
| BL-618 | - - - | Varian | 0.90 | Cobalt 60 | Required | Exempt | III |
| BL-619 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-62/6378 | 296-5932 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-620 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-622 | 583-4175 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-623 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-624 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-625 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-628 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-629 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-63 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-630 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-631 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-632 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-633 | - - - | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-634 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-635 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-636 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-637 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-638 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-639 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-640 | - - - | Varian | 0.256 | Cobalt 60 | Required | Exempt | III |
| BL-640 | 688-9031 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-640B | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-641 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-642 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-643/6906 | 682-8864 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-644 | 503-4436 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-645 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-646 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-647 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-648 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-649 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-650 | 892-3323 | Varian | 0.203 | Cobalt 60 | Required | Exempt | III |
| BL-650/6890 | 892-3323<br>553-3519 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-651 | 543-0693 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-651H | 801-6865 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-652 | 608-1960 | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-654 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-655 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-658 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-659 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-66 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-660 | 553-7337<br>962-3345 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-661 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-662 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-664 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-665A | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-665/6962 | 646-4719 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-666 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-667 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-668 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-669 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-67/6650 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-670 | 553-7338 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-672 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-673 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-674 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-678 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-68 | 060-6608 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-681 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-682 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-683 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-684 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-685 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-686 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-69 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-690 | 806-8155 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-693 | 799-9471 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-693/7309 | 633-6304 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-696 | 615-5824 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-697 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-698 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-699 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-70 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-700 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - *Continued*

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-702 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-703 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-706 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-71/6564 | 060-6628<br>841-2609 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-714 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-716 | 727-7374 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-717 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-718 | 519-9961 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-72 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-724 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-726 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-73 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-735 | 552-1759 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-74 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-744 | 538-3862 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-745 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-749 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-75 | 636-3613 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BL-752 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-757 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL76 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-762 | 984-6113 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-763 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-77 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-772 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-774 | 800-3537 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-776 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-778 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-78 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-781 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-783 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-784 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-785 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-786 | 615-0842 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-787 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-79 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-792 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-799 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |



MIL-HDBK-600
SECTION C

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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-8 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-80 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-81/6643 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-82 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-83 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-84 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-84H | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-85 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-86/6641 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-87/6636 | 800-0594 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-88 | - - - | Varian | 0.50 | Cobalt 60 | Required | Exempt | III |
| BL-89 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-90/6634 | - - - | Varian | 0.90 | Cobalt 60 | Required | Exempt | III |
| BL-901 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-902 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-904 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-905 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-907 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-908 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-909 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-910 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-911 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-912 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-913 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-914 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-915 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-916 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-917 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-918 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-919H | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-92 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-920 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-924 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-925 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-926 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-927 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-928 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-929 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-931 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-932 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-933/7166 | 681-9793 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL933/7166 | 681-9793 | Microwave | 0.42 | Cobalt 60 | Required | Exempt | III |
| BL-934 | 835-2576 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-934A | 835-2574 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-934B | 835-2574 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-935 | - - - | Varian | 0.80 | Cobalt 60 | Required | Exempt | III |
| BL-938 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-94/6631 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-939 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-941 | 812-2113 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-942 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-943 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-944 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-945 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-946 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-947 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-948 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| L-949 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-95/6644 | 843-8063 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-95/6644 | 752-5259<br>752-5239 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-95H/6645 | 503-4497<br>073-9277 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-951 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-952 | 583-4641 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-953 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-954 | 800-3535 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-955 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-956 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-957 | 804-2215 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-959 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-96/6605 | 503-4493 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-96A/6605 | 581-5535 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-960A | 754-9177 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-961 | 503-4493 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-962 | 661-0340 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-963 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BL-964 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-965 | 661-0339<br>811-6773 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-966 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-967 | 835-2575 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-967A | 924-9606 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-969 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-97 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-970 | 824-8668 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-971 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-972 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-973 | 681-8037 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-974 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| BL-975 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| BL-976 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-977 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-978 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-979/7381 | 984-6164 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-98 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-981 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| F -982 | 681-8036 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-983 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-99/6638 | 569-9508 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-990 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-991-1 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-992 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-994 | 882-1994 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-994A | 903-2852 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-995 | 903-2853 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-995B/8313 | 903-2853 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BL-996 | 051-3188 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BL-997 | - - - | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| BL-998 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BL-991-1 | 079-5942 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| BP-840 | 553-7338 | Bendix Radio | 1.0 | Cobalt 60 | Required | Exempt | III |
| BS-101/5962 | 188-6592 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| BS-10-/5962 | 188-6592 | Eon Corp. | 4.0 | Nickel 63 | Required | Exempt | IV |
| BS-101/5962 | 188-6592 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| BS-101/5962 | 188-6592 | Electronic Products, Inc. | 3.0 | Nickel 63 | Required | Exempt | IV |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| BS-101/5962 | 188-6592 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| BS-101/5962 | 188-6592 | Eon Corp. | 0.1 | Nickel 63 | Required | Exempt | IV |
| BTR-121 | 760-1786 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-140 | 807-2032<br>739-7922 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-141 | 070-8080 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BTR-147 | 847-3445 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BTR-148A | 947-7292 | Varian | 0.5 | Cobalt 60 | Required | Exempt | III |
| BTR-165 | 897-6964 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| BTR-183 | 762-6682<br>943-4392 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| BTR-537 | 075-6770 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-540 | 850-0788 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-541 | 868-6140 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| BTR-546 | 788-9889 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| BTR-548 | 902-0502 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-608 | 762-7692 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-614 | 464-4093 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-614A | 320-6631 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| BTR-632A | 403-1162 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| B107A | 593-9860 | Hartman Elec. | 1.0 | Radium 60 | Required | Exempt | I |
| B300D | - - - | Amperex | 100 mg | U02 enriched | Required | Exempt | III |
| B4021AL | 811-6838 | Burrough | <1.0 | Krypton 85 | Required | Exempt | III |
| Cem-306 | - - - | Central Elec. | 0.50 | Cobalt 60 | Required | Exempt | III |
| Cem-332 | - - - | Central Elec. | 1.0 | Cobalt 60 | Required | Exempt | III |
| CH-1067H | 552-1501 | Chatham (Tung-Sol) | 0.0012 | Krypton 85 | Not Required | Exempt | III |
| CH-1145 | - - - | Chatham (Tung-Sol) | 0.05 | Radium 226 | Required | Exempt | I |
| CK1097-7 | 062-6692 | Raytheon | 0.9 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| CK5829 | 272-8546 | Raytheon | - - - | - - - | Not Required | Exempt | II |
| DOD-004 | 181-0310 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| DOD-005 | 011-8325 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| E-35/5956 | 890-6617 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-37B/5957 | 806-0292 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-38/8370 | 067-9364 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-39 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-40/5958 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-40 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| E-41/5959 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| ECC81/12AT7 | 166-7662 | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| EF95/6AK5 | 188-5545 | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| EP68U/7615 | 825-4999 | Elec. Prod. | - - - | - - - | Required | Exempt | II |
| EP-92A/7619 | 825-4562 | Elec. Prod. | - - - | - - - | Required | Exempt | II |
| ET-415 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| EVM-1000 | 269-2736 | Eon Corp. | <0.1 | Nickel 63 | Required | Exempt | IV |
| EVM-3500 | 593-2555 | Eon Corp. | <0.1 | Nickel 63 | Required | Exempt | IV |
| EVM-3800 | - - - | Eon Corp. | <0.1 | Nickel 63 | Required | Exempt | IV |
| F-175D5 | - - - | Amprex Elec. | 100 mg | U02 enriched | Required | Exempt | III |
| F-2700 | - - - | ITT | 1.0 | Nickel 63 | Required | Exempt | IV |
| F-2701 | 808-6977 | ITT | 1.5 | Cesium 137 | Required | Exempt | III |
| F-2702 | - - - | ITT | 1.5 | Cesium 137 | Required | Exempt | III |
| F-2703 | - - - | ITT | 1.5 | Cesium 137 | Required | Exempt | III |
| F-2704-3 | 882-9042 | ITT | 1.5 | Cesium 137 | Required | Exempt | III |
| F-2705 | - - - | ITT | 1.0 | Nickel 63 | Required | Exempt | IV |
| F-2750 | - - - | ITT | 0.08 | Krypton 85 | Not Required | Exempt | III |
| F-2753 | - - - | ITT | 0.165 | Krypton 85 | Required | Exempt | III |
| F-2755 | - - - | ITT | 0.08 | Krypton 85 | Not Required | Exempt | III |
| F-2813C | - - - | ITT | 0.17 | Krypton 85 | Required | Exempt | III |
| F-2824A | - - - | ITT | 0.2 | Krypton 85 | Required | Exempt | III |
| F-2865 | - - - | ITT | 0.04 | Krypton 85 | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| F-2887 | - - - | ITT | 0.20 | Krypton 85 | Required | Exempt | III |
| F-2897 | - - - | ITT | 0.26 | Krypton 85 | Required | Exempt | III |
| F-2899 | - - - | ITT | 0.40 | Krypton 85 | Required | Exempt | III |
| GE 12661 | - - - | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| GE 14501 | - - - | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| GE 16841 | - - - | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| GE 20111 | - - - | General Electric | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| GL-5513 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-5556 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-5674 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-5740 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-5973 | 912-9285 | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| Gl-6019 | 912-9285 | General Electric | 0.005 | Thorium 232 | Not Required | Exempt | III |
| GL-6039 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| GL-6182 | - - - | General Electric | 0.02 | Thorium 232 | Not Required | Exempt | III |
| GL-6251 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-6410 | - - - | General Electric | 0.60 | Thorium 232 | Required | Exempt | III |
| GL-6942 | 834-5522 | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-7C29 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-7D21 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-7085/356 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GL-851 | - - - | General Electric | 0.03 | Thorium 232 | Not Required | Exempt | III |
| GV-3A | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3A-1200 | 878-6554 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3A-1400 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3A-1500 | 675-3439 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3A-1800 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3A-2000 | 812-6904 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3A-2500 | 850-6168 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| GV3A-950 | 060-3441<br>617-3128 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3B | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3B-400 | 824-7834 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3B-400/ VTR400 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3B-400/ VT-600 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3B-450 | 578-1141 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3B-600 | 686-6756 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3B-600/ VTR600 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3B-700 | 846-4538 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV3H | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV3S | - - - | Victoreen | 0.265 | Nickel 63 | Required | Exempt | IV |
| GV4S-2200 | 855-8789 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |
| GV4S-2800 | 855-8790 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |
| GV4S-3250 | 855-8791 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |
| GV4S-410 | 897-8418 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |
| GV5A | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-1200 | 617-5825 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| GV5A-1300 | 858-4074 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-1400 | 892-0975 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-1600 | 811-7500 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-2000 | 617-4105 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-2200 | 828-0613 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-2500 | 679-8153 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-3000 | 682-8649 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5A-3300 | 617-3482 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5B | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5C | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV5C-3300 | 617-3482 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV5C-3800 | 850-1522 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV5C-4000 | 617-4449 | - - - | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV5C-4300 | 852-7334 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6A | - - - | Victoreen | 0.20 | Nickel 63 | Required | Exempt | IV |
| GV6A-1000 | 880-5499 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6A-1200 | 808-4217 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6A-2000 | 729-5602 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6A-2100 | 835-1517 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| GV6A-2100 | 810-7348 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6A-700 | 826-1280 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6C | - - - | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| GV6C-3800 | 840-0766 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6R-1400 | 617-5647 | - - - | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV6R-1400A | 687-1034 | - - - | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV9A | - - - | Victoreen | 0.40 | Nickel 63 | Required | Exempt | IV |
| GV9A-1420 | 838-8881 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV9A-2000 | 613-6796 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV9A-600 | 933-1702 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV9A-900 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| GV9A-900 | 839-9525 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| HF-300 | 188-8648 | Electronic Interprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| KU-14/6777 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-16 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-17 | 924-5255 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-21 | 899-9964 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |





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Table C-ID - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| KU-22 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-23 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-24 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-25/8424 | 390-5208 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-26 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-27 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-28 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-29/8488 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-34 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-401/7603 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-402/8329 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-51 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-52/8264 | 852-5655 078-9500 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| KU-80 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-81 | - - - | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KU-82/7583 | 925-2111 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| KX-642 | 501-0866<br>581-8178 | Westinghouse | 0.3 | Krypton 85 | Required | Exempt | III |
| MA-306 | 630-2941 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-306B/7379 | 617-5696 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-310 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3101 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3102 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3103A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3104 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3106A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3106B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3107 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3108 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3109 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-311 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3112 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| A3113A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3113B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3114 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3115 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3116 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3117 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3118 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3119 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3120 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3121 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3122 | 842-7692 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3123 | 842-7693 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3124S | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3126 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3127 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3129 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3131 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3132 | 877-6026 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3133 | 856-9805 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA-3134 | 866-5427 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3136 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3136A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3136B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3137 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3138 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3148 | 464-4093 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3151 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3152 | 857-3083 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3152/3130 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3152/3135 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3153S | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3154 | 990-4631 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3157 | 914-6268 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3157D-8576 | 760-1786 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3158A | 918-3721 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA316/6232 | 780-8151 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3160 | 948-5806 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3161S | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA 3164/3162 | 947-7424 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3164/3163 | 145-3188 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3165 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3178 | - - - | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| MA-3192 | 942-6407 | Microwave | 0.42 | Cobalt 60 | Required | Exempt | III |
| MA-320 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3215-1 | 822-2494 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-324B | 617-3129 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-325 | 725-4068 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3254-1 | 478-4198 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-326 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-327 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-331A | 765-4539 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-331B | 765-4540 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-332 | 620-1435 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-332B | 628-1435 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-333 | 553-7337 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA-335B | 712-3891 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-336-7166 | 679-6507 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-336A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-336B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-336C | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-337, 337A | 688-9031 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-337B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-337C | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-337D | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-338A/7381 | 895-0802 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-338B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-339/6962 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-340B | 847-3752 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-340D | 811-6773 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-340F | 893-9153 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-340K | 843-0702 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-340T | 079-5341 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-341 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA-346 | 581-5535 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-347S | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-348 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-349 | 620-4528 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-349B | 897-6637 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-350 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-351A, B, C | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-352A, B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-354 | 765-4524 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-354A, B, C, D, E | - - - | Microwave | 0.40 | Cobalt | Required | Exempt | III |
| MA-355A | 051-0095 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356 | 707-0816 849-3494 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356A | 729-1269 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356C | 825-1000 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356E | 081-9255 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356L | 707-0816 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-356P | 503-8046 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-357 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA-358 | 824-8668 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-360/7309 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-361S | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-362A, B, C, D, E | 056-8225 (C) | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| | 762-7692 (E) | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-363 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-365/7562 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-365A, B, C, D | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-366 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-366-2 | 835-0057 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-366-3 | 835-0058 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-366-4 | 835-0059 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-366C | 835-0058 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-367 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-368A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-370A, AS | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-37019X | 469-7363 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3702 | 801-6865 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



MIL-HDBK-600
SECTION C

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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| MA-371A | 056-8223 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-372 | 822-8875 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3722 | 893-0151 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-373 | 005-1194 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3743 | 661-0340 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-375-7563 | 762-6682 | Microwave | 0.42 | Cobalt 60 | Required | Exempt | III |
| MA-375A/6560 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3753 | 486-9161 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA376/6636 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3763 | 479-1624 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-377 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3770 | 112-9482 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-378A | 610-8486 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3784 | 140-6535 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3785 | 139-3204 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-379 | 262-0174 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-379A, B, C, DS | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3799 | 003-7978 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| MA-38GI | 959-7413 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-382A | 056-8227 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-38155 | 763-8807 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-382A | 856-8227 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3821Z | 938-4359 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-383 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3830X | 422-5376 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3847 | 105-3737 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3848Z | 105-8594 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3855K | 226-4306 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3879X | 577-3250 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3886Z | 160-2183 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| MA-3887Z | 160-2186 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| MA-3895X | 495-9113 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| MA-3899Z | 449-7994 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| MA-390 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3919Z | 009-8082 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| MA-393 | 850-9920 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-394 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-395 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-3959Z | 01-013-0079 | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MA-396 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-397 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MA-399 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| MCC-17A | 788-9889 | Metcom (Omni-wave) | 0.20 | Cobalt 60 | Required | Exempt | III |
| MCT 14 | 295-3589 | Metcom (Omni-wave) | 0.40 | Cobalt 60 | Required | Exempt | III |
| MCT 20 | 503-4436 | Metcom (Omni-Wave) | 0.40 | Cobalt 60 | Required | Exempt | III |
| MDS-56/ 8311 | 064-7943 | Metcom (Omni-Wave) | 0.40 | Cobalt 60 | Required | Exempt | II |
| MD 100X2 | 134-0912 | Micro Dynamics | 3000.0 | Tritium (H3) | Required | Exempt | IV |
| MD 100X3 | 134-0913 | Micro Dynamics | - - - | Tritium (H3) | Required | Exempt | IV |
| MD 100X5 | 857-3083 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD70C1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C2 | 262-0174 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C3 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C4 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C5 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C6 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD70C7 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70C8 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K10 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K201 | 937-9459 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K3 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K4 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K5 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K6 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K7 | 835-2575 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K8 | 762-7692 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70K9 | 822-8875 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70L1 | 707-0816 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70L2 | 942-6407 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70S1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70S2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70S3 | 226-2035 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70V1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD70V2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70V3 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70V4 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70V5 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70V6 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X1 | 188-3534 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X10 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X11 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X12 | 073-9277 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X13 | 760-1786 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X14 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X15 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X16 | 897-6637 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X17 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X18 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X19 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X20 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X201 | 897-6637 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD70X21 | 893-0151 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X22 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X3 | 583-0613 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X4 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X5 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X6 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X7 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X8 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD70X9 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD72L1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD72L2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD72L3 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD73C1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73C2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73C3 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73C4 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K3 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |





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Table C-19 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD73J4 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K5 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K6 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K7 | 762-6682 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73K8 | - - - | Micro Dynamics | 1.0 | Cpbalt 60 | Required | Exempt | III |
| MD73S1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73V1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73V2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X10 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X11 | 617-3129 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X12 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X13 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X14 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X15 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X16 | 947-7292 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X3 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X4 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD73X5 | 947-7424 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X6 | 801-6865 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X7 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X8 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD73X9 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD74S1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD79C1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79K1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79K2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79S1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79U1 | 866-5263 | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79V1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD79X1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exmept | III |
| MD79X2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD80C1 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C10 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C11 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD70C12 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C13 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | UCC and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80C14 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C15 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C16 | - - - | Micro Dynamics | 36000 | Tritium (H2) | Required | Exempt | IV |
| MD80C17 | 106-0423 | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C19-1 | 106-0416 | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C18 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C19 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C2 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C20 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C21 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C22 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C23 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C24 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C25 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C26 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C3 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C4 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C5 | 404-8546 | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C6 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C7 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80C8 | 225-3419 | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80C9 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD80K1 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K7 | 449-7994 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K10 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K11 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K12 | 160-2186 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K13 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K14 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K15 | 160-2186 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K16 | - - - | Micro Dynamcis | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K17 | 009-8087 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K18 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K19 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K2 | 105-3757 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K20 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K21 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K22 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K23 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K24 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |





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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD90K25 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K26 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K27 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K28 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K29 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K3 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K30 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K31 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K32 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K33 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K4 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K5 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K6 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K7 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K8 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80K9 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80L1 | - - - | Micro Dynamics | 18000 | Tritian (H3) | Required | Exempt | IV |
| MD80L2 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S1 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80S10 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S11 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S12 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S13 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S2 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S3 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S4 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S5 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S6 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S7 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S8 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80S9 | - - - | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD80X1 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X10 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X11 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X12 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X13 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X14 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X15 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |





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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80X16 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X17 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X18 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X19 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X2 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X20 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X21 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X22 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X23 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X24 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X25 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X26 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X27 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X28 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X29 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X3 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X30 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X31 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X32 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80X33 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X34 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X35 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X36 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X37 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X38 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X39 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X4 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X40 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X41 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X42 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X43 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X44 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X45 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X46 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X47 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X5 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X6 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X7 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |





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Table C-19 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD80X8 | - - - | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD80X9 | 495-9113 | Micro Dynamics | 3000 | Tritium (H3) | Required | Exempt | IV |
| MD81C1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81K1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81X1 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81X2 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81X3 | - - - | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81X4 | 422-5376 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD81X5 | 469-7363 | Micro Dynamics | 0.5 | Cobalt 60 | Required | Exempt | III |
| MD82L1 | 729-1269 | Micro Dynamics | 18000 | Tritium (H3) | Required | Exempt | IV |
| MD83K1 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD84C1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD84K1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD84X1 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD84X2 | - - - | Micro Dynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| MD85C1 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD85C2 | - - - | Micro Dynamics | 36000 | Tritium (H3) | Required | Exempt | IV |
| MD85K1 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K10 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD85K11 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K12 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K13 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K14 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K15 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K16 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K17 | 009-8082 | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K18 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K2 | 105-8594 | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K3 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K4 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K5 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K6 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K7 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K8 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85K9 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X1 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X2 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X3 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MD8[illegible]4 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X5 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X6 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MD85X7 | - - - | Micro Dynamics | 6000 | Tritium (H3) | Required | Exempt | IV |
| MKT-11/6282 | 365-2548 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| MLT-13/7823 | 729-1269 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| ML-199-8094 | 948-6413 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| ML-212E | - - - | Machlett | 0.00068 | Thorium 232 | Not Required | Exempt | III |
| ML-241B | - - - | Machlett | 0.00068 | Thorium 232 | Not Required | Exempt | III |
| ML-279 | 188-3541 | Machlett | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| ML-356/5771 | - - - | Machlett | 0.036 | Thorium 232 | Not Required | Exempt | III |
| ML-357B | 237-0108 | Machlett | 0.0011 | Thorium 232 | Not Required | Exempt | III |
| ML-379 | - - - | Machlett | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| ML-5530 | - - - | Machlett | 0.0096 | Thorium 232 | Not Required | Exempt | III |
| ML-5531 | - - - | Machlett | 0.029 | Thorium 232 | Not Required | Exempt | III |
| ML-5541 | - - - | Machlett | 0.014 | Thorium 232 | Not Required | Exempt | III |
| ML-5681 | - - - | Machlett | 0.086 | Thorium 232 | Not Required | Exempt | III |
| ML-5682 | 755-9109 | Machlett | 0.20 | Thorium 232 | Required | Exempt | III |
| ML-5736 | - - - | Machlett | 0.006 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| ML-6256 | - - - | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| ML-6257 | 755-9108 | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| ML-6258 | 809-3372 | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| ML-6420 | - - - | Machlett | 0.0091 | Throium 232 | Not Required | Exempt | III |
| ML-6421 | - - - | Machlett | 0.0091 | Throium 232 | Not Required | Exempt | III |
| ML-6422 | - - - | Machlett | 0.0091 | Thorium 232 | Not Required | Exempt | III |
| ML-6423 | 816-1386 | - - - | 0.0091 | Thorium 232 | Not Required | Exempt | III |
| ML-6424 | - - - | Machlett | 0.016 | Thorium 232 | Not Required | Exempt | III |
| ML-6425 | - - - | Machlett | 0.016 | Thorium 232 | Not Required | Exempt | III |
| ML-6426 | - - - | Machlett | 0.03 | Thorium 232 | Not Required | Exempt | III |
| ML-6427 | - - - | Machlett | 0.03 | Thorium 232 | Not Required | Exempt | III |
| ML-6544 | 561-8230 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| ML-6577 | - - - | Machlett | 0.036 | Thorium 232 | Not Required | Exempt | III |
| ML-6623 | - - - | Machlett | 0.006 | Thorium 232 | Not Required | Exempt | III |
| ML-6696 | 763-7806 | Machlett | 0.053 | Thorium 232 | Not Required | Exempt | III |
| ML-6908 | - - - | Machlett | 0.0066 | Thorium 232 | Not Required | Exempt | III |
| ML-7002 | - - - | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| ML-7003 | 681-8017 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| ML-7003A | 020-0074 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |




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Table C-19 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| ML-[illegible]07 | - - - | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| ML-7120 | - - - | Machlett | 0.0091 | Thorium 232 | Not Required | Exempt | III |
| ML-7121 | - - - | Machlett | 0.0091 | Thorium 232 | Not Required | Exempt | III |
| ML-7124 | - - - | Machlett | 0.016 | Thorium 232 | Not Required | Exempt | III |
| ML-7125 | - - - | Machlett | 0.016 | Thorium 232 | Not Required | Exempt | III |
| ML-7248 | 615-5607 | Machlett | 0.0014 | Throium 232 | Not Required | Exempt | III |
| ML-7482/7560 | - - - | Machlett | 0.1078 | Thorium 232 | Required | Exempt | III |
| ML-7715 | 137-7692 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| ML-7845 | - - - | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| ML-8094/199 | 984-6413 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| ML-8317 | 942-3980 | Machlett | 0.1078 | Thorium 232 | Required | Exempt | III |
| MPT-12/7821 | 799-9471 | Omni-Wave (Metcom) | - - - | Cobalt 60 | Required | Exempt | III |
| MPT-13/7901 | 689-1922 882-1994 | Omni-Wave (Metcom) | 1.0 | Cobalt 60 | Required | Exempt | III |
| MPT-16 | 842-9233 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| MPT-22 | 897-5744 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| MPT-23 | 894-7127 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| MPT-24 | 894-7128 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| MSA-10 | 553-7337 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| MSA-14 | - - - | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| MST-60 | 845-6417 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| MXT-44 | 060-6825 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| MXT-61B | 073-9277 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| MXT-91 | 583-4641 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| MXT-94 | 772-6061 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| PJ-4 (MX) | - - - | General Elec. | 0.03 | Thorium 232 | Not Required | Exempt | III |
| PL-195 | - - - | Penta Labs | 0.0029 | Thorium 232 | Not Required | Exempt | III |
| PL-8295/172 | 993-5167 | Penta Labs | 0.0029 | Thorium 232 | Not Required | Exempt | III |
| PL-8295A | - - - | Penta Labs | 0.0029 | Thorium 232 | Not Required | Exempt | III |
| P5A | - - - | Electronics, Inc. | 0.02 | Krypton 85 | Not Required | Exempt | III |
| QKS-1347 | - - - | Raytheon | 0.35 | Thorium 232 | Required | Exempt | III |
| QKS-1350 | - - - | Raytheon | 0.01 | Thorium 232 | Not Required | Exempt | III |
| QKS-1380 | - - - | Raytheon | 0.37 | Thorium 232 | Required | Exempt | III |
| QKS-1442 | - - - | Raytheon | 0.01 | Thorium 232 | Not Required | Exempt | III |
| QKS-1443 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QKS-1452 | - - - | Raytheon | 1.0 | Thorium 232 | Required | Exempt | III |
| QKS-1484 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QKS-1517 | - - - | Raytheon | 0.01 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| QKS-1539 | - - - | Raytheon | 0.06 | Thorium 232 | Not Required | Exempt | III |
| QKS-1540 | - - - | Raytheon | 0.38 | Thorium 232 | Required | Exempt | III |
| QKS-1541 | - - - | Raytheon | 0.38 | Thorium 232 | Required | Exempt | III |
| QKS-1543 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QKS-1590 | - - - | Raytheon | 0.91 | Thorium 232 | Required | Exempt | III |
| QKS-1594 | - - - | Raytheon | 0.38 | Thorium 232 | Required | Exempt | III |
| QKS-1595 | - - - | Raytheon | 0.38 | Thorium 232 | Required | Exempt | III |
| QKS-1611 | - - - | Raytheon | 0.03 | Thorium 232 | Not Required | Exempt | III |
| QKS-1629 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QKS-1648 | - - - | Raytheon | 0.03 | Thorium 232 | Not Required | Exempt | III |
| QKS-1655 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| QKS-1672 | - - - | Raytheon | 1.0 | Thorium 232 | Required | Exempt | III |
| QKS-1697 | - - - | Raytheon | 0.04 | Thorium 232 | Not Required | Exempt | III |
| QK-1000 | - - - | Raytheon | 0.013 | Thorium 232 | Not Required | Exempt | III |
| QK-1002 | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| QK-1102 | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| QK-1110 | - - - | Raytheon | 0.03 | Thorium 232 | Not Required | Exempt | III |
| QK-1122 | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| QK-1130 | - - - | Raytheon | 0.03 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| QK-1104 | - - - | Raytheon | 0.55 | Thorium 232 | Required | Exempt | III |
| QK-1220 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| QK-1243 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| QK-1244 | - - - | Raytheon | 0.01 | Thorium 232 | Not Required | Exempt | III |
| QK-1267 | - - - | Raytheon | 0.35 | Thorium 232 | Required | Exempt | III |
| QK-1286 | - - - | Raytheon | 0.37 | Thorium 232 | Required | Exempt | III |
| QK-172/6959 | 262-3768<br>951-6292 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QK-221/6002 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| QK-235/6344 | - - - | Raytheon | 0.49 | Thorium 232 | Required | Exempt | III |
| QK-241 | - - - | Raytheon | 0.169 | Thorium 232 | Required | Exempt | III |
| QK-254/6518 | 061-0485 | Raytheon | 0.40 | Thorium 232 | Required | Exempt | III |
| QK-283A | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| QK-324C/7452 | - - - | Raytheon | 0.17 | Thorium 232 | Required | Exempt | III |
| QK-327/7529 | 682-2771 | Raytheon | 0.19 | Thorium 232 | Required | Exempt | III |
| QK-338A/641CA | - - - | Raytheon | 0.46 | Thorium 232 | Required | Exempt | III |
| QK-349/6249B | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| QK-390 | - - - | Raytheon | 0.12 | Thorium 232 | Required | Exempt | III |
| QK-410A | - - - | Raytheon | 0.008 | Thorium 232 | Not Required | Exempt | III |
| QK-427 | - - - | Raytheon | 0.003 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| QK-428A/6406A | - - - | Raytheon | 0.35 | Thorium 232 | Required | Exempt | III |
| QK-437/6402B | - - - | Raytheon | 0.07 | Thorium 232 | Not Required | Exempt | III |
| QK-447/6843 | 284-6000 | Raytheon | 0.61 | Thorium 232 | Required | Exempt | III |
| QK-494 | - - - | Raytheon | 0.05 | Thorium 232 | Not Required | Exempt | III |
| QK-496 | - - - | Raytheon | 0.04 | Thorium 232 | Not Required | Exempt | III |
| QK-506 | - - - | Raytheon | 0.04 | Thorium 232 | Not Required | Exempt | III |
| QK-622/8129 | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| QK-624/7718 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| QK-665 | - - - | Raytheon | 0.81 | Thorium 232 | Required | Exempt | III |
| QK-665 | - - - | Raytheon | 3.50 | Thorium 232 | Required | Exempt | III |
| QK-666 | - - - | Raytheon | 3.50 | Thorium 232 | Required | Exempt | III |
| QK-668 | - - - | Raytheon | 0.008 | Thorium 232 | Not Required | Exempt | III |
| QK-727 | - - - | Raytheon | 0.04 | Thorium 232 | Not Required | Exempt | III |
| QK-783 | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| QK-846/7798 | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| QK-871 | - - - | Raytheon | 2.20 | Thorium 232 | Required | Exempt | III |
| QK-883 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| QK-898 | - - - | Raytheon | 0.46 | Thorium 232 | Required | Exempt | III |
| QK-904 | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| QK-912 | - - - | Raytheon | 2.30 | Thorium 232 | Required | Exempt | III |
| QK-942 | - - - | Raytheon | 1.25 | Thorium 232 | Required | Exempt | III |
| QK-943 | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| QK-976 | - - - | Raytheon | 2.20 | Thorium 232 | Required | Exempt | III |
| QK-990 | - - - | Raytheon | 0.38 | Thorium | Required | Exempt | III |
| QS-1212 | - - - | English Elec. Valve Company | 0.10 | Uranium (UO2) | Required | Exempt | III |
| QS-1213 | - - - | English Elec. Valve Company | 0.10 | Uranium (UO2) | Required | Exempt | III |
| QS-1215 | - - - | English Elec. Valve Company | 0.10 | Uranium (UO2) | Required | Exempt | III |
| RK-6002 | - - - | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| RK-6249B | - - - | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| RK-6344 | - - - | Raytheon | 0.49 | Thorium 232 | Required | Exempt | III |
| RK-6402B | - - - | Raytheon | 0.07 | Thorium 232 | Not Required | Exempt | III |
| RK-6406A | - - - | Raytheon | 0.03 | Thorium 232 | Not Required | Exempt | III |
| RK-6410A | - - - | Raytheon | 0.19 | Thorium 232 | Required | Exempt | III |
| RK-6518 | - - - | Raytheon | 0.40 | Thorium 232 | Required | Exempt | III |
| RK-6543 | - - - | Raytheon | 0.61 | Thorium 232 | Required | Exempt | III |
| RK-6959 | 951-6292 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| RK-7452 | - - - | Raytheon | 0.17 | Thorium 232 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| RK-/529 | - - - | Raytheon | 0.19 | Thorium 232 | Required | Exempt | III |
| RK-7718 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| RK-7798 | - - - | Raythoen | 0.19-0.48 | Thorium 232 | Required | Exempt | III |
| RK-8129 | - - - | Raytheon | 0.19-0.48 | Thorium 232 | Required | Exempt | III |
| R1130B/1B59 | 729-1719 | - - - | - - - | - - - | Required | Exempt | II |
| R1131C | 038-1663 | Sylvania | - - - | - - - | Required | Exempt | II |
| R4340 | - - - | Sylvania | - - - | - - - | Required | Exempt | II |
| R4410 | 665-3150 | Sylvania | - - - | - - - | Required | Exempt | II |
| TD-19 | - - - | Bendix | 0.03 | Radium 226 | Required | Exempt | I |
| TD-36 | 855-8680 | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| TD-36A | 866-5264 | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| TD-38 | - - - | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-39 | 471-9818 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-40 | 081-0192 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-41 | 813-5616 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-42 | 968-5229 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-43 | - - - | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-44/7988 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-49 | - - - | Bendix | 0.60 | Radium 226 | Required | Exempt | I |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TD-51 | - - - | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| TD-52 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-53 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-55 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-56 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-58 | 828-8004 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-61 | 855-8682 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-63 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-64 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-65 | - - - | Bendix | 0.03 | Radium 226 | Required | Exempt | I |
| TD-66 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-67 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-68 | - - - | Bendix | 0.03 | Radium 226 | Required | Exempt | I |
| TD-69 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-72/8059 | 844-5182<br>969-4162 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-74 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-78 | 852-5546 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-81 | 054-3205 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TD-9A | - - - | Bendix | 0.03 | Radium 226 | Required | Exempt | I |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TE-31 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TE-32 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TE-33 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TE-34 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TE-72 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TE-73 | - - - | Signalite | 0.5 - 1.0 mg | Thorium 232 Thorium 228 | Not Required | Exempt | I |
| TG-100 | (5841-989-2364) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-101 | (5920-629-2731) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-102 | (5841-774-9480) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-103 | (5841-117-4090, 5840-251-8620) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-104 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-105 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-106 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-107 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-108 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-109 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-11 | - - - | Signalite | 3 - 5.0 | Cesium 137 | Required | Exempt | III |
| TG-110 | (See 5920-856-5965, 1430-896-5689) | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-111 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-112 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-114 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-115 | (See 5920-953-1924) | | | | | | |
| TG-116 | - - - | Signalite | 1 - 2.00 | Cesium 137 | Required | Exempt | III |
| TG-117 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-118 | (1270-673-6060) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-119 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-12 | - - - | Signalite | 3 - 5.0 | Cesium 137 | Required | Exempt | III |
| TG-120 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-121 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-122 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-123 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-124 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-125 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-126 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-127 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-13 | - - - | Signalite | 3 - 5.00 | Cesium 137 | Required | Exempt | III |
| TG-131 | - - - | Signalite | <1.0 | Cesium 137 | Required | *Exempt | III |
| TG-132 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-133 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-134 | (See 5920-950-0902) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-135 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-136 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-139 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-14 | (See 5920-163-4107) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-140 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-141 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-142 | - - - | Signalite | 0.09 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-143 thru TG-146 | - - - | Signalite | 0.06 | Cesium 137 | Required | Exempt | III |
| TG-148 | 077-1828 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-149 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-15 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-151 | (See 5920-789-7979) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-153 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-155 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-156 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-157 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-16 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-160 | (5840-791-5302) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-161 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-162 | (5840-880-5304, 5840-760-1298) | - - - | - - - | - - - | - - | - - - | - - - |
| TG-163 | (5840-880-5303) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-164 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-165 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-166 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-167 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-168 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-169 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-17 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-170 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-171 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-172 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-173 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-174 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-175 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-176 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-178 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-18 | - - - | Signalite | 0.1 - 0.2 | Cesium 137 | Required | Exempt | III |
| TG-183 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-184 | (See 5920-163-4106) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-186 | - - - | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-187 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-188 | - - - | Signalite | 1.0 | Cesium 137 | Required | Exempt | III |
| TG-189 | (1430-613-9185) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-19 | (5840-755-4307) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-190 | (5841-978-2492) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-191 | (See 5920-011-4995) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-192 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-194 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-196 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-197 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-198 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-2 | 171-9725 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-20A | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-21 | - - - | Signalite | 0.1 - 0.2 | Cesium 137 | Required | Exempt | III |
| TG-22 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-24 | (1270-673-5107) | - - - | - - - | - - - | - - - | - - - | - - - |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| T -249 | (5840-003-6050) | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-25 | (See 5920-578-3088) | Signalite | 1.0 | Cesium 137 | Required | Exempt | III |
| TG-26A | (5840-812-1976, 5821-400-5429) | - - - | - - - | - - - | - - - | Exempt | III |
| TG-27A | (See 5920-822-2329) | - - - | - - - | - - - | - - - | Exempt | III |
| TG-28A | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-29 | (5840-819-0036) 878-1839 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-30 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-31 | (See 5920-787-7036) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-32 | (5840-557-5680) 853-0056 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-33 | 853-0056 | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-34 | (See 5920-738-8074) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-35 | 821-6134 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-36 | 808-6977 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |



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Table C-19 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-137 | (5840-581-2170) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-38 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-39 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-40 | (See 5920-027-4320) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-41A | 800-4409 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-42A | (See 5920-858-0215) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-43A | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-44 | (See 5920-027-8691, 5920-892-9060, 5895-056-9194) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-45 | (See 5920-578-9265) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-46 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-47 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-48 | (See 5020-727-0595) 077-1828 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-49 | (See 5020-062-3993, 5920-062-5995) | - - - | - - - | - - - | - - - | - - - | - - - |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-50 | 805-3626 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-51 | (See 5920-813-4039) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-52 | (See 5020-822-2326) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-52/633B | 865-4959 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-53 | (See 5920-827-5751, 5920-679-3827) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-54 | (See 5920-682-7301) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-55 | (See 5920-827-5752) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-56 | (See 5920-806-1090, 5920-839-5538) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-57 | (See 5920-608-2142) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-58 | (See 5920-827-5753) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-59 | (See 5920-613-0114) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-6 | - - - | Signalite | < 1.0 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-60 | (5840-131-6679, 5840-010-7649) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-61 | (See 5920-107-5371) 882-9042 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-62 | 882-9042 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-63 | 882-9042 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-64 | (See 5840-010-7647) 435-1784 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-65 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-66 | (See 5840-881-9675) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-67 | (See 5840-010-7651) 882-9042 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-68 & 69 | - - - | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-7 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-70 & 71 | - - - | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-72 | 966-9469 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-73 | 774-6338 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-74 | 774-6338 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |





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Table C-10 - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-75 | 774-6338 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| TG-76 | (See 5920-713-4191) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-77 | 774-6338 | Signalite (Bendix) | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-78 | (See 5920-020-9745) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-79 | - - - | Signalite (Bendix) | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-8 | - - - | Signalite (Bendix) | < 1.0 | Cesium 137 | Required | Exempt | III |
| TG-80 to 82 | - - - | Signalite (Bendix) | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-83 | (See 5920-687-2067) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-84 | (See 5920-687-2066) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-85 | - - - | Signalite (Bendix) | 0.90 | Cesium 137 | Required | Exempt | III |
| TG-86 | (See 5920-822-2327) | - - - | - - - | - - - | - - - | - - - | - - - |
| TG-87 | (See 5920-856-5966, 5920-087-6337) | - - - | - - - | - - - | - - - | - - - | - - - |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TG-88 | - - - | Signalite (Bendix) | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-89 | - - - | Signalite Bendix | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-9 | - - - | Signalite (Bendix) | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-90 | 972-8990 | Signalite (Bendix) | <1.0 | Cesium 137 | Required | Exempt | III |
| TG-91 thru TG-99 | - - - | Signalite (Bendix) | 1.0 - 2.0 | Cesium 137 | Required | Exempt | III |
| TG-98 | (5840-754-7507) | - - - | - - - | - - - | - - | - - - | - - - |
| TG-99 | - - - | Signalite (Bendix) | 0.90 | Cesium 137 | Required | Exempt | III |
| TN-11 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-12 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-13 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-5 0 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-6 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-7 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-8 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TN-9 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| TR331/5925 | 262-1693 267-1693 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| TR361/5865 | 262-0174 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| TR361/5865 | 262-0174 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| TR368/5853 | 193-5147 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| TR368/5853 | 193-5147 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| TR368/5853 | 193-5147 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| TR432/6232 | 390-5241 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| TR610/BLT033 | 817-6181 | Varian | 0.9 | Cobalt 60 | Required | Exempt | III |
| T236A/7416 | 866-5264 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| UE2-450 | 581-9594 | United Electronics | 0.0015 | Thorium 232 | Not Required | Exempt | III |
| USA6605 | 581-5535 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Required | Exempt | III |
| USN8370/E38 | 067-9364 | - - - | - - - | - - - | Required | Exempt | II |
| UX6653 | 636-2217 | - - - | - - - | - - - | Required | Exempt | II |
| VDC1025 | 404-8546 | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| VDC-1029 | 106-0423 | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| VDK-1000A | - - - | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |
| VDS-1011 | 186-1964 | Varian | 70.0 | Tritium (H3) | Required | Exempt | III |
| VDX-1015 | - - - | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| VDS-1020 | - - - | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| VDS-1032 | - - - | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| VDX-1001A | - - - | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| VDX1003 | 469-7363 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| VDX1017 | 134-0912 | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |
| VDX1017E | 009-7572 | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |
| VDX1031 | 134-0913 | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |
| VDX1033 | - - - | Varian | 30.0 | Promethium 147 | Required | Exempt | IV |
| VDX1035 | 009-2771 | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |




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Table C-10 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| VTNC204 | 725-2234 | Varian | - - - | - - - | Required | Exempt | II |
| VTR400/GV3B400 | - - - | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| VTR600/GV3B600 | - - - | Victoreen | 1.0 | Nickel 63 | Not Required | Exempt | IV |
| VXR130 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR1500 | 731-1742 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR16000 | 296-4142 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR1800 | 578-8701 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR2500 | 816-3759 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR2501 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR2700 | 501-0865 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR2701 | 557-6401 | Victoreen | 1.00 | Nickel 63 | Required | Exempt | IV |
| VXR3002 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR3500 | 593-2555 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR3501 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR4000 | 617-4449 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR4001 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR5000 | - - - | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| VXR7001 | 661-0267 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| WD160 | 134-0913 | Westinghouse | 10.0 | Promethium 147 | Required | Exempt | IV |



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*Table C-1D - Continued*

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| WD160 | 134-0913 | Westinghouse | 45.0 | Tritium (H3) | Required | Exempt | IV |
| WD191 | 425-6246 | Westinghouse | 10.0 | Promethium 147 | Required | Exempt | IV |
| WD191 | 425-6246 | Westinghouse | 45.0 | Tritium (H3) | Required | Exempt | IV |
| WL-4404 | - - - | Westinghouse | 0.8 | Krypton 85 | Required | Exempt | III |
| WL-7188 | - - - | Westinghouse | 0.19 | Uranium 235 | Required | Exempt | III |
| WL-759 | 284-7229 | Westinghouse | 0.3 | Krypton 85 | Required | Exempt | III |
| WX-3891A, B, C, D | - - - | Westinghouse | 0.8 | Krypton 85 | Required | Exempt | III |
| XAS-1117 | - - - | General Elec. | 6.0 | Americium 241 | Required | Exempt | I |
| XB818 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XB819 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XB846 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XD708 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XD751 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XD764 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XD765 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XD860 | - - - | Bendix | 50.0 | Radium 226 | Required | Exempt | I |
| XD972 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XG-721 | - - - | Bendix | 1.0 - 2.0 | Cesium 137 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| [illegible]-722 | - - - | Bendix | 1.0 - 2.0 | Cesium 137 | Required | Exempt | III |
| XG-739 | - - - | Bendix | 0.80 | Cesium 137 | Required | Exempt | III |
| XG-768 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-769 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-828 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-836 | - - - | Bendix | 0.09 | Cesium 137 | Not Required | Exempt | III |
| XG-845 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-901 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-919 | - - - | Bendix | 0.09 | Cesium 137 | Not Required | Exempt | III |
| XG-946 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-949 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-953 | - - - | Bendix | 0.09 | Cesium 137 | Not Required | Exempt | III |
| XG-954 | - - - | Bendix | 0.09 | Cesium 137 | Not Required | Exempt | III |
| XG-962 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-964 | - - - | Bendix | 0.09 | Cesium 137 | Not Required | Exempt | III |
| XG-975 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-987 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-988 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-989 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| XG-990 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XG-992 | - - - | Bendix | 0.90 | Cesium 137 | Required | Exempt | III |
| XN-757 | - - - | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| XTT-4253 | - - - | Sylvania | 0.03 | Krypton 85 | Not Required | Exempt | III |
| X6007/5791 | 220-6526 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| X7108/5790 | 220-6525 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| X7109/5863 | 261-8671 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| X7109/5863 | 261-8671 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| Y-1938 | - - - | General Elec. | 0.0000001 | Rhenium 187 | Not Required | Exempt | IV |
| Y-1939 | - - - | General Elec. | 0.0000001 | Rhenium 187 | Not Required | Exempt | IV |
| Y-300 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| Y-4075 | - - - | General Elec. | 0.0000001 | Rhenium 187 | Not Required | Exempt | IV |
| Y-4102 | - - - | General Elec. | 0.0000001 | Rhenium 187 | Not Required | Exempt | IV |
| Y-4108 | - - - | General Elec. | 0.0000001 | Rhenium 187 | Not Required | Exempt | IV |
| Y-4109 | - - - | - - - | - - - | - - - | - - - | - - - | - - - |
| 0A2 | 503-4880<br>557-6883 | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 0A2WA | 503-4880<br>769-0869<br>617-6367 | CBS Hytron | 0.02 | Cobalt 60 | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 0A2WA | 503-4880 | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 0A2WA | 503-4880 | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 0A2WA | 503-4880 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 0A2WA | - - - | English Elec. Valve Company | 0.10 | Uranium Oxide (UO2) | Required | Exempt | III |
| 0A3A | - - - | Raytheon | 0.05 | Krypton 85 | Not Required | Exempt | III |
| 0B2 | 166-7648<br>557-6885 | Raytheon | 0.04 | Krypton 85 | Not Required | Exempt | III |
| 0B2WA | 624-4718 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 0B2WA | 624-4718 | Raytheon | 0.04 | Krypton 85 | Not Required | Exempt | III |
| 0B2WA | 624-4718 | CBS Hytron | 0.05 | Nickel 63 | Not Required | Exempt | IV |
| 0B2WA/6627 | 624-4718 | Jack & Heintz | 6.0 | Radium 226 | Required | Exempt | IV |
| 0B2WA | 624-4718 | Tung-Sol (Chatham) | 0.005 | Radium 226 | Not Required | Exempt | I |
| 0B2WA | - - - | English Elec. Valve Company | 0.10 | Uranium Oxide (UO2) | Required | Exempt | III |
| 0B3A | - - - | Raytheon | 0.08 | Krypton 85 | Not Required | Exempt | III |
| 0C2 | 800-0556 | Raytheon | 0.025 | Krypton 85 | Not Required | Exempt | III |
| 0C3A | - - - | Raytheon | 0.05 | Krypton 85 | Not Required | Exempt | III |
| 0C3W | 188-0968 | Raytheon | 0.05 | Krypton 85 | Not Required | Exempt | III |
| 0D3A | - - - | Raytheon | 0.067 | Krypton 85 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 0D3W | 193-5085 | Raytheon | 0.067 | Krypton 85 | Not Required | Exempt | III |
| 0G3 | 836-6273 | Tung-Sol (Chatham) | 0.004 | Krypton 85 | Not Required | Exempt | III |
| 1AD2 | 715-1833 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 1AL2 | - - - | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 1B22 | 193-5088 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 1B23 | 355-7250<br>188-3535 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B23 | 188-3535 | General Elec. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B23 | 188-3535 | Central Sales | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B24 | 188-3533 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 1B24A | 193-5091 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B24A | 193-5091 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B24A | 193-5091 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 1B24A | 193-5091 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B24A | 193-5091 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 1B26 | 262-0137 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B27 | 193-5092 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B29 | 261-9155 | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 1B3GT | 193-5087 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1B3[illegible] | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 1B32 | 188-6582 | Westinghouse | - - - | - - - | Required | Exempt | II |
| 1B35A | 262-1356 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B35A | 262-1356 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B35A | 262-1356 | Omni-Wave (Metcom) | 0.2 | Cobalt 60 | Required | Exempt | III |
| 1B36 | 262-0122 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 1B36A | 755-4210 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 1B37 | 188-6581 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B37A | 220-6892 | General Elec. | 0.950 | Cobalt 60 | Required | Exempt | III |
| 1B37A | 220-6892 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B37A | 220-6892 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B38 | 237-2413 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 1B38 | 237-2413 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B40 | 237-2414 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 1B40 | 237-2414 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B41 | 262-0123 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 1B41 | 262-0123 | Westinghouse | 1.0 | Radium 226 | Required | Exempt | I |
| 1B42 | 262-0124 | Machlett | 1.0 | Radium 226 | Required | Exempt | I |
| 1B44 | 193-5093 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1B44 | 193-5093 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B44 | 193-5093 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B45 | 262-0125 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 1B45 | 262-0125 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 1B49 | - - - | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 1B50 | 256-9985 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B51 | 256-9988 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B52 | 262-0191 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B53 | 245-9561<br>262-0192 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B54 | 295-9561 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B55 | 060-3475<br>262-0230 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 1B56 | 193-5094 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B56 | 193-5094 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B56 | 193-5094 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B57 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 1B58 | 193-5095 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 1B58 | 193-5095 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B58 | 193-5095 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1b58A | 665-1057 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 1B58A | 665-1057 | Microwave | 0.42 | Cobalt 60 | Required | Exempt | III |
| 1B59/R1130B | 729-1719 | - - - | - - - | - - - | Required | Exempt | II |
| 1B60 | 853-6468<br>193-5091 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B60 | 193-5096 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 1B63 | - - - | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B63A | 735-4210<br>188-3534 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 1B63A | 188-3534 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 1B63A | 188-3534 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 1B63A | 188-3534 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 1B63B | 365-6183 | Varian | .15 | Cobalt 60 | Required | Exempt | III |
| 1DNS | - - - | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 1G3GT | 193-5087<br>752-5270 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 1J3 | - - - | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 1K3 | 676-7979 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 1R5 | 188-3954 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 1S5 | 188-3952 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 1T4 | 188-3595 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1U4 | 188-3593 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 1U5 | 188-0844 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 100IG | - - - | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 100R | - - - | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 100TH | 556-1236<br>116-9979 | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 100TL | - - - | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 100D1 | - - - | Eimac | 0.00056 | Thorium 232 | Not Required | Exempt | III |
| 1010/6476 | 864-2984 | Raytheon | 0.90 | Cesium 137 | Required | Exempt | III |
| 1022 | - - - | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 1037/6437 | 800-5552 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 1038 | - - - | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 1039/6438 | - - - | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 1056 | - - - | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 1058/1058A | - - - | Raytheon | 0.02 | Nickel 63 | Not Required | Exempt | IV |
| 1061/5787A | 262-3771 | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1068/5783A | 284-7166<br>230-5253 | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1069/6542A | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1071 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1072 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |




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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1073 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1075/6213A | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1077 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1078 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1079 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1080 | - - - | Raytheon | 0.9 | Cesium 137 | Required | Exempt | III |
| 1096 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 1097 | - - - | Raytheon | 0.9 | Cesium 137 | Required | Exempt | III |
| 1098 | - - - | Raytheon | 0.9 | Cesium 137 | Required | Exempt | III |
| 12AT7 | 179-4446<br>166-7662 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AT7WA | 087-4931<br>262-0167 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AT7WB | 752-5892 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AT7WC | 179-4446 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AT8 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AU7A | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12AX7 | 841-2352 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 12DT8 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 1332-0-40Y | - - - | ITE Circuit | 10.0 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1500T | 556-1232<br>615-5568 | Eimac | 0.0013 | Thorium 232 | Not Required | Exempt | III |
| 152TH | 189-5976 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 172/PL8295 | 993-5167 | Penta Labs | 0.0029 | Thorium 232 | Not Required | Exempt | III |
| 175A | 855-1853 | Eimac | 0.0008 | Thorium 232 | Not Required | Exempt | III |
| 177A | - - - | Eimac | 0.0011 | Thorium 232 | Not Required | Exempt | III |
| 177WA | 950-6085 | Eimac | 0.0011 | Thorium 232 | Not Required | Exempt | III |
| 19EZ8 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 199/ML8094 | 984-6413 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 2-150D | 507-3122 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 2-2000A | - - - | Eimac | 0.0018 | Thorium 232 | Not Required | Exempt | III |
| 2-240A | 508-1044 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 2-25A | 319-0568 | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 2-240A | 581-9594 | Eimac | 0.0012 | Thorium 232 | Not Required | Exempt | III |
| 2-50A | 284-6120 | Eimac | 0.000111 | Thorium 232 | Not Required | Exempt | III |
| 2DF4 | 843-4580 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 2X2A | 556-1231 | RCA | 0.0000176 | Thorium 232 | Not Required | Exempt | III |
| 2X3000F | 729-3424 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 2X3000F3 | 729-3424 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 2041 | 740-8350 | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| 2054 | 043-4743 | RCA | 0.580 | Thorium 232 | Required | Exempt | III |
| 211W | 166-7643 | U. E. | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 211W | 166-7643 | RCA | 0.000125 | Thorium 232 | Not Required | Exempt | III |
| 2150A | 507-3122 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 22000A | - - - | Eimac | 0.0018 | Thorium 232 | Not Required | Exempt | III |
| 2240A | 508-1044 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 22754 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 22868 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 22869 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 22923 | - - - | Westinghouse | 0.000191 | Thorium (nat) | Not Required | Exempt | III |
| 22947 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 22948 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 22949 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 23033 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23044 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 23051 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 23065 | - - - | Westinghouse | 14.464 | Uranium 235 | Required | Exempt | III |
| 23070 | - - - | Westinghouse | 22.60 | Uranium 235 | Required | Exempt | III |
| 23071 | - - - | Westinghouse | 121.475 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 23072 | - - - | Westinghouse | 44.41 | Uranium 235 | Required | Exempt | III |
| 23073 | - - - | Westinghouse | 0.271 | Uranium 235 | Required | Exempt | III |
| 23074 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23075 | - - - | Westinghouse | 23.73 | Uranium 235 | Required | Exempt | III |
| 23076 | - - - | Westinghouse | 0.559 | Uranium depleted | Required | Exempt | III |
| 23077 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23088 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23089 | - - - | Westinghouse | 0.154 | Uranium 235 | Required | Exempt | III |
| 23090 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 23091 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 23092 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 23093 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 23094 | - - - | Westinghouse | 0.001 | Uranium 235 | Not Required | Exempt | III |
| 23110 | - - - | Westinghouse | 23.50 | Uranium 235 | Required | Exempt | III |
| 23149 | - - - | Westinghouse | 46.90 | Uranium 235 | Required | Exempt | III |
| 23150 | - - - | Westinghouse | 47.46 | Uranium 235 | Required | Exempt | III |
| 23170 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 23172 | - - - | Westinghouse | 0.015 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 2' 195 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 23202 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23234 | - - - | Westinghouse | 0.932 | Uranium 235 | Required | Exempt | III |
| 23238 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23254 | - - - | Westinghouse | 0.153 | Uranium 235 | Required | Exempt | III |
| 23261 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 23263 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 23264 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 23265 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 23266 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23284 | - - - | Westinghouse | 0.079 | Uranium 235 | Required | Exempt | III |
| 23287 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 23292 | - - - | Westinghouse | 0.006 | Uranium 235 | Not Required | Exempt | III |
| 23430 | - - - | Westinghouse | 10.06 | Uranium 235 | Required | Exempt | III |
| 23432 | - - - | Westinghouse | 3.19 | Uranium 235 | Required | Exempt | III |
| 23434 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 23435 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23436 | - - - | Westinghouse | 0.0904 | Uranium 235 | Required | Exempt | III |
| 23448 | - - - | Westinghouse | 0.071 | Uranium 234 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 23465 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23466 | - - - | Westinghouse | 1.13 | Uranium 235 | Required | Exempt | III |
| 23467 | - - - | Westinghouse | 1.13 | Uranium 235 | Required | Exempt | III |
| 23468 | - - - | Westinghouse | 1.13 | Uranium 235 | Required | Exempt | III |
| 23470 | - - - | Westinghouse | 205.66 | Uranium 235 | Required | Exempt | III |
| 23477 | - - - | Westinghouse | 23.56 | Uranium 235 | Required | Exempt | III |
| 23478 | - - - | Westinghouse | 7.23 | Uranium 235 | Required | Exempt | III |
| 23485 | - - - | Westinghouse | 0.141 | Uranium 235 | Required | Exempt | III |
| 23489 | - - - | Westinghouse | 154.25 | Uranium 235 | Required | Exempt | III |
| 23604 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23610 | - - - | Westinghouse | 0.046 | Uranium 235 | Required | Exempt | III |
| 23612 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23621 | - - - | Westinghouse | 0.046 | Uranium 235 | Required | Exempt | III |
| 23622 | - - - | Westinghouse | 0.077 | Uranium 235 | Required | Exempt | III |
| 23623 | - - - | Westinghouse | 0.057 | Uranium 235 | Required | Exempt | III |
| 23625 | - - - | Westinghouse | 88.14 | Uranium 235 | Required | Exempt | III |
| 23630 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23637 | - - - | Westinghouse | 0.185 | Uranium 235 | Required | Exempt | III |
| 23639 | - - - | Westinghouse | 0.0001 | Uranium depleted | Not Required | Exempt | III |




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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 23640 | - - - | Westinghouse | 0.0001 | Uranium depleted | Not Required | Exempt | III |
| 23642 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 23643 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23644 | - - - | Westinghouse | 0.0003 | Uranium depleted | Not Required | Exempt | III |
| 23650 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23651 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23652 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23653 | - - - | Westinghouse | 0.046 | Uranium 235 | Required | Exempt | III |
| 23660 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23669 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23670 | - - - | Westinghouse | 0.678 | Uranium 235 | Required | Exempt | III |
| 23671 | - - - | Westinghouse | 1.02 | Uranium 235 | Required | Exempt | III |
| 23684 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23698 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23700 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 23718 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23723 | - - - | Westinghouse | 121.48 | Uranium 235 | Required | Exempt | III |
| 23724 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 23733 | - - - | Westinghouse | 43.62 | Uranium 235 | Required | Exempt | III |
| 23734 | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |
| 23735 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23747 | - - - | Westinghouse | 0.000191 | Thorium (nat) | Not Required | Exempt | III |
| 23761 | - - - | Westinghouse | 0.103 | Carbon - 14 | Required | Exempt | IV |
| 23767 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23771 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 23777 | - - - | Westinghouse | 3.96 | Uranium 235 | Required | Exempt | III |
| 23782 | - - - | Westinghouse | 0.000191 | Thorium (nat) | Not Required | Exempt | III |
| 23789 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 23790 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 23792 | - - - | Westinghouse | 0.226 | Uranium 235 | Required | Exempt | III |
| 23798 | - - - | Westinghouse | 0.057 | Uranium 235 | Required | Exempt | III |
| 23814 | - - - | Westinghouse | 0.187 | Uranium 235 | Required | Exempt | III |
| 23816 | - - - | Westinghouse | 0.226 | Uranium 235 | Required | Exempt | III |
| 2450A | 581-9594 | Eimac | 0.0012 | Thorium 232 | Not Required | Exempt | III |
| 25T | 319-0569 | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 250R | 116-9944 | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 250TH | 617-8569<br>114-3826 | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 25TL | 617-8563<br>116-9982 | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 253 | 615-5666<br>508-1046 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 254W | - - - | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 264 | 759-9336 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 284 | 518-1688 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 294 | - - - | Eimac | 0.00009 | Thorium 232 | Not Required | Exempt | III |
| 3-1000Z | 954-4825 | Eimac | 0.00073 | Thorium 232 | Not Required | Exempt | III |
| 3-200A3/592 | 781-1808 | Eimac | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 3-200A3 | - - - | Eimac | 0.00032 | Thorium 232 | Not Required | Exempt | III |
| 3-400Z | 781-1808 | Eimac | 0.0008 | Thorium 232 | Not Required | Exempt | III |
| 3-500Z | - - - | Eimac | 0.0008 | Thorium 232 | Not Required | Exempt | III |
| 3B24WA | 060-3449<br>082-3306 | Tung-Sol (Chatham) | 0.0005 | Uranium 232 | Not Required | Exempt | I |
| 3B24WB | 060-3449 | United Elec. | $7 \times 10^{-6}$ | Thorium (nat) | Not Required | Exempt | III |
| 3CV1500A7 | - - - | Eimac | 0.0015 | Thorium (nat) | Not Required | Exempt | III |
| 3CV30000A1 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CV30000A3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CV30000H3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |



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Table C-10 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 3CV35000A | - - - | Eimac | 0.0034 | Thorium 232 | Not Required | Exempt | III |
| 3CV50000A7 | - - - | Eimac | 0.0175 | Thorium 232 | Not Required | Exempt | III |
| 3CW10000A3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW10000H3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW20000A | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CW20000A1 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CW20000A3 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CW20000A7 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CW20000H3 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CW25000A3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CW30000H3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CW30000H7 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CW40000H3 | - - - | Eimac | 0.01 | Thorium 232 | Not Required | Exempt | III |
| 3CW5000A1 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW5000A3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW5000F1 | 679-1791 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW5000F3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CW5000H3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX1000A7 | - - - | Eimac | 0.0015 | Thorium 232 | Not Required | Exempt | III |





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Table C-10 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 3CX1000A1 | 988-1583 | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CX10000A3 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CX10000A7 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CX10000H3 | - - - | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 3CX1500A7 | - - - | Eimac | 0.0015 | Thorium 232 | Not Required | Exempt | III |
| 3CX15000A3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CX15000A7 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CX15000H3 | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 3CX20000A3 | - - - | Eimac | 0.01 | Thorium 232 | Not Required | Exempt | III |
| 3CX20000A7 | - - - | Eimac | 0.01 | Thorium 232 | Not Required | Exempt | III |
| 3CX20000H3 | - - - | Eimac | 0.01 | Thorium 232 | Not Required | Exempt | III |
| 3CX2500A3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX2500F3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX2500H3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX3000A1 | 893-1184 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX3000A7 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX3000F1 | 957-0951 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX3000F7 | 989-9810 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3CX5000A3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 3CX5000H3 | - - - | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 3C24 | 116-9931 | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 3C45W | 713-7014 | ITT | 0.08 | Thorium 232 | Not Required | Exempt | III |
| 3S4 | 100-5163 | General Elec. | 0.00002 | Uranium 235 | Not Required | Exempt | III |
| 3V4 | 188-3524 | General Elec. | 0.00001 | Uranium 235 | Not Required | Exempt | III |
| 3W5000A1 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3W5000A3 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3W5000F1 | 679-1791 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3W5000F3 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X2500A3 | 230-5298 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X2500F3 | 617-4891 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X3000A1 | 893-1184 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X3000A7 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X3000F1 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 3X3000F7 | - - - | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 30AG11 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 30045 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 30055 | - - - | Westinghouse | 5.424 | Uranium 235 | Required | Exempt | III |
| 30084 | - - - | Westinghouse | 0.154 | Uranium 235 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 30224 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 30301 | 114-4858 | Westinghouse | 0.559 | Uranium depleted | Required | Exempt | III |
| 30303 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 304TH | 114-4858, 061-1209, 100-7152 | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 304TL | 023-9497 100-7152 | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 30501 | - - - | Westinghouse | 0.054 | Uranium 235 | Required | Exempt | III |
| 30548 | - - - | Westinghouse | 10.06 | Uranium 235 | Required | Exempt | III |
| 30549 | - - - | Westinghouse | 0.38 | Uranium depleted | Required | Exempt | III |
| 30572 | - - - | Westinghouse | 6.10 | Uranium 235 | Required | Exempt | III |
| 30573 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30574 | - - - | Westinghouse | 0.001 | Uranium depleted | Required | Exempt | III |
| 30585 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 30588 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30604 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 30606 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 30609 | - - - | Westinghouse | 0.249 | Uranium 235 | Required | Exempt | III |
| 30612 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 30626 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30627 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30655 | - - - | Westinghouse | 1.164 | Uranium 235 | Required | Exempt | III |
| 30656 | - - - | Westinghouse | 1.164 | Uranium 235 | Required | Exempt | III |
| 30665 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 30666 | - - - | Westinghouse | 13.56 | Uranium 235 | Required | Exempt | III |
| 30673 | - - - | Westinghouse | 4.435 | Uranium 235 | Required | Exempt | III |
| 30683 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30689 | - - - | Westinghouse | 0.034 | Uranium 235 | Required | Exempt | III |
| 30691N | - - - | Westinghouse | 0.684 | Thorium (nat) | Required | Exempt | III |
| 30711 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30713 | - - - | Westinghouse | 0.452 | Uranium 235 | Required | Exempt | III |
| 30715 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30727 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 30748 | - - - | Westinghouse | 0.559 | Uranium depleted | Required | Exempt | III |
| 30758 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 30760 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30797 | - - - | Westinghouse | 0.09 | Uranium 235 | Required | Exempt | III |
| 30817 | - - - | Westinghouse | 1.774 | Uranium 235 | Required | Exempt | III |
| 30820 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30863 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 30868 | - - - | Westinghouse | 0.071 | Uranium 235 | Required | Exempt | III |
| 30869 | - - - | Westinghouse | 0.057 | Uranium 235 | Required | Exempt | III |
| 30874 | - - - | Westinghouse | 0.000384 | Thorium (nat) | Not Required | Exempt | III |
| 30881 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 30883 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 30904 | - - - | Westinghouse | 0.09 | Uranium 235 | Required | Exempt | III |
| 30933 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30942 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 30943 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 30944 | - - - | Westinghouse | 0.695 | Uranium 235 | Required | Exempt | III |
| 30960 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30961 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 30964 | - - - | Westinghouse | 0.0005 | Uranium depleted | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 30965 | - - - | Westinghouse | 0.002 | Uranium depleted | Not Required | Exempt | III |
| 30969 | - - - | Westinghouse | 0.780 | Uranium 235 | Required | Exempt | III |
| 30970 | - - - | Westinghouse | 0.780 | Uranium 235 | Required | Exempt | III |
| 30971 | - - - | Westinghouse | 0.780 | Uranium 235 | Required | Exempt | III |
| 30977 | - - - | Westinghouse | 0.170 | Uranium 235 | Required | Exempt | III |
| 30980 | - - - | Westinghouse | 0.0005 | Uranium depleted | Not Required | Exempt | III |
| 30981 | - - - | Westinghouse | 0.002 | Uranium depleted | Not Required | Exempt | III |
| 30988 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31002 | - - - | Westinghouse | 0.001 | Uranium 235 | Not Required | - - - | - - - |
| 31093 | - - - | Westinghouse | 342.39 | Uranium 235 | Required | Exempt | III |
| 31095 | - - - | Westinghouse | 46.895 | Uranium 235 | Required | Exempt | III |
| 31100 | - - - | Westinghouse | 47.46 | Uranium 235 | Required | Exempt | III |
| 31152 | - - - | Westinghouse | 0.144 | Uranium 235 | Required | Exempt | III |
| 31168 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 31169 | - - - | Westinghouse | 0.166 | Uranium 235 | Required | Exempt | III |
| 31183 | - - - | Westinghouse | 0.045 | Uranium depleted | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 31186 | - - - | Westinghouse | 0.0021 | Thorium (nat) | Not Required | Exempt | III |
| 31234 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31246 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 31261 | - - - | Westinghouse | 0.932 | Uranium 235 | Required | Exempt | III |
| 31291 | - - - | Westinghouse | 0.002 | Uranium 235 | Not Required | Exempt | III |
| 31293 | - - - | Westinghouse | 0.678 | Uranium 235 | Required | Exempt | III |
| 313C | 801-5578 116-9975 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 313C | 801-5578 116-9975 | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 313C | 116-9973 801-5578 | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 313CA | 230-5311 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 313CA | 230-5311 | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 313CA | 230-5311 | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 313CB | - - - | Cetron | 0.50 | Krypton 85 | Required | Exempt | III |
| 313CB | - - - | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 313CC | 800-0578 230-5293 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 313CC | 800-0578 230-5293 | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 313CC | 800-0578 | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 313CC | - - - | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 313CCJW | 800-0578 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 313CD | 188-3964 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 313CJW | - - - | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 3120/MA3152 & 3135/MA3152 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 31335 | - - - | Westinghouse | 0.015 | Uranium 235 | Required | Exempt | III |
| 31354 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 31356 | - - - | Westinghouse | 0.078 | Uranium depleted | Not Required | Exempt | III |
| 31369 | - - - | Westinghouse | 204.53 | Uranium 235 | Required | Exempt | III |
| 31384 | - - - | Westinghouse | 204.53 | Uranium 235 | Required | Exempt | III |
| 31417 | - - - | Westinghouse | 7.063 | Uranium 235 | Required | Exempt | III |
| 31418 | - - - | Westinghouse | 2.204 | Uranium 235 | Required | Exempt | III |
| 31419 | - - - | Westinghouse | 1.243 | Uranium 235 | Required | Exempt | III |
| 31424 | - - - | Westinghouse | 0.002 | Uranium 235 | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 31434 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31438 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31440 | - - - | Westinghouse | 0.678 | Uranium 235 | Required | Exempt | III |
| 31441 | - - - | Westinghouse | 0.678 | Uranium 235 | Required | Exempt | III |
| 31442 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 31456 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 31458 | - - - | Westinghouse | 0.003 | Uranium depleted | Not Required | Exempt | III |
| 31463 | - - - | Westinghouse | 2.825 | Uranium 235 | Required | Exempt | III |
| 31465 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31471 | - - - | Westinghouse | 0.678 | Uranium 235 | Required | Exempt | III |
| 31474 | - - - | Westinghouse | 0.232 | Uranium 232 | Required | Exempt | III |
| 31479 | - - - | Westinghouse | 0.0005 | Uranium depleted | Not Required | Exempt | III |
| 31480 | - - - | Westinghouse | 0.002 | Uranium depleted | Not Required | Exempt | III |
| 31491 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31493 | - - - | Westinghouse | 0.170 | Uranium 235 | Required | Exempt | III |
| 31496 | - - - | Westinghouse | 0.170 | Uranium 235 | Required | Exempt | III |
| 31517 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 31542 | - - - | Westinghouse | 0.000191 | Thorium (nat) | Not Required | Exempt | III |
| 31543 | - - - | Westinghouse | 115.83 | Uranium 235 | Required | Exempt | III |
| 31546 | - - - | Westinghouse | 13.56 | Uranium 235 | Required | Exempt | III |
| 31552 | - - - | Westinghouse | 0.170 | Uranium 235 | Required | Exempt | III |
| 31553 | - - - | Westinghouse | 0.187 | Uranium 235 | Required | Exempt | III |
| 31554 | - - - | Westinghouse | 0.006 | Uranium 235 | Required | Exempt | III |
| 31563 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 31568 | - - - | Westinghouse | 46.61 | Uranium 235 | Required | Exempt | III |
| 31573 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31574 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31575 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31576 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31578 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 31579 | - - - | Westinghouse | 0.037 | Uranium 235 | Required | Exempt | III |
| 3162/MA3164 & 3163/MA3164 | - - - | Microwave | 0.40 | Uranium 235 | Required | Exempt | III |
| 31640 | - - - | Westinghouse | 1.61 | Uranium 235 | Required | Exempt | III |
| 31656 | - - - | Westinghouse | 3.955 | Uranium 235 | Required | Exempt | III |
| 31671 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 1687 | - - - - | Westinghouse | 0.0002 | Uranium depleted | Not Required | Exempt | III |
| 31688 | - - - | Westinghouse | 0.0002 | Uranium depleted | Not Required | Exempt | III |
| 31721 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 31749 | - - - | Westinghouse | 166.675 | Uranium 235 | Required | Exempt | III |
| 31756 | - - - | Westinghouse | 2.32 | Uranium 235 | Required | Exempt | III |
| 31771 | - - - | Westinghouse | 3.955 | Uranium 235 | Required | Exempt | III |
| 31772 | - - - | Westinghouse | 3.955 | Uranium 235 | Required | Exempt | III |
| 31790 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 31830 | - - - | Westinghouse | 46.895 | Uranium 235 | Required | Exempt | III |
| 31854 | - - - | Westinghouse | 0.000384 | Thorium (nat) | Not Required | Exempt | III |
| 31875 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 31942 | - - - | Westinghouse | 1.446 | Uranium 235 | Required | Exempt | III |
| 31943 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 31964 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 32172 | - - - | Westinghouse | 14.464 | Uranium 235 | Required | Exempt | III |
| 32173 | - - - | Westinghouse | 0.009 | Uranium 235 | Not Required | Exempt | III |
| 32174 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 32253 | - - - | Westinghouse | 0.046 | Uranium 235 | Required | Exempt | III |



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Table C-10 - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 32254 | - - - | Westinghouse | 0.057 | Uranium 235 | Required | Exempt | III |
| 346A | 984-8065 | Western Elec. | 1.98 | Krypton 85 | Required | Exempt | III |
| 346C | 849-3769 | Cetron | 4.5 | Krypton 85 | Required | Exempt | III |
| 346C | 849-3769 | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 346CJW | 849-3769 | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 35T | 116-9980 | Eimac | 0.000096 | Thorium 232 | Not Required | Exempt | III |
| 35TG | 114-4871 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 3514 | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |
| 3527 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 353A | 937-9870 | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 353A | 188-8594 937-9870 | Western Elec. | 0.01 | Radium 226 | Required | Exempt | I |
| 353A | 188-8594 | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 356/5771/6576 | - - - | Machlett | 0.02 | Thorium 232 | Not Required | Exempt | III |
| 356/CL7085 | - - - | General Elec. | 0.03 | Thorium 232 | Not Required | Exempt | III |
| 358A | - - - | Western Elec. | 0.05 | Krypton 85 | Not Required | Exempt | III |
| 3586 | - - - | Westinghouse | 4.92 | Uranium 235 | Required | Exempt | III |
| 359A | 188-8633 439-5705 | Western Elec. | 1.20 | Krypton 85 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 359AJW | 188-8633 | Western Elec. | 1.1 | Krypton 85 | Required | Exempt | III |
| 3595 | - - - | Westinghouse | 23.17 | Uranium 235 | Required | Exempt | III |
| 3709 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 371B | 116-9960 | Electronic Enterprises | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 371B | - - - | United Elec. | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 375/7563 | 762-6682 | Microwave | 0.84 | Cobalt 60 | Required | Exempt | III |
| 3755 | - - - | Westinghouse | 46.90 | Uranium 235 | Required | Exempt | III |
| 3756 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 376B | - - - | Western Elec. | 4.0 | Krypton 85 | Required | Exempt | III |
| 376C | 537-3959 | Western Elec. | 4.0 | Krypton 85 | Required | Exempt | III |
| 376C | 537-3959 | Cetron | 4.0 | Krypton 85 | Required | Exempt | III |
| 3786 | - - - | Westinghouse | 23.17 | Uranium 235 | Required | Exempt | III |
| 3787 | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |
| 3806 | - - - | Westinghouse | 104.53 | Uranium 235 | Required | Exempt | III |
| 3833 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 3871 | - - - | Westinghouse | 22.60 | Uranium 235 | Required | Exempt | III |
| 3872 | - - - | Westinghouse | 22.60 | Uranium 235 | Required | Exempt | III |
| 3880 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 3881 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 3884 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 3905 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 3906 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 3916 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 3930 | - - - | Westinghouse | 104.53 | Uranium 235 | Required | Exempt | III |
| 3931 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 3933 | - - - | Westinghouse | 23.73 | Uranium 235 | Required | Exempt | III |
| 3935 | - - - | Westinghouse | 9.492 | Uranium 235 | Required | Exempt | III |
| 3936 | - - - | Westinghouse | 9.492 | Uranium 235 | Required | Exempt | III |
| 3937 | - - - | Westinghouse | 0.492 | Uranium 235 | Required | Exempt | III |
| 3941 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 395A | 556-0920 230-5291 | Western Elec. | 1.6 | Krypton 85 | Required | Exempt | III |
| 395AJW | - - - | Western Elec. | 1.6 | Krypton 85 | Required | Exempt | III |
| 3958 | - - - | Westinghouse | 120.91 | Uranium 235 | Required | Exempt | III |
| 3968 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 3996 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 4-1000A | 189-5968, 615-5689, 864-7480, 578-1524 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Par. Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4-125A/4D21 | 723-4226<br>188-0921 | Eimac | 0.00029 | Thorium232 | Not Required | Exempt | III |
| 4-250A | 188-0948,<br>845-2633,<br>887-2618 | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 4-400A | 556-1759,<br>617-8660<br>243-5018 | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 4-400C | - - - | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 4-500A | - - - | Eimac | 0.0015 | Thorium 232 | Not Required | Exempt | III |
| 4-65A | 985-1110<br>243-5017 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 4CV100000C | - - - | Eimac | 0.04 | Thorium 232 | Not Required | Exempt | III |
| 4CV20000A | - - - | Eimac | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 4CV250000A | - - - | Eimac | 0.1214 | Thorium 232 | Required | Exempt | III |
| 4CV35000A | - - - | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 4CV50000E | - - - | Eimac | 0.0175 | Thorium 232 | Not Required | Exempt | III |
| 4CV50000J | - - - | Eimac | 0.0175 | Thorium 232 | Not Required | Exempt | III |
| 4CV75000A | - - - | Eimac | 0.04 | Thorium 232 | Not Required | Exempt | III |
| 4CW8000A | - - - | Eimac | 0.0023 | Thorium 232 | Not Required | Exempt | III |
| 4CW10000A | 845-5743 | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 4CW100000D | - - - | Eimac | 0.04 | Thorium 232 | Not Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4CW100000E | - - - | Eimac | 0.0012 | Thorium 232 | Not Required | Exempt | III |
| 4CW25000A | 478-7692 | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 4CW250000A | - - - | Eimac | 0.1214 | Thorium 232 | Required | Exempt | III |
| 4CW50000E | - - - | Eimac | 0.0175 | Thorium 232 | Not Required | Exempt | III |
| 4CW50000J | - - - | Eimac | 0.0175 | Thorium 232 | Not Required | Exempt | III |
| 4CX10000D | 054-1987 | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 4CX10000J | - - - | Eimac | 0.0037 | Thorium 232 | Not Required | Exempt | III |
| 4CX1500A | - - - | Eimac | 0.0025 | Thorium 232 | Not Required | Exempt | III |
| 4CX15000A | 905-9126 | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 4CX15000J | 479-1377 | Eimac | 0.0057 | Thorium 232 | Not Required | Exempt | III |
| 4CX3000A | 078-7091 | Eimac | 0.0023 | Thorium 232 | Not Required | Exempt | III |
| 4CX35000C | - - - | Eimac | 0.04 | Thorium 232 | Not Required | Exempt | III |
| 4CX5000A | 996-5975<br>052-4112 | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 4CX5000J | 479-1378 | Eimac | 0.0037 | Thorium 232 | Not Required | Exempt | III |
| 4CX5000R | - - - | Eimac | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 4C35 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 4C35A/6268 | 552-8277 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 4D21 | 188-0921 | Eimac | 0.00029 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4D21A | - - - | Eimac | 0.00029 | Thorium 232 | Not Required | Exempt | III |
| 4E27A/5-125B | 262-1358 | Eimac | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 4PR65A | 080-8751 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 4PR1000A | 080-8750 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 4PR1000B | - - - | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 4PR125A | 078-1643 | Eimac | 0.00029 | Thorium 232 | Not Required | Exempt | III |
| 4PR250C | 958-3154 | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 4PR400A | 476-7964<br>080-8748 | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 4PR65A | 080-8751 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 4W20000A | - - - | Eimac | 0.021 | Thorium 232 | Not Required | Exempt | III |
| 4W20000A | 993-0695 | Eimac | 0.0300 | Thorium 232 | Not Required | Exempt | III |
| 4X500A | 865-0259 | Eimac | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 4014 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4021A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 4021AL | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 4022AL | 811-6837 | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 4023A | - - - | Burroughs | - - - | - - - | Required | Exempt | II |
| 4024A | - - - | Burroughs | - - - | - - - | Required | Exempt | II |
| 4025A | - - - | Burroughs | - - - | - - - | Required | Exempt | II |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4026A | - - - | Burroughs | < 1.0 | Krypton 85 | Required | Exempt | III |
| 402A | - - - | Western Elec. | 1.6 | Krypton 85 | Required | Exempt | III |
| 4031 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 4031A | - - - | Burroughs | < 1.0 | Krypton 85 | Required | Exempt | III |
| 4032A | - - - | Burroughs | < 1.0 | Krypton 85 | Required | Exempt | III |
| 40330A | - - - | Burroughs | < 1.0 | Krypton 85 | Required | Exempt | III |
| 40331A | - - - | Burroughs | < 1.0 | Krypton 85 | Required | Exempt | III |
| 4036 | - - - | Westinghouse | 106.22 | Uranium 235 | Required | Exempt | III |
| 4093 | - - - | Westinghouse | 26.56 | Uranium 235 | Required | Exempt | III |
| 4094 | - - - | Westinghouse | 23.73 | Uranium 235 | Required | Exempt | III |
| 41000A | 189-5968 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 4107 | - - - | Westinghouse | 120.91 | Uranium 235 | Required | Exempt | III |
| 4110 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 413 | 578-1140 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 413B | - - - | Western Elec. | 4.4 | Krypton 85 | Required | Exempt | III |
| 413B | - - - | Cetron | 4.0 | Krypton 85 | Required | Exempt | III |
| 414 | 228-8961 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 4152 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 4162 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 417 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4174 | - - - | Westinghouse | 46.90 | Uranium 235 | Required | Exempt | III |
| 4182 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 420A | 272-9197 | Cetron | 2.0 | Krypton 85 | Required | Exempt | III |
| 4214 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 423A | 248-3065 | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 423C | 813-0750 | Western Elec. | 4.50 | Krypton 85 | Required | Exempt | III |
| 4245A | - - - | Westinghouse | 46.90 | Uranium 235 | Required | Exempt | III |
| 425 | - - - | Cetron | 2.0 | Krypton 85 | Required | Exempt | III |
| 425A | 844-7209 | Western Elec. | 2.10 | Krypton 85 | Required | Exempt | III |
| 426A | - - - | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 426A | - - - | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 426A | - - - | Western Elec | 2.0 | Krypton 85 | Required | Exempt | III |
| 427A | 333-3540 | Western Elec | 4.00 | Krypton 85 | Required | Exempt | III |
| 430-B | 849-3768 | Cetron | 4.5 | Krypton 85 | Required | Exempt | III |
| 430-B | 849-3768 | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 430-C | - - - | Western Elec. | 15.00 | Krypton 85 | Required | Exempt | III |
| 4314 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 432B | 849-6864 | Western Elec. | 4.50 | Krypton 85 | Required | Exempt | III |
| 4320 | - - - | Westinghouse | 0.019 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4336 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 434 | - - - | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 4343 | - - - | Westinghouse | 0.559 | Uranium depleted | Required | Exempt | III |
| 4344 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 435 | - - - | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 4359 | - - - | Westinghouse | 0.117 | Uranium 235 | Required | Exempt | III |
| 436 | - - - | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 439A | 547-2928 | Western Elec. | 3.00 | Krypton 85 | Required | Exempt | III |
| 4392 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 4396 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4414 | - - - | Westinghouse | 44.41 | Uranium 235 | Required | Exempt | III |
| 4420 | - - - | Westinghouse | 106.22 | Uranium 235 | Required | Exempt | III |
| 4422 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 4423 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 443 | - - - | Cetron | 2.0 | Krypton 85 | Required | Exempt | III |
| 443A | - - - | Western Elec. | 2.00 | Krypton 85 | Required | Exempt | III |
| 446A | 107-7607 | Western Elec. | 0.10 | Krypton 85 | Not Required | Exempt | III |
| 447A | 849-3721 | Western Elec. | 0.10 | Krypton 85 | Not Required | Exempt | III |




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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4475 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 4491 | - - - | Westinghouse | 0.441 | Uranium 235 | Required | Exempt | III |
| 450TH | 116-9962<br>615-5570 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 450TL | 116-9963 | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 451A | 194-1474 | Western Elec. | 2.10 | Krypton 85 | Required | Exempt | III |
| 4516 | - - - | Westinghouse | 120.91 | Uranium 235 | Required | Exempt | III |
| 453A | - - - | Western Elec. | 0.30 | Krypton 85 | Required | Exempt | III |
| 4559 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4576 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 458A | - - - | Western Elec. | 1.00 | Radium 226 | Required | Exempt | I |
| 4585 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4628 | 770-1927 | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 4645 | - - - | Westinghouse | 0.232 | Uranium 235 | Required | Exempt | III |
| 4647 | - - - | RCA | 0.0707 | Thorium 232 | Not Required | Exempt | III |
| 4648 | - - - | RCA | 0.136 | Thorium 232 | Required | Exempt | III |
| 4652 | - - - | Westinghouse | 92.10 | Uranium 235 | Required | Exempt | III |
| 4652A | - - - | Westinghouse | 92.10 | Uranium 235 | Required | Exempt | III |
| 4661 | - - - | Westinghouse | 0.141 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 4676 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4677 | - - - | Westinghouse | 0.559 | Uranium depleted | Required | Exempt | III |
| 469A | - - - | Western Elec. | 1.00 | Radium 226 | Required | Exempt | I |
| 4690 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 4706 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 4712 | - - - | Westinghouse | 0.000384 | Thorium (nat) | Not Required | Exempt | III |
| 4757 | - - - | Westinghouse | 0.218 | Uranium 235 | Required | Exempt | III |
| 476 | 615-5822 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 4765 | - - - | Westinghouse | 121.47 | Uranium 235 | Required | Exempt | III |
| 4766 | - - - | Westinghouse | 12.15 | Uranium 235 | Required | Exempt | III |
| 4767 | - - - | Westinghouse | 1.20 | Uranium 235 | Required | Exempt | III |
| 477 | 816-3759 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 483 | 824-7834 | Lionel Elec. | 0.25 | Nickel 63 | Required | Exempt | IV |
| 4863 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 487 | 721-2495 | Anton Labs | 1.0 | Carbon 14 | Required | Exempt | IV |
| 4925 | - - - | Westinghouse | 0.311 | Uranium 235 | Required | Exempt | III |
| 4948 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 5-125B | - - - | Eimac | 0.00032 | Thorium 232 | Not Required | Exempt | III |
| 5-500A | - - - | Eimac | 0.0015 | Throium 232 | Not Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5CX1500A | - - - | Eimac | 0.0025 | Thorium 232 | Not Required | Exempt | III |
| 5CX3000A | 156-0297 | Eimac | 0.0023 | Thorium 232 | Not Required | Exempt | III |
| 5C22/6279 | 116-9969 | Kuthe Labs (ITT) | 0.080 | Uranium (nat) | Not Required | Exempt | III |
| 5003 | - - - | Westinghouse | 111.87 | Uranium 235 | Required | Exempt | III |
| 5016A | - - - | Brurroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5022 | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |
| 5027 | - - - | Westinghouse | 0.001 | Uranium depleted | Not Required | Exempt | III |
| 5030 | - - - | Westinghouse | 0.622 | Uranium 235 | Required | Exempt | III |
| 5031A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 50344A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 50346A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 50348A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5046 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 5092A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5406 | - - - | Westinghouse | 0.271 | Uranium 235 | Required | Exempt | III |
| 5449 | - - - | Westinghouse | 23.73 | Uranium 235 | Required | Exempt | III |
| 545 | 557-5105 | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5589(376B) | - - - | Western Elec. | 1.0 | Radium 226 | Required | Exempt | III |
| 5644 | 262-0184 | Raytheon | 0.006 | Cobalt 60 | Not Required | Exempt | III |
| 5651 | 617-8685, 262-0286, 167-0389 | Raytheon (see 5651 WA) | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 5651A | - - - | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 5651WA | 262-0286 | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 5651WA | 262-0286 | Tung-Sol | 0.005 | Radium 226 | Not Required | Exempt | I |
| 5651WA | 262-0286 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 5651WA | 262-0286 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 5651WA | 262-0286 | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 5654 | 134-6031, 262-1357 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5654W | 045-8639 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5663 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5670 | 134-5994 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5670W | 188-6584 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5671 | - - - - | RCA | 0.0892 | Thorium 232 | Not Required | Exempt | III |
| 5672 | 188-6588 | - - - | - - - | Thorium 232 | Required | Exempt | II |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5681 | - - - | Machlett | 0.039 | Thorium 232 | Not Required | Exempt | III |
| 5682 | - - - | Machlett | 0.088 | Thorium 232 | Not Required | Exempt | III |
| 5682 | 847-6738 | General Elec. | 0.20 | Thorium 232 | Required | Exempt | III |
| 5685 | 262-0174 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| 5696 | 617-8677 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5725 | 134-6064, 237-6917 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5725W | 134-6064 237-6917 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5736 | 682-2777 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5751 | 082-4139 193-5145 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 576A | 296-9997 | Electronic Ent. | 0.0004 | Thorium 232 | Not Required | Exempt | III |
| 5762/7C24 | 889-7697 262-0204 | RCA | 0.0071 | Thorium 232 | Not Required | Exempt | III |
| 577 | 262-0198 | United Elec. | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 577W | 878-6030 | United Elec. | 0.00002 | Thorium 232 | Not Required | Exempt | III |
| 5770 | - - - | RCA | 0.0892 | Thorium 232 | Not Required | Exempt | III |
| 5771/356/6576 | - - - | Machlett | 0.02 | Thorium 232 | Not Required | Exempt | III |
| 5771 | 061-1127 501-0906 | RCA | 0.2511 | Thorium 232 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5783 | 230-5253<br>284-7166 | Raytheon | 0.0033 | Krypton 85 | Not Required | Exempt | III |
| 5783A/1068 | 230-5253<br>284-7166 | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 5783WA | 284-7166 | Tung-Sol (Chatham) | 0.0013 | Krypton 85 | Not Required | Exempt | III |
| 5783WA | 284-7166 | Raytheon | 0.0033 | Krypton 85 | Not Required | Exempt | III |
| 5786 | 689-1896 | RCA | 0.0051 | Thorium 232 | Not Required | Exempt | III |
| 5787 | 272-8545<br>262-3771 | Raytheon | 0.0067 | Krypton 85 | Not Required | Exempt | III |
| 5787A/1061 | 262-3771 | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 5787A(CK1061) | - - - | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 5787WA | 617-8876 | Tung-Sol (Chatham) | 0.0028 | Krypton 85 | Not Required | Exempt | III |
| 5787WA | 272-8545 | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 5787WA | 272-8545 | Raytheon | 0.0067 | Krypton 85 | Not Required | Exempt | III |
| 5790/X7108 | 220-6525 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5791/X6007 | 220-6526 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5792/ATR345 | 193-5141 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 5792/ATR345 | 193-5141 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5792 | 193-5141 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5793/ATR346 | 193-5142 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 5793/ATR346 | 193-5142 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5814A | 262-0210 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5823 | 248-3078 | Tung-Sol (Chatham) | 0.0018 | Krypton 85 | Not Required | Exempt | III |
| 5825 | 262-0211 | RCA | 0.0000077 | Thorium 232 | Not Required | Exempt | III |
| 5841 | 819-1154<br>262-0213 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 5841 | 262-0213<br>819-1154 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 5853/TR368 | 193-5147 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| 5853/TR368 | 193-5147 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| 5853/TR368 | 193-5147 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5863 | 261-8671 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 5863/X7109 | 261-8671 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| 5863/X7109 | 261-8671 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5864/ATR321 | 549-0045<br>842-6660 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 5864/ATR321 | 549-0045 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5865/HA379 | 262-0174 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 5865/TR361 | 262-0174 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5865/TR361 | 262-0174 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 5867A | - - - | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 5883 | 262-0225 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 592/3-200A3 | 781-1808 | Eimac | 0.00032 | Thorium 232 | Not Required | Exempt | III |
| 5921 | 248-1271 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 5922 | 248-1270 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 5925/TR331 | 262-1693<br>267-1693 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 5927 | 237-0103 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 5939 | 636-1812<br>262-0176 | Westinghouse | 3.30 | Radium 226 | Required | Exempt | I |
| 5939A | 038-1660<br>636-1812 | Varian | 3.30 | Cobalt 60 | Required | Exempt | III |
| 5939A | 038-1660<br>636-1812 | Westinghouse | 3.30 | Radium 226 | Required | Exempt | I |
| 5948A | 752-5384 | Tung-Sol | - - - | - - - | Required | Exempt | II |
| 5950 | 800-2357 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 5956 | 890-6617<br>179-4746 | Kuthe Labs (ITT) | 0.080 | Uranium (nat) | Not Required | Exempt | III |
| 5957 | 806-0292 | Kuthe Labs (ITT) | 0.080 | Uranium (nat) | Not Required | Exempt | III |
| 5958/E40 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 5959/E41 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 5961A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5962 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 5962/BS-101 | 188-6592 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 5962/BS-101 | 188-6592 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 5962/BS-101 | 188-6592 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 5962/BS-101 | 188-6592 | Electronic Products | 3.0 | Nickel 63 | Required | Exempt | IV |
| 5962/BS-101 | 188-6592 | Eon Corp. | <0.1 | Nickel 63 | Required | Exempt | IV |
| 5965A | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 5991A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5991A | - - - | National Electronics (Varian) | 0.21 | Krypton 85 | Required | Exempt | III |
| 59917A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 59918A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 59919A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 5992A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 59920A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 59921A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6AB4 | 262-0190 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6AK5/EF95 | 188-3545 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6AK5WB | 134-6037, 503-0607 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6C21 | 810-8310 114-3831 | Eimac | 0.00056 | Thorium 232 | Not Required | Exempt | III |
| 6DT8 | 728-1775 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6D10 | 938-2670 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6EZ8 | 726-8926 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6GY8 | 166-7662 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6002 | 261-8678 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| 6022/ATR332 | 262-1690 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6024/ATR387 | 842-6661 272-9195 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6033 | 262-6356 548-3803 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6034 | 220-6892 548-3802 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6035 | 264-2996 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6072 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6072A | 892-3809 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6081/ATR407 | 262-0289 262-0287 060-6626 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 608 /ATR407 | 262-0289<br>262-0289 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6081 | 060-6626 | Omni-Wave (Metcom) | 0.32 | Cobalt 60 | Required | Exempt | III |
| 6091A | - - - | Burroughs | <1.0 | Krypton 85 | Required | Exempt | III |
| 6117 | 272-9196 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6117 | 272-9196 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 6117 | 272-9196 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 6117 | 272-9196 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6119 | 319-0527 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 6135 | 617-4911 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6140 | 561-8233<br>553-3770 | Western Elec. | 4.50 | Krypton 85 | Required | Exempt | III |
| 6141/427A | 188-3531<br>284-6554 | Western Elec. | 4.00 | Krypton 85 | Required | Exempt | III |
| 6143 | 842-6659 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 6143 | 842-6659 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 6155 | 905-8020 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 6156 | - - - | Eimac | 0.0003 | Thorium (nat) | Not Required | Exempt | III |
| 6162/ATR388 | 272-9197 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6163 | 060-6664 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 6163 | 264-2998<br>543-1968 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6163 | 264-2998<br>543-1968 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6164 | 583-0613<br>577-1537 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6164 | 583-0613 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| 6166 | 902-7423<br>669-6865 | RCA | 0.0276 | Thorium 232 | Not Required | Exempt | III |
| 6166A/7007 | 883-7030 | RCA | 0.0276 | Thorium 232 | Not Required | Exempt | III |
| 6167 | 264-3000 | Western Elec. | 3.00 | Krypton 85 | Required | Exempt | III |
| 6199 | 813-0755 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | III |
| 6201 | 179-4446<br>296-3354 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6213 | 682-9452<br>851-7848 | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 6213A | - - - | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 6214 | 608-2774 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6214 | 608-2774 | Tracerlab, Inc | 0.1 | Cobalt 60 | Required | Exempt | III |
| 6232/ATR427 | 780-8151<br>390-5241 | Sylvania | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6232/TR427 | 390-5241 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6232/ATR427 | 390-5241 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6232/MA316 | 780-8151 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| 6249A | 543-1142<br>752-0641 | Westinghouse | 0.016 | Radium 226 | Required | Exempt | I |
| 6249B | 451-5824<br>752-0641 | Raytheon | 0.006 | Thorium 232 | Not Required | Exempt | III |
| 626 | - - - | Victoreen | 0.25 | Nickel 63 | Required | Exempt | IV |
| 626A | - - - | Victoreen | 0.25 | Nickel 63 | Required | Exempt | IV |
| 6260 | 272-9201 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 6260 | 272-9201 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6265 | 646-4613 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6268/4C35A | 552-8277 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 6276 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6279 | 116-9969 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 6282/BL-11 | 365-2548 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6282/MKT-11 | 365-2548 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 6284 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6303 | 804-0954 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 6303A | 892-3413 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 6304/BL43 | 617-5810 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6305 | 804-0954 | Klystronics | 0.000309 | Thorium (nat) | Not Required | Exempt | III |
| 6312 | 519-6175 | Klystronics | 0.15 | Thorium (nat) | Required | Exempt | II |
| 6322 | 296-7326 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 6322/BL25 | 296-7326<br>263-9910<br>415-1625<br>577-3205 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 633B/TG52 | 865-4959 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| 6334 | 578-5483 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6334 | 574-5483<br>752-0594 | Nuclear | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6334/BL27 | 759-5483<br>578-5483<br>504-7571 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6334 | 752-0594 | Omni-Wave (Metcom) | 0.20 | Cobalt 60 | Required | Exempt | III |
| 6334/BL27 | 578-5483 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6334/BL27 | 578-5483<br>759-5483 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6344 | - - - | Raytheon | 0.49 | Thorium 232 | Required | Exempt | III |
| 6357 | 850-6633 | - - - | - - - | - - - | Required | Exempt | II |
| 6368 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6369 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6376 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 6376A | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6378 'BL62 | 548-9734<br>296-5932 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6386 | 636-2244 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6388 | - - - | Western Elec. | 2.00 | Krypton 85 | Required | Exempt | III |
| 6393/BL68 | 060-6608 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6396 | 296-5509 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6396 | 296-5509 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6402B | - - - | Raytheon | 0.07 | Thorium 232 | Not Required | Exempt | III |
| 6406A | - - - | Raytheon | 0.35 | Thorium 232 | Required | Exempt | III |
| 6410A | - - - | Raytheon | 0.46 | Thorium 232 | Required | Exempt | III |
| 6414 | 054-3946<br>581-9404<br>812-3745 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6414W | 054-3946 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 642 | - - - | Westinghouse | 0.3 | Krypton 85 | Required | Exempt | III |
| 6420/6421 | - - - | Machlett | 0.008 | Thorium 232 | Not Required | Exempt | III |
| 6421/6420 | - - - | Machlett | 0.008 | Thorium 232 | Not Required | Exempt | III |
| 6422 | - - - | Machlett | 0.008 | Thorium 232 | Not Required | Exempt | III |
| 6423 | 926-7263 | Machlett | 0.008 | Thorium 232 | Not Required | Exempt | III |
| 6424/6425 | - - - | Machlett | 0.014 | Thorium 232 | Not Required | Exempt | III |
| 6425/6424 | - - - | Machlett | 0.014 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6426/6427 | - - - | Machlett | 0.026 | Thorium 232 | Not Required | Exempt | III |
| 6427/6426 | - - - | Machlett | 0.026 | Thorium 232 | Not Required | Exempt | III |
| 6437 & 6438 | 800-5552 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 6448 | - - - | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| 6455/BL61 | 727-5618 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6501 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6518/QK254 | 061-0485 | Raytheon | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6542 | 553-7091 | Tung-Sol (Chatham) | 0.0025 | Krypton 85 | Not Required | Exempt | III |
| 6542 | 754-9916<br>553-7091<br>578-5241 | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 6542 | 754-9916<br>578-5247<br>553-7091 | Raytheon | 0.006 | Cobalt 60 | Not Required | Exempt | III |
| 6542A/1069 | - - - | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 6544 | 561-8230 | Machlett | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 6545 | 752-0185 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6546 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6560 | 501-1066 | Omni-Wave (Metcom) | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6560/BL35 | 296-1065<br>501-1066<br>296-2698 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 65 0/MA375A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 654/BL71 | 060-6628 841-2609 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6565/BL313 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6566 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6568/BL28 | 538-2793 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6569 | 060-6570 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 6576/356/5771 | - - - | Machlett | 0.02 | Thorium 232 | Not Required | Exempt | III |
| 6580 | - - - | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 6587 | 553-7010 | Kuthe Labs (ITT) | 0.0080 | Uranium (nat) | Not Required | Exempt | III |
| 6591 | 975-2250 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6592/BL309 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6593/BL310 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6594/BL311 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6595/BL316 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6596/BL317 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6597/BL320 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6599/BL322 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6601/BL327 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6602/BL329 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 6604/BL509 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6605/BL96A | 556-8812<br>581-5535 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 6605/MA346 | 581-5535 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6605/BL96 | 503-4493<br>834-0571<br>501-5535 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 6609/BL327 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6613/BL324 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6614/BL314 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6615/BL312 | 646-4803 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6616/BL326 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6620 | 060-6573<br>390-5239 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6624 | 975-2251 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6626/0A2WA | 557-6926<br>503-4880<br>769-0869 | CBS Hytron | 0.01 | Krypton 85 | Not Required | Exempt | III |
| 6626/0A2WA | 262-0964 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 6626/0A2WA | 557-6926<br>617-8973 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 6626/0A2WA | 557-6926 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6627/0B2WA | 624-4718 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 6627/0B2WA | 624-4718 | Tung-Sol (Chatham) | 0.005 | Radium 226 | Not Required | Exempt | I |
| 6627/0B2WA | 624-4718 800-0554 | Jack & Heintz, Inc. | 6.0 | Radium 226 | Required | Exempt | I |
| 6627/0B2WA | 624-4718 | CBS Hytron | 0.05 | Nickel 63 | Not Required | Exempt | IV |
| 6628/BL38 | - - - | Tung-Sol (Chatham) | 0.055 | Radium 226 | Required | Exempt | IV |
| 6629/BL54 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6630/BL55 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6631/BL94 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6632/BL18 | 060-6630 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 6633 | 850-0925 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6633/BL37 | 646-4720 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 6634/BL90 | - - - | Varian | 0.90 | Cobalt 60 | Required | Exempt | III |
| 6635/BL57 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6636/BL87 | 800-0594 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 6636/MA376 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6637/BL31 | 719-5454 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6638/BL99 | 569-9508 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6639/BL46 | 295-3589 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6640/BL60 | 543-0143 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6641/BL86 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6642/BL600 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6643/BL81 | - - - | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6644 | 752-5239 | Tracerlab, Inc | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 6644/BL95 | 843-8063<br>752-5239 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6645 | 080-9194<br>503-4497 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6645 | 080-9194 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 6645 | 080-9194 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 6646/BL604 | 905-7498 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6647/BL604H | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6648/BL615 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6649/BL56 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6650/BL67 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6664 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6679 | 179-4446<br>878-2510 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6680 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |




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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6631 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6685/BL-616 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6696 | 932-0426<br>935-0171 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 6696A | 135-0130 | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 6696A | 179-3104 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 6697 | 932-0426<br>935-0171 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 6697A | 935-0171<br>932-0426 | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 6697A | 932-0426 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 6697AW | - - - | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 6777/KU-14 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 6785 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 6795 | 941-2639 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6795 | 941-2639 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 6795A | 089-7831 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6796 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6797 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6805/BL625 | 880-8596 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6806 | - - - | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| 6829 | 646-4756 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 6843 | - - - | Raytheon | 0.61 | Thorium 232 | Required | Exempt | III |
| 6844A | 833-5515 | National Electronics (Varian) | 0.06 | Krypton 85 | Not Required | Exempt | III |
| 6890/BL650 | 892-3323 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 6899/MA349 | 581-8645<br>620-4528 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6904/BL348 | - - - | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6905/BL613 | 682-8544<br>729-3648 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 6906/BL643 | 682-8864<br>729-0285 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 6941 | - - - | Westinghouse | 23.165 | Uranium 235 | Required | Exempt | III |
| 6941A | - - - | Westinghouse | 23.165 | Uranium 235 | Required | Exempt | III |
| 6949 | - - - | RCA | 0.249 | Thorium 232 | Required | Exempt | III |
| 6949V1 | - - - | RCA | 0.249 | Thorium 232 | Required | Exempt | III |
| 6950 | 752-5426 | RCA | 0.249 | Thorium 232 | Required | Exempt | III |
| 6959 | 951-6292 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| 6962/BL665 | 646-4719 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 6962/MA339 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 6971 | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 6971A | - - - | Westinghouse | 20.34 | Uranium 235 | Required | Exempt | III |
| 7C24/5762 | 889-7697<br>262-0204 | RCA | 0.0071 | Thorium 232 | Not Required | Exempt | III |
| 7002 | - - - | Machlett | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 7003A | 020-0074 | Machlett | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 7007 | - - - | Machlett | 0.012 | Thorium 232 | Not Required | Exempt | III |
| 7009A | - - - | National Electronics (Varian) | 0.06 | Krypton 85 | Not Required | Exempt | III |
| 705WA | 900-3508 | United Elec. | 0.00002 | Thorium (nat) | Not Required | Exempt | III |
| 705WA | 900-3508 | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |
| 7077 | 688-6706 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 7099 | 754-5782 | Tung-Sol (Chatham) | 0.0025 | Radium 226 | Not Required | Exempt | I |
| 7114 | - - - | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 7115/BL915 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 7131 | - - - | Nuclear | 1.0 | Cobalt 60 | Required | Exempt | III |
| 7152/BL612 | 834-0571<br>807-7374<br>615-5047 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 7152/MA348A | 834-0571 | Microwave | 0.80 | Cobalt 60 | Required | Exempt | III |
| 7159 | 685-9806 | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |
| 7161 | 168-7809<br>883-4793 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7161 | 168-7809<br>883-4793 | Westinghouse | 0.7 | Uranium 235 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 7162 | 816-3759 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7166/BL933 | 681-9793<br>834-2902 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 7166/MA366 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7186 | - - - | Westinghouse | 0.151 | Uranium 235 | Required | Exempt | III |
| 7187 | - - - | Westinghouse | 46.895 | Uranium 235 | Required | Exempt | III |
| 7188 | - - - | Westinghouse | 4.916 | Uranium 235 | Required | Exempt | III |
| 7205 | - - - | CBS Hytron | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 721A | 295-9829<br>188-3913 | Western Elec. | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 721A | 188-3913 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 721A | - - - | Sylvania | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 721B | 188-3913 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 724B | 193-5125 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 7229 | - - - | CBS Hytron | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 7230 | - - - | CBS Hytron | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 7231 | - - - | CBS Hytron | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 7232 | - - - | CBS Hytron | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 724A | 193-5125 | Western Elec. | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 724B | 193-5125 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |





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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 727A | - - - | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 7286 | 082-4112 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7309/BL693 | 633-6304 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 7309/MA360 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7310 | 688-2414 | United Elec. | 0.0002 | Thorium (nat) | Not Required | Exempt | III |
| 7323 | 988-0874 | Tung-Sol (Chatham) | 0.33 | Krypton 85 | Required | Exempt | III |
| 7324/BL931 | - - - | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 7369 | - - - | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 7379/MA306B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7380/MA332 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7381/BL979 | 984-6164 688-3756 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 7381 | 984-6164 | Microwave | 0.30 | Cobalt 60 | Required | Exempt | III |
| 7381/MA338A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7387 | - - - | Westinghouse | 1.90 | Uranium 235 | Required | Exempt | III |
| 7400 | - - - | Tung-Sol (Chatham) | 0.0015 | Krypton 85 | Not Required | Exempt | III |
| 7401 | - - - | Tung-Sol (Chatham) | 0.0008 | Krypton 85 | Not Required | Exempt | III |
| 7402 | - - - | Klystronics | 0.00309 | Throium (nat) | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 7402 | 914-0064 | United Elec. | 0.0001 | Thorium (nat) | Not Required | - - - | III |
| 7416/TZ36A | 866-5264 | Signalite | 0.90 | Cesium 137 | Required | Exempt | III |
| 7421 | 825-7752 | Anton | 0.25 | Nickel 63 | Required | Exempt | IV |
| 744 | - - - | Victoreen | 2.05 | Nickel 63 | Required | Exempt | IV |
| 7447/BL352A | 615-0719 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 7452 | - - - | Raytheon | 0.17 | Thorium 235 | Required | Exempt | III |
| 7462 | 752-5460 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 7479 | - - - | Machlett | 0.026 | Throium 232 | Not Required | Exempt | III |
| 748 | - - - | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 7480 | - - - | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 7482 | - - - | Machlett | 0.089 | Thorium 232 | Not Required | Exempt | III |
| 7486 | 883-0971 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 75TH | 617-9056 | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 75TL | 615-5565<br>188-8571<br>853-5846 | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 750TL | 556-1238<br>188-3900<br>615-5565 | Eimac | 0.001 | Thorium 232 | Not Required | Exempt | III |
| 7529 | - - - | Raytheon | 0.19 | Thorium 232 | Required | Exempt | III |
| 7560/ML7482 | - - - | Machlett | 0.1078 | Thorium 232 | Required | Exempt | III |





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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 7560V | 179-8054 | Machlett | 0.089 | Thorium 232 | Not Required | Exempt | III |
| 7562/MA365 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7562A/MA365A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7563/375 | 762-6682 | Microwave | 0.84 | Cobalt 60 | Required | Exempt | III |
| 7563/MA375 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7564/MA378A | 610-8486 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7583/KU82 | 778-3857 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 759 | - - - | Westinghouse | 6.60 | Radium 226 | Required | Exempt | I |
| 7603/KU401 | - - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 7615/EP680 | 825-4999 | Elec. Prod. | 3.0 | Nickel 63 | Required | Exempt | IV |
| 7615 | 825-4999 | Eon Corp. | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7617 | 825-4562 | Eon Corp. | <0.17 | Nickel 63 (No radioactive material used after 1975) | Required | Exempt | IV |
| 7652/MA362A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7657 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 7676 | - - - | Westinghouse | 0.000384 | Thorium (nat) | Not Required | Exempt | III |
| 7715 | - - - | Machlett | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 7718 | - - - | Raytheon | 0.10 | Thorium 232 | Not Required | Exempt | III |
| 7720 | - - - | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 7743/MA363 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 7753762P001 | - - - | General Elec. | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7798 | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| 7813 | - - - | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 7821 | 799-9471 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 7821/MPT12 | 799-9471 | Omni-Wave (Metcom) | - - - | Thorium 232 | Required | Exempt | III |
| 7823/MLT-13 | 729-1269 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 7823/MA356A | 729-1269 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 7835 | 843-7356 | RCA | 0.580 | Thorium 232 | Required | Exempt | III |
| 7859 | 105-1481 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7894 | 991-1566 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 7901/MPT-13 | 882-1994 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 7902/BLT004A | 754-5431 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 7903 | - - - | Westinghouse | 0.000191 | Thorium 232 | Not Required | Exempt | III |
| 7910 | 020-0071 | General Elc. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 7977A | - - - | National Electronics (Varian) | 0.06 | Krypton 85 | Not Required | Exempt | III |
| 7988/TD-44 | 707-3336 | Signalite | 0.06 | Radium 226 | Required | Exempt | I |
| 7996 | - - - | Raytheon | 0.02 | Nickel 63 | Not Required | Exempt | IV |
| 8000 | 188-3936 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 8005 | 615-5619<br>116-9988 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 801A | 581-8177<br>116-9985 | RCA | 0.00002 | Thorium 232 | Not Required | Exempt | III |
| 8013A | 726-9305 | RCA | 0.000058 | Thorium 232 | Not Required | Exempt | III |
| 8020 | 808-4522<br>615-9871 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 8020 | 808-4522<br>114-3846 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 8020W | 808-4522 | Cetron | 0.0016 | Thorium oxide | Not Required | Exempt | III |
| 8020W | 808-4522 | United Elec. | 0.00002 | Thorium oxide | Not Required | Exempt | III |
| 803 | 116-9984 | RCA | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 8038 | - - - | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 804 | 114-4881 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 8040 | - - - | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 8041 | - - - | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 8048/BL601 | 701-7582 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 805 | 114-4841 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 805 | 114-4881 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 805 | 114-4881 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 8059 | 854-8643 | Bendix | 1.2 | Cesium 137 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 8059 | 854-8643 | Hewlett Packard | 1.2 | Cesium 137 | Required | Exempt | III |
| 8059/TD-72 | 969-4162 | Bendix Corp. | 0.06 | Radium 226 | Required | Exempt | I |
| 8059/TD-72 | 844-5182 | Signalite | 0.06 | Radium 226 | Required | Exempt | I |
| 8060/BLT041 | 894-7127 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8061/BLT059 | 894-7128 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 8073 | - - - | Westinghouse | 94.92 | Uranium 235 | Required | Exempt | III |
| 808 | 114-3850 | RCA | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 8082 | 908-7622 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 8083 | 990-5914 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 8089 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 809 | 114-4867<br>615-5562 | RCA | 0.000029 | Thorium 232 | Not Required | Exempt | III |
| 8090 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 8094/ML199 | 984-6413 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 810 | 114-3863<br>615-5561<br>567-4415 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 810 | 114-3863 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 811A | 556-1454<br>789-5973 | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 811A | 189-5973 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 8112/BL640 | 688-9031 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 812A | 552-1107<br>030-1901 | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 812A | 030-1901 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 8129 | - - - | Raytheon | 0.02 | Thorium 232 | Not Required | Exempt | III |
| 813 | 556-1455<br>114-3843 | RCA | 0.000245 | Thorium 232 | Not Required | Exempt | III |
| 813-V1 | - - - | RCA | 0.000245 | Thorium 232 | Not Required | Exempt | III |
| 814 | 617-8653<br>114-4877 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 8152/TD-40 | 810-0192 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| 816 | - - - | Victoreen | 2.0 | Nickel 63 | Required | Exempt | IV |
| 8252W | 177-1716 | Eimac | < 5.0 | Thorium 232 | Required | Exempt | III |
| 8256 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 8257 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 8264/KU52 | 078-9500 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 8265/MA355A | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8265 | 702-5028 | Microwave | 0.42 | Cobalt 60 | Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 827R | 682-8629 | RCA | 0.00256 | Thorium 232 | Not Required | Exempt | III |
| 8271/MA356D | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8272/MA356E | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8273/MA356F | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 828 | 170-4586<br>553-4582<br>114-4876 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 829 | - - - | Victoreen | 2.0 | Nickel 63 | Required | Exempt | IV |
| 830B | 114-4860,<br>615-5559 | RCA | 0.000049 | Thorium 232 | Not Required | Exempt | III |
| 8311/MDS-56 | 064-7943 | Omni-Wave (Metcom) | 3000.0 | - - - | Required | Exempt | II |
| 8312/BL960B | 754-9177 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 8312 | 754-9177<br>897-9177 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 8313/BL995B | 903-2853 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 8317 | 942-3980 | Machlett | 0.089 | Thorium 232 | Not Required | Exempt | III |
| 8329/KU402 | 116-9965 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 833A | 617-8539 | RCA | 0.00049 | Thorium 232 | Not Required | Exempt | III |
| 834 | 617-9037<br>538-4709 | RCA | 0.000081 | Thorium 232 | Not Required | Exempt | III |
| 8359/MA365B | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8370/E38 | 067-9364 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |





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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 838 | 114-4888 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 8385/MA325 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8419/MA362 | - - - | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 8424/KU-25 | 390-5208 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 8437 | 948-2686 | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8438-4-400A | 243-5018 | - - - | - - - | - - - | Required | Exempt | II |
| 845 | 114-4705<br>873-8884 | RCA | 0.000125 | Thorium 232 | Not Required | Exempt | III |
| 845W | 873-8884<br>193-5127 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 8470/BL994A | 903-2852 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 8488/KU29 | - - | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 8501 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8514 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 8515 | 945-1822 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 8516 | - - - | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 861 | 118-3927 | RCA | 0.001 | Thorium 232 | Not Required | Exempt | III |
| 8612 | 226-6861 | Victoreen | 0.2 | Nickel 63 | Required | Exempt | IV |
| 8618 | - - - | Machlett | 0.041 | Thorium 232 | Not Required | Exempt | III |



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Table C-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| Tube Type or Part Number | NSN 5960-00- | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | DOT Label (Radioactive) | DOT Transport Group |
| 8793 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8794 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8795 | - - - | Machlett | 0.088 | Thorium 232 | Not Required | Exempt | III |
| 8806 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8807 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 885 | - - - | Victoreen | 0.25 | Nickel 63 | Required | Exempt | IV |
| 8890 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8891 | - - - | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 8991 | - - - | Victoreen | 0.2 | Nickel 63 | Required | Exempt | IV |
| 9C25 | - - - | RCA | 0.043 | Thorium 232 | Not Required | Exempt | III |
| 999/445 | 980-6749 | Microwave | 0.80 | Cobalt 60 | Required | Exempt | III |





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# SECTION D

RADIOACTIVE COMMODITIES - GP59

FIIN SEQUENCE

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# SECTION D - RADIOACTIVE COMMODITIES - FIIN SEQUENCE

## D-1 INTRODUCTION

The tables included in this section are identical to the section C tables, except the commodities are identified to an NSN in FIIN (Federal Item Identification Number) sequence.



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TABLE D-1A

RADIOACTIVE COMMODITIES - FSC 5920

FUSES AND LIGHTNING ARRESTERS

FIIN SEQUENCE



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Table D-1A
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5920-00- | TYPE OR PART NUMBER | MANUFACTURER | QUANTITY (MICROCURIES) | RADIOISOTOPE | MARKING AND LABELING REQUIREMENTS | | |
|---|---|---|---|---|---|---|---|
| | | | | | USE AND STORAGE LABEL | DOT LABEL (RADIOACTIVE) | DOT TRANSPORT GROUP |
| 011-499 | TG191 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 020-9745 | TG78 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 027-4320 | TG40 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 027-8691 | TG44 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 062-3993 | TG49 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 062-5995 | TG49 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 087-6337 | TG87 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 107-5371 | TG61 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 163-4106 | TG184 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 163-4107 | TG14 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 578-3088 | TG25 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 578-9265 | TG45 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 608-2142 | TG-57 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 613-0114 | TG59 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 629-2731 | TG101 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 679-3827 | TG53 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 682-7301 | TG54 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 687-2066 | TG84 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |



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Table D-1A - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5920-00- | TYPE OR PART NUMBER | MANUFACTURER | QUANTITY (MICROCURIES) | RADIOISOTOPE | MARKING AND LABELING REQUIREMENTS | | |
|---|---|---|---|---|---|---|---|
| | | | | | USE AND STORAGE LABEL | DOT LABEL (RADIOACTIVE) | DOT TRANSPORT GROUP |
| 687-2067 | TG83 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 713-4191 | TG76 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 727-0595 | TG48 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 738-8074 | TG34 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 787-7036 | TG31 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 789-7979 | TG151 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 806-1090 | TG56 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 813-4039 | TG51 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 822-2326 | TG52 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 822-2327 | TG86 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 822-2329 | TG27A | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 827-5751 | TG53 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 827-5752 | TG55 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 827-5753 | TG58 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 839-5538 | TG56 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 856-5965 | TG110 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 856-5966 | TG87 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |




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Table D-1A - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5920-00- | TYPE OR PART NUMBER | MANUFACTURER | QUANTITY (MICROCURIES) | RADIOISOTOPE | MARKING AND LABELING REQUIREMENTS | | |
|---|---|---|---|---|---|---|---|
| | | | | | USE AND STORAGE LABEL | DOT LABEL (RADIOACTIVE) | DOT TRANSPORT GROUP |
| 858-0215 | TG42A | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 892-9060 | TG44 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 896-5549 | TG110 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 922-4902 | TG57 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 950-0902 | TG134 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 953-1924 | TG115 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |



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TABLE D-1B

RADIOACTIVE COMMODITIES - FSC 5925

CIRCUIT BREAKERS

FIIN SEQUENCE



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Table D-1B
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5925-00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 027-2102 | 271-1290P9 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 109-2211 | 3H900-80 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| 144-0438 | C6363-1-25 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0443 | C6363-2-10 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| 144-0451 | C6363-2-25 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0453 | C6363-2-30 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0459 | D6364-1-120 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0462 | D6364-1-60 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0464 | D6364-1-70 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0466 | D6364-1-80 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0473 | D6364-2-120 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 144-0482 | D6364-2-100 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |



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Table D-1B - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5925-00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 155-8455 | C6363-1-2LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 155-8456 | C6363-1-5 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 155-8463 | C6363-1-20LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 159-0414 | 8751K4 | Cutler-Hammer | - - - | Radium 226 | Required | Exempt | I |
| 173-7627 | D6751-1-50 | Metals and Controls, Inc. | - - - | Radium | Required | Exempt | I |
| 175-2580 | C6363-2-5LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 175-2587 | C6363-1-10 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 175-5222 | AN3161P35 | Mechanical Prod., Inc. | 20 mi/hr at surface with toggles out | Radium 226 | Required | Exempt | I |
| 175-5227 | D6364-1-100 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 258-7121 | C6363-1-5LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 265-2259 | AM1513F2 | Heinemann Elec. Company | 0.05 | Radium 226 | Required | Exempt | I |
| 271-8658 | 8725K4 | Cutler-Hammer | 0.05 | Radium 226 | Required | Exempt | I |




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Table D-1B - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5925-00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 279-1029 | AM1513F35 | Heinemann Elec. Company | 0.05 | Radium 226 | Required | Exempt | I |
| 279-1031 | AM1614-25 | Heinemann Elec. Company | 0.05 | Radium 226 | Required | Exempt | I |
| 279-1350 | 1163SM6-5A125V | Heinemann Elec. Company | - - - | - - - | - - - | - - - | - - - |
| 279-2930 | D6364-1-90 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 296-8123 | C6363-1-15 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 299-1677 | C6363-2-3LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 501-6168 | C6363-1-8 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 501-7729 | D6364-1-40 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 502-7420 | 271-1289P6 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 503-5847 | C6363-1-12-5 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 506-4365 | 271-1289P5 | Raytheon | - - - | - - - | - - - | - - - | - - - |
| 506-4367 | 271-1289P19 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 506-4368 | 271-1289P18 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |



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Table D-1B - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5925-00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 506-4370 | 271-1289P4 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 506-4372 | 271-1290P24 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 506-4382 | 271-1290P6 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 506-4384 | 271-1290P7 | Raytheon | - - - | - - - | - - - | - - - | - - - |
| 506-4388 | 271-1290P19 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 506-4389 | 271-1290P20 | Raytheon | 0.05 | Radium 226 | Required | Exempt | I |
| 581-0376 | PAM1014MG3 | Heinemann Elec. Company | 0.05 | Radium 226 | Required | Exempt | I |
| 583-0590 | C6363-1-35 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 583-8292 | C6363-1-6 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 586-6129 | D6364-2-60 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 679-5834 | ET5943 | ITE Imperial Corp | - - - | - - - | - - - | - - - | - - - |
| 813-2093 | 6750-2-35 | Metals and Controls, Inc. | 5.0 | Radium 226 | Required | Exempt | I |
| 889-0768 | C6363-2-12-1/2LT | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |
| 893-4299 | D6364-1-50 | Metals and Controls, Inc. | 60.0 | Radium | Required | Exempt | I |





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TABLE D-1C

RADIOACTIVE COMMODITIES - FSC 5930

SWITCHES

FIIN SEQUENCE



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Table D-1C
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 593( -00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 112-5288 | 8803K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 180-2541 | 8708 | Cutler-Hammer | - - - | - - - | - - - | - - - | - - - |
| 187-3741 | 8201K11 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 201-2545 | 8207K3 | Cutler-Hammer | 0.2 | Radium 226 | Required | Exempt | I |
| 232-6537 | B7A | - - - | - - - | - - - | - - - | - - - | - - - |
| 259-8350 | 8822K1 | Cutler-Hammer | - - - | - - - | - - - | - - - | - - - |
| 259-8386 | 8212K11 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 272-1368 | 8800K12 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 296-5899 | 8851K3 | Cutler-Hammer | - - - | - - - | - - - | - - - | - - - |
| 296-9017 | 8715K2 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 296-9664 | 8823K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 320-9934 | 24-1164 | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 473-9099 | 8916K504 | Cutler-Hammer | - - - | - - - | Required | Exempt | I |
| 486-9058 | 8826K3 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 504-8679 | A9503C30 | Controls Company of America | - - - | - - - | - - - | - - - | - - - |
| 518-3135 | 8905K520 | Cutler-Hammer | - - - | Radium 226 | Required | Exempt | I |
| 518-3174 | A9503C28 | Controls Company of America | 20 mi/hr at at surface | Radium 226 | Required | Exempt | I |



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Table D-1C - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5930-00- | Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 549-0116 | 8816K8 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 565-9501 | 8905K514 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 615-7880 | 8830K3 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 654-4356 | C6363-6A | Metals and Controls, Inc. | - - - | - - - | - - - | - - - | - - - |
| 655-1514 | 8803K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 655-1523 | 8809K12 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 655-1582 | 8825K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 655-1582 | 8825K9 | Cutler-Hammer | 0.32 | Radium 226 | Required | Exempt | I |
| 669-7711 | 8802K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 687-1097 | 8905K526 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 760-2622 | C6363-12 1/2A | Metals and Controls, Inc. | - - - | - - - | - - - | - - - | - - - |
| 810-6128 | 8821K9 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 820-0566 | 8801K12 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 950-7445 | 8824K11 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |
| 991-9287 | 8820K13 | Cutler-Hammer | 0.15 | Radium 226 | Required | Exempt | I |





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TABLE D-1D

RADIOACTIVE COMMODITIES - FSC 5960

ELECTRON TUBES

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Table D-1D
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 003-7978 | MA3799 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 005-1194 | MA3743 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 009-2771 | VDX-1035 | Varian | 35.0 | Tritium (H3) | Required | Exempt | IV |
| 009-8082 | MA3919Z | Microwave | ≤50 | Tritium (H3) | Required | Exempt | IV |
| 009-8082 | MD85K17 | Micro Dynamics | 6,000.0 | Tritium (H3) | Required | Exempt | IV |
| 009-8087 | MD85K17 | Micro Dynamics | ≤50 | Tritium (H3) | Required | Exempt | IV |
| 011-8325 | DOD-005 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 020-0071 | 7910 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 020-0074 | ML7003A | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| 023-9497 | 304TL | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 030-1901 | 812A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 030-1901 | 812A | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 038-1660 | 5939A | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 038-1660 | 5939A | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 038-1663 | R1131C | Sylvania | --- | --- | Required | Exempt | II |
| 043-4743 | 2054 | RCA | 0.580 | Thorium 232 | Required | Exempt | III |
| 045-8639 | 5654 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 051-0095 | MA355A | Microwave | 0.84 | Cobalt 60 | Required | Exempt | III |
| 051-3188 | BL996 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 054-1987 | 4CX10000D | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 054-3205 | TD-81 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| 054-3946 | 6414W | General Elec. | 0.00001 | Rhenium 187 | Not required | Exempt | IV |
| 056-7033 | BL-P-040D | Varian | 0.8 | Cobalt 60 | Required | Exempt | III |
| 056-8223 | MA-371A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 056-8225 | MA-362C | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 056-8227 | MA-382A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 060-3441 | GV3A-950 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 060-3449 | 3B24WB | United Elec. | $7 \times 10^{-6}$ | Thorium (nat) | Not Required | Exempt | III |
| 060-3478 | 1B55 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 060-6570 | 6569 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 060-6604 | 6163 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 060-6604 | 6163 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Required | Exempt | III |
| 060-6608 | 6393/BL68 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 060-6626 | 6081/ATR407 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 060-6626 | 8081/ATR407 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 060-6626 | 6081 | Omni-Wave (Metcom) | 0.32 | Cobalt 60 | Required | Exempt | III |
| 060-6628 | 6564/BL71 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 060-6629 | BL28/6568 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 060-6630 | 6632/BL18 | Varian | 0.70 | Cobalt 60 | Required | Exampt | III |
| 060-6825 | BL-367A, B, C | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 061-1127 | 5771 | RCA | 0.2511 | Thorium 232 | Required | Exempt | III |
| 061-1209 | 304TH | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 062-6692 | CK1097-7 | Raytheon | 0.9 | Cesium 137 | Required | Exempt | III |
| 064-0204 | 5751 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 067-9364 | 8370/E38 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 070-8080 | BES-033 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 070-8080 | BTR-141 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 073-9277 | 102663(BL95H) | Varian | 0.15 | Cobalt 60 | Required | Exempt | VI |
| 075-6770 | BTR-537 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 077-1828 | TG-148 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 078-1643 | 4PR-125A | Eimac | 0.00029 | Thorium 232 | Not Required | Exempt | III |
| 078-7091 | 4CX3000A | Eimac | 0.0023 | Thorium 232 | Not Required | Exempt | III |
| 078-9500 | 8264/KU52 | ITT | 0.08 | Uranium (nat) | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 079-5341 | MA340T | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 079-5942 | BL-999-1 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 080-8748 | 4PR400A/8188 | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 080-8751 | 4PR65A/8187 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 080-9194 | 6645 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 080-9194 | 6645 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 080-9194 | 6645 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 081-0192 | TD-40 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| 081---- | BL-T-056 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 081-9255 | MA-356E | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 082-3306 | 3B24WBM | United Elec. | $7 \times 10^{-6}$ | Thorium (nat) | Not Required | Exempt | III |
| 082-4112 | 7286 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 082-4139 | 5751 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 087-9010 | 5957 | Kuthe Labs (ITT) | 0.080 | Uranium (nat) | Not Required | Exempt | III |
| 089-5851 | BL-A-007 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 089-5851 | BL-A-007-1 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 100-5163 | 3S4 | General Elec. | 0.00002 | Uranium 238 | Not Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 100-7152 | 304TL | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 105-1481 | 7859 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 105-3737 | MA3847 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 105-8594 | MD85K2 | Microdynamics | 6,000.0 | Tritium ($H^3$) | Required | Exempt | IV |
| 105-8594 | MA3848Z | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 106-0416 | MD80C17-1 | Microdynamics | $\leq$50 | Tritium ($H^3$) | Required | Exempt | IV |
| 106-0423 | MD80C17 | Microdynamics | $\leq$50 | Tritium ($H^3$) | Required | Exempt | IV |
| 106-0423 | VDC-1029 | Varian | 35.0 | Tritium ($H^3$) | Required | Exempt | IV |
| 107-7607 | 446A | Western Elec | 0.10 | Krypton 85 | Not Required | Exempt | III |
| 112-9482 | MA3770 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 144-3826 | 250TH | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 114-3831 | 6C21 | Eimac | 0.00056 | Thorium 232 | Not Required | Exempt | III |
| 114-3843 | 813 | RCA | 0.000245 | Thorium 232 | Not Required | Exempt | III |
| 114-3846 | 8020 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 114-3850 | 808 | RCA | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 114-3863 | 810 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 114-3863 | 810 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued

RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 114-4705 | 845 | RCA | 0.000125 | Thorium 232 | Not Required | Exempt | III |
| 114-4841 | 805 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 114-4849 | 2X2A | RCA | 0.0000176 | Thorium 232 | Not Required | Exempt | III |
| 114-4858 | 304TH | Eimac | 0.00084 | Thorium 232 | Not Required | Exempt | III |
| 114-4860 | 830B | RCA | 0.000049 | Thorium 232 | Not Required | Exempt | III |
| 114-4867 | 809 | RCA | 0.000029 | Thorium 232 | Not Required | Exempt | III |
| 114-4871 | 35TG | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 114-4876 | 828 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 114-4877 | 814 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 114-4880 | 811A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 114-4880 | 811A | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 114-4881 | 805 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 114-4881 | 805 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 114-4888 | 838 | RCA | 0.00013 | Thorium 232 | Not Required | Exempt | III |
| 116-9931 | 3C24 | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 116-9944 | 250R | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 116-9950 | GL-851 | General Elec. | 0.03 | Thorium 232 | Not Required | Exempt | III |





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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 116-9960 | 371B | Electronic Enterprises | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 116-9962 | 450TH | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 116-9963 | 450TL | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 116-9965 | 833A | RCA | 0.00049 | Thorium 232 | Not Required | Exempt | III |
| 116-9969 | 5C22/6279 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 116-9975 | 313C | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 116-9975 | 313C | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 116-9975 | 313C | Western Elec. | 0.48 | Krypton 85 | Required | Exempt | III |
| 116-9975 | 313C | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 116-9979 | 100TH | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 116-9980 | 35T | Eimac | 0.000096 | Thorium 232 | Not Required | Exempt | III |
| 116-9982 | 250TL | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 116-9984 | 803 | RCA | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 116-9985 | 801A | RCA | 0.00002 | Thorium 232 | Not Required | Exempt | III |
| 116-9988 | 8005 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 134-6031 | 5654 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 135-0130 | 6696A | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 137-7692 | ML-7715 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| 139-3204 | MA3785 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 140-6535 | MA3784 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 145-3128 | MA3164-3163 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 156-0297 | 5CX3000A | Eimac | 0.0023 | Thorium 232 | Not Required | Exempt | III |
| 160-2183 | MA3886Z | Microwave | $\leq$50 | Tritium ($H^3$) | Required | Exempt | IV |
| 160-2186 | MD80K15 | Microdynamics | 3,000.0 | Tritium ($H^3$) | Required | Exempt | IV |
| 160-2186 | MA3887Z | Microwave | $\leq$50 | Tritium ($H^3$) | Required | Exempt | IV |
| 166-7643 | 211W | United Elec. | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 166-7643 | 211W | RCA | 0.000125 | Thorium 232 | Not Required | Exempt | III |
| 166-7648 | OB2 | Raytheon | 0.04 | Krypton 85 | Not Required | Exempt | III |
| 166-7662 | 12AT7 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 167-0389 | 5651 | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 168-7809 | 7161 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 168-7809 | 7161 | Westinghouse | 0.7 | Uranium 235 | Required | Exempt | III |
| 177-1716 | 825W | Eimac | <5.0 | Thorium 232 | Required | Exempt | III |
| 179-4446 | 12AT7WC | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 179-8054 | 7560V | Machlett | 0.089 | Thorium 232 | Not Required | Exempt | III |
| 181-0310 | DOD-004 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 186-1964 | VDS-1011 | Varian | 70.0 | Tritium ($H^3$) | Required | Exempt | IV |
| 188-0844 | 1U5 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-0921 | 4D21 | Eimac | 0.000029 | Thorium 232 | Not Requried | Exempt | III |
| 188-3524 | 3V4 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-3531 | 6141/427A | Western Elec. | 4.00 | Krypton 85 | Required | Exempt | III |
| 188-3534 | 1B63A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 188-3534 | 1B63A | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 188-3534 | 1B63A | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 188-3534 | 1B63A | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 188-3535 | 1B23 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 188-3535 | 1B23 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 188-3535 | 1B23 | Central Sales | 1.0 | Cobalt 60 | Required | Exempt | III |
| 188-3545 | 6AK5/EF95 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 188-3567 | 0B3 | Raytheon | 0.08 | Krypton 85 | Not Required | Exempt | III |
| 188-3593 | 1U4 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-3595 | 1T4 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-3900 | 750TL | Eimac | 0.001 | Thorium 232 | Not Required | Exempt | III |
| 188-3913 | 721B | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 188-3936 | 8000 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 188-3952 | 1S5 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-3954 | 1R5 | General Elec. | 0.00001 | Uranium 238 | Not Required | Exempt | III |
| 188-3964 | 313CD | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 188-6575 | 1960 | RCA | - - - | - - - | Required | Exempt | II |
| 188-6581 | 1B37 | Varian | 0.30 | Cobalt 60 | Required | Exempt | II |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 188-6582 | 1B32 | Westinghouse | - - - | - - - | Required | Exempt | II |
| 188-6584 | 5670 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 188-6588 | 5672 | - - - | - - - | - - - | Required | Exempt | II |
| 188-6592 | 5962/BS101 | Anton/Lionel (now Eon Corp.) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 188-6592 | 5962/BS101 | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 188-6592 | 5962/BL101 | Victoreen | 0.001 | Nickel 63 | Not Requried | Exempt | IV |
| 188-6592 | 5962/BS101 | Electronic Products | 3.0 | Nickel 63 | Required | Exempt | IV |
| 188-6592 | BS101/5962 | Varian | 0.001 | Nickel 63 | Not required | Exempt | IV |
| 188-6592 | 5962/BS101 | Eon Corp. | <0.1 | Nickel 63 | Not required | Exempt | IV |
| 188-6593 | 5696 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 188-8571 | 75TL | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 188-8594 | 353A | Western Elec | 0.01 | Radium 226 | Required | Exempt | I |
| 188-8594 | 353A | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 188-8633 | 359A | Western Elec. | 1.20 | Krypton 85 | Required | Exempt | III |
| 188-8648 | HF-300 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 189-5968 | 4-1000A | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 189-5973 | 811A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 189-5973 | 811A | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 189-5976 | 152TH | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 193-5085 | 0D3W | Raytheon | 0.067 | Krypton 85 | Not Required | Exempt | III |
| 193-5087 | 1B3GT | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 193-5088 | 1B22 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 193-5091 | 1B24A | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 193-5091 | 1B24A | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 193-5091 | 1B24A | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 193-5091 | 1B24A | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 193-5092 | 1B27 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 193-5093 | 1B44 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 193-5093 | 1B44 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 193-5093 | 1B44 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 193-5094 | 1B56 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 193-5094 | 1B56 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 193-5094 | 1B56 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 193-5095 | 1B58 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 193-5095 | 1B58 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 193-5095 | 1B58 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-( )- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 193-5125 | 724B | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 193-5127 | 845W | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 193-5141 | 5792/ATR345 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 193-5141 | 5792/ATR345 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 193-5142 | 5793/ATR346 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 193-5142 | 5793/ATR346 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 193-5145 | 5751 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 193-5147 | 5853/TR368 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| 193-5147 | 5853/TR368 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 193-5147 | 5853/TR368 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 194-1474 | 451A | Western Elec. | 2.10 | Krypton 85 | Required | Exempt | III |
| 220-6525 | 5790/X7108 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 220-6526 | 5791/X6007 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 220-6892 | 1B37A | General Elec. | 0.950 | Cobalt 60 | Required | Exempt | III |
| 220-6892 | 1B37A | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 220-6892 | 1B37A | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 222-1388 | BES-050 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 222-1389 | BLA-011 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 222-1397 | BLA-012 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 225-3419 | MD80C8 | Micro Dynamics | ≤50 | Tritium ($H^3$) | Required | Exempt | IV |
| 226-4306 | MA3855K | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 226-6861 | 8612 | Victoreen | 0.20 | Nickel 63 | Required | Exempt | IV |
| 228-8961 | 414 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 230-5253 | 5783 | Raytheon | 0.0033 | Krypton 85 | Not Required | Exempt | III |
| 230-5253 | 5783/1064 | Raytheon | 0.70 | Nickel 63 | Required | Exempt | IV |
| 230-5291 | 395A | Western Elec. | 1.6 | Krypton 85 | Required | Exempt | III |
| 230-5293 | 313CC | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 230-5293 | 313CC | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 230-5293 | 313CC | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 230-5298 | 3X2500A3 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 230-5311 | 313CA | Western Elec. | 0.50 | Krypton 85 | Required | Exempt | III |
| 230-5311 | 313CA | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 230-5311 | 313CA | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 237-0103 | 5927 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 237-0108 | ML-357B | Machlett | 0.0011 | Thorium 232 | Not Required | Exempt | III |
| 237-2413 | 1B38 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 237-2413 | 1B38 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 237-2414 | 1B40 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 237-2414 | 1B40 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 237-6917 | 5725 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 243-5017 | 4-65A | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 243-5018 | 8438-4-400A | - - - | - - - | - - - | Required | Exempt | II |
| 244-8041 | 3B24WA | Tung-Sol (Chatham) | 0.0005 | Uranium 232 | Not Required | Exempt | I |
| 248-1270 | 5922 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 248-1271 | 5921 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 248-3065 | 423A | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 248-3078 | 5823 | Tung-Sol (Chatham) | 0.0018 | Krypton 85 | Not Required | Exempt | III |
| 248-8504 | ATR387 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 256-9985 | 1B50 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 256-9988 | 1B51 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 261-8671 | 5863 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 261-8671 | 5863/X7109 | Microwave | 0.50 | Cobalt 60 | Required | Exempt | III |
| 261-8671 | 5863/X7109 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 261-8678 | 6002 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 261-9155 | 1B29 | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 262-0122 | 1B36 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-0123 | 1B41 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 262-0123 | 1B41 | Westinghouse | 1.0 | Radium 226 | Required | Exempt | I |
| 262-0125 | 1B45 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 262-0125 | 1B45 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 262-0137 | 1B26 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 262-0167 | 12AT7WA | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 262-0174 | 5865/MA379 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 262-0174 | 5865/TR361 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 262-0174 | 5865/TR361 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 262-0176 | 5939 | Westinghouse | 3.30 | Radium 226 | Required | Exempt | I |
| 262-0184 | 5644 | Raytheon | 0.006 | Cobalt 60 | Not Required | Exempt | III |
| 262-0190 | 6AB4 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 262-0191 | 1B52 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-0192 | 1B53 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-0198 | 577 | United Elec. | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 262-0204 | 5762/7C24 | RCA | 0.0071 | Thorium 232 | Not Required | Exempt | III |





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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 262-0210 | 5814A | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 262-0211 | 5825 | RCA | 0.0000077 | Thorium 232 | Not Required | Exempt | III |
| 262-0213 | 5841 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 262-0213 | 5841 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 262-0225 | 5883 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-0230 | 1B55 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 262-0286 | 5651WA | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 262-0286 | 5651WA | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 262-0286 | 5651WA | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 262-0286 | 5651WA | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 262-0286 | 5651WA | Tung-Sol (Chatham) | 0.005 | Radium 226 | Not Required | Exempt | I |
| 262-0289 | 6081/ATR407 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-0289 | 6081/ATR407 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 262-0289 | 6081 | Omni-Wave (Metcom) | 0.32 | Cobalt 60 | Required | Exempt | III |
| 262-1356 | 1B35A | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-1356 | 1B35A | General Elec. | 0.95 | Cobalt 60 | Required | Exempt | III |
| 262-1356 | 1B35A | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 262-1356 | 1B35A | Omni-Wave (Metcom) | 0.2 | Cobalt 60 | Required | Exempt | III |
| 262-1357 | 5654 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 262-1358 | 4E27A/5-125B | Eimac | 0.0003 | Thorium 232 | Not Required | Exempt | III |
| 262-1693 | 5925/TR331 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 262-1696 | 6022/ATR332 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 262-3771 | 5787 | Raytheon | 0.0067 | Krypton 85 | Not Required | Exempt | III |
| 263-9910 | BL25/6322 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 264-2996 | 6035 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 264-2998 | 6163 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 264-2998 | 6163 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| 264-3000 | 6167 | Western Elec. | 3.00 | Krypton 85 | Required | Exempt | III |
| 269-2736 | EVM-1000 | Eon Corp. | <0.1 | Nickel 63 | Not Required | Exempt | IV |
| 272-8545 | 5787WA | Tung-Sol (Chatham) | 0.0028 | Krypton 85 | Not Required | Exempt | III |
| 272-8545 | 5787WA | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 272-8545 | 5787WA | Raytheon | 0.0067 | Krypton 85 | Not Required | Exempt | III |
| 272-9195 | 6024/ATR387 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 272-9196 | 6117 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 272-9195 | 6117 | General Elec. | 0.475 | Cobalt 60 | Required | Exempt | III |
| 272-9196 | 6177 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 272-9196 | 6117 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 272-9197 | 6162/ATR388 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 272-9197 | 420 | Cetron | 2.0 | Krypton 85 | Required | Exempt | III |
| 272-9194 | 420A | Cetron | 2.0 | Krypton 85 | Required | Exempt | III |
| 272-9199 | 6213 | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 272-9201 | 6260 | Westinghouse | 2.0 | Radium 226 | Required | Exempt | I |
| 272-9201 | 6260 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 284-6000 | QK-447 | Raytheon | 0.61 | Thorium 232 | Required | Exempt | III |
| 284-6120 | 2-50A | Eimac | 0.000111 | Thorium 232 | Not Required | Exempt | III |
| 284-6554 | 6141/427A | Western Elec. | 4.0 | Krypton 85 | Required | Exempt | III |
| 284-7166 | 5783WA | Tung-Sol (Chatham) | 0.0013 | Krypton 85 | Not Required | Exempt | III |
| 284-7166 | 5783WA | Raytheon | 0.0033 | Krypton 85 | Not Required | Exempt | III |
| 284-7229 | WL759 | Westinghouse | 0.166 | Krypton 85 | Required | Exempt | III |
| 295-3589 | BL46/6639 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 295-9561 | 1B54 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 296-0507 | 6143 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 296-0507 | 6143 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 296-2698 | 6560/BL35 | Varian | 0.25 | Cobalt 60 | Required | Exempt | IV |
| 296-3354 | 6201 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 296-4142 | VXR16000 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 296-5509 | 6396 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 296-5932 | 6378/BL62 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 296-7326 | 6322 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 296-7326 | 6322/BL25 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 296-9997 | 576A | Electronic Enterprises | 0.0004 | Thorium 232 | Not Required | Exempt | III |
| 319-0527 | 6119 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 319-0568 | 2-25A | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 319-0569 | 25T | Eimac | 0.000083 | Thorium 232 | Not Required | Exempt | III |
| 320-6631 | BTR-614A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 333-3540 | 427A | Western Elec. | 4.00 | Krypton 85 | Required | Exempt | III |
| 348-2498 | BL-A-030 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 355-7250 | 1B23 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 355-7290 | QK-254/6518 | Raytheon | 0.40 | Thorium 232 | Required | Exempt | III |
| 365-2548 | BL11/6282 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 365-6183 | 1B63B | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 390-5208 | KU25/8424 | Kuthe Labs (ITT) | 0.08 | Uranium (nat.) | Not Required | Exempt | III |
| 390-5239 | 6620 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 390-5241 | 6232/ATR427 | Sylvania | 0.15 | Cobalt 60 | Required | Exempt | III |
| 403-1162 | BTR632A | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| 405-8367 | BES-052 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 422-5376 | MA3830X | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 434-0229 | BES-045 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 435-1784 | TG-64 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 439-5705 | 359A | Western Elec. | 1.20 | Krypton 85 | Required | Exempt | III |
| 449-7994 | MD80K7 | Micro Dynamics | $\leq 50$ | Tritium ($H_3$) | Required | Exempt | IV |
| 449-7994 | MA3899Z | Microwave | $\leq 50$ | Tritium ($H_3$) | Required | Exempt | IV |
| 464-4093 | MA3148 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 469-7363 | VDX-1003 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 469-7363 | MA37019X | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 471-9818 | TD-39 | Bendix | 0.60 | Radium 226 | Required | Exempt | I |
| 472-9435 | BLT-003 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 478-4198 | MA3254-1 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 478-7692 | 4CW25000A | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 479-1378 | 4CX5000J | Eimac | 0.0037 | Thorium 232 | Not Required | Exempt | III |
| 479-1624 | MA3763 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 486-9161 | MA3753 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 495-9113 | MA3895X | Microwave | $\leq$50 | Tritium ($H_3$) | Required | Exempt | IV |
| 501-0865 | VXR2700 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 501-0866 | KX-642 | Westinghouse | 0.3 | Krypton 85 | Required | Exempt | III |
| 501-1066 | 6560 | Omni-Wave (Metcom) | 0.40 | Cobalt 60 | Required | Exempt | III |
| 501-5535 | 6605/BL96 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 503-0607 | 6AK4WB | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 503-3115 | 812A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 503-4436 | BL-644 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 503-4493 | BL96/6605 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 503-4497 | BL-95H/6645 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 503-4880 | OA2WA | CBS Hytron | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 503-4880 | OA2WA | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 503-4880 | OA2WA | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 503-4880 | OA2WA | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 503-4880 | OA2 | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 503-8046 | MA356P | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 504-7571 | 6334/BL27 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 507-3122 | 2-150D | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 508-1044 | 2-240A | Eimac | 0.00042 | Thorium 232 | Not Required | Exempt | III |
| 508-1046 | 253 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 509-3171 | 5651 | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 518-1688 | 284 | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 519-6175 | 6312 | - - - | 0.15 | - - - | Required | Exempt | II |
| 537-3959 | 376C | Western Elec. | 4.0 | Krypton 85 | Required | Exempt | III |
| 537-3959 | 376C | Cetron | 4.0 | Krypton 85 | Required | Exempt | III |
| 538-2793 | 6568/BL28 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 538-3862 | BL-744 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 538-4709 | 834 | RCA | 0.000081 | Thorium 232 | Not Required | Exempt | III |
| 543-0143 | 6640/BL60 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 543-0693 | BL-651 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 543-1142 | 6249A | Westinghouse | 0.016 | Radium 226 | Required | Exempt | I |
| 547-2928 | 439A | Western Elec. | 3.0 | Krypton 85 | Required | Exempt | III |
| 548-3802 | 6034 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 548-3803 | 6033 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 548-9734 | BL62/6378 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 549-0045 | 5864/ATR321 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 552-1107 | 812A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 552-1501 | CH1067H | Tung-Sol (Chatham) | 0.0012 | Krypton 85 | Not Required | Exempt | III |
| 552-1759 | BL-735 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 552-8277 | 4C35A/6268 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 553-3519 | BL-650/6890 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 553-3770 | 6140 | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 553-4582 | 828 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 553-7010 | 6587 | Kuthe Labs (ITT) | 0.080 | Uranium (nat) | Not Required | Exempt | III |
| 553-7091 | 6542 | Tung-Sol (Chatham) | 0.0025 | Krypton 85 | Not Required | Exempt | III |
| 553-7338 | BL-670 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 556-1231 | 2X2A | RCA | 0.0000176 | Thorium 232 | Not Required | Exempt | III |
| 556-1232 | 1500T | Eimac | 0.0013 | Thorium 232 | Not Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 556-1236 | 100TH | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 556-1238 | 750TL | Eimac | 0.001 | Thorium 232 | Not Required | Exempt | III |
| 556-1454 | 811A | RCA | 0.000056 | Thorium 232 | Not Required | Exempt | III |
| 556-1454 | 811A | Elec. Enter. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 556-1455 | 813 | RCA | 0.000245 | Thorium 232 | Not Required | Exempt | III |
| 557-5105 | 545 | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |
| 557-6883 | 0A2 | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 557-6885 | 0B2 | Raytheon | 0.04 | Krypton 85 | Not Required | Exempt | III |
| 557-6926 | 0A2WA/6626 | CBS Hytron | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 557-6926 | 0A2WA/6626 | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 557-6926 | 0A2WA | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 557-6926 | 0A2WA/6626 | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 561-8230 | ML-6544 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| 567-4415 | 810 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 567-4415 | 810 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 557-3205 | 6322 | Omni-Wave (Metcom) | 0.10 | Cobalt 60 | Not Required | Exempt | III |
| 577-1537 | 6164 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 577-3205 | 6322/BL25 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 577-3520 | MA3879X | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 578-1140 | 413 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 578-114V | GV3B450 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 578-1524 | 4-1000A | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | IV |
| 578-8701 | VXR1800 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | III |
| 578-8724 | BL-357 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 581-5535 | 6605/BL96A | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 581-5535 | 6605/BL96 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 581-8178 | KX-642 | Westinghouse | 0.3 | Krypton 85 | Required | Exempt | III |
| 581-8645 | 6899/MA349 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 581-9031 | BL-350 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 581-9404 | 6414 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 581-9594 | 2-450A | Eimac | 0.0012 | Thorium 232 | Not Required | Exempt | III |
| 583-0613 | 6164 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 583-4175 | BL-622 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 593-2555 | EVM-3500 | EON Corp. | <0.1 | Nickel 63 | Not Required | Exempt | IV |
| 605-9192 | 6829 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 608-1960 | BL-652 | Varian | 0.8 | Cobalt 60 | Required | Exempt | III |
| 608-2774 | 6214 | Tracerlab, Inc. | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 608-2774 | 6214 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| 610-8486 | BLT-086 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| 610-8486 | 7564/MA-378A | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 613-6796 | GV9A-2000 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 615-0719 | BL-352A | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 615-5047 | 7152/BL612 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 615-5528 | 12AT7WB | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 615-5559 | 830B | RCA | 0.000049 | Thorium 232 | Not Required | Exempt | III |
| 615-5561 | 810 | RCA | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 615-5561 | 810 | United Elec. | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 615-5562 | 809 | RCA | 0.000029 | Thorium 232 | Not Required | Exempt | III |
| 615-5565 | 750TL | Eimac | 0.001 | Thorium 232 | Not Required | Exempt | III |
| 615-5568 | 1500T | Eimac | 0.0013 | Thorium 232 | Not Required | Exempt | III |
| 615-5607 | 724B | Varian | 0.15 | Colbalt 60 | Required | Exempt | III |
| 615-5619 | 8005 | RCA | 0.00012 | Thorium 232 | Not Required | Exempt | III |
| 615-5666 | 253 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 615-5689 | 4-1000A | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 615-5822 | 476 | Anton (Lionel) | 1.0 | Carbon 14 | Required | Exempt | IV |
| 615-5824 | BL-696 | Varian | 0.15 | Colbalt 60 | Required | Exempt | III |
| 615-9618 | BL-A-034 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 615-9871 | 8020 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 615-9871 | 8020 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 615-9871 | 8020W | Cetron | 0.35 | Thorium Oxide | Required | Exempt | III |
| 615-9871 | 8020W | United Elec. | 0.00002 | Thorium Oxide | Not Required | Exempt | III |
| 617-3129 | BLT-028 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 617-3129 | MA-324 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 617-3129 | MA-324B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 617-3482 | GV5A-3300 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 617-3482 | GV5C-3300 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 617-3736 | BL-362 | Varian | 0.25 | Cobalt | Required | Exempt | III |
| 617-4105 | GV5A-2000 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 617-4449 | GV5C-4000 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 617-4891 | 3X2500F3 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 617-4911 | 6135 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 617-5696 | MA-306B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 617-5806 | BL-367 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 617-5806 | BL-367A, B, C | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 617-5807 | BL-397 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 617-5810 | BL43/5810 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 617-5825 | GV5A-1200 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 596- 00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 617-6367 | OA2WA | CBS Hytron | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 617-6367 | OA2WA | Raytheon | 0.03 | Krypton 85 | Not Required | Exempt | III |
| 617-6367 | OA2WA | Sylvania | 0.013 | Krypton 85 | Not Required | Exempt | III |
| 617-6367 | OA2WA | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 617-8539 | 833A | RCA | 0.00049 | Thorium 232 | Not Required | Exempt | III |
| 617-8563 | 250TL | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 617-8569 | 250TH | Eimac | 0.0006 | Thorium 232 | Not Required | Exempt | III |
| 617-8677 | 5696 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 617-8685 | 5651 | Raytheon | 0.0125 | Krypton 85 | Not Required | Exempt | III |
| 617-8876 | 5787WA | Tung-Sol (Chatham) | 0.0028 | Krypton 85 | Not Required | Exempt | III |
| 617-8876 | 5787WA | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 617-8876 | 5787WA | Raytheon | 0.0067 | Krypton 85 | Not Required | Exempt | III |
| 617-8973 | 6626/OA2WA | RCA | 0.0105 | Nickel 63 | Not Required | Exempt | IV |
| 617-9037 | 834 | RCA | 0.000081 | Thorium 232 | Not Required | Exempt | III |
| 617-9056 | 75TH | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 620-1435 | MA-332 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 620-1435 | MA-332B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 620-4528 | BLT-080 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 620-4528 | MA-349 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 620-4528 | 6899 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 624-4718 | OB2WA | RCA | 0.01 | Nickel 63 | Not Required | Exempt | IV |
| 624-4718 | OB2WA | Raytheon | 0.04 | Krypton 85 | Not Required | Exempt | III |
| 624-4718 | OB2WA | CBS Hytron | 0.05 | Nickel 63 | Not Required | Exempt | IV |
| 624-4718 | OB2WA/6627 | Jack & Heintz Inc. | 6.0 | Radium 226 | Required | Exempt | I |
| 624-4718 | OB2WA | Tung-Sol (Chatham) | 0.005 | Radium 226 | Not Required | Exempt | I |
| 630-2941 | MA306 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 633-6304 | BL693/7309 | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 636-1812 | 5939A | Varian | 0.30 | Cobalt 60 | Required | Exempt | III |
| 636-1812 | 5939A | Westinghouse | 3.30 | Radium 226 | Required | Exempt | I |
| 636-2217 | UX6653 | - - - | - - - | - - - | Required | Exempt | II |
| 636-2244 | 6386 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 636-3613 | BL-75 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 646-4613 | 6265 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960 00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 646-4719 | 6962/BL665 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 646-4720 | 6633/BL37 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| 646-4756 | 6829 | General Elec. | 0.00001 | Rhenium 187 | - - - | - - - | - - - |
| 661-0339 | BL965 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 661-0340 | MA-373 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 665-1057 | 1B58A | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 665-1057 | 1B58A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 665-3150 | R4410 | Sylvania | - - - | - - - | Required | Exempt | II |
| 669-6865 | 6166 | RCA | 0.0276 | Thorium 232 | Not Required | Exempt | III |
| 672-3836 | BL-366 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 675-3439 | GV3A-1500 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 676-7979 | 1K3 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 679-1791 | 3CW5000F1 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 679-6507 | MA336-7166 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 679-8153 | GV5A-2500 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 681-8017 | ML-7003 | Machlett | 0.0055 | Thorium 232 | Not Required | Exempt | III |
| 681-8036 | BL-982 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 681-8037 | BL-973 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 681-9793 | 7166/BL933 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| 681-9793 | 7166/BL933 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 682-2771 | QK-327/7529 | Raytheon | 0.19 | Thorium 232 | Required | Exempt | III |
| 682-2777 | 5736 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 682-8544 | 6905/BL613 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 682-8629 | 827R | RCA | 0.00256 | Thorium 232 | Not Required | Exempt | III |
| 682-8649 | GV5A-3000 | Victoreen | 0.10 | Nickel 63 | Not Required | Exempt | IV |
| 682-8864 | 6906/BL643 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 682-9452 | 6213 | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 685-8706 | 7159 | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |
| 686-6756 | GV3B-600 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 687-1034 | GV6R-1400A | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 688-2414 | 7310 | United Elec. | 0.0002 | Thorium (nat) | Not Required | Exempt | III |
| 688-3756 | 7381/BL979 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 688-6706 | 7077 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 688-9031 | MA337 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 688-9031 | MA-337A | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 689-1896 | 5786 | RCA | 0.0051 | Thorium 232 | Not Required | Exempt | III |
| 701-7582 | BL-601 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 701-7582 | BL-601/8048 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 702-5028 | 8265 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 707-0816 | MA-356 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 707-3336 | 7988/TD44 | Signalite | 0.06 | Radium 226 | Required | Exempt | I |
| 709-2031 | BL-29 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 712-3891 | MA-335B | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 713-7014 | 3CH5W | ITT | 0.08 | Thorium 232 | Not Required | Exempt | III |
| 715-1833 | 1AD2 | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 719-5454 | BL-31/6637 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 719-7508 | BLT-40 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 721-2495 | 487 | Anton Labs | 1.0 | Carbon 14 | Required | Exempt | IV |
| 725-4068 | MA-325 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 726-9826 | 6EZ8 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 727-5618 | 6455/BL61 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 728-1775 | 6DT8 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 729-0285 | 6906/BL643 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 729-1269 | 7823/MA-356A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 729-1269 | 7823/MLT13 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 729-1719 | R1130B/1B59 | - - - | - - - | - - - | Required | Exempt | II |
| 729-3424 | 2X3000F3 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 729-3424 | 2X3000F | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 729-3648 | BL613/6905 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 729-5602 | GV6A-2000 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 731-1742 | VXR1500 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 735-4210 | 1B63A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 739-7922 | BTR-140 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-( )- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 740-8350 | 2041 | RCA | 0.032 | Thorium 232 | Not Required | Exempt | III |
| 752-0185 | 6545 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 752-0594 | 6334 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 752-0594 | 6334 | Omni-Wave (Metcom) | 0.2 | Cobalt 60 | Required | Exempt | III |
| 752-5239 | 6644 | Tracerlab, Inc. | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 752-5239 | 6644/BL95 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 752-5270 | 1B3GT/1G3GT | General Elec. | 0.000001 | Rhenium 187 | Not Required | Exempt | IV |
| 752-5384 | 5948A | Tung-Sol (Chatham) | - - - | - - - | Required | Exempt | II |
| 752-5426 | 6950 | RCA | 0.249 | Thorium 232 | Required | Exempt | III |
| 752-5460 | 7462 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 752-5892 | 12AT7WB | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 754-5431 | BLT-004 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 754-5431 | BLT-004A/7902 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 754-5782 | 7099 | Tung-Sol (Chatham) | 0.0025 | Radium 226 | Not Required | Exempt | I |
| 754-9177 | BL-960A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 754-9916 | 6542 | Tung-Sol (Chatham) | 0.0025 | Krypton 85 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 754-9916 | 6542 | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 754-9916 | 6542 | Raytheon | 0.006 | Cobalt 60 | Not Required | Exempt | III |
| 755-4210 | 1B36A | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 755-9108 | ML-6257 | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| 755-9109 | ML-5682 | Machlett | 0.20 | Thorium 232 | Required | Exempt | III |
| 760-1786 | BTR-121 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 760-1786 | MA3157D/8576 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 762-6682 | BTR-183 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 762-6682 | MA-375 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 762-6682 | 7563/MA375 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 762-7692 | BTR-608 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 762-7692 | MA-362E | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 762-7692 | MD-70K8 | Microdynamics | 0.5 | Cobalt 60 | Requried | Exempt | III |
| 763-7806 | 6696A | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 763-8807 | MA38155 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 765-4524 | MA-354 | Microdynamics | 0.40 | Cobalt 60 | Required | Exempt | III |
| 765-4539 | MA-331A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 765-4540 | MA-331B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 769-0869 | 6626/OA2WA | CBS Hytron | 0.01 | Krypton 85 | Not Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 770-1927 | 4628 | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 774-6338 | TG-77 | Bendix | <1.0 | Cesium 137 | Required | Exempt | III |
| 778-3857 | KU82/7583 | Kuthe Labs(ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 780-7856 | BLT-055 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 780-8151 | 6232/ATR427 | Sylvania | 0.15 | Cobalt 60 | Required | Exempt | III |
| 780-8151 | 6232/TR427 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 780-8151 | 6232/ATR427 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 780-8151 | 6232/MA316 | Microwave | 0.5 | Cobalt 60 | Required | Exempt | III |
| 781-1808 | 3-400Z | Eimac | 0.0008 | Thorium 232 | Not Required | Exempt | III |
| 781-8963 | BLT-058 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 788-9889 | BTR-546 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 788-9889 | MCC-17A | Omni-Wave (Metcom) | 0.2 | Cobalt 60 | Required | Exempt | III |
| 799-9471 | 7821/MPT12 | Omni-Wave (Metcom) | - - - | Thorium 232 | Required | Exempt | III |
| 799-9471 | 7821 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 800-0554 | 6627/OB2WA | Raytheon | 0.02 | Cobalt 60 | Not Required | Exempt | III |
| 800-0554 | 6627/OB2WA | Jack & Heintz, Inc. | 6.0 | Radium 226 | Required | Exempt | I |
| 800-0554 | 6627/OB2WA | CBS Hytron | 0.05 | Nickel 63 | Not Required | Exempt | IV |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 800-0556 | OC2 | Raytheon | 0.025 | Krypton 85 | Not Required | Exempt | III |
| 800-0578 | 313CC | Western Elec. | 0.5 | Krypton 85 | Required | Exempt | III |
| 800-0578 | 313CC | Western Elec. | 0.01 | Radium 226 | Not Required | Exempt | I |
| 800-0578 | 313CC | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 800-0594 | BL87/6636 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| 800-2357 | 5950 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 800-3535 | BL-954 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 800-3536 | BL-377 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 800-3537 | BL-374 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 800-4409 | TG41A | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 801-5578 | 313C | Western Elec. | 0.5 | Krypton 85 | Required | Exempt | III |
| 801-5578 | 313C | Tung-Sol (Chatham) | 0.05 | Radium 226 | Required | Exempt | I |
| 801-5578 | 313C | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 801-6865 | BL-651H | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 801-6865 | MA3702 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 801-6865 | MD73X6 | Microdynamics | 1.0 | Cobalt 60 | Required | Exempt | III |
| 804-0954 | 6303 | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 804-2215 | BL-957 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 805-3626 | TG-50 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 806-0292 | 5957 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 806-8155 | BL-690 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 807-7374 | 7152/BL612 | Varian | 0.60 | Cobalt 60 | Required | Exempt | III |
| 808-4217 | GV6A-1200 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 808-4522 | 8020 | Electronic Enterprises | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 808-4522 | 8020 | Eimac | 0.0002 | Thorium 232 | Not Required | Exempt | III |
| 808-4522 | 8020W | Cetron | 0.0016 | Thorium Oxide | Not Required | Exempt | III |
| 808-4522 | 8020W | United Elec. | <0.00002 | Thorium Oxide | Not Required | Exempt | III |
| 808-6977 | F-2701 | ITT | 1.5 | Cesium 137 | Required | Exempt | III |
| 808-6977 | TG-36 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 808-9472 | BL-N-003 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 809-3372 | ML-6258 | Machlett | 0.013 | Thorium 232 | Not Required | Exempt | III |
| 809-7515 | BLS-016 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 810-8310 | 6C21 | Eimac | 0.00056 | Thorium 232 | Not Required | Exempt | III |
| 811-6773 | MA-340D | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 811-6773 | BL-965 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 811-6838 | B4021AL | Burrough | 1.0 | Krypton 85 | Required | Exempt | III |



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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 811-7500 | GV5A-1600 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 812-2113 | BL-941 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 812-3745 | 6414 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 812-3745 | 6414W | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 812-6904 | GV3A-2000 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 813-0750 | 423C | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 813-0755 | 6199 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 813-5616 | TD41 | Bendix | 0.6 | Radium 226 | Required | Exempt | I |
| 816-1386 | 6423 | Machlett | 0.008 | Thorium 232 | Not Required | Exempt | III |
| 816-3759 | 7162 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 817-6181 | BLT-033 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 817-6181 | BLT033/TR610 | Varian | 0.9 | Cobalt 60 | Required | Exempt | III |
| 819-1154 | 5841 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 819-1154 | 5841 | Vihtoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 821-5031 | 6386 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 821-6134 | TG-35 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 822-2494 | MA3215-1 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 822-8875 | MA-372 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |




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RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 824-7834 | GV3B-400 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 824-7834 | 483 | Lionel Elec. | 0.25 | Nickel 63 | Required | Exempt | IV |
| 824-8668 | BL-970 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 824-8668 | MA-358 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 825-1000 | MA-356C | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 825-4562 | 7617 | Eon Corp. | <0.17 (no radioactive material used after 1975) | Nickel 63 | Required | Exempt | IV |
| 825-4999 | 7615/EP680 | Electronic Products | 3.0 | Nickel 63 | Required | Exempt | IV |
| 825-4999 | 7615 | Eon Corp. | <0.1 (no radioactive material used after 1975) | Nickel 63 | Not Required | Exempt | IV |
| 826-1280 | GV6A-700 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 828-8004 | TD-58 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| 833-5515 | 6844A | National Electronics (Varian) | 0.06 | Krypton 85 | Not Required | Exempt | III |
| 834-0571 | 7152/BL612 | Varian | 0.6 | Cobalt 60 | Required | Exempt | III |
| 834-2902 | 7166/BL933 | Varian | 0.70 | Cobalt 60 | Required | Exempt | III |
| 834-5522 | GL-6942 | General Elec. | 0.03 | Thorium 232 | Not Required | Exempt | III |
| 835-0057 | MA366-2 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 835-0058 | MA366C | Microwave | 0.40 | Cobalt | Required | Exempt | III |
| 835-0058 | MA366-3 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 835-0059 | MA366-4 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 835-1517 | GV6A-2100 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 835-2574 | BL-934A | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 835-2575 | BL-967 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 835-2575 | MD70K7 | Micro Dynamics. | 0.50 | Cobalt 60 | Required | Exempt | III |
| 835-2576 | BL-934 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 836-6273 | OG3 | Tung-Sol (Chatham) | 0.004 | Krypton 85 | Not Required | Exempt | III |
| 838-8881 | GV9A-1420 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 839-9525 | GV9A-900 | Victoreen | 0.001 | Nickel 63 | Not Required | Exempt | IV |
| 840-0766 | GV6C-3800 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 841-2352 | 12AX7 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 841-2609 | BL-71/6564 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 842-6659 | 6143 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 842-6660 | 5864/ATR321 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 842-6660 | 5864/ATR321 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |
| 842-6661 | 6024/ATR387 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 842-7692 | MA3122 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 843-0702 | MA340K | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 843-8063 | BL95/6644 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 844-5183 | BLT-042 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 844-5183 | MA362 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 845-2633 | 4-250A | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 845-5743 | 4CW10000A | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 846-4538 | GV3B-700 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 847-1319 | A15030 | RCA | 0.498 | Thorium 232 | Required | Exempt | III |
| 847-3445 | BTR-147 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 847-3752 | BLT-110 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 847-3752 | MA-340B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 847-6738 | 5682 | General Elec. | 0.2 | Thorium 232 | Required | Exempt | III |
| 848-8575 | BLT-024 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 849-3494 | MA-356 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 849-3494 | BL-T-097 | Varian | 0.7 | Cobalt 60 | Required | Exempt | III |
| 849-3721 | 447A | Western Elec. | 0.1 | Krypton 85 | Not Required | Exempt | III |
| 849-3768 | 430-B | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 849-3769 | 346C | Cetron | 4.5 | Krypton 85 | Required | Exempt | III |
| 849-3769 | 346C | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 849-3769 | 346CJW | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 849-6864 | 432B | Western Elec. | 4.5 | Krypton 85 | Required | Exempt | III |
| 850-0788 | BTR-540 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 850-0925 | 6633 | Microwave | 0.25 | Cobalt 60 | Required | Exempt | III |
| 850-1522 | GV5C-3800 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 850-6633 | 6357 | - - - | - - - | - - - | Required | Exempt | II |
| 850-9920 | MA-393 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 851-7848 | 6213 | Raytheon | 0.7 | Nickel 63 | Required | Exempt | IV |
| 852-1514 | BL-T-086 | Varian | 0.40 | Cobalt 60 | Required | Exempt | III |
| 852-5546 | TD-78 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| 852-5655 | KU52/8264 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 852-7334 | GV5C-4300 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 853-5846 | 75TL | Eimac | 0.00021 | Thorium 232 | Not Required | Exempt | III |
| 853-6468 | 1B60 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 855-1853 | 175A | Eimac | 0.0008 | Thorium 232 | Not Required | Exempt | III |
| 855-8680 | TD-36 | Bendix | 0.9 | Cesium 137 | Required | Exempt | III |
| 855-8682 | TD-61 | Bendix | 0.06 | Radium 226 | Required | Exempt | I |
| 855-8790 | GV4S-2800 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 856-8853 | BES-048 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 856-9805 | MA3133 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 857-3083 | MA-3152 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 858-4074 | GV5A-1300 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 864-7480 | 10106946 | Raytheon | 0.90 | Cesium 137 | Required | Exempt | III |
| 864-7480 | 4-1000A | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 865-0259 | 4X500A | Eimac | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 866-5263 | BLT-036 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 866-5264 | TD-36A | Bendix | 0.9 | Cesium 137 | Required | Exempt | III |
| 866-5264 | 7416/TZ36A | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 866-5427 | MA3134 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 873-8884 | 845W | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 877-6026 | MA3132 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 878-1838 | TG-25 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 878-2510 | 6679 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 878-2935 | 6396 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 878-2935 | 6396 | Sylvania | 1.0 | Cobalt 60 | Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 878-6030 | 577 | United Elec. | 0.0005 | Thorium 232 | Not Required | Exempt | III |
| 878-6030 | 577W | United Elec. | 0.00002 | Thorium 232 | Not Required | Exempt | III |
| 878-6554 | GV3A-1200 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | III |
| 880-8596 | 6805/BL625 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 882-1994 | BL-994 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 882-1994 | BL-994A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 882-1998 | BLW-015 | Varian | 0.4 | Cobalt 60 | Required | Exempt | III |
| 882-9042 | TG-67 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 883-0971 | 7486 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 883-4793 | 7161 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 883-4793 | 7161 | Westinghouse | 0.7 | Uranium 235 | Required | Exempt | III |
| 883-7030 | 6166 | RCA | 0.276 | Thoriumm 232 | Not required | Exempt | III |
| 883-7030 | 6166A/7007 | RCA | 0.0276 | Thorium 232 | Not Required | Exempt | III |
| 887-2618 | 4-250A | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 889-7697 | 5762 | RCA | 0.0071 | Thorium 232 | Not Required | Exempt | III |
| 890-6617 | 5956 | Kuthe Labs | 0.08 | Uranium (nat) | Not Required | Exempt | III |





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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 596C-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 892-0862 | SS501 | ITT | - - - | - - - | Required | Exempt | II |
| 892-0975 | GV5A-1400 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 892-3323 | BL-650 | Varian | 0.203 | Cobalt 60 | Required | Exempt | III |
| 892-3413 | 6303A | United Elec. | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 892-3809 | 6072A | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 893-0151 | MA-3722 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 893-1184 | 3CX3000A1 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 893-1184 | 3X3000A1 | Eimac | 0.0017 | Thorium 232 | Not Required | Exempt | III |
| 893-1757 | BL-335 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 893-9153 | MA340F | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 894-7127 | MPT-23 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 894-7127 | 8060/BLT041 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 894-7127 | BL-T-041 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 894-7128 | BLT-059/8061 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 894-7128 | MPT-24 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |
| 895-0802 | MA338A | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 897-5744 | MPT-22 | Omni-Wave (Metcom) | 0.1 | Cobalt 60 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 897-6637 | BLT-072 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 897-6637 | MA-349B | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 897-6964 | 7381/BL979 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 897-6964 | 7381 | Microwave | 0.30 | Cobalt 60 | Required | Exempt | III |
| 897-8418 | GV4S-410 | Victoreen | 1.5 to 2.0 | Nickel 63 | Required | Exempt | IV |
| 897-9177 | 8312/BL960B | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 899-8118 | BLT-015 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 899-9964 | KU-21 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 900-3508 | 705WA | United Elec. | 0.0001 | Thorium (nat) | Not Required | Exempt | III |
| 900-3508 | 705WA | United Elec. | 0.00002 | Thorium (nat) | Not Required | Exempt | III |
| 902-0502 | BTR-548 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 902-0941 | BES-033 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 903-2852 | MPT13 | Omni-Wave (Metcom) | 1.0 | Cobalt 60 | Required | Exempt | III |
| 903-2852 | 8470/BL994A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 903-2853 | 8313/BL995B | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 903-3308 | BL-A-026 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 903-5816 | BLT-077 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 905-7498 | 6646/BL604 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 905-8020 | 6155 | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 905-9126 | 4CX1500A | Eimac | 0.007 | Thorium 232 | Not Required | Exempt | III |
| 908-7622 | 8082 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 912-3721 | MA3158A | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 913-3128 | BL-N-045 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 914-6268 | BLT-099 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 914-6268 | MA-3157 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 943-4392 | BTR-183 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 916-3669 | BES-048 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 916-3670 | BES-046 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 924-5255 | KU-17 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 924-9606 | BL-967A | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 925-2111 | KU82/7583 | Kuthe Labs (ITT) | 0.08 | Uranium (nat) | Not Required | Exempt | III |
| 932-0426 | 6697 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 932-0426 | 6697A | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 932-0426 | 6697A | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 932-3067 | BLT-005B | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 935-0171 | 6697 | Machlett | 0.042 | Thorium 232 | Not Required | Exempt | III |
| 935-0171 | 6697A | Eimac | 0.054 | Thorium 232 | Not Required | Exempt | III |
| 937-9870 | 353A | Cetron | 0.5 | Krypton 85 | Required | Exempt | III |
| 937-9870 | 353A | Western Elec. | 0.01 | Radium 226 | Required | Exempt | I |
| 938-2670 | 6D10 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 938-4359 | MA3821Z | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 941-2639 | 6795 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 941-2639 | 6795 | Nuclear Corp. | 1.0 | Cobalt 60 | Required | Exempt | III |
| 942-3980 | 8317 | Machlett | 0.089 | Thorium 232 | Not Required | Exempt | III |
| 942-6407 | MA-3192 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 943-4392 | BTR-183 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 945-1833 | 8515 | Victoreen | 0.1 | Nickel 63 | Not Required | Exempt | IV |
| 947-7424 | MA3164/3162 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 948-2686 | 8437 | RCA | 0.003 | Thorium 232 | Not Required | Exempt | III |
| 948-5806 | MA-3160 | Microwave | 0.4 | Cobalt 60 | Required | Exempt | III |
| 948-9472 | BL-N-012 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 950-6085 | [illegible]7WA | Eimac | 0.0011 | Thorium 232 | Not Required | Exempt | III |
| 951-6292 | QK172/6959 | Raytheon | 0.22 | Thorium 232 | Required | Exempt | III |
| 954-4825 | 3-1000Z | Eimac | 0.00073 | Thorium 232 | Not Required | Exempt | III |




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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 956-2318 | BLA-009 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 957-0951 | 3CX3000F1 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 958-3154 | 4PR-250C | Eimac | 0.00066 | Thorium 232 | Not Required | Exempt | III |
| 958-9146 | 4-65A | Eimac | 0.0001 | Thorium 232 | Not Required | Exempt | III |
| 958-9147 | 4-1000A | Eimac | 0.0007 | Thorium 232 | Not Required | Exempt | III |
| 959-7413 | MA38G1 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 962-3345 | BL-660 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 966-9469 | TG-72 | Signalite | 0.9 | Cesium 137 | Required | Exempt | III |
| 968-0500 | BLS-029 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 968-5229 | TD-42 | Bendix | 0.6 | Radium 226 | Required | Exempt | I |
| 972-8990 | TG-90 | Signalite | <1.0 | Cesium 137 | Required | Exempt | III |
| 973-7957 | BL-P-027D | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 975-2250 | 6591 | Varian | 0.3 | Cobalt 60 | Required | Exempt | III |
| 975-2251 | 6624 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 983-5684 | BL397A, B, C | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 984-6164 | 7381/BL979 | Varian | 0.25 | Cobalt 60 | Required | Exempt | III |
| 984-6164 | 7381 | Microwave | 0.30 | Cobalt 60 | Required | Exempt | III |
| 984-6413 | 8094/ML199 | Machlett | 0.003 | Thorium 232 | Not Required | Exempt | III |



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Table D-1D - Continued
RADIOACTIVE COMMODITIES MARKING AND LABELING REQUIREMENTS

| NSN 5960-00- | Tube Type or Part Number | Manufacturer | Quantity (Microcuries) | Radioisotope | Marking and Labeling Requirements | | |
|---|---|---|---|---|---|---|---|
| | | | | | Use and Storage Label | Dot Label (Radioactive) | Dot Transport Group |
| 988-0874 | 7323 | Tung-Sol (Chatham) | 0.33 | Krypton 85 | Required | Exempt | III |
| 988-1583 | 3CX10000A1 | Eimac | 0.0041 | Thorium 232 | Not Required | Exempt | III |
| 989-9810 | 3CX3000F7 | Eimac | 0.002 | Thorium 232 | Not Required | Exempt | III |
| 990-3077 | BL-398 | Varian | 0.15 | Cobalt 60 | Required | Exempt | III |
| 990-5914 | 8083 | General Elec. | 0.00001 | Rhenium 187 | Not Required | Exempt | IV |
| 993-0695 | 4W20000A | Eimac | 0.03 | Thorium 232 | Not Required | Exempt | III |
| 993-1702 | GV9A-600 | Victoreen | 1.0 | Nickel 63 | Required | Exempt | IV |
| 996-5975 | 4CX5000A | Eimac | 0.0031 | Thorium 232 | Not Required | Exempt | III |
| 980-6749 | 9994445 | Microwave | 0.80 | Cobalt 60 | Required | Exempt | III |
| 990-4631 | MA3154 | Microwave | 0.40 | Cobalt 60 | Required | Exempt | III |
| 01-013-0079 | MA3959Z | Microwave | $\leq$50 | Tritium (H3) | Required | Exempt | IV |





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# APPENDIX 1

## GUIDELINES FOR RADIOLOGICAL MONITORS

### 10. SCOPE

10.1 This appendix describes the procedures for radiological monitoring for personnel protection, decontamination, storage, and disposition of radioactive commodities.

### 20. PURPOSE

20.1 The purpose of this appendix is to assure that all personnel who work with radioactive commodities are completely aware of the hazards, or potential hazards, that are introduced into their working environment by inclusion of radioactive materials and of the applicable directives and procedures by which they may conduct their work safely. Applicable directives and safety procedures (e.g. control documentation and standard operating procedures) should be provided for each radioactive commodity or group such as receiving, handling, warehousing, maintenance and inspection. Each individual should receive instructions in his specific operation and all safety procedures before being exposed to the potential radiation hazards.

### 30. PERSONNEL PROTECTION

30.1 Aspects of radiation. Radiation is silent, unseen, and unfelt; its presence can be detected only by means of instruments. Some of the damaging effects on the human system may not develop until after many years of exposure. Therefore, radiation hazards cannot be treated like some of the common hazards. The lay mind consequently may have developed a fear that in some cases is inordinate. In most cases, the matter of accidents is well threshed out; most accidents are sudden and only predictable on a probability basis. On the other hand, many radiation injuries are cumulative and gradual, and the hazard measureable.

30.2 Potential radiation hazards. Potential radiation hazards depend on the type or radiation being emitted, e.g., alpha, beta, gamma, etc. (See figure 1-1 and table B-1, column II, for types of radiations emitted for radioactive commodities included herein.)

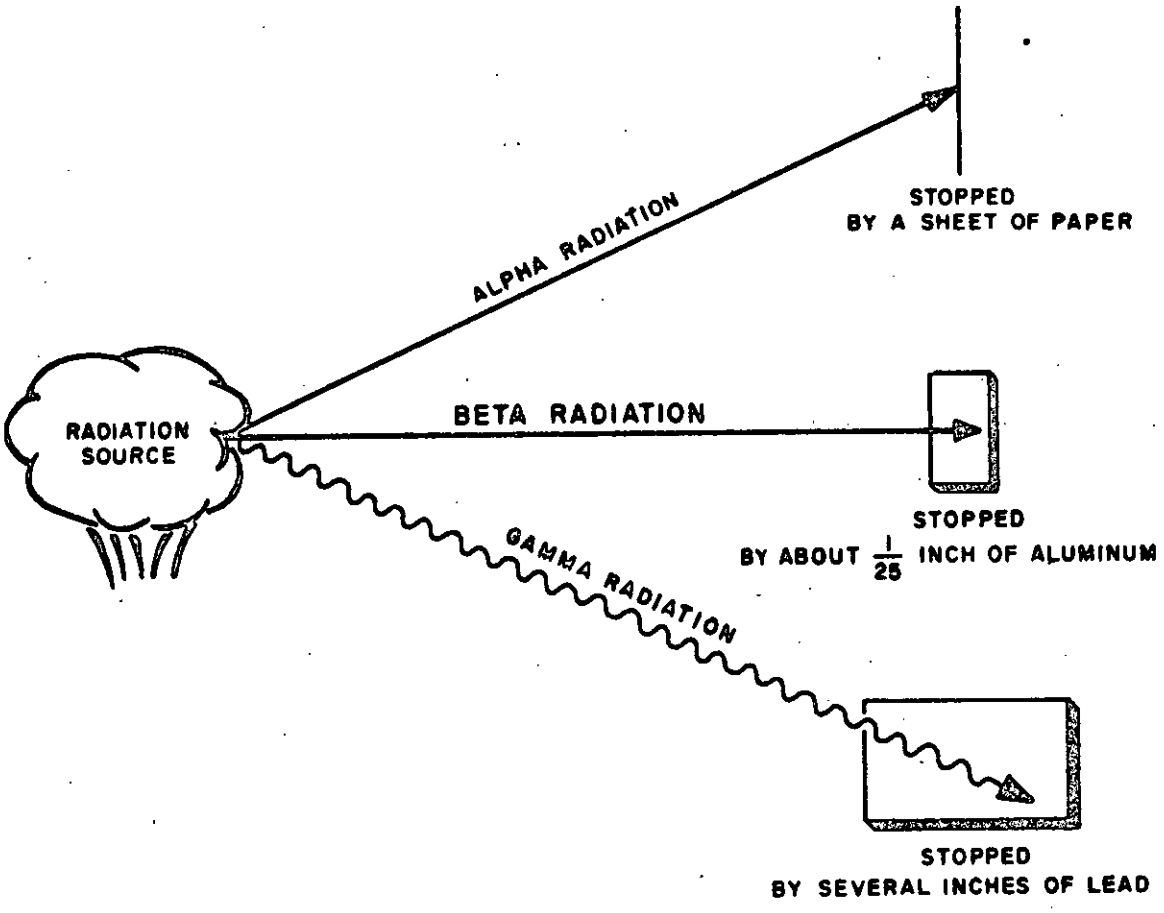
30.2.1 Alpha radiation. Alpha radiation is not an external hazard because it is not capable of penetrating the skin; however, alpha radiation is a serious internal hazard. Every precaution should be taken to prevent the entry of alpha emitting particles into the body by ingestion, inhalation, or open skin wounds; inhalation being by far the most serious.

30.2.2 Beta radiation. Beta emitting contamination, within the concept of this handbook, is considered an internal hazard; i.e., the contamination must enter the body before it can do appreciable harm. Contamination may gain entry into the body through the mouth, nose, or open wounds. If the contamination is extremely concentrated (on the order of 2000 rems/hr), physical contact with the contaminated surface may produce skin burns. The wearing of cotton gloves will, in most cases, be sufficient to nullify this hazard.

30.2.3 Gamma rays. Gamma rays present the greatest external hazard. They have no mass, travel at the speed of light, and are highly penetrating, their effective range depending on their energy. The effect of air on gamma rays is so small that it is not practical to measure the range, but its penetrating power is measured in terms of the amount of material required to reduce it to some fraction of its original value. Gamma rays can penetrate to an extreme depth in the body, destroying tissues and inflicting serious burns quite rapidly. Therefore, gamma rays consitute a radiation hazard, for the entire body.



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| Emission | Description | Charge | Approx range in air | Relative penetration | Relative ionization | Hazard |
|---|---|---|---|---|---|---|
| Alpha $\alpha$ | Helium nucleus | +2 | 4 inch | 1 | 10,000 | Internal |
| Beta $\beta$ | High velocity electron | −1 | 15 – 20 feet | 100 | 100 | Internal |
| Gamma $\gamma$ | Electromagnetic wave of energy | Nuetral | * | 10,000 | 1 | External |

* Dependent upon half-value thickness

FIGURE 1-1. The penetrating power of radiation.



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30.3 Maximum permissible exposure levels. Based upon considerable experience by many workers in the field of biological effects of ionizing radiations, it has been determined that an individual may be safely exposed to certain maximal dosages of radiation, either intermittently or over a life span. In setting these dosage levels, it is considered improbable that the cumulative exposure will be sufficient to shorten life, to increase the incidence of pathologic disorders, or to impair vitality and well being. The permissible dosages given herein are for observance during peacetime operations at all installations dealing with ionizing radiation or radioactive materials. They are particularly designed for those constantly and periodically exposed to ionizing radiation in the course of their routine work. The following exposure doses are maximums; however, every effort should be made to minimize the exposure of individuals to radiation as far below these standards as practicable.

30.3.1 Equivalent doses. For the purpose of this handbook, any one of the following is considered to be equivalent to a dose of one rem:

a. A dose of one roentgen due to X- or gamma radiation.

b. A dose of one rad due to X-, gamma, or beta radiation.

c. A dose of 0.1 rad due to neutrons or high-energy protons.

30.3.2 Exposure of individuals to radiation in restricted areas. Except as provided in 30.5.2.1, no person shall possess, use, or transfer licensed material in such a manner as to cause any individual in a restricted area to receive, in any one calendar quarter from radioactive material and other sources of radiation in the person's possession or control, a dose in excess of the limits specified in the following table (10 CFR 20):

| | Rems per calendar quarter |
|---|---|
| Whole body; head and trunk, active blood-forming organs; lens of eyes; or gonads | 1 1/4 |
| Hands and forearms; feet and ankles | 18 3/4 |
| Skin of whole body | 7 1/2 |

30.3.2.1 Exemptions. An individual in a restricted area may be permitted to receive a dose to the whole body greater than those permitted under 30.3.2 provided:

a. During any calendar quarter the dose to the whole body shall not exceed 3 rems; and

b. The dose to the whole body, when added to the accumulative occupational dose to the whole body, shall not exceed 5 (N-18) rems where "N" equals the individual's age in years at his last birthday; and

c. Adequate records are maintained of past and current exposures which show that the addition of such a dose will not cause the individual to exceed the amount authorized in this subparagraph.

30.3.2.2 Exposure of minors to radiation. No person shall permit an individual within a restricted area who is under 18 years of age to receive in any period of one calendar quarter, a dose in excess of the following:

a. Radioactive materials and other sources - - 10 percent of the limits specified in 30.3.2 above.

b. Airborne radioactive material - - - - - - Limits of 10CFR20, appendix B, table II. (The concentration may be average over periods not greater than a week.)



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30.3.3 Exposure of individuals to airborne radioactive material. No person shall possess, use, or transfer radioactive material in such a manner as to cause any individual within a restricted area to be exposed to airborne radioactive material in an average concentration in excess of the limits specified in appendix B, table I of 10 CFR 20. "Expose" means that the individual is present in an airborne concentration. No allowance shall be made for protective clothing or equipment, or particle size, except when authorized by the NRC. The limits given in appendix B, table I, are for exposure to the concentrations specified for 40 hours in any workweek of 7 consecutive days. In any such period where the number of hours of exposure is less than 40, the limits specified in the table may be increased proportionately. In any such period where the number of hours of exposure is greater than 40, the limits specified in the table shall be decreased proportionately.

30.3.4 Permissible levels of radiation in unrestricted areas. Except as authorized by the NRC, no person shall possess, use or transfer radioactive material in such a manner as to create in any unrestricted area from radioactive material and other sources of radiation in his possession:

a. Radiation levels which, if an individual was continuously present in the area, could result in his receiving a dose in excess of two millirems in any one hour, or 100 millirems in any seven consecutive days, or

b. Release to an unrestricted area radioactive material in concentrations which exceed the limits specified in appendix B, table II of 10 CFR 20, unless authorized by the NRC.

30.3.5 Service/agency radiation protection standards. Radiation protection standards adopted by the Department of Army (DA) and the Defense Supply Agency (DSA) for the control of occupational exposures to ionizing radiation are included in: AR 40-14, DSAR 4145.24 "Control and Recording Procedures For Occupational Exposure To Ionizing Radiation."

30.3.6 Maximum permissible contamination levels (See DSAM 4145.8, AR 700-64, NAVSUPINST 400.34, AFM 67-8, MCO P4400-105A).

30.3.6.1 Personnel, equipment, and material. Contamination of personnel, equipment, and material will be kept to the absolute minimum. Wipe tests of surfaces of material containers and worktables will not be allowed to indicate any appreciable concentration of alpha, beta, or gamma activity. Radiation levels caused by combined beta-gamma will not exceed twice the normal background level. Body surfaces will be decontaminated when any detectable radioactivity is found and will be continued until the acceptable levels that have been established by the installation or agency are reached.

30.3.6.2 Air and water. Unless otherwise specified, the maximum permissible concentrations of unknown radioisotopes in air and water are as follows (expressed in microcuries per milliliter). (If the identity of the radioisotope is known, refer to Title 10, Code of Federal Regulations, Part 20):

| Medium | Beta and gamma emitters | Alpha emitters |
|---|---|---|
| Air | $0.2 \times 10^{-9}$ | $0.2 \times 10^{-11}$ |
| Water | $10^{-7}$ | $0.4 \times 10^{-7}$ |

30.3.7 Radiological safety committee. A radiological safety committee should be established at each installation where personnel are engaged in work involving regular exposure to ionizing radiation or handling radioactive materials. The committee should include a medical officer, radiological protection officer, safety officer, and other members as considered necessary by the commander of the installation or activity.



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30.3.7.1 Responsibilities. The radiological safety committee will serve as staff advisory council to the commanding officer. As such, it will be responsible for the establishment and review of the general technical policies relating to radiation hazard control. The committee should:

a. Meet quarterly to review radiological safety conditions and to recommend guidelines to the radiological protection officer.

b. Recommend solutions to unsolved problems relating to radiation hazard control.

c. Recommend changes in the radiological safety program, when required.

d. Perform other duties and functions relative to radiation safety as considered necessary.

30.3.7.2 Medical and radiological safety officers. The medical and radiological safety officers should be the action agents for the radiological safety committee. As such, they will be responsible for radiological policies and procedures relative to personnel work safety and health protection. The officers should be guided by the latest regulations and established procedures pertaining to radiation safety. These officers will:

a. Indoctrinate personnel relative to the radiological safety program.

b. Act in a consultant and advisory capacity, on radiation hazard control, to personnel working with or planning to work with radioactive materials.

c. Provide a personnel dosimetry service, including film badges.

d. Designate personnel required to wear dosimetric devices.

e. Designate personnel that will require periodic physical examination.

40. PRECAUTIONARY MEASURES

40.1 Precautions. The following precautions are in addition to those normally observed by personnel engaged in regular storing and handling of equipment and commodities. Special problems and specific information with regard to handling and disposition of radioactive items, not covered in this handbook, should be referred to the medical service or the radiological protection officer.

40.1.1 Personnel requirements. Radioisotopes and contaminated material will be monitored and handled by authorized personnel only. Receiving, storing, and shipping activities will have assigned at least two radiological monitors (see section 50). Monitors will be selected from personnel assigned to the areas (preferably inspection personnel) and trained to accomplish this function. Personnel selected to handle radioisotopes during shipment or storage will be indoctrinated by qualified radiological monitors or by medical personnel on the precautionary measures to be observed. Personnel involved in the actual use of radioisotopes must have formal training in their use, in the precautionary measures to be observed, or on-the-job training under the close supervision of qualified personnel (such as medical service personnel, radiological protection officer, or a qualified radiological monitor).

40.2 Personnel monitoring equipment. "Personnel monitoring equipment" means devices designed to be worn or carried by an individual for the purpose of measuring the dose received. Each installation or agency shall provide appropriate monitoring equipment, such as film badges, pocket chambers, pocket dosimeters, or film rings (see section 60), to, and shall require the use of such equipment by, all personnel working with radioactive materials, or entering radiation areas as described below.



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40.2.1 Individuals in restricted areas. Each individual who enters a restricted area under such circumstances that the individual receives, or is likely to receive, a dose in any calendar quarter in excess of 25 percent of the applicable value specified in 30.3.2 herein.

40.2.1.1 Individual under 18 years of age. Each individual under 18 years of age who enters a restricted area under such circumstances that the individual receives, or is likely to receive, a dose in any calendar quarter in excess of 5 percent of the applicable value specified in 30.3.2 herein.

40.2.1.2 Radiation area. "Radiation area" means any area accessible to personnel where radiation exists at such levels that a major portion of the body could receive in any one hour a dose in excess of 5 millirem, or in any 5 consecutive days, a dose in excess of 100 millirem.

40.2.1.3 High radiation area. "High radiation area" means any area accessible to personnel where radiation exists at such levels that a major portion of the body could receive in any one hour a dose in excess of 100 millirem. Each high radiation area shall be conspicuously posted with signs or placards as described in Section 70.4.

40.2.1.4 Airborne radioactivity area. See 70.4.2 herein.

40.2.2 Entry into a high radiation area. Positive measures, such as locked or guarded gates or doors, in addition to placards, shall be established to prevent unauthorized entry into high radiation areas, i.e., an area in which in individual might receive a radiation dose equivalent to 100 milliroentgens in one hour. Entry into a high radiation area, or exposure of a source which would make such a dose possible, shall energize a conspicuous visible or audible alarm signal designed to alert personnel present (see 70.4.1).

40.3 Removal or radioactive material. Personnel will not tamper with or in any way expose or remove an item containing radioactive material from the unit package or shipping container unless required to do so in the performance of duty. Under no conditions will radioactive items be carried in the pockets of clothing worn by personnel.

40.4 Accidentally broken or damaged items. Should an item containing radioactive material be accidentally broken or its container damaged, direct contact with the radioactively-contaminated item will be avoided. The radiological monitor on duty, the medical service, the supervisor, and other concerned personnel will be notified immediately. Thereupon, the following rules and procedures will be observed:

a. Personnel having come in direct contact with broken or damaged radioactive items or containers or radioactively contaminated items will, if possible, have other personnel notify proper authorities; remove self from the immediate area (but will not migrate to other areas) for subsequent monitoring and decontamination, if necessary, by attending radiological monitors.

b. The immediate area will be roped off, monitored, and decontaminated, if necessary (see sections 70 and 80).

c. Broken or damaged items will be monitored and safely packaged under surveillance of the radiological monitor and disposal made in accordance with instructions in this handbook, or other applicable instructions (see section 90).

d. Radioactively contaminated material should not be allowed to come in contact with the body at any time. Rubber or plastic gloves should be worn while handling broken items or damaged containers of radioactive material.



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e. Adherence to good personal hygienic practices (such as washing of hands and face before eating and smoking) is mandatory.

f. Food or drink will not be brought into the contaminated area, nor will smoking be permitted.

40.5 Theft or loss. Strict accounting of radioactive commodities and materials is mandatory at all times. Theft and loss of radioactive items shall be reported immediately to the radiological protection officer, medical officer, and other cognizant personnel.

40.6 Fire department provisions. The fire department will be provided with storage locations of all radioactive materials, so that proper precautions may be observed in case of fire (see 80.5).

40.7 Protective clothing. Protective clothing should be available for use by personnel at each installation or agency where radioactive materials or commodities are handled or stored. The type clothing provided will vary with the local situations, but may include:

a. Plastic or hard hat, or other head covering as appropriate.

b. Safety glasses.

c. Suitable washable, or disposable outer garments.

d. Appropriate footwear.

e. Appropriate gloves.

f. Self-contained respirators.

40.7.1 Contaminated clothing. Protective clothing which is contaminated so that it emits a radiation level of 1.0 mR/hr or more of beta-gamma, or is contaminated with any alpha, shall be decontaminated or disposed (see section 80).

40.8 Radiological surveys. Each installation or agency shall conduct surveys as necessary in order to comply with the requirements of 10 CFR 20, and 29 CFR 1910.96 (see section 70). "Survey" means an evaluation of the radiation hazards incident to the production, use, release, disposal, or presence of radioactive material or other sources of radiation under s specific set of conditions. When appropriate, such evaluation includes a physical survey of the location of materials and equipment, and measurements of the levels of radiation or concentrations of radioactive material present.

40.9 Records and reports of surveys and exposure. Each installation or agency shall prepare and maintain records showing the radiation exposures of all individuals occupationally exposed to ionizing radiation under the conditions of 40.2 herein. These records shall be kept on forms provided by NRC, or DD Form 1141 (see AR 40-14, DSAR 4145-24). Each installation or agency shall maintain records in the same units as specified in 30.3.2 herein and appendix B to 10 CFR 20. Each individual shall be advised of his individual exposure on at least an annual basis.





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50. PERSONNEL TRAINING, DUTIES, AND REQUIREMENTS

50.1 General requirements. All personnel involved with radioactive materials, or commodities should be made aware of the hazards, or potential hazards of these materials. They should be aware of the contents of applicable directives and procedures relative to this subject. Directives and safety procedures should be prepared and distributed to involved personnel for each radioactive commodity or group of related commodities and for each particular operation such as receiving, handling, maintenance, inspection, etc. Involved personnel should receive training and instructions prior to being exposed or assigned to a work area that is involved with radioactive materials or commodities.

50.2 Instruction courses for radiological monitors. Personnel assigned radiological monitoring duties should be provided training in this assigned area. Personnel should have received a minimum of 40 hours of instruction covering the subjects outlined below before being considered qualified as a radiological monitor. This instruction shall be provided by qualified personnel. A suggested outline of the radiological monitor course is as follows:

| | |
|---|---|
| Introduction to Radiation | 2 hours |
| Basic Nuclear Physics | 2 hours |
| Effects of Nuclear Explosions | 2 hours |
| Medical Aspects | 2 hours |
| Detection Equipment (General) | 4 hours |
| Monitoring Procedures | 6 hours |
| Decontamination | 2 hours |
| Use of Instruments | 10 hours |
| Exposure Control | 8 hours |
| Review and Test | 2 hours |
| TOTAL | 40 hours |

50.2.1 Additional training requirements. In addition to the general subject matter listed above, monitors shall have on-the-job training to acquaint them with the specific materials and conditions with which they will be concerned.

50.2.2 Refresher requirements. Monitors having satisfied the training requirements shall receive a minimum of eight hours of refresher training each year thereafter. The latest information on the topics specified in 50.2 shall be reviewed.

50.3 Duties of radiological monitors. Radiological monitors shall perform the following:

a. Monitor known or suspected radioactive and radioactively contaminated material, completing and attaching the appropriate radiological markers and placards (see 70.4) and notifying proper authorities of radiation levels present.

b. Monitor radioactive storage areas every 30 days or when new shipments are placed in such areas. Where the permissible level of radioactivity is being exceeded, the stocks of items where excessive levels are found should be broken into smaller quantities or moved to other areas in an attempt to reduce the radiation level to within the permissible level. In the event the excessive level cannot be reduced, request immediate guidance from the radiological protection officer or the medical officer.

c. Monitor personnel assigned to work in areas where possible radiation exposure exists.



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d. Insure that personnel working such areas observe necessary precautions and adhere to applicable regulations and directives.

e. Provide guidance and assist in the performance of decontamination, and disposal operations, when required.

NOTE

The monitors assigned to operating units are not authorized to change permissible exposure dose limits or deviate from standard precautionary procedures. The radiological protection officer and the medical service will assist or advise the radiological monitors when necessary.

60. RADIOLOGICAL MONITORING EQUIPMENT

60.1 Detection methods. Radiological instruments detect the interaction of radiation with some type of matter. The different principles of radiation detection are characterized by the nature of the interaction of the radiation with the detecting or sensing element. Several types operate by virtue of the ionization produced in them by the passage of charged particles. In other detectors, excitation and sometimes molecular disassociation play important roles.

60.2 Radiological monitoring systems. Radiological monitoring systems have become an integral part of every program where radiation is a threat to health. These systems include instruments to measure exposure levels and devices to warn when permissible levels are exceeded. Warning devices, such as ringing bells, flashing lights, clicking sounds, and fluctuating dials, may be attached to monitoring instruments to give notice of the presence of radiation since radiation penetrating the body causes no pain and one can be injured severly without realizing it. Physical examinations and clinical tests are another phase of monitoring to determine the exposure sustained and the amounts that are permissible.

60.3 Radic instruments. There are two distinct groups of instruments employed in the detection and measurement of ionizing radiation; those that measure total accumulated exposure dose, and those that measure the instantaneous rate of exposure. The first is referred to as a MEASUREMENT (Dosimeters) and the second as a RATE MEASUREMENT (Survey Meters). The two groupings are called "radiac" instruments, derived from the words "radiation, detection, indicator and computation".

60.4 Dosimeters. Dosimeters are divided into several classes depending on their principle of operation. The more common classes are: photographic emulsion, scintillation media, ionization chamber, chemical decomposition, and radiophotoluminscent media.

60.4.1 Film badge. The film badge worn on the outer clothing is the primary dosimetric device and the most common personnel radiation monitor. It is used mainly to detect X- or gamma-radiation and the more energetic beta radiation; it may be used also for neutron detection. The typical badge consists of a metal or plastic jacket with an identification photo or nameplate insert. An index number is usually engraved on the back of the jacket or on the nameplate (see figure 1-2).



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60.4.1.1 Film pack. The film pack may contain one, two, or three films with varying sensitivites. One film emulsion, for example, may measure up to 20 roentgens, another up to 50 roentgens, and a third up to 600 roentgens. Each film is coded with the index number. With the film badge shown in figure 1-2, this is done by passing X-radiation through the perforated number at the base of the front filter (other systems for coding films, however, may be used). The coding process is performed for identification purposes. After the film badge is worn for a period of time, usually 1 week, the film is developed to determine the quantity of radiation received by the wearer of the badge. A densitometer, which measures the density of a photographic image, determines the relative density on the film. This density is compared with densities of known quantities of radiation of given energies and is then converted into appropriate units.

60.4.2 Film badge service. Film badge service for the Department of Army (DA) and the Defense Supply Agency (DSA) shall be provided as specified in AR 40-14, DSAR 4145.24, and for other services activities as specified by the directives of these activities.

60.5 Survey meters. The ionization chamber, proportional counter, and Geiger-Mueller counter are common portable monitoring instruments. (They may also be used in fixed applications.) The essential features of the three types are:

(1) A positive and negative electrode to attract negative and positive ions; and

(2) An appropriate gas or mixture of gases in an ionization chamber through which the radiation must pass; and

(3) An applied voltage to the chamber.

60.6 Recommended monitoring equipment. The following dosimeters and survey instruments are recommended for monitoring purposes at most activities. However, since there are many variations, the actual equipment required at each installation or agency should be determined by the responsible service officer(s).

TABLE 1-1. Recommended monitoring equipment.

| NSN | Nomenclature | Range | Purpose |
|---|---|---|---|
| 6665-00-561-5787 | Radiac set AN/PDR-27 ( ) | 0-500 mR/hr | Gamma measuring, beta-gamma detection, personnel monitoring |
| 6665-00-526-5336 | Radiac set AN/PDR-39 | 0-50 R/hr | Gamma (high intensity), area monitoring |
| 6665-00-106-7554 | Radiac set AN/PDR-43E | 0-500 R/hr | Gamma measuring, beta-detecting |
| 6665-00-581-2227 | Radiac set PAC-2GA | 0-100,000 cpm | Alpha detection |
| *6665-00-526-7174 | Radiacmeter | 0-200 mR/hr | X- and gamma detecting |
| *6665-00-526-7169 | Radiac detector (dosimeter) | 0-200 mR | X- and gamma detecting |
| *6665-00-526-8230 | Charger, radiac detector | | Dosimeter charging |

*These items to be used only if considered necessary by the installation or agency.

NOTE
Film badges will be obtained through the medical service.



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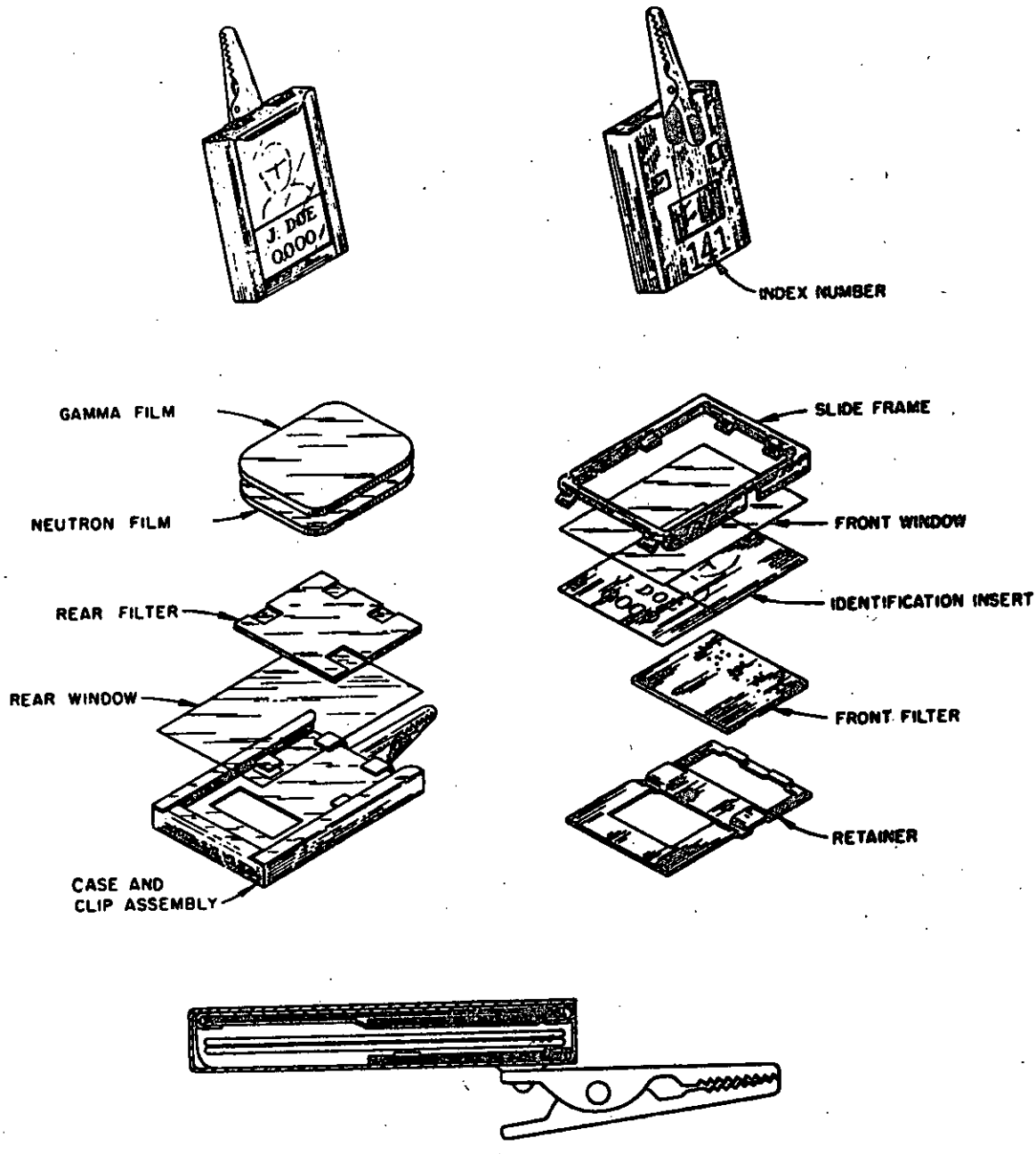



FIGURE 1-2. Film badge.



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60.6.1 Calibration and certification of instruments. All survey instruments should be maintained in a serviceable condition at all times, calibrated and certified by qualified personnel in qualified laboratories at intervals specified by service or Agency directives.

70. RADIOLOGICAL MONITORING TECHNIQUES AND OPERATIONS

70.1 Importance of monitoring. The radiological program for personnel protection and safe handling of radioactive materials must be based on facts. Radiological monitoring provides the facts required by medical and safety personnel in case of emergency, or other operations. The information provided by qualified radiological monitors is the key to decisions on what to do and how to do it.

70.2 Purpose of monitoring. Radiological monitoring is defined as the detection and measurement of radiation. With the information gained through monitoring we can: (1) determine extent and location of radiation; (2) validate predictions about radiation; and (3) decide on a course of action.

70.3 Types of monitoring. There are two types of radiation monitoring: personnel and area. Personnel monitoring is the use of instruments to determine the exposure received by the individual. Area monitoring is the use of instruments to determine the levels of radiation in an area and to give alarm when permissible levels are exceeded.

70.3.1 Personnel monitoring. There are two types of personnel monitoring instruments: instruments that can be read directly and instruments that require supplemental processes or supplemental readings and are a measure of exposure-levels over a period of time (see sections 40 and 60).

70.3.1.1 Personnel-monitoring procedures. Procedures for personnel monitoring are as follows:

a. Use the AN/PDR-27 or equivalent. Attach the headphones as this allows the monitor to visually follow and, thus better control the position of the probe while monitoring. The headphone also responds more quickly to changes in radiation levels than the meter.

b. Check the operability of the instrument.

c. Place the probe in a light plastic bag or cover of light-weight material to prevent contamination. This is desirable, but not mandatory.

d. Select a reception location for conducting the monitoring operation. Precautions should be taken to prevent contamination of the area.

e. Determine the background radiation level periodically at the location where the monitoring is to take place. If the meter indication is above 50 mR/hr with the probe shield open, find a better shielded location that will bring the meter indication below 30 mR/hr. This might be done by selecting a different location, or by sweeping the area several times to reduce possible contamination.

f. Open the shield on the probe of the AN/PDR-27 or equivalent and put on the headphone.



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g. Place the probe about two inches from the person's body being careful not to touch him. Starting at top of the head, move the probe downward on one side of the neck, collar, shoulder, arm, wrist, hand, underarm, armpit, side, leg, cuff, and shoe. Monitor the insides of the legs and the other side of the body in the same fashion. Monitor the front and back of the body. Pay particular attention to the feet, seat, elbows, armpit, and hary or moist areas (see figure 1-3). Record the exposure (see 40.9 herein).

h. Decontaminate personnel found to be contaminated (see section 80).

i. Monitor all personnel after decontamination to determine that contamination has been effectively removed. Repeat decontamination procedures if required.

70.3.1.2 Monitoring radioactive commodities. Personnel monitoring procedures are also applicable for monitoring commodities that exhibit low intensities, or for contamination detection.

70.3.2 Area monitoring. Area monitoring is used to locate zones of radiation and contamination and to determine the dose rates in these zones. The monitor shall: (1) be advised by the radiological protection officer of the allowable dose rate and the importance of the mission; (2) wear a protective mask, gloves, head covering, and sufficient clothing to cover the skin area when dusty conditions prevail; and (3) have paper, pencil, maps, and contamination signs, if areas are to be marked.

70.3.2.1 Equipment required. The following equipment will be required:

a. AN/PDR-39 or equivalent. If the dose rates are expected to be below 500 mR/hr, also carry the AN/PDR-27 or equivalent.

b. PAC-2GA or equivalent for alpha detection.

c. Film badge and the appropriate self-reading dosimeter (see section 60).

70.3.2.2 Area monitoring procedures. The procedures for area monitoring are as follows:

a. Zero the dosimeter and place in a pocket to protect it from possible contamination.

b. Check the operability of the AN/PDR-39 and AN/PDR-27, if it is to be used.

c. Take readings at about three feet (belt high) above the ground or floor.

d. Record the dose rate, the time and location for each reading. If readings are taken within a vehicle, this should be noted in the report.

e. Post markers, if required by the mission. The marker should face away from the restricted area. Write the date, time, and dose rate on the back of the marker (see 70.4).

f. Read the pocket dosimeter at frequent intervals to determine when to depart the area.



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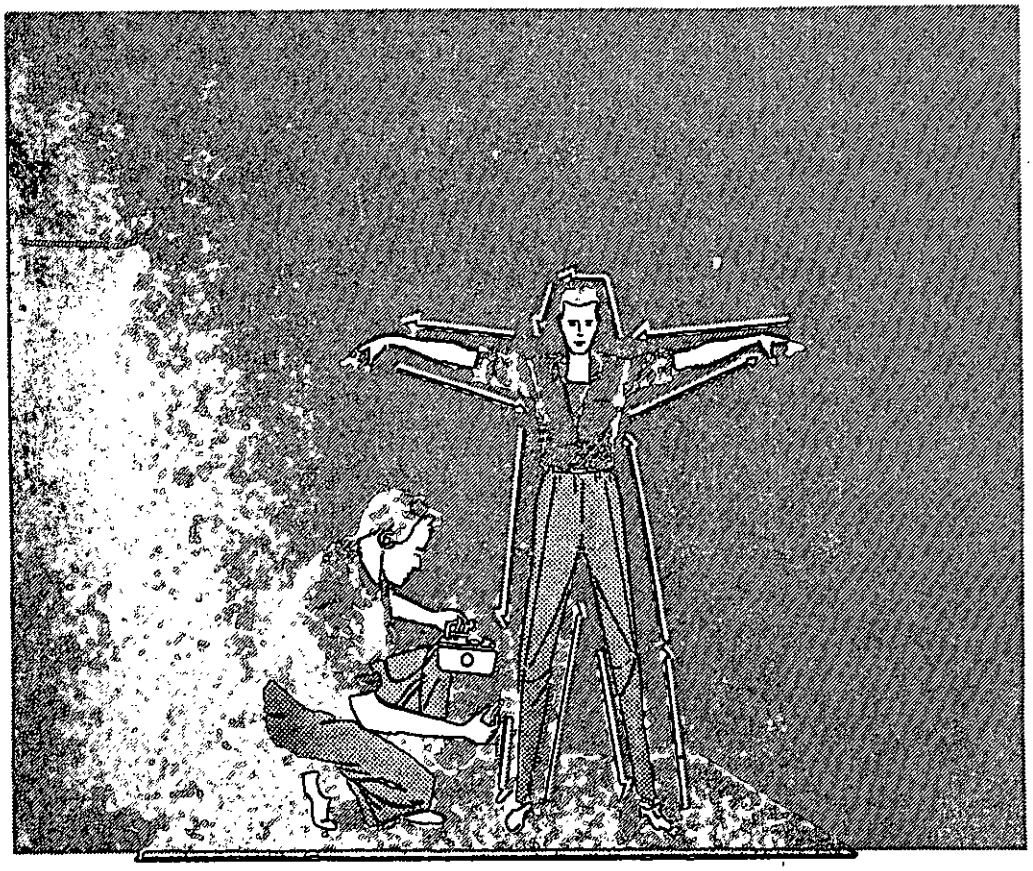


FIGURE 1-3. Personnel monitoring.



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g. Remove outer clothing on return and check all personnel for contamination. Decontaminate, if required (see section 80).

h. Record the radiation exposure and report results of the survey (see 40.9).

70.4 Area identification (caution signs, labels, signals and controls. Appropriate signs or placards shall be conspicuously posted at each entrance to a radiation area or at intervals of not less than 20 feet around the perimeter of an open area where concentrations of radioactive materials are located. The signs or placards can be constructed if preprinted ones are not available (see figure I-4) and shall bear the Nuclear Regulatory commission (NRC) standard radiation symbol and the words as appropriate:

CAUTION (or DANGER)

RADIATION AREA (or HIGH RADIATION AREA or AIRBORNE RADIOACTIVITY AREA or RADIOACTIVE MATERIALS)

70.4.1 High radiation area. Each high radiation area shall be equipped with a control device which shall either cause the level of the radiation to be reduced below that at which an individual might receive a dose of 100 millirems in 1 hour upon entry into the area or shall energize a conspicuously visible or audible alarm signal in such a manner that the individual entering and the guard or supervisor of the activity are made aware of the entry. In the case of a high radiation area established for a period of 30 days or less, such control device is not required.

70.4.2 Airborne radioactivity area. Each airborne radioactivity shall be posted as specified in 70.4. "Airborne radioactivity" as used in this section, means:

a. Any room, enclosure, or operating area in which airborne radioactive materials, composed wholly or partly of radioactive material in contration exceeding the limits specified in 10 CFR 20, appendix B, table 1, or
b. Any room enclosure or operating area in which airborne radioactive material exist in concentrations which when averaged over the number of hours in any week during which individuals are in the area, exceeds 25 percent of the values specified in 10 CFR 20, appendix B, table 1.

70.4.3 Additional requirements. Each room or area shall be posted as specified in 70.4 where radioactive material (excluding natural uranium or thorium) is used or stored in any amount exceeding 10 times the quantity of such material specified in table A-1, column IV herein, or in which natural uranium or thorium is used or stored in quantities exceeding 100 times the quantity of such material specified in table A-1, column IV herein.

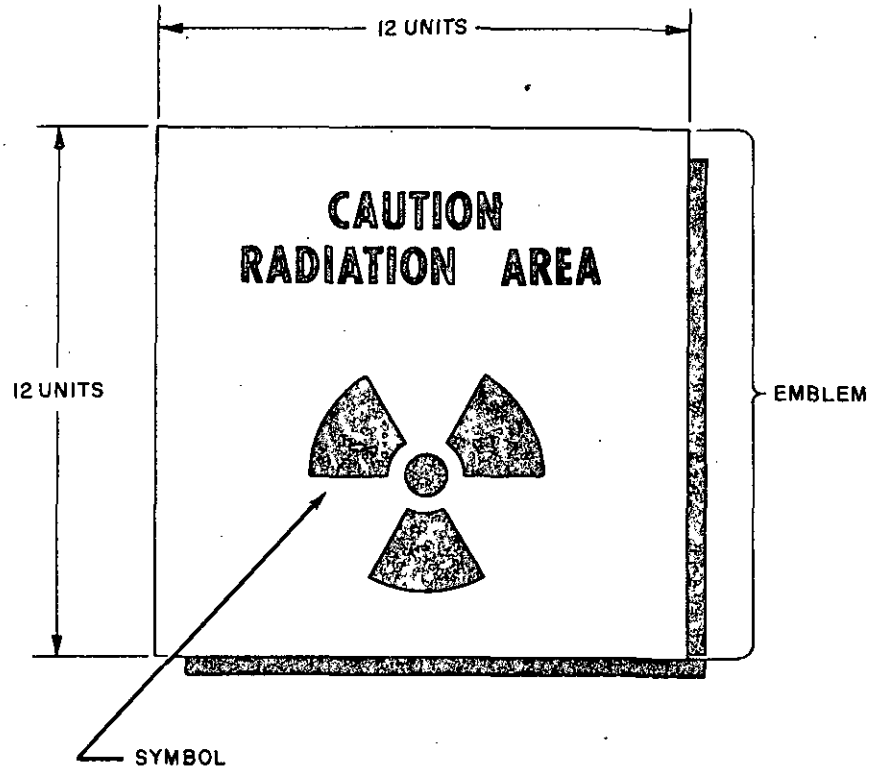
70.4.4 Container labels. See 11.2.1 herein.

80. DECONTAMINATION

80.1 Cooperation requirement. Closely allied with the problem of monitoring, discussed in section 70, is the problem of decontamination. The two are, in fact, closely linked and cooperation between monitors and decontamination personnel (if different) is a basie requirement if wasted effort or unnecessary hazard is to be avoided.



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THE PROPORTIONS OF THIS EMBLEM ARE BASED UPON THE VALUE OF ONE UNIT.
ONE UNIT EQUALS THE DIAMETER OF THE CENTER CIRCLE OF THE SYMBOL.
THE EMBLEM IS TWELVE UNITS HIGH BY TWELVE UNITS WIDE.
THE SYMBOL IS CENTERED IN THE LOWER TWO-THIRDS OF THE EMBLEM.
THE SYMBOL IS CONSTRUCTED OF THREE CONCENTRIC CIRCLES.
THE OUTER CIRCLES BEING DIVIDED INTO SIX SECTORS, ALTERNATE SECTORS COLORED.
THE CIRCLES HAVE RADII IN THE PROPORTIONS OF ONE TO ONE AND ONE-HALF AND ONE TO FIVE.
THE LETTERS ARE OF BLOCK TYPE, ONE UNIT HIGH, AND CENTERED IN THE UPPER ONE-THIRD OF THE EMBLEM.
BACKGROUND COLOR OF THE EMBLEM IS YELLOW.
SYMBOL COLOR IS MAGENTA OR PURPLE.
LETTERS ARE BLACK.

FIGURE 1-4. Standard radiation hazard warning emblem.



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80.2 Contamination behavior. Contamination results when radioisotopes attach themselves to dust particles or water droplets. It behaves physically like any other dirt or moisture. In fact, the phenomena of radioactive contamination is exactly the same as getting "dirty" except this dirt is radioactive and only small quantities are required to produce a hazard.

80.3 Purpose of decontamination. The objective of radiological decontamination is to reduce the contamination to an acceptable level with the least possible exposure of personnel to radiation. Radioactivity cannot be destroyed or neutralized, but in the event of accident, the radiation hazard can be reduced or eliminated by removing radioactive particles from a contaminated surface and safely disposing of them, by covering the contaminated surface with shielding material, such as earth, or by isolating the contaminated object and waiting for its radiation to decrease through the process of natural radioactive decay.

80.4 Decontamination procedures. In the event of accidents involving radioactive materials, immediately notify the radiological protection officer and the medical service of the incident. The following procedures for decontamination will proceed under their guidance:

NOTE

All decontamination personnel will wear protective gloves, and other protective apparatus as prescribed by the radiological safety officer and the medical officer (see 40.7 herein).

a. Pick up the large fragments with forceps and place them in a metal shipping container, NSN 8110-00-254-5715, or similar item for subsequent disposal. If forceps are not available, use rubber or cotton gloves.

WARNING

Do not stir up dust as this may cause the radioactive materials to permeate the atmosphere and produce an airborne hazard.

b. Carefully clean the area with an industrial type vacuum cleaner equipped with an exhaust filter. The exhaust stream will be analyzed by the medical service to insure that an airborne hazard is not being produced. Upon completion, the exhaust filter and collecting bag will be placed in the container with the fragments for subsequent monitoring.

c. Using a cloth dampened with water, wipe across the contaminated area making one swipe at a time. (Do not rub the radioactive material into the surface by the use of pressure or a back-and-forth motion.) The cloth will be folded in half after each swipe. Upon completion or when it becomes too small, the cloth will be placed in the waste container with the fragments.

d. Monitor the area with a AN/PDR-27 or equivalent to determine the effectiveness of the decontamination and to insure that no contamination has been spread beyond the breakage area. If levels or radiation are below 1.0 mR/hr, the area will be considered clean and returned to normal use. If the levels or radiation are above 1.0 mR/hr, repeat the above procedures as outlined until the radiation level present is acceptable.



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e. All benches and floors in the decontamination area will be monitored, and decontaminated if necessary. Rags and mops soaked in cleaning solvent will generally do a thorough job of cleaning benches and floors. The rags and mops may be rinsed in cleaning solvent to remove any contamination.

f. Personnel performing the decontamination, and their work clothing will be monitored to determine if they have become contaminated (see 70.3.1). If contamination above 1,000 counts per minute (CPM) or 1 mR/hr is found on the clothing, the clothing will be removed and all exposed skin surfaces and underclothes will be surveyed. If underclothes are contaminated above permissible levels, they will be removed and the person will be required to take a shower. After showering, the person will be surveyed for contamination. Areas where radiation remains will be washed until clean. The contaminated clothing will be placed in a separate container for subsequent decontamination.

g. The forceps, gloves, or other equipment which have been used for decontamination will be monitored. If contaminated, the forceps and other equipment may be wiped with a damp cloth until the contaminants have been removed. If the gloves are contaminated, they will be placed in the same container with the contaminated clothing for subsequent decontamination or disposal.

h. The waste container will then be monitored for external contamination by use of wipe samples. If contaminated, a damp cloth will be used to wipe off the contaminates. The cloth will then be placed in the container and the container will be sealed and marked for subsequent disposal (see section 90).

80.5 Fire control. Fire in any area where radioactive commodities are handled or stored will be brought under control by using established fire fighting procedures, and if possible, under direct guidance of the fire protection activity.

80.5.1 Monitoring of fire-damaged area. All fire damaged areas will be monitored immediately after the fire is under control to determine the radiation levels present, and the necessity for decontamination. Firefighting operation may disperse radioactive materials into areas that initially were not a part of the incident site. The drainage of liquid from firefighting operations should be noted and avoided by personnel unless they are wearing appropriate protective clothing. Additional measures should be taken as necessary to cope with any resulting hazards.

80.5.2 Isolation of fire-damaged area. The area will be roped off and isolated for decontamination, if required. The ropes will be placed at positions where the level of radiation intensity is less than 1 milliroentgen per hour.

80.5.3 Monitoring of personnel and equipment. Personnel and equipment will be monitored and decontaminated, if necessary, utilizing the procedures specified in 80.4.

80.5.4 Additional monitoring and decontamination. Personnel and equipment will be returned to the fire station for additional monitoring and decontamination (showers, etc), if required.

90. DISPOSITION OF RADIOACTIVE MATERIALS

90.1 Radioactive waste. Radioactive waste will be disposed of as indicated in individual Nuclear Regulatory Commission licenses, authorizations, or instructional directives, and:

a. Defense supply Agency. DSAR 4145.30, Disposal of Unwanted Radioactive Material.



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b. Department of the Army. AR 755-15, Disposal of Unwanted Radioactive Material.

c. Department of the Navy. NAVMED P5055, Radiation Health Protection Manual.

d. Department of the Air Force. T. O. 00-110N-2, Radioactive Waste Disposal.

e. Marine Corps. MCO P4400.19, Marine Corps Supply Manual, Volume I.

90.2 Provisions for disposal. Waste disposal will be accomplished by the installations division in accordance with instructions from the medical service. Disposal by means of sewers will be avoided, unless specifically authorized by the medical service. In such cases, where sewer disposal is authorized, necessary laboratory analysis will be performed to determine that the level of activity discharged is below the prescribed limits of the Nuclear Regulatory Commission, and local and state authorities. Samples of waste water at various locations in the sewage system should be collected by the medical service, and the radioactivity determined to insure that compliance with standards has been met. Liquid waste may be allowed to decay to acceptable levels (10 microcuries per milliliter) and disposed of as normal waste. Alternate methods of disposal will be to: (1) dispose of the liquid after it has been diluted to the acceptable level; or (2) meter the liquid into normal waste disposal channels at a rate such that the acceptable levels are not exceeded. For additional information of disposal of solid or liquid waste, reference should be made to 10 CFR 20 and to local and state laws pertinent to the subject.

90.3 Metal container for solid-radioactive wastes. The following metal containers will be used to ship or store solid-radioactive wastes. Activities will choose a container of the appropriate size for the amount of solid radioactive waste they normally generate.

| NSN | Volume |
|---|---|
| 8110-00-254-5715 | 1 cu ft |
| 8110-00-254-5717 | 2 cu ft |
| 8110-00-254-5718 | 2.5 cu ft |
| 8110-00-285-6125 | 4 cu ft |
| 8110-00-281-8966 | 5.7 cu ft |
| 8110-00-281-8967 | 7 cu ft |

90.3.1 Warning labels for metal containers. Attach two radiation warning labels (see figure 1-4) on opposite sides of the metal container. Place the container in a convenient location, where it is readily accessible; however, keep it at least 6 feet away from work areas such as desks, work benches, etc.

90.3.2 Generated waste. Always place radioactive waste in the metal container as it is generated. Keep the container covered with its lid.

NOTE

Due to the high cost of radioactive waste disposal, nonradioactive waste will not be placed in these containers. However, if radioactive waste in the minimum size container must be disposed of, wadding may be used to fill the void (see 90.3.4).



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90.3.3 Periodic monitoring of waste containers. The waste containers will be monitored periodically to determine the intensity buildup of the radioactive waste contents. Use an AN/PDR-27 or equivalent. The radiation levels should not exceed 200 mR/hr at the outside surface of the container, or 10 mR/hr at 3 feet from the surface. If these levels are exceeded, see 90.3.4 for instructions. When monitoring the container, wipe-samples will be taken also to determine if there is any contamination of the outside surface. One-inch discs may be cut from filter paper or paper towels and these discs used for the purpose of taking wipe-samples. Exterior surfaces of containers should not be contaminated to levels greater than 0.1 mR/hr per 100 $cm^2$ removable beta-gamma contamination by wipe-test or 50 d/m per 100 $cm^2$ removable alpha contamination. If these limits are exceeded, wipe container outside surfaces with a rag dampened with solvent (such as P-S-661) and place the rag inside the container; repeat wipe-sampling procedure. The wipe-samples may be read with the AN/PDR-27, or equivalent, to determine beta-gamma contamination and PAC-1S or PAC-2GA radiac meter to determine the amount of alpha contamination. Monitoring of radioactive waste disposal containers will be accomplished by assigned radiological monitors, medical service personnel, or the radiological safety officer.

90.3.4 When to seal the container. If an intensity of 200 mR/hr on the outside surface of the container or 10 mR/hr at 3 feet from the surface is reached before the container is full, pack the void with wadding and seal the container. This will apply only to the minimum size container (1 cu ft). When using containers larger than the minimum size and the limits specified above are reached before the container is full, the contents will be transferred to a smaller container of appropriate size but no smaller than the 1 cubic foot container.

90.3.5 Sealing the container. When the metal shipping container is filled with radioactive waste, seal the container, insuring that the rubber gasket is in place to provide an airtight seal, then perform the following:

a. Monitor the outside surfaces as outlined in 90.3.3.

b. The previously attached radiation warning labels will be properly filled out with the required information as specified by the label.

90.3.6 Disposition of radioactive waste container. Forward the radioactive waste container to the transportation officer upon receiving instructions from authorized officials. This may be accomplished by notifying the installations engineer to pick up and deliver the container, or the container may be delivered by unit personnel, whichever is deemed appropriate.

90.3.7 Information to be forwarded with container. Forward the following information to the transportation officer along with the container:

Type of radioactive waste - This information will include the nomenclature, part number, quantity(s) (if applicable), radioisotope and total ratio activity of the waste in microcuries.

100. RECEIVING

100.1 Notification of user. Upon receipt of commodities containing radioactive material, the individual user or appropriate storage activity will be notified immediately, in order that preparations for storing and handling can be made. Procedures for pickup, receipt, opening, monitoring, recording, and reporting radioactive commodity shipment shall be in accordance with 10 CFR 80.



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100.2 Inspection and identification. The containers will be monitored by a radiological monitor or the medical service. The intensity of radiation on the external surfaces of the container will be checked and noted on the required forms which will be attached to the container in a conspicuous place. The detachable stub on the form will be filled in with the same data, detached and forwarded to the medical service. A visual inspection of the container will be made by the monitor to insure that the seals (when applicable) have not been broken or tampered with and that the container has not been damaged in transit.

100.3 Forwarding commodities. Properly packaged and tagged radioactive commodities will be forwarded immediately to the using activity or the appropriate storage area, whichever is applicable.

100.4 Damaged shipments. In the event shipments are received damaged with seals broken or tampered with, or with intensity readings above Nuclear Regulatory Commission limitations, the container will be isolated and the radiological protection officer and the medical service contacted to assist in determining subsequent action. Possible accidental exposures will be thoroughly investigated.

100.5 Radioactively contaminated items. When items found to be or known to be contaminated are delivered to receiving personnel, the containers will be monitored and the information on the attached labels will be verified or corrected. The intended recipient and authorized service personnel will be called to determine action to be taken (see 80.4).

110. STORAGE

110.1 Properly packaged, marked, and identified. Commodities properly packaged, marked and identified shall be stored in accordance with instructions contained in this handbook, and instructions from the medical officer or radiological protection officer (see 70.4). Maximum permissible exposure will be the same as in 30.3.

110.2 Storage precautions. Radioactive materials shall be stored or kept in such a manner as to insure that the dose rate therefrom shall not exceed the appropriate limits specified in this handbook, or other applicable directives. Radioactive materials stored in a nonradiation area shall be secured against unauthorized removal from the place of storage.

110.3 Normal storage provisions. Radioactive items can be kept in normal storage provided the intensity at 1 foot from the container does not exceed 1 milliroentgen per hour (unless otherwise directed by the installation or agency directives) and the containers are properly marked.

110.4 Storage requirements. Radioactive items exhibiting intensities greater than 1 milliroentgen per hour at 1 foot must be stored in marked areas, storerooms, or buildings. The areas, storerooms, or buildings must be controlled to prevent entry of unauthorized personnel. The exterior of each such area shall be marked with the appropriate warning symbols. The area must be remonitored every 30 days. The reading at 1 foot from the exterior of the storage area must not exceed 1 milliroentgen per hour. If this level is exceeded, the radioactive material will be rearranged to bring the radiation level within limits, or the size of the storage area increased.

WARNING

Fragile commodities will not be stored loose in boxes or containers, as this increases the possibility of breakage and subsequent contamination of storage facilities.



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110.5 Workroom requirements. Radioactive materials in a workroom or other location where persons are regularly or frequently present shall be enclosed in containers of such thickness, material, and construction, or otherwise shielded in such thickness, material, and construction, or otherwise shielded in such manner, that no person will be exposed to radiation in amounts greater than those indicated in section 30.3 of this appendix.

110.6 Ventilation of vaults or storage rooms. Vaults or rooms used for storing materials that may emit radioactive gases shall be suitably ventilated in such a manner that the gases do not constitute a radiation hazard.

110.7 Secondary tray or catchment. When there is any possibility that chemical, radiation, or other action might weaken or rupture the container of radioactive material sufficiently to cause leakage therefrom, the container shall be provided with a suitable secondary tray or catchment adequate to retain the entire amount of radioactive material.

110.8 Identification of material and user. Each container of radioactive material in storage shall, in addition to the standard radiation-hazard symbol, be labeled in such manner that the kind and quantity of material, the date of measurement, and the name of the person responsible for the material can be easily and quickly determined.

110.9 Material in excess of 1 curie. Storage containers for radioactive material in excess of 1 curie shall be designed to be resistant to fire and earthquake damage, and to maintain reasonable temperatures. Containers shall be structurally sound over the period of intended use with due regard to corrosion, radiation, and temperature effects that may develop.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

*DATE*

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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